

GSB518 Notes

Kevin Ross

Sep 11, 2025

Table of contents

Preface	8
1 Randomness and Probability	9
2 Working with Probabilities	15
2.1 Equally Likely Outcomes and Uniform Probability Measures	17
3 Conditional Probability	20
3.1 Bayes Rule	24
4 Independence of Events	29
5 Discrete Random Variables	34
5.1 Simulating from a Distribution	35
5.2 Probability Mass Functions	37
5.3 Matching Problem	38
6 Common Discrete Distributions	44
6.1 Poisson Distributions	44
6.2 Binomial Distributions	46
7 Expected Value	53
8 Variance and Standard Deviation	59
8.1 Standardization	63
9 Continuous Random Variables: Probability Density Functions	66
9.1 Uniform Distributions	66
9.2 Probability Density Functions	68
9.3 Exponential Distributions	71
9.4 Expected Values	74
9.5 Law of the Unconscious Statistician	77
10 Percentiles	79
10.1 Cumulative Distribution Functions	79
10.2 Quantile functions	82

11 Normal Distributions	85
12 Joint Distributions	94
12.1 Joint probability density functions	99
12.2 Independence of random variables	100
13 Covariance and Correlation	102
13.1 Covariance	102
13.2 Correlation	104
14 Conditional Distributions	107
14.1 Conditional Expected Value	113
15 Joint Normal Distributions	116
16 Expected Values of Linear Combinations of Random Variables	121
16.1 Linear rescaling	121
16.2 Linearity of expected value	121
16.3 Variance of linear combinations of random variables	122
17 From Probability to Statistics	126
18 Maximum Likelihood Estimation: Principles	132
19 Maximum Likelihood Estimation: Calculus	142
20 Bias of Estimators	145
21 Variance of Estimators	151
22 Mean Square Error of an Estimator	156
23 Central Limit Theorem	160
24 Confidence Intervals	164
25 Bootstrap Confidence Intervals	174
26 Null Hypothesis Testing	182
27 Some issues with hypothesis testing	192
28 Comparing multiple means: ANOVA F procedures	198
Homework 1	209

Homework 2	212
Problem 1	212
Problem 2	213
Problem 3	213
Problem 4	213
Problem 5	214
Problem 6	214
Problem 7	215
 Homework 3	 216
Problem 1	216
Problem 2	217
Problem 3	217
Problem 4	217
Problem 5	218
 Homework 4	 219
Problem 1	219
Problem 2	219
Problem 3	220
Problem 4	220
Problem 5	220
 Homework 5	 223
Problem 1	223
Problem 2	223
Problem 3	224
Problem 4	225
Problem 5	225
 Homework 6	 227
Problem 1	227
Problem 2	228
 Homework 7	 229
Problem 1	229
Problem 2	229
Problem 3	230
 Homework 8	 231
Problem 1	231
Problem 2	231
Problem 3	232
Problem 4	233

Homework 9	234
Problem 1	234
Problem 2	235
Problem 3	235
Problem 4	235
Problem 5	236
Problem 6	237
 Homework 1 Solutions	 238
Problem 1	238
Solution	238
Problem 2	238
Solution	239
Problem 3	239
Solution	240
Problem 4	241
Solution	242
Problem 5	242
Solution	243
Problem 6	243
Solution	243
 Homework 2 Solutions	 244
Problem 1	244
Solution	245
Problem 2	247
Solution	247
Problem 3	248
Solution	248
Problem 4	249
Solution	250
Problem 5	250
Solution	251
Problem 6	253
Solution	253
Problem 7	256
Solution	256
 Homework 3 Solutions	 258
Problem 1	258
Solution	259
Problem 2	260
Solution	260

Problem 3	261
Solution	262
Problem 4	263
Solution	264
Homework 4 Solutions	266
Problem 1	266
Solution	266
Problem 2	269
Solutions	269
Problem 3	271
Solution	272
Problem 4	272
Solution	272
Problem 5	272
Solution	274
Homework 5 solutions	282
Problem 1	282
Solution	282
Problem 2	284
Solution	285
Problem 3	286
Solution	286
Problem 4	287
Solution	287
Problem 5 part 1	291
Solution	291
Problem 5 part 2	295
Solution	295
Homework 6 solutions	299
Problem 1	299
Solutions	300
Problem 2	302
Solution	302
Homework 7 solutions	304
Problem 1	304
Solution	304
Problem 2	305
Solution	306

Problem 3	307
Solution	307
Simulation	308
For $\mu = 2.3$	308
For all μ	309
Bias function	310
Squared bias function	311
Variance function	311
MSE function	312
Different n	313
Homework 8 solutions	318
Problem 1	318
Solution	318
Problem 2	319
Solution	320
Problem 3	321
Solution	322
Problem 4	322
Solution	323
Homework 9	324
Problem 1	324
Solution	325
Problem 2	325
Solution	326
Problem 3	326
Solution	326
Problem 4	327
Solution	327
Problem 5	328
Solution	329
Problem 6	330
Solution	330
Summary of Some Common Distributions	332
References	333

Preface

This is a selection of Handouts for Cal Poly GSB 518, Essential Statistics for Business Analytics, covering topics in Probability and Statistics.

1 Randomness and Probability

Probability comes up in a wide variety of situations. Consider just a few examples.

1. The probability you win the next [Powerball lottery](#) if you purchase a single ticket, 4-8-15-16-42, plus the Powerball number, 23.
2. The probability that a “randomly selected” Cal Poly student is a California resident.
3. The probability that the high temperature in San Luis Obispo next Friday is above 90 degrees F.
4. The probability that the San Francisco 49ers win the next Superbowl.
5. The probability that extraterrestrial life currently exists somewhere in the universe.
6. The probability that you ate an apple on April 17, 2009.

Example 1.1. How are the situations above similar, and how are they different? What is one feature that all of the situations have in common? Is the interpretation of “probability” the same in all situations? The goal here is to just think about these questions, and not to compute any probabilities (or to even think about how you would).

- A phenomenon is **random** if there are multiple potential outcomes, and there is **uncertainty** about which outcome will occur.
- Uncertainty is understood in broad terms, and in particular does not only concern future occurrences.
- Many phenomena involve physical randomness, like flipping a coin or drawing powerballs at random from a bin, or in statistical applications of random sampling or random assignment.
- But in many other situations, randomness just vaguely reflects uncertainty.
- Random does *not* mean haphazard. In a random phenomenon, while individual outcomes are uncertain, there is a *regular distribution of outcomes over a large number of (hypothetical) repetitions*.
- Also, random does *not* necessarily mean equally likely. In a random phenomenon, certain outcomes or events might be more or less likely than others.

- The **probability** of an event associated with a random phenomenon is a number in the interval $[0, 1]$ measuring the event's likelihood or degree of uncertainty. A probability can take any value in the continuous scale from 0% to 100%.
- There are two main interpretations of probability.
 - **Long run relative frequency.** The probability of an event can be interpreted as the proportion of times that the event would occur in a very large number of hypothetical repetitions of the random phenomenon.
 - **Subjective probability.** There are many situations where the outcome is uncertain, but it does not make sense to consider the situation as repeatable. In such situations, a subjective (a.k.a., personal) probability describes the degree of likelihood a given individual ascribes to a certain event. Think of subjective probabilities as measuring relative degrees of likelihood rather than long run relative frequencies.
- Fortunately, the mathematics of probability work the same way regardless of the interpretation. In either case, the same basic logical consistency requirements must be satisfied.
- A **simulation** involves an artificial recreation of the random phenomenon, usually using a computer. The probability of an event can be approximated by simulating the random phenomenon a large number of times and determining the proportion of simulated repetitions on which the event occurred out of the total number of repetitions in the simulation.

Example 1.2. One of the oldest documented problems in probability is the following: If three fair six-sided dice are rolled, what is more likely: a sum of 9 or a sum of 10?

1. Explain how you could conduct a *simulation* to investigate this question.
2. Use the simulation results to approximate the probability that the sum is 9; repeat for a sum of 10.
3. It can be shown that the theoretical probability that the sum is 9 is $25/216 = 0.116$. Write a clearly worded sentence interpreting this probability as a long run relative frequency.

4. It can be shown that the theoretical probability that the sum is 10 is $27/216 = 0.125$. How many times more likely is a sum of 10 than a sum of 9?

- A probability can be interpreted as a theoretical long run relative frequency.
- A probability can be approximated by a relative frequency from a large number of simulated repetitions, but there is some simulation margin of error, due to natural variability in the simulation.
- The margin of error when approximating a single probability based on a simulated relative frequency is roughly on the order $1/\sqrt{N}$, where N is the *number of independently simulated values used to calculate the relative frequency*. For example, if $N = 10000$ then the margin of error is roughly $1/\sqrt{10000} = 0.01$. (But be careful when approximating *conditional* probabilities.)

Example 1.3. As of Jun 13, [FanGraphs](#) listed the following probabilities for who will win the 2025 MLB World Series.

Team	Probability
Dodgers	21%
Yankees	17%
Tigers	10%
Mets	10%
Phillies	7%
Other	

According to FanGraphs (as of Jun 13):

1. Are the above percentages relative frequencies or subjective probabilities? Why?
2. What must be the probability that a team other than the above five teams wins the championship? That is, what value goes in the “Other” row in the table?

3. The Dodgers are how many times more likely than the Phillies to win?

4. What must be the probability that the Dodgers do *not* win the championship? How many times more likely are the Dodgers to not win than to win (this ratio is the “odds against” the Dodgers winning).

5. How could you construct a circular spinner (like from a kids game) to simulate the World Series champion according to these probabilities? According to this model, what would you expect the results of 10000 repetitions of a simulation of the champion to look like?

- We will use the two interpretations — long run relative frequencies and subjective probabilities — interchangeably.
- With subjective probabilities it is often helpful to consider what might happen in a simulation.
- It is also useful to consider long run relative frequencies in terms of relative degrees of likelihood.
- Fortunately, the mathematics of probability work the same way regardless of the interpretation.
- A probability takes a value in the sliding scale from 0 to 100%.
- Don’t just focus on computation; always remember to properly interpret probabilities.

Example 1.4. In each of the following parts, which of the two probabilities, a or b, is larger, or are they equal? You should answer conceptually without attempting any calculations. Explain your reasoning.

1. Consider a Cal Poly student who frequently has blurry, bloodshot eyes, generally exhibits

slow reaction time, always seems to have the munchies, and disappears at 4:20 each day. Which of the following events, A or B , has a *higher* probability? (Assume the two probabilities are not equal.)

- a. The student has a GPA above 3.0.
 - b. The student has a GPA above 3.0 and smokes marijuana regularly.
2. Randomly select a man.
 - b. The probability that a randomly selected man is greater than six feet tall.
 - c. The probability that a randomly selected man who plays in the NBA is greater than six feet tall.
3. Randomly select a man.
 - a. The probability that a randomly selected man who is greater than six feet tall plays in the NBA.
 - b. The probability that a randomly selected man who plays in the NBA is greater than six feet tall.
4. Flip a coin *which is known to be fair* 10 times.
 - a. The probability that the results are, in order, HHHHHHHHHH.
 - b. The probability that the results are, in order, HHTHTTTTHHT.
5. Flip a coin which is known to be fair 10 times.
 - a. The probability that all 10 flips land on H.
 - b. The probability that exactly 5 flips land on H.
6. In the [Powerball lottery](#) there are roughly 300 million possible winning number combinations, all equally likely.
 - a. The probability you win the next Powerball lottery if you purchase a single ticket, 4-8-15-16-42, plus the Powerball number, 23
 - b. The probability you win the next Powerball lottery if you purchase a single ticket, 1-2-3-4-5, plus the Powerball number, 6.
7. Continuing with the Powerball
 - a. The probability that the numbers in the winning number are not in sequence (e.g., 4-8-15-16-42-23)
 - b. The probability that the numbers in the winning number are in sequence (e.g., 1-2-3-4-5-6)
8. Continuing with the Powerball

- a. The probability that you win the next Powerball lottery if you purchase a single ticket.
 - b. The probability that someone wins the next Powerball lottery. (FYI: especially when the jackpot is large, there are hundreds of millions of tickets sold.)
- Warning! Your psychological judgment of probabilities is often inconsistent with the mathematical logic of probabilities.
- When interpreting probabilities, consider the conditions under which the probabilities were computed, in the proper direction
- When interpreting probabilities, be careful not to confuse “the particular” with “the general”.
 - **“The particular:”** A very specific event, surprising or not, often has low probability.
 - **“The general:”** While a very specific event often has low probability, if there are many like events their combined probability can be high.
- Even if an event has extremely small probability, given enough repetitions of the random phenomenon, the probability that the event occurs on *at least one* of the repetitions is often high.

2 Working with Probabilities

- Due to the wide variety of types of random phenomena, an **outcome** can be virtually anything. In particular, an outcome does *not* have to be a number.
- The **sample space** is the set of all possible outcomes of the random phenomenon..
- An event is something that could happen. That is, an *event* is a collection of outcomes that satisfy some criteria. If the sample space outcomes are represented by rows in a spreadsheet, then an event is a subset of rows that satisfies some criteria
- Events are typically denoted with capital letters near the start of the alphabet, with or without subscripts (e.g. A , B , C , A_1 , A_2). Events can be composed from others using [basic set operations](#) like unions ($A \cup B$), intersections ($A \cap B$), and complements (A^c).
 - Read A^c as “not A ”.
 - Read $A \cap B$ as “ A and B ”
 - Read $A \cup B$ as “ A or B ”. Note that unions ($\cup$, “or”) are always inclusive. $A \cup B$ occurs if A occurs but B does not, B occurs but A does not, or both A and B occur.
- A collection of events $A_1, A_2, \dots$ are **disjoint** (a.k.a. mutually exclusive) if $A_i \cap A_j = \emptyset$ for all $i \neq j$. That is, multiple events are disjoint if none of the events have any outcomes in common.
- A **probability measure**, typically denoted P , assigns probabilities to *events* to quantify their relative likelihoods according to the assumptions of the model of the random phenomenon.
- The probability of event A , computed according to probability measure $P(A)$, is denoted $P(A)$.
- A valid probability measure P must satisfy the following three logical consistency “axioms”.
 - For any event A , $0 \leq P(A) \leq 1$.
 - If Ω represents the sample space then $P(\Omega) = 1$.
 - (*Countable additivity.*) If $A_1, A_2, A_3, \dots$ are disjoint then

$$P(A_1 \cup A_2 \cup A_3 \cup \dots) = P(A_1) + P(A_2) + P(A_3) + \dots$$

- Additional properties of a probability measure follow from the axioms
 - *Complement rule.* For any event A , $P(A^c) = 1 - P(A)$.
 - *Subset rule.* If $A \subseteq B$ then $P(A) \leq P(B)$.

- *Addition rule for two events.* If A and B are any two events

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- *Law of total probability.* If $C_1, C_2, C_3 \dots$ are disjoint events with $C_1 \cup C_2 \cup C_3 \cup \dots = \Omega$, then

$$P(A) = P(A \cap C_1) + P(A \cap C_2) + P(A \cap C_3) + \dots$$

- The axioms of a probability measure are minimal logical consistent requirements that ensure that probabilities of different events fit together in a valid, coherent way.
- A single probability measure corresponds to a particular set of assumptions about the random phenomenon.

Example 2.1. The probability that a randomly selected U.S. household has a pet dog is 0.47. The probability that a randomly selected U.S. household has a pet cat is 0.25. (These values are based on the 2018 [General Social Survey \(GSS\)](#).)

1. Represent the information provided using proper symbols.
2. Donny Don't says: "the probability that a randomly selected U.S. household has a pet dog OR a pet cat is $0.47 + 0.25 = 0.72$." Do you agree? What must be true for Donny to be correct? Explain. (Hint: for the remaining parts it helps to consider two-way tables.)
3. What is the *smallest possible* value of the probability that a randomly selected U.S. household has a pet dog AND a pet cat? Describe the (unrealistic) situation in which this extreme case would occur.

4. What is the *largest possible* value of the probability that a randomly selected U.S. household has a pet dog AND a pet cat? Describe the (unrealistic) situation in which this extreme case would occur. What would be the probability that a randomly selected U.S. household has a pet dog OR a pet cat in this scenario?

5. Donny Don't says: "I remember hearing once that in probability OR means add and AND means multiply. So the probability that a randomly selected U.S. household has a pet dog AND a pet cat is $0.47 \times 0.25 = 0.1175$." Do you agree? Explain.

6. According to the GSS, the probability that a randomly selected U.S. household has a pet dog AND a pet cat is 0.15. Compute the probability that a randomly selected U.S. household has a pet dog OR a pet cat.

7. Compute and interpret $P(C \cap D^c)$.

2.1 Equally Likely Outcomes and Uniform Probability Measures

- For a sample space Ω with finitely many possible outcomes, assuming **equally likely outcomes** corresponds to a probability measure P which satisfies

$$P(A) = \frac{|A|}{|\Omega|} = \frac{\text{number of outcomes in } A}{\text{number of outcomes in } \Omega} \quad \text{when outcomes are equally likely}$$

Example 2.2. Roll a fair four-sided die twice, and record the result of each roll in sequence.

1. How many possible outcomes are there? Are they equally likely?
 2. Compute $P(A)$, where A is the event that the sum of the two dice is 4.
 3. Compute $P(B)$, where B is the event that the sum of the two dice is at most 3.
 4. Compute $P(C)$, where C the event that the larger of the two rolls (or the common roll if a tie) is 3.
 5. Compute and interpret $P(A \cap C)$.
- The continuous analog of equally likely outcomes is a **uniform probability measure**. When the sample space is uncountable, size is measured continuously (length, area, volume) rather than discretely (counting).

$$P(A) = \frac{|A|}{|\Omega|} = \frac{\text{size of } A}{\text{size of } \Omega} \quad \text{if } P \text{ is a uniform probability measure}$$

Example 2.3. Regina and Cady are meeting for lunch. Suppose they each arrive uniformly at random at a time between noon and 1:00, independently of each other. Record their arrival times as minutes after noon, so noon corresponds to 0 and 1:00 to 60.

Table 2.1: Table representing the sample space of two rolls of a four-sided die. The outcomes in orange comprise the event A , the sum is equal to 4.

First roll	Second roll	Sum is 4?
1	1	no
1	2	no
1	3	yes
1	4	no
2	1	no
2	2	yes
2	3	no
2	4	no
3	1	yes
3	2	no
3	3	no
3	4	no
4	1	no
4	2	no
4	3	no
4	4	no

1. Draw a picture representing the sample space.
2. Compute the probability that the first person to arrive has to wait at most 15 minutes for the other person to arrive. In other words, compute the probability that they arrive within 15 minutes of each other.
3. Compute the probability that the first person to arrive arrives before 12:15.

3 Conditional Probability

Example 3.1. Continuing Example 2.1. The probability that a randomly selected U.S. household has a pet dog is 0.47. The probability that a randomly selected U.S. household has a pet cat is 0.25. the probability that a randomly selected U.S. household has a pet dog and a pet cat is 0.15.

1. Compute the probability that a randomly selected U.S. household that has a pet dog also has a pet cat. Is it 0.25?
2. Compute the probability that a randomly selected U.S. household that does not have a pet dog has a pet cat. Is it the same as the previous part? Is it one minus the previous part?
3. Describe in words the probability that results from subtracting the answer to the first part from 1.

- The **conditional probability of event A** given event B , denoted $P(A|B)$, is defined as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- The conditional probability $P(A|B)$ represents how the likelihood or degree of uncertainty of event A should be updated to reflect information that event B has occurred.

- Conditional probabilities are probabilities. There are analagous conditional versions of the properties of probability, e.g. $P(A^c|B) = 1 - P(A|B)$.
- In general, knowing whether or not event B occurs influences the probability of event A . That is,

$$\text{In general, } P(A|B) \neq P(A)$$

- Be careful: order is essential in conditioning. That is,

$$\text{In general, } P(A|B) \neq P(B|A)$$

- Within the context of two events, we have joint, conditional, and marginal probabilities.
 - Joint: unconditional probability involving both events, $P(A \cap B)$.
 - Conditional: conditional probability of one event given the other, $P(A|B)$, $P(B|A)$.
 - Marginal: unconditional probability of a single event $P(A)$, $P(B)$.
- The relationship $P(A|B) = P(A \cap B)/P(B)$ can be stated generically as

$$\text{conditional} = \frac{\text{joint}}{\text{marginal}}$$

- When dealing with probabilities (or proportions or percentages) be sure to ask “probability of *what?*” Thinking in fraction terms, be careful to identify the correct reference group which corresponds to the denominator.

Example 3.2. Are Americans in favor of free tuition at public colleges and universities? According to a [study conducted by the Pew Research Center in January 2020](#)

- 83% of Democrats are in favor of free tuition
- 60% of Independents are in favor of free tuition
- 39% of Republicans are in favor of free tuition

Also suppose that¹

- 32% of Americans are Democrats
- 42% of Americans are Independents
- 26% of Americans are Republicans

We’ll use this information to investigate the following questions, as well as a few others.

- What is the probability that a randomly selected American is in favor of free college tuition?
- What is the probability that a random selected American who is in favor of free college tuition is a Democrat?

¹These values are based on surveys by [Gallup](#), but the values change somewhat over time.

1. Represent the given information using proper notation.
2. Construct a hypothetical two-way table of counts representing this scenario.
3. What is the probability that an American who is in favor of free college tuition is a Democrat? (For this and the remaining parts, in addition to computing, represent the probability with proper notation.)
4. What is the probability that an American who is a Democrat is in favor of free college tuition?
5. What is the probability that an Americans is a Democrat and in favor of free college tuition?

6. What is the probability that an American is in favor of free college tuition?

- Rearranging the definition of conditional probability we get the **Multiplication rule**: the probability that two events both occur is

$$\begin{aligned}P(A \cap B) &= P(A|B)P(B) \\ &= P(B|A)P(A)\end{aligned}$$

- The multiplication rule says that you should think “multiply” when you see “and”. However, be careful about *what* you are multiplying: to find a joint probability you need an unconditional and an appropriate conditional probability.
- You can condition either on A or on B , provided you have the appropriate marginal probability; often, conditioning one way is easier than the other.
- Be careful: the multiplication rule does *not* say that $P(A \cap B)$ is the same as $P(A)P(B)$.

Example 3.3. Continuing Example 3.2. Explain in detail how you could conduct a simulation, using only the information provided in the set up, and how you would use the results to approximate the probabilities we computed in Example 3.2.

- Conditional probabilities can be approximated by filtering (or subsetting) based on the condition
- To approximate $P(A|B)$, simulate the random phenomenon for a set number of repetitions (say 10000), *discard those repetitions on which B does not occur*, and compute the relative frequency of A among the remaining repetitions (on which B does occur).
- Be careful! The margin of error is based on only the number of repetitions used to compute the relative frequency. So if you perform 10000 repetitions but B occurs only on 2000, then the margin of error for estimate $P(A|B)$ is roughly on the order of $1/\sqrt{2000} = 0.022$ (rather than $1/\sqrt{10000} = 0.01$). Especially if $P(B)$ is small, the margin of error could be large resulting in an imprecise estimate of $P(A|B)$. (Advantage: not computationally intensive.)

3.1 Bayes Rule

- *Bayes' rule* describes how to update uncertainty in light of new information, evidence, or data.

Example 3.4. A recent survey of American adults asked: “Based on what you have heard or read, which of the following two statements best describes the scientific method?”

- 70% selected “The scientific method produces findings meant to be continually tested and updated over time”. (We’ll call this the “iterative” opinion.)
- 14% selected “The scientific method identifies unchanging core principles and truths”. (We’ll call this the “unchanging” opinion).
- 16% were not sure which of the two statements was best.

How does the response to this question change based on education level? Suppose education level is classified as: high school or less (HS), some college but no Bachelor’s degree (college), Bachelor’s degree (Bachelor’s), or postgraduate degree (postgraduate). The education breakdown is

- Among those who agree with “iterative”: 31.3% HS, 27.6% college, 22.9% Bachelor’s, and 18.2% postgraduate.
- Among those who agree with “unchanging”: 38.6% HS, 31.4% college, 19.7% Bachelor’s, and 10.3% postgraduate.
- Among those “not sure”: 57.3% HS, 27.2% college, 9.7% Bachelor’s, and 5.8% postgraduate

1. Use the information to construct an appropriate two-way table.
2. Overall, what percentage of adults have a postgraduate degree? How is this related to the values 18.2%, 10.3%, and 5.8%?
3. What percent of those with a postgraduate degree agree that the scientific method is “iterative”? How is this related to the values provided?

- **Bayes' rule for events** specifies how a prior probability $P(H)$ of event H is updated in response to the evidence E to obtain the posterior probability $P(H|E)$.

$$P(H|E) = \frac{P(E|H)P(H)}{P(E)}$$

- Event H represents a particular hypothesis (or model or case)
- Event E represents observed evidence (or data or information)
- $P(H)$ is the unconditional or **prior probability** of H (prior to observing evidence E)
- $P(H|E)$ is the conditional or **posterior probability** of H after observing evidence E .
- $P(E|H)$ is the **likelihood** of evidence E given hypothesis (or model or case) H

Example 3.5. Continuing the previous example. Randomly select an American adult.

1. Consider the conditional probability that a randomly selected American adult agrees that the scientific method is “iterative” given that they have a postgraduate degree. Identify the prior probability, hypothesis, evidence, likelihood, and posterior probability, and use Bayes' rule to compute the posterior probability.
2. Find the conditional probability that a randomly selected American adult with a postgraduate degree agrees that the scientific method is “unchanging”.
3. Find the conditional probability that a randomly selected American adult with a postgraduate degree is not sure about which statement is best.
4. How many times more likely is it for an *American adult* to have a postgraduate degree and agree with the “iterative” statement than to have a postgraduate degree and agree with the “unchanging” statement?

5. How many times more likely is it for an *American adult with a postgraduate degree* to agree with the “iterative” statement than to agree with the “unchanging” statement?
6. What do you notice about the answers to the two previous parts?
7. How many times more likely is it for an *American adult* to agree with the “iterative” statement than to agree with the “unchanging” statement?
8. How many times more likely is it for an American adult to have a postgraduate degree when the adult agrees with the iterative statement than when the adult agree with the unchanging statement?
9. How many times more likely is it for an *American adult with a postgraduate degree* to agree with the “iterative” statement than to agree with the “unchanging” statement?
10. How are the values in the three previous parts related?

- Bayes rule is often used when there are multiple hypotheses or cases. Suppose $H_1, \dots, H_k$ is a series of distinct hypotheses which together account for all possibilities, and E is any event (evidence).
- Combining Bayes' rule with the law of total probability,

$$\begin{aligned} P(H_j|E) &= \frac{P(E|H_j)P(H_j)}{P(E)} \\ &= \frac{P(E|H_j)P(H_j)}{\sum_{i=1}^k P(E|H_i)P(H_i)} \end{aligned}$$

$$P(H_j|E) \propto P(E|H_j)P(H_j)$$

- The symbol $\propto$ is read “is proportional to”. The relative *ratios* of the posterior probabilities of different hypotheses are determined by the product of the prior probabilities and the likelihoods, $P(E|H_j)P(H_j)$. The marginal probability of the evidence, $P(E)$, in the denominator simply normalizes the numerators to ensure that the updated probabilities sum to 1 over all the distinct hypotheses.
- **In short, Bayes' rule says**

$$\text{posterior} \propto \text{likelihood} \times \text{prior}$$

Example 3.6. Now suppose we want to compute the posterior probabilities for an American adult's perception of the scientific method given that the randomly selected American adult has some college but no Bachelor's degree (“college”).

1. Before computing, make an educated guess for the posterior probabilities. In particular, will the changes from prior to posterior be more or less extreme given the American has some college but no Bachelor's degree than when given the American has a postgraduate degree? Why?
2. Construct a Bayes table and compute the posterior probabilities. Compare to the posterior probabilities given postgraduate degree from the previous examples.

- Bayesian analysis is often an iterative process.
- Posterior probabilities are updated after observing some information or data. These probabilities can then be used as prior probabilities before observing new data.
- Posterior probabilities can be sequentially updated as new data becomes available, with the posterior probabilities after the previous stage serving as the prior probabilities for the next stage.
- The final posterior probabilities only depend upon the cumulative data. It doesn't matter if we sequentially update the posterior after each new piece of data or only once after all the data is available; the final posterior probabilities will be the same either way. Also, the final posterior probabilities are not impacted by the order in which the data are observed.

4 Independence of Events

Example 4.1. Consider the following hypothetical data.

	Democrat (D)	Not Democrat (D^c)	Total
Loves puppies (L)	180	270	450
Does not love puppies (L^c)	20	30	50
Total	200	300	500

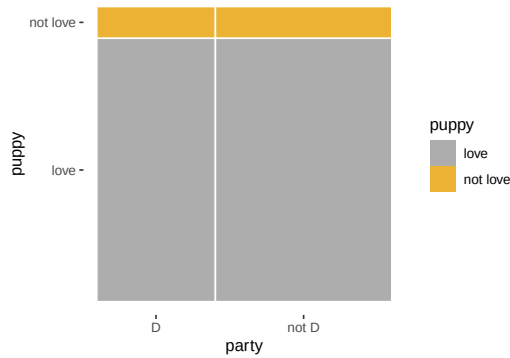
Suppose a person is randomly selected from this group. Consider the events

$$L = \{\text{person loves puppies}\}$$

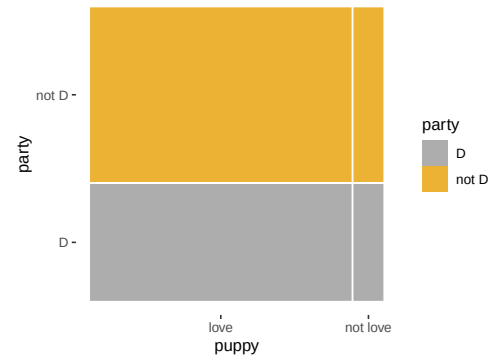
$$D = \{\text{person is a Democrat}\}$$

1. Compute and interpret $P(L)$.
2. Compute and interpret $P(L|D)$.
3. Compute and interpret $P(L|D^c)$.
4. What do you notice about $P(L)$, $P(L|D)$, and $P(L|D^c)$?

5. Compute and interpret $P(D)$.
6. Compute and interpret $P(D|L)$.
7. Compute and interpret $P(D|L^c)$.
8. What do you notice about $P(D)$, $P(D|L)$, and $P(D|L^c)$?
9. Compute and interpret $P(D \cap L)$.
10. What is the relationship between $P(D \cap L)$ and $P(D)$ and $P(L)$?
11. When randomly selecting a person from this particular group, would you say that events D and L are independent? Why?



(a) Mosaic plot of puppy love given party



(a) Mosaic plot of party given puppy love

- Events A and B are **independent** if the knowing whether or not one occurs does not change the probability of the other.
- For events A and B (with $0 < P(A) < 1$ and $0 < P(B) < 1$) the following are equivalent. That is, if one is true then they all are true; if one is false, then they all are false.

A and B are independent

$$P(A \cap B) = P(A)P(B)$$

$$P(A^c \cap B) = P(A^c)P(B)$$

$$P(A \cap B^c) = P(A)P(B^c)$$

$$P(A^c \cap B^c) = P(A^c)P(B^c)$$

$$P(A|B) = P(A)$$

$$P(A|B) = P(A|B^c)$$

$$P(B|A) = P(B)$$

$$P(B|A) = P(B|A^c)$$

Example 4.2. A certain system consists of four identical components. Suppose that the probability that any particular component fails is 0.1, and failures of the components occur independently of each other. Find the probability that the system fails if:

1. The components are connected in *parallel*: the system fails only if *all* of the components fail.

2. The components are connected in *series*: the system fails whenever *at least one* of the components fails.

3. Donny Don't says the answer to the previous part is $0.1 + 0.1 + 0.1 + 0.1 = 0.4$. Explain the error in Donny's reasoning.

- Events $A_1, A_2, A_3, \dots$ are **independent** if:
 - any pair of events $A_i, A_j, (i \neq j)$ satisfies $P(A_i \cap A_j) = P(A_i)P(A_j)$,
 - and any triple of events A_i, A_j, A_k (distinct i, j, k) satisfies $P(A_i \cap A_j \cap A_k) = P(A_i)P(A_j)P(A_k)$,
 - and any quadruple of events satisfies $P(A_i \cap A_j \cap A_k \cap A_m) = P(A_i)P(A_j)P(A_k)P(A_m)$,
 - and so on.
- Intuitively, a collection of events is independent if knowing whether or not any combination of the events in the collection occur does not change the probability of any other event in the collection.
- When events are independent, the multiplication rule simplifies greatly.

$$P(A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n) = P(A_1)P(A_2)P(A_3) \dots P(A_n) \quad \text{if } A_1, A_2, A_3, \dots, A_n \text{ are independent}$$

- When a problem involves independence, you will want to take advantage of it. Work with “and” events whenever possible in order to use the multiplication rule. For example, for problems involving “at least one” (an “or” event) take the complement to obtain “none” (an “and” event).

Example 4.3. In the Powerball lottery, a player picks five different whole numbers between 1 and 69, and another whole number between 1 and 26 that is called the Powerball. In the drawing, the 5 numbers are drawn without replacement from a “hopper” with balls labeled 1 through 69, but the Powerball is drawn from a separate hopper with balls labeled 1 through 26. The player wins the jackpot if both the first 5 numbers match those drawn, in any order, and the Powerball is a match. Under this set up, there are 292,201,338 possible winning numbers.

1. What is the probability the next winning number is 6-7-16-23-26, plus the Powerball number, 4.

2. What is the probability the next winning number is 1-2-3-4-5, plus the Powerball number, 6.

3. The Powerball drawing happens twice a week. Suppose you play the same Powerball number, twice a week, every week for over 50 years. Let's say you purchase a ticket for 6000 drawings in total. What is the probability that you win at least once?

4. Instead of playing for 50 years, you decide only to play one lottery, but you buy 6000 tickets, each with a different Powerball number. What is the probability that at least one of your tickets wins? How does this compare to the previous part? Why?

5. Each ticket costs 2 dollars, but the jackpot changes from drawing to drawing. Suppose you buy 6000 tickets for a single drawing. How large does the jackpot need to be for your "expected" profit to be positive? To be \$100,000? (We're ignoring inflation, taxes, transaction costs, and any changes in the rules.)

5 Discrete Random Variables

- Roughly, a *random variable* assigns a number measuring some quantity of interest to each outcome of a random phenomenon.
- Mathematically, a **random variable (RV)** X is a *function* that takes an outcome in the sample space as input and returns a real number as output
- The random variable itself is typically denoted with a capital letter (X); possible values of that random variable are denoted with lower case letters (x).
 - Think of the capital letter X as a label standing in for a formula like “the number of heads in 4 flips of a coin” and
 - x as a dummy variable standing in for a particular value like 3.
- **Discrete random variables** take at most countably many possible values (e.g., $0, 1, 2, \dots$). They are often counting variables (e.g., the number of Heads in 10 coin flips).
- **Continuous random variables** can take any real value in some interval (e.g., $[0, 1]$, $[0, \infty)$, $(-\infty, \infty)$). That is, continuous random variables can take uncountably many different values. Continuous random variables are often measurement variables (e.g., height, weight, income).
- A function of a random variable is also a random variable: if X is a RV then so is $g(X)$
- Sums and products, etc., of random variables *defined on the same sample space* are random variables. If X and Y are RVs defined on the same sample space then so are $X + Y$, $X - Y$, XY
- If the sample space outcomes are represented by rows in a spreadsheet, then random variables correspond to columns.
- Expressions like $X = x$ or $\{X = x\}$ represent *events*: for which outcomes is the value of the random variable X equal to the value x . (Remember, if the sample space outcomes are represented by rows in a spreadsheet, then an event corresponds to a subset of rows (outcomes) that satisfies some criteria.)
- The **(probability) distribution** of a collection of random variables identifies the possible values that the random variables can take and their relative likelihoods.
- We will see many ways of describing a distribution, depending on how many random variables are involved and their types (discrete or continuous).

Example 5.1. Roll a four-sided die twice, and record the result of each roll in sequence. Let X be the sum of the two dice, and let Y be the larger of the two rolls (or the common value if both rolls are the same). Table 5.1 represents the possible outcomes and the values of the random variables X and Y . Assume the die is fair and the rolls are independent.

Table 5.1: Table representing the sum (X) and larger (Y) of two rolls of a four-sided die

Outcome (First roll, second roll)	X (sum)	Y (max)
(1, 1)	2	1
(1, 2)	3	2
(1, 3)	4	3
(1, 4)	5	4
(2, 1)	3	2
(2, 2)	4	2
(2, 3)	5	3
(2, 4)	6	4
(3, 1)	4	3
(3, 2)	5	3
(3, 3)	6	3
(3, 4)	7	4
(4, 1)	5	4
(4, 2)	6	4
(4, 3)	7	4
(4, 4)	8	4

1. Identify the event $\{X = 4\}$ and compute its probability. Then interpret the probability both as a long relative frequency and as a relative likelihood.
2. Construct a table and plot of $P(X = x)$ for each possible value x of X .
3. Construct a table and plot of $P(Y = y)$ for each possible value y of Y .

5.1 Simulating from a Distribution

Example 5.2. Continuing Example 5.1.

1. Describe how you could simulate a single value of Y .

Table 5.2: [“The marginal distribution of X , the sum of two rolls of a fair four-sided die.”, “The marginal distribution of Y , the larger of two rolls of a fair four-sided die.”]

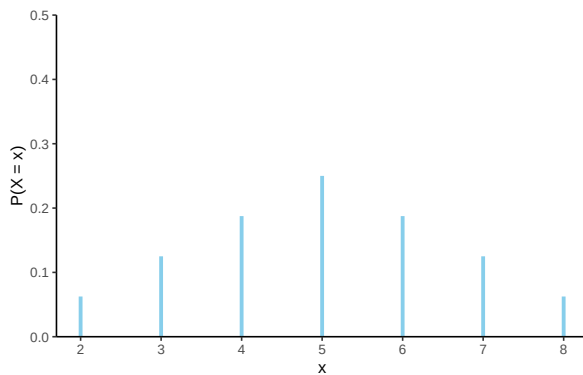


Figure 5.1: The marginal distribution of X , the sum of two rolls of a fair four-sided die.

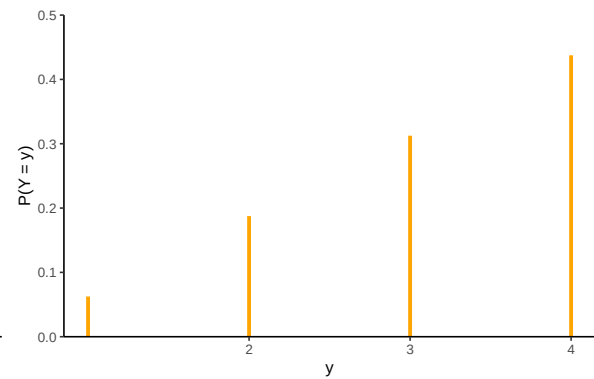


Figure 5.2: The marginal distribution of Y , the larger of two rolls of a fair four-sided die.

2. Construct a spinner (like from a kids game) to represent the distribution of Y .
 3. Describe another way to simulate a single value of Y .
 4. Describe how you could use simulation to approximate the distribution of Y .
- Don't confuse a random variable with its distribution!
 - A random variable measures a numerical quantity which depends on the outcome of a random phenomenon
 - The distribution of a random variable specifies the long run pattern of variation of values of the random variable over many repetitions of the underlying random

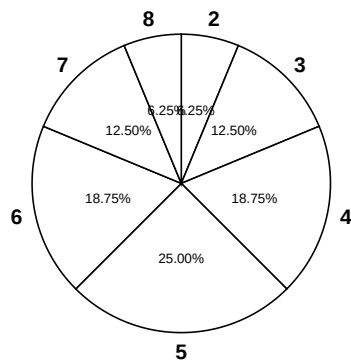


Figure 5.3: Spinner representing the marginal distribution of X , the sum of two rolls of a fair four-sided die.

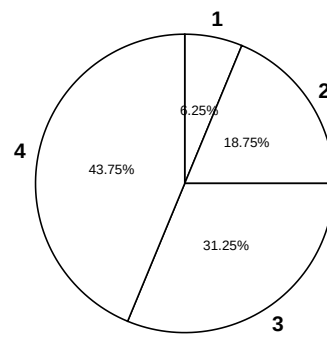


Figure 5.4: Spinner representing the marginal distribution of Y , the larger of two rolls of a fair four-sided die.

phenomenon.

- Any marginal distribution can be represented by a single spinner (like from a kids game).
- In principle, there are always two ways of simulating a value x of a random variable X .
 1. *Simulate from the probability space.* Simulate an outcome ω from the underlying probability space and set $x = X(\omega)$.
 2. *Simulate from the distribution.* Construct a spinner corresponding to the distribution of X and spin it once to generate x .
- The second method requires that the distribution of X is known. However, as we will see in many examples, *it is common to specify the distribution of a random variable directly without defining the underlying probability space.*

5.2 Probability Mass Functions

- The probability distribution of a single discrete random variable X is often displayed in a table containing the probability of the event $\{X = x\}$ for each possible value x .
- In some cases, a distribution has a “formulaic” shape. For a discrete random variable X , the **probability mass function (pmf)** p_X expresses $P(X = x)$ as a function of x : $p_X(x) = P(X = x)$.
- Be sure to specify the possible values of the random variable!
- Certain common distributions have special names and properties.

Example 5.3. Continuing Example 5.1.

1. Verify that the pmf of X is

$$p_X(x) = \frac{4 - |x - 5|}{16}, \quad x = 2, 3, \dots, 8$$

2. Verify that the pmf of Y is

$$p_Y(y) = \frac{2y - 1}{16}, \quad y = 1, 2, 3, 4$$

5.3 Matching Problem

The “matching problem” involves n distinct objects labeled $1, \dots, n$ which are placed in n distinct boxes labeled $1, \dots, n$, with exactly one object placed in each box. Suppose the objects are placed in the boxes uniformly at random, so that any possible arrangement is equally likely. Let X be the number of matches, that is, the number of objects for which their label matches the label of the box in which they are placed.

Example 5.4. Consider the matching problem with $n = 4$. There are $4! = 24$ possible outcomes, displayed in Table 5.4.

1. Complete Table 5.4 to identify the value of X for each outcome.
2. Identify the event $\{X = 0\}$ and compute its probability. Then interpret the probability both as a long relative frequency and as a relative likelihood.

Table 5.4: Table representing the sample space in the matching problem with $n = 4$.

Spot 1	Spot 2	Spot 3	Spot 4	X (number of matches)
1	2	3	4	
1	2	4	3	
1	3	2	4	
1	3	4	2	
1	4	2	3	
1	4	3	2	
2	1	3	4	
2	1	4	3	
2	3	1	4	
2	3	4	1	
2	4	1	3	
2	4	3	1	
3	1	2	4	
3	1	4	2	
3	2	1	4	
3	2	4	1	
3	4	1	2	
3	4	2	1	
4	1	2	3	
4	1	3	2	
4	2	1	3	
4	2	3	1	
4	3	1	2	
4	3	2	1	

Table 5.5: Theoretical distribution of X (number of matches) in the matching problem with $n = 4$.

x	$P(X=x)$
0	0.3750
1	0.3333
2	0.2500
4	0.0417

3. Construct a table, plot, and spinner representing the distribution of X .

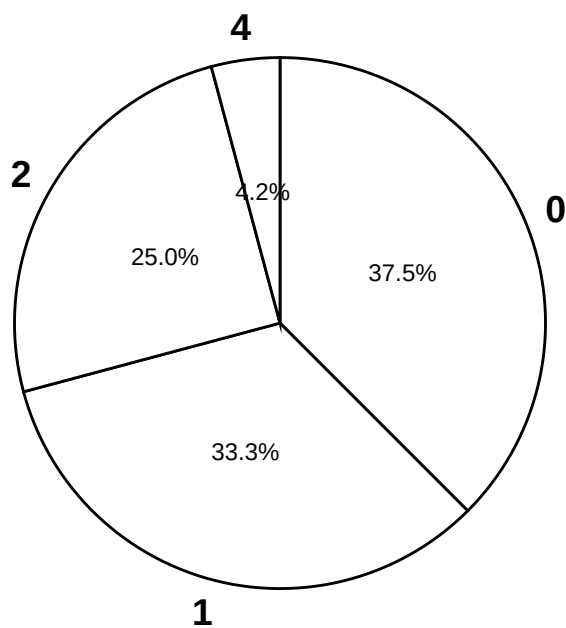


Figure 5.5: Spinner representing the theoretical distribution of X , the number of matches in the matching problem when $n = 4$.

Table 5.6: Approximate distribution of number of matches in the matching problem for general n

x	p_x
0	0.367879
1	0.367879
2	0.183940
3	0.061313
4	0.015328
5	0.003066
6	0.000511
7	0.000073

Example 5.5. Now consider the matching problem with general n . Finding the distribution of X by enumerating the $n!$ possible outcomes as we did for $n = 4$ is not feasible for general n . But we can approximate the distribution of X with simulation. Simulation shows that the distribution of the number of matches X is approximately the same for all n (unless n is very small, i.e., $n \leq 5$). In particular, for any n , the probability mass function of X is approximately

$$p_X(x) = \frac{e^{-1}}{x!}, \quad x = 0, 1, 2, \dots$$

1. Use this pmf to approximate the probability of at least one match, and compare to the simulation results for general n .
2. Interpret the probability both as a long relative frequency and as a relative likelihood.
3. Construct a table, plot, and spinner corresponding to the above pmf. Compare to the simulation results for general n .

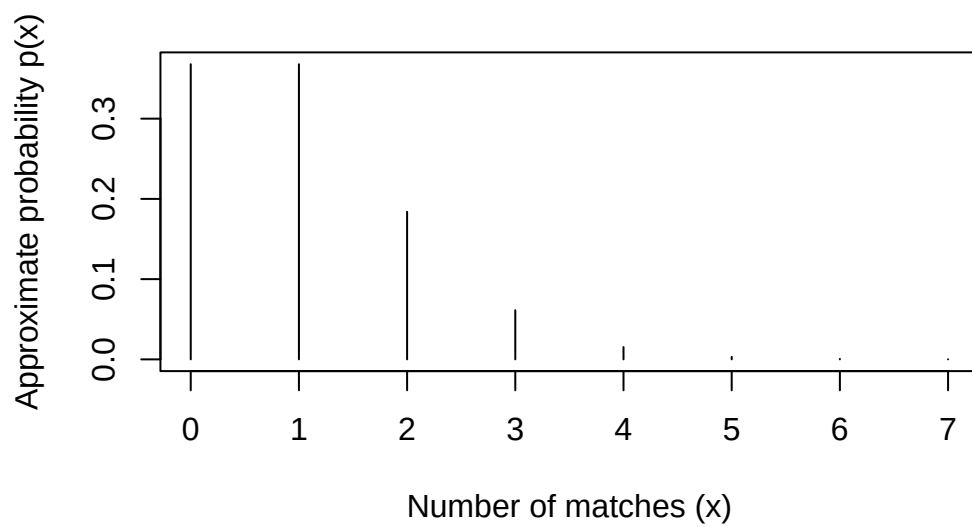


Figure 5.6: Approximate distribution of number of matches in the matching problem for general n

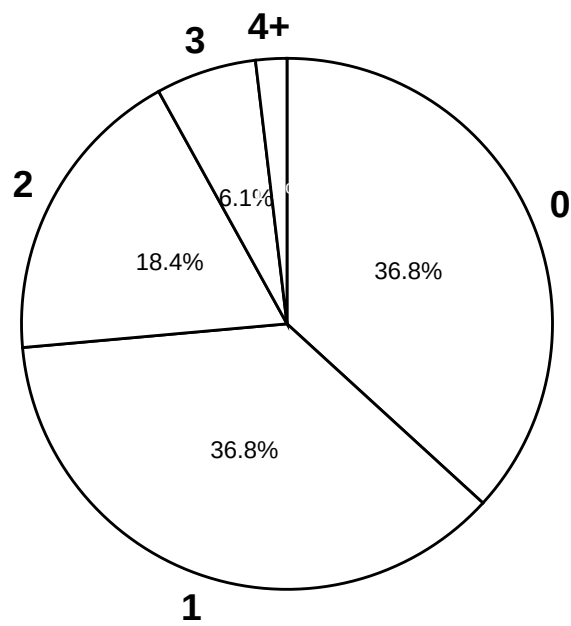


Figure 5.7: Spinner representing the approximate distribution of number of matches in the matching problem for general n

6 Common Discrete Distributions

6.1 Poisson Distributions

Example 6.1. Let X be the number of home runs hit (in total by both teams) in a randomly selected Major League Baseball game. Technically, there is no fixed upper bound on what X can be, so mathematically it is convenient to consider $0, 1, 2, \dots$ as the possible values of X . Assume that the pmf of X is

$$p_X(x) = e^{-2.3} \frac{2.3^x}{x!}, x = 0, 1, 2, \dots$$

This distribution is called the Poisson(2.3) distribution.

1. Compute $P(X = 3)$. Then interpret the value as a long run relative frequency, and a relative likelihood.
2. Construct a table, plot, and spinner corresponding to the distribution of X .
3. Compute and interpret $P(X \leq 2)$.
4. Compute and interpret $P(X \leq 13)$. (The most home runs ever hit in a baseball game is 13.)

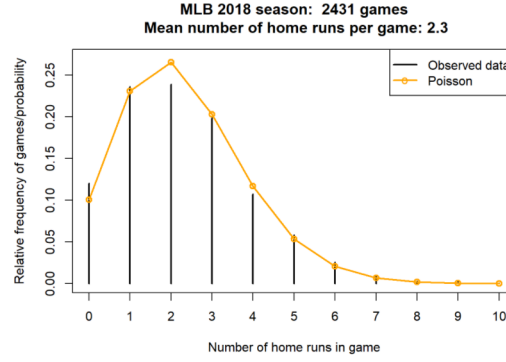


Figure 6.1: Relative frequencies of home runs per game in the 2018 MLB season, compared with the Poisson(2.3) distribution.

Table 6.1: Table representing the Poisson(2.3) probability mass function.

x	$p(x)$	Value
0	$e^{-2.3} \frac{2.3^0}{0!}$	0.100259
1	$e^{-2.3} \frac{2.3^1}{1!}$	0.230595
2	$e^{-2.3} \frac{2.3^2}{2!}$	0.265185
3	$e^{-2.3} \frac{2.3^3}{3!}$	0.203308
4	$e^{-2.3} \frac{2.3^4}{4!}$	0.116902
5	$e^{-2.3} \frac{2.3^5}{5!}$	0.053775
6	$e^{-2.3} \frac{2.3^6}{6!}$	0.020614
7	$e^{-2.3} \frac{2.3^7}{7!}$	0.006773
8	$e^{-2.3} \frac{2.3^8}{8!}$	0.001947
9	$e^{-2.3} \frac{2.3^9}{9!}$	0.000498
10	$e^{-2.3} \frac{2.3^{10}}{10!}$	0.000114
11	$e^{-2.3} \frac{2.3^{11}}{11!}$	0.000024
12	$e^{-2.3} \frac{2.3^{12}}{12!}$	0.000005
13	$e^{-2.3} \frac{2.3^{13}}{13!}$	0.000001
14	$e^{-2.3} \frac{2.3^{14}}{14!}$	0.000000

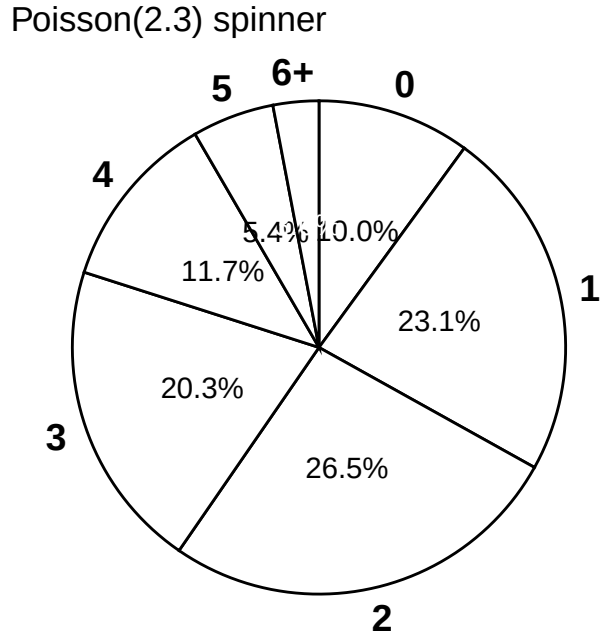


Figure 6.2: Spinner corresponding to the Poisson(2.3) distribution (values 6, 7, ... lumped into one 6+ category).

- A discrete random variable X has a **Poisson distribution** with parameter $\mu > 0$ if its probability mass function p_X satisfies

$$p_X(x) = \frac{e^{-\mu} \mu^x}{x!}, \quad x = 0, 1, 2, \dots$$

- The function $\mu^x/x!$ defines the shape of the pmf. The constant $e^{-\mu}$ ensures that the probabilities sum to 1.
- Poisson distributions are often used to model random variables which count the number of “relative rare” events that occur.
- If X has a Poisson(μ) distribution, $P(X = 0) = e^{-\mu}$ and the probabilities of possible values follow the

$$\text{Poisson}(\mu) \text{ pattern: } x \text{ is } \frac{\mu}{x} \text{ as likely as } x - 1 \quad (\text{for } x = 1, 2, 3, \dots)$$

6.2 Binomial Distributions

Example 6.2. Consider an extremely simplified model for the daily closing price of a certain stock. Every day the price either goes up or goes down, and the movements are independent

from day-to-day. Assume that the probability that the stock price goes up on any single day is 0.25. Let X be the number of days in which the price goes up in the next 5 days.

1. Compute and interpret $P(X = 0)$.

2. Compute the probability that the price goes up on the first day and then down on the following four days.

3. Why is $P(X = 1)$ different from the probability in the previous part? Compute and interpret $P(X = 1)$.

4. Suggest a general formula for the probability mass function of X .

- A discrete random variable X has a **Binomial distribution** with parameters n , a nonnegative integer, and $p \in [0, 1]$ if its probability mass function is

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n$$

- The *binomial coefficient* (read “ n choose x ”)

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

counts the number of success/failure sequences of length n in which there are exactly x successes. (Remember: by definition $0! = 1$.)

Example 6.3. Continuing Example 6.2.

1. Construct a table, plot, and spinner representing the distribution of X .

2. Compute and interpret $P(X \leq 2)$.

Table 6.2: Table representing the Binomial(5, 0.25) probability mass function.

x	$p(x)$	Value
0	$\binom{5}{0}0.25^0(1 - 0.25)^{5-0}$	0.237305
1	$\binom{5}{1}0.25^1(1 - 0.25)^{5-1}$	0.395508
2	$\binom{5}{2}0.25^2(1 - 0.25)^{5-2}$	0.263672
3	$\binom{5}{3}0.25^3(1 - 0.25)^{5-3}$	0.087891
4	$\binom{5}{4}0.25^4(1 - 0.25)^{5-4}$	0.014648
5	$\binom{5}{5}0.25^5(1 - 0.25)^{5-5}$	0.000977

Binomial(5, 0.25) spinner

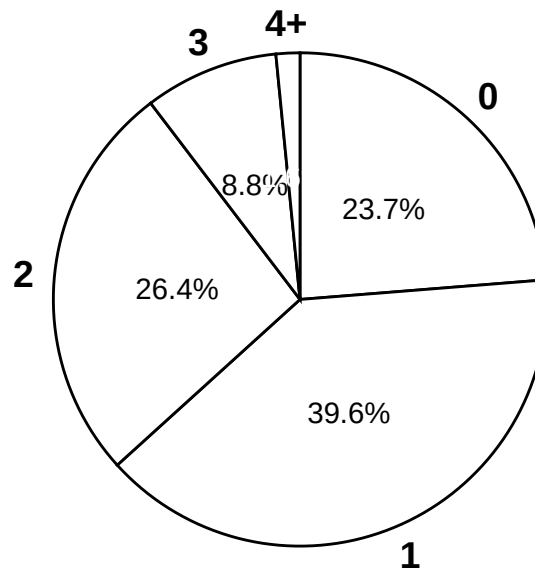


Figure 6.3: Spinner corresponding to the Binomial(5, 0.25) distribution (4 and 5 lumped into 4+).

Example 6.4. Continuing Example 6.2.

1. What does the random variable $5 - X$ represent? What is its distribution?
2. Suppose that the price is currently \$100 and each it either moves up \$2 or down \$2. Let S be the stock price after 5 days. How does S relate to X ? Does S have a Binomial distribution?
3. Recall that X is the number of days on which the price goes up in the next five days. Suppose that Y is the number of days on which the price goes up in the ten days after that (days 6-15). What is the distribution of $X + Y$? (Continue to assume independence between days, with probability 0.25 of an up movement on any day.)

- Imagine a box containing tickets
 - Each ticket is labeled either 1 (“success”) or 0 (“failure”)
 - p is the proportion of tickets in the box labeled 1 (“success”); the rest are labeled 0 (“failure”).
 - Randomly select n tickets from the box *with replacement* and let X be the number of tickets in the sample that are labeled 1.
 - Then X has a Binomial(n, p) distribution.
 - Since the tickets are labeled 1 and 0, the random variable X which counts the number of successes is equal to the sum of the 1/0 values on the tickets.
- The above situation involves a sequence of **Bernoulli**(p) trials.
 - There are only two possible outcomes, “success” (1) and “failure” (0), on each trial.
 - The unconditional/marginal probability of success is the same on every trial, and equal to p
 - The trials are independent
 - * If sampling with replacement, or
 - * If sampling without replacement but the population size (number of tickets in the box) is much larger than the sample size n
- If X counts the number of successes in a *fixed number*, n , of Bernoulli(p) trials then X has a Binomial(n, p) distribution.

Example 6.5. In each of the following situations determine whether or not X has a Binomial distribution. If so, specify n and p . If not, explain why not.

1. Roll a die 20 times; X is the number of times the die lands on an even number.
2. Roll a die 20 times; X is the number of times the die lands on 6.

3. Roll a die until it lands on 6; X is the total number of rolls.
4. Roll a die 20 times; X is the sum of the numbers rolled.
5. Shuffle a standard deck of 52 cards (13 hearts, 39 other cards) and deal 5 *without replacement*; X is the number of hearts dealt. (Hint: be careful about why.)
6. Roll a fair six-sided die 10 times and a fair four-sided die 10 times; X is the number of 3s rolled (out of 20).
7. Randomly select a sample of 35 Cal Poly students; X is the number of students in the sample who are CA residents.

Example 6.6. Donny Dont is thoroughly confused about the distinction between a random variable and its distribution. Help him understand by providing a simple concrete example of *two different random variables* X and Y that have the *same distribution*. Can you think of X and Y that have the same distribution but for which $P(X = Y) = 0$? (Hint: think coin flipping.)

7 Expected Value

- The distribution of a random variable specifies the possible values and the probability of any event that involves the random variable.
- The distribution of a random variable contains all the information about its long run behavior. It is also useful to summarize some key features of a distribution.
- One summary characteristic of a distribution is the **long run average value** of the random variable.
- We can approximate the long run average value by simulating many values of the random variable and computing the average (mean) in the usual way: sum the simulated values and divide by the number of simulated values.

Example 7.1. Recall the matching problem with $n = 4$. Table 7.1 displays the value of X , the number of matches, for 10 simulated repetitions.

1. How could you use the simulation results to approximate the long run average value of X ? How could you get a better approximation of the *long run* average?
2. Rather than adding the 10 values and dividing by 10, how could you simplify the calculation in the previous part?
3. Table 7.2 below summarizes 24000 simulated values of X . Approximate the long run average value of X .

Table 7.1: The value of X , the number of matches, for 10 simulated repetitions of the matching problem with $n = 4$.

Repetition	X (number of matches)
1	0
2	1
3	0
4	0
5	2
6	0
7	1
8	1
9	4
10	2

4. Recall the distribution of X from Example 5.4. What would be the corresponding mathematical formula for the theoretical long run average value of X ? This number is called the “expected value” of X .

5. Is the expected value the most likely value of X ?

6. Is the expected value of X the “value that we would expect” on a *single* repetition of the matching problem?

7. Explain in what sense the expected value is “expected”.

Table 7.2: Summary of 24000 simulated values of X , the number of matches, in the matching problem with $n = 4$.

Number of Matches (X)	Frequency
0	8973
1	7958
2	6034
4	1035
Total	24000

Table 7.3: Running average of 10 simulated values of X in the matching problem with $n = 4$.

Repetition	Value of X	Running average of X
1	0	0.000
2	1	0.500
3	0	0.333
4	0	0.250
5	2	0.600
6	0	0.500
7	1	0.571
8	1	0.625
9	4	1.000
10	2	1.100

- The **expected value** (a.k.a. *expectation* a.k.a. *mean*), of a random variable X is a number denoted $E(X)$ representing the probability-weighted average value of X .
- The expected value of a discrete random variable with pmf p_X is defined as

$$E(X) = \sum_x xp_X(x) = \sum \text{value} \times \text{probability}$$

- Note well that $E(X)$ represents *a single number*.
- The expected value is the “balance point” (center of gravity) of a distribution.
- The expected value of a random variable X is defined by the probability-weighted average according to the underlying probability measure. But the expected value can also be interpreted as the **long-run average value**, and so can be approximated via simulation.
- Read the symbol $E(\cdot)$ as
 - Simulate lots of values of what’s inside $(\cdot)$
 - Compute the average. This is a “usual” average; just sum all the simulated values and divide by the number of simulated values.

Example 7.2. Continuing Example 6.2. Consider an extremely simplified model for the daily closing price of a certain stock. Every day the price either goes up or goes down, and the

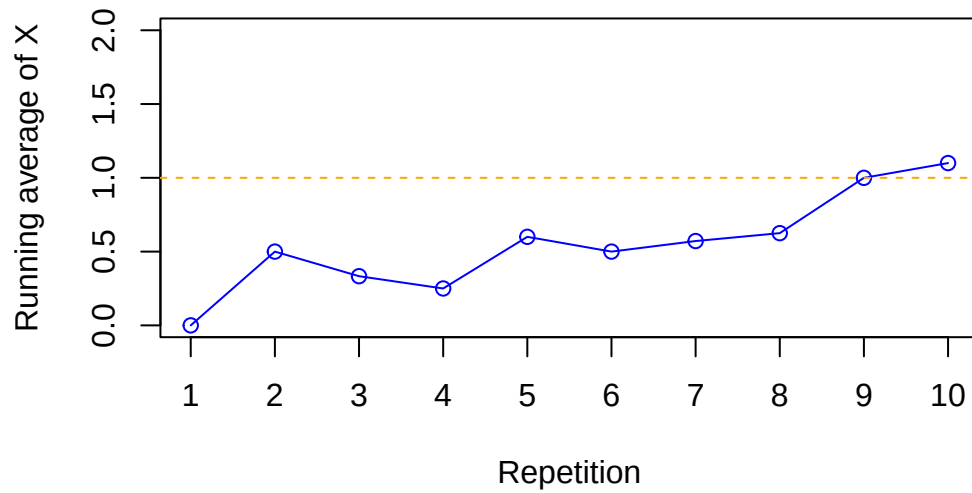


Figure 7.1: Running average of 10 simulated values of X in the matching problem with $n = 4$.

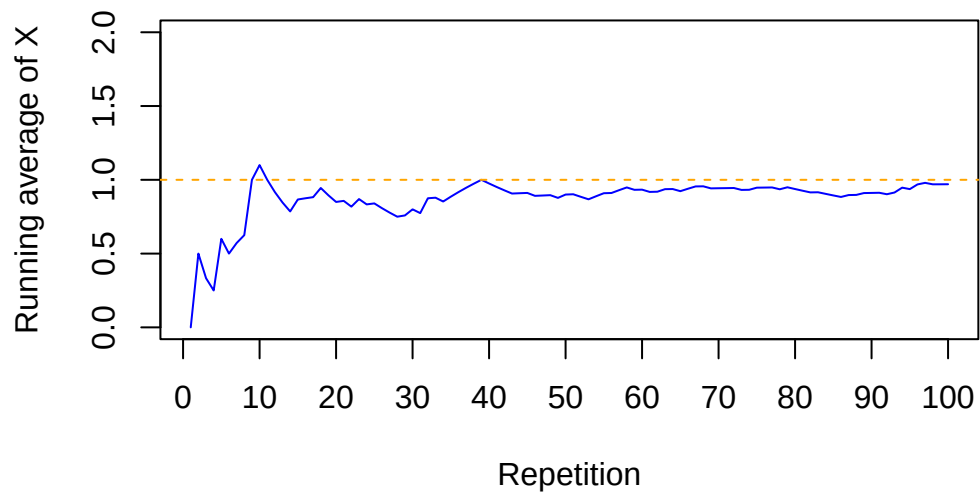


Figure 7.2: Running average of 100 simulated values of X in the matching problem with $n = 4$.

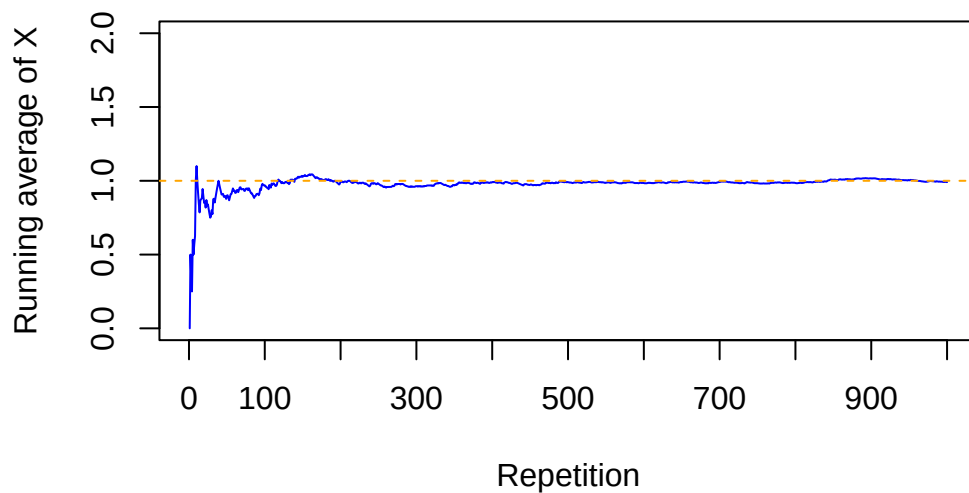


Figure 7.3: Running average of 1000 simulated values of X in the matching problem with $n = 4$.

movements are independent from day-to-day. Assume that the probability that the stock price goes up on any single day is 0.25. Let X be the number of days in which the price goes up in the next 5 days.

1. Suggest a shortcut formula for $E(X)$.

2. Compute $E(X)$ using the distribution of X from Example 6.2. Did the shortcut formula work?

3. Interpret $E(X)$.

- If X has a Binomial(n, p) distribution then $E(X) = np$.

Example 7.3. Continuing Example 6.1. Assume that X , the number of home runs hit (in total by both teams) in a randomly selected Major League Baseball game, follows a Poisson(2.3) distribution.

1. Compute $E(X)$.

2. Interpret $E(X)$ in context.

- If X has a Poisson(μ) distribution then $E(X) = \mu$.

8 Variance and Standard Deviation

- The long run average value is just one feature of a distribution.
- Random variables vary, and the distribution describes the entire pattern of variability.
- Roughly, standard deviation measures overall degree of variability in a single number, as the average distance from the mean.
- Technically, the **variance** is the long run average squared distance from the mean, and the **standard deviation** is the square root of the variance.

Example 8.1. A roulette wheel has 18 black spaces, 18 red spaces, and 2 green spaces, all the same size and each with a different number on it. Guillermo bets \$1 on black. If the wheel lands on black, Guillermo wins his bet back plus an additional \$1; otherwise he loses the money he bet. Let W be Guillermo's net winnings (net of the initial bet of \$1.)

1. Find the distribution of W .
2. Compute $E(W)$.
3. Interpret $E(W)$ in context.
4. An expected profit *for the casino* of 5 cents per \$1 bet seems small. Explain how casinos can turn such a small profit into billions of dollars.

5. The random variable $(W - E(W))^2$ represents the squared deviation from the mean. Find the distribution of this random variable and its expected value.

6. Standard deviation is the square root of the variance. Why would we want to take the square root of the variance? Compute and interpret the standard deviation of W .

7. Compute $E(W^2)$. (For this W you should be able to compute $E(W^2)$ without any calculations; why?) Then compute $E(W^2) - (E(W))^2$; what do you notice?

- The **variance** of a random variable X is

$$\begin{aligned}\text{Var}(X) &= E\left((X - E(X))^2\right) \\ &= E(X^2) - (E(X))^2\end{aligned}$$

- The **standard deviation** of a random variable is

$$\text{SD}(X) = \sqrt{\text{Var}(X)}$$

- Variance is the long run average squared deviation from the mean.
- Standard deviation measures, roughly, the long run average distance from the mean. The measurement units of the standard deviation are the same as for the random variable itself.
- The definition $E((X - E(X))^2)$ represents the concept of variance. However, variance is usually computed using the following equivalent but slightly simpler formula.

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

- That is, variance is the expected value of the square of X minus the square of the expected value of X .

- Variance has many nice theoretical properties. Whenever you need to compute a standard deviation, first find the variance and then take the square root at the end.

Example 8.2. Continuing with roulette, Nadja bets \$1 on number 7. If the wheel lands on 7, Nadja wins her bet back plus an additional \$35; otherwise she loses the money she bet. Let X be Nadja's net winnings (net of the initial bet of \$1.)

1. Find the distribution of X .
2. Compute $E(X)$.
3. How do the expected values of the two \$1 bets — bet on black versus bet on 7 — compare? Explain what this means.
4. Are the two \$1 bets — bet on black versus bet on 7 — identical? If not, explain why not.
5. Before doing any calculations, determine if $SD(X)$ is greater than, less than, or equal to $SD(W)$. Explain.
6. Compute $Var(W)$ and $SD(W)$.

7. Which \$1 bet — betting on black or betting on 7 — is “riskier”? How is this reflected in the standard deviations?

Example 8.3. The plots in Figure 8.1 summarize hypothetical distributions of quiz scores in six classes. All plots are on the same scale. Each quiz score is a whole number between 0 and 10 inclusive.

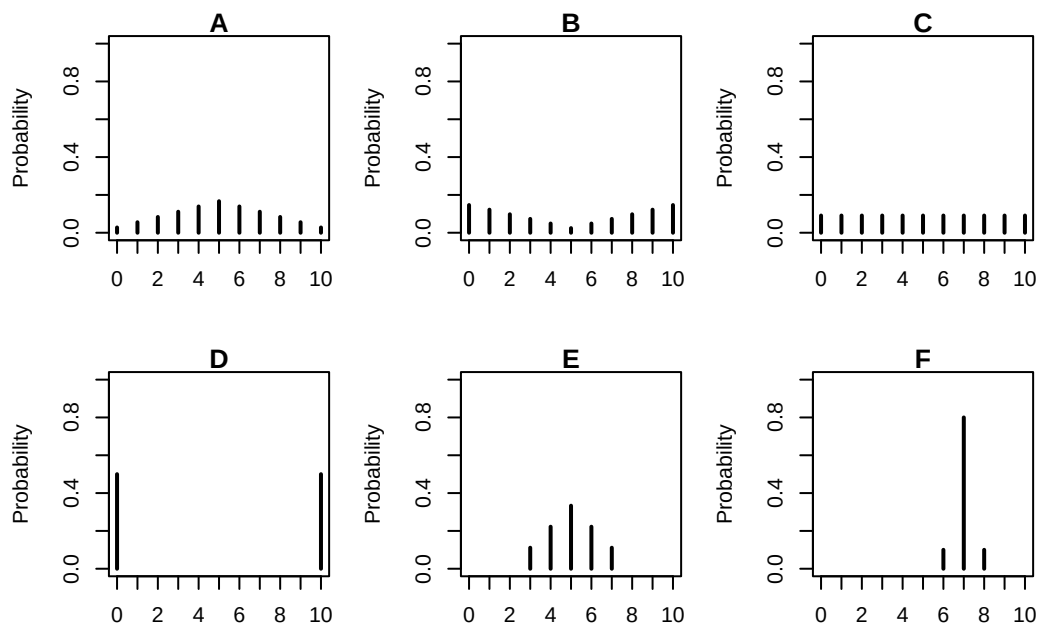


Figure 8.1: A few discrete distributions.

1. Donny Dont says that C represents the smallest SD, since there is no variability in the heights of the bars. Do you agree that C represents “no variability”? Explain.

2. What is the smallest *possible* value the SD of quiz scores could be? What would need to be true about the distribution for this to happen? (This scenario might not be represented by one the plots.)
3. Without doing any calculations, arrange the classes in order based on their SDs from smallest to largest.
4. In one of the classes, the SD of quiz scores is 5. Which one? Why?
5. Is the SD in F greater than, less than, or equal to 1? Why?
6. Provide a ballpark estimate of SD in each case.

8.1 Standardization

- Standardization measures values in terms of “standard deviations away from the mean”

- This idea is particularly useful when comparing random variables with different measurement units but whose distributions have similar shapes.

$$\text{Standardized value} = \frac{\text{Value} - \text{Mean}}{\text{Standard deviation}}$$

- If X is a random variable with expected value $E(X)$ and standard deviation $SD(X)$, then the **standardized random variable** is

$$Z = \frac{X - E(X)}{SD(X)}$$

Example 8.4. SAT scores have, approximately, a symmetric bell-shaped distribution with a mean of 1050 and a standard deviation of 200. ACT scores have, approximately, a symmetric bell-shaped distribution with a mean of 21 and a standard deviation of 5.5. Darius's score on the SAT is 1500. Alfred's score on the ACT is 31. Who scored relatively better on their test?

1. Compute and interpret the standardized value for Darius's SAT score.
2. Compute and interpret the standardized value for Alfred's ACT score
3. Who scored relatively better on their test?

SAT scores: Normal distribution with mean 1050 and SD 2

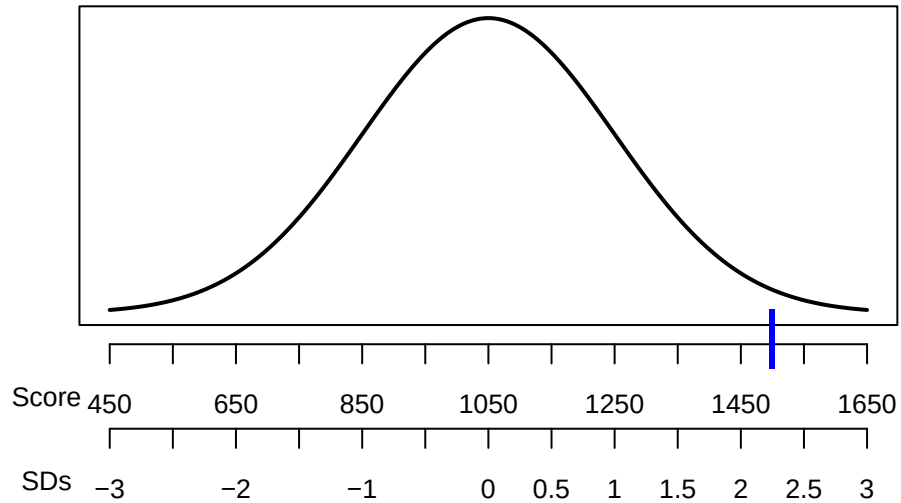


Figure 8.2: Comparison of the distributions in Example 8.4. The blue mark indicates Darius's SAT score and the orange mark indicates Alfred's ACT score.

ACT scores: Normal distribution with mean 21 and SD 5.

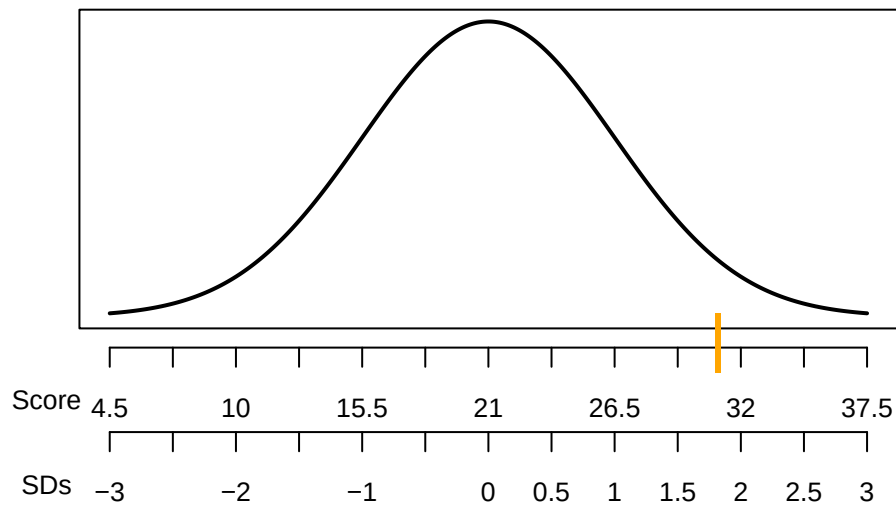


Figure 8.3: Comparison of the distributions in Example 8.4. The blue mark indicates Darius's SAT score and the orange mark indicates Alfred's ACT score.

9 Continuous Random Variables: Probability Density Functions

- Simulated values of a continuous random variable are usually plotted in a **histogram** which groups the observed values into “bins” and plots densities or frequencies for each bin. (R: `hist`).
- In a histogram *areas* of bars represent relative frequencies; the axis which represents the height of the bars is called “density” (R: `hist` with `freq = FALSE`).
- The distribution of a continuous random variable can be described by a **probability density function (pdf)**, for which *areas under the density curve determine probabilities*.

9.1 Uniform Distributions

Example 9.1. In the meeting problem assume that Regina’s arrival time X (minutes after noon) follows a $\text{Uniform}(0, 60)$ distribution. Figure 9.1 displays a histogram of 10000 simulated values of X .

1. Sketch a spinner to simulate values of X .
2. Sketch a plot of the pdf of X .

Repetition	X
1	34.39460
2	16.88999
3	55.56485
4	38.52262
5	42.61946
6	28.38516

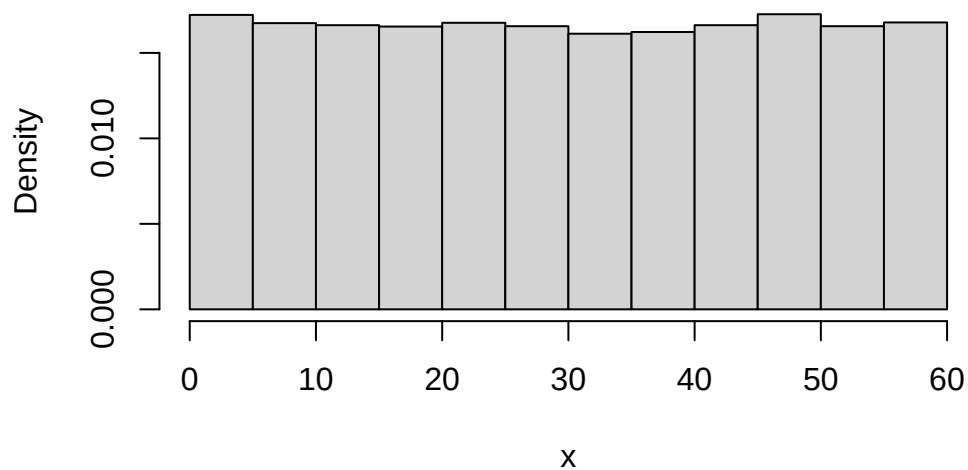


Figure 9.1: Simulated values from a $\text{Uniform}(0, 60)$ distribution

3. Use the pdf to find the probability that Regina arrives before 12:15.
4. Use the pdf to find the probability that Regina arrives after 12:45.
5. Use the pdf to find the probability that Regina arrives between 12:15:00 and 12:16:00.
6. Use the pdf to find the probability that Regina arrives between 12:15:00 and 12:15:01.

7. Compute $P(X = 15)$, the probability that Regina arrives at the exact time 12:15:00 (with infinite precision).

- **The probability that a continuous random variable X equals any particular value is 0.** That is, if X is continuous then $P(X = x) = 0$ for all x .
- Even though any specific value of a continuous random variable has probability 0, *intervals* still can have positive probability.
- In practical applications involving continuous random variables, “equal to” really means “close to”, and “close to” probabilities correspond to intervals which can have positive probability.
- A continuous random variable X has a **Uniform distribution** with parameters a and b , with $a < b$, if its probability density function f_X satisfies

$$f_X(x) \propto \text{constant}, \quad a < x < b \quad (9.1)$$

$$= \frac{1}{b-a}, \quad a < x < b. \quad (9.2)$$

- If X has a $\text{Uniform}(a, b)$ distribution then

$$E(X) = \frac{a+b}{2} \quad (9.3)$$

$$\text{Var}(X) = \frac{|b-a|^2}{12} \quad (9.4)$$

$$\text{SD}(X) = \frac{|b-a|}{\sqrt{12}} \quad (9.5)$$

9.2 Probability Density Functions

- The continuous analog of a probability mass function (pmf) is a *probability density function* (pdf).
- However, while pmfs and pdfs play analogous roles, they are different in one fundamental way; namely, a pmf outputs probabilities directly, while a pdf does not.
- A pdf of a continuous random variable must be *integrated* to find probabilities of related events.

- The **probability density function (pdf)** (a.k.a. density) of a *continuous* RV X is the function f_X which satisfies

$$P(a \leq X \leq b) = \int_a^b f_X(x)dx, \quad \text{for all } -\infty \leq a \leq b \leq \infty \quad (9.6)$$

- For a continuous random variable X with pdf f_X , the probability that X takes a value in the interval $[a, b]$ is the *area under the pdf over the region* $[a, b]$.
- The axioms of probability imply that a valid pdf must satisfy

$$f_X(x) \geq 0 \quad \text{for all } x, \quad (9.7)$$

$$\int_{-\infty}^{\infty} f_X(x)dx = 1 \quad (9.8)$$

Example 9.2. Continuing Example 9.1, we will now we assume Regina's arrival time has pdf

$$f_X(x) = \begin{cases} cx, & 0 \leq x \leq 60, \\ 0, & \text{otherwise.} \end{cases}$$

where c is an appropriate constant.

1. Sketch a plot of the pdf. Generally, what does this say about Regina's arrival time?
2. Find the value of c and specify the pdf of X .
3. Find the probability that Regina arrives before 12:15.
4. Find the probability that Regina arrives after 12:45. How does this compare to the previous part? What does that say about Regina's arrival time?

5. Find $P(X = 15)$, the probability that Regina arrives at the exact time 12:15 (with infinite precision).
 6. Find $P(X = 45)$, the probability that Regina arrives at the exact time 12:45 (with infinite precision).
 7. Find the probability that Regina arrives between 12:15 and 12:16.
 8. Find the probability that Regina arrives between 12:45 and 12:46. How does this compare to the probability for 12:15 to 12:16? What does that say about Regina's arrival time?
- **The probability that a continuous random variable X equals any particular value is 0.** That is, if X is continuous then $P(X = x) = 0$ for all x .
 - For continuous random variables, it doesn't really make sense to talk about the probability that the random value is *equal to* a particular value. However, we can consider the probability that a random variable is *close to* a particular value.
 - The density $f_X(x)$ at value x is *not* a probability.
 - Rather, the density $f_X(x)$ at value x is related to the probability that the RV X takes a value "close to x " in the following sense

$$P\left(x - \frac{\epsilon}{2} \leq X \leq x + \frac{\epsilon}{2}\right) \approx f_X(x)\epsilon, \quad \text{for small } \epsilon$$

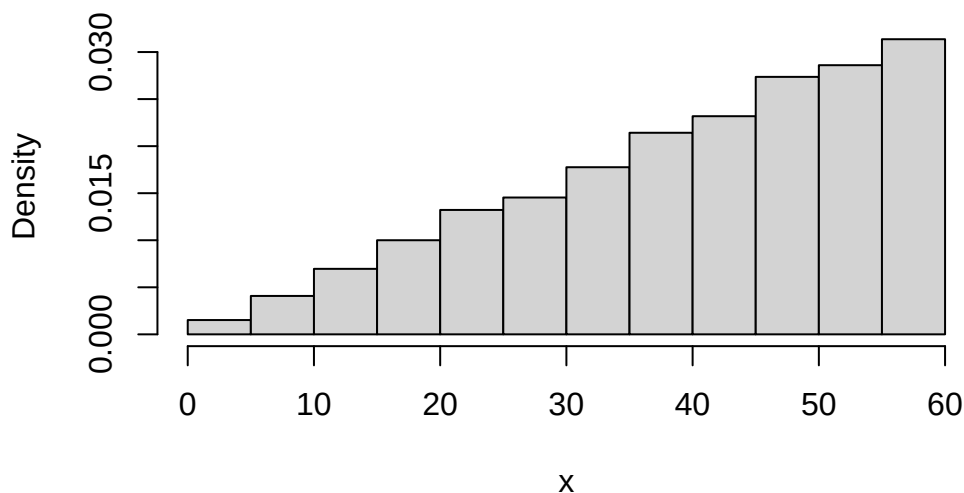


Figure 9.2: Simulated values of X with the pdf in Example 9.2. (We’ll see how this simulation works soon.)

- The quantity ϵ is a small number that represents the desired degree of precision. For example, rounding to two decimal places corresponds to $\epsilon = 0.01$.
- What’s important about a pdf is *relative* heights. For example, if $f_X(x_2) = 2f_X(x_1)$ then X is roughly “twice as likely to be near x_2 than to be near x_1 ” in the above sense.

$$\frac{f_X(x_2)}{f_X(x_1)} = \frac{f_X(x_2)\epsilon}{f_X(x_1)\epsilon} \approx \frac{\mathbb{P}(x_2 - \frac{\epsilon}{2} \leq X \leq x_2 + \frac{\epsilon}{2})}{\mathbb{P}(x_1 - \frac{\epsilon}{2} \leq X \leq x_1 + \frac{\epsilon}{2})}$$

9.3 Exponential Distributions

Example 9.3. Suppose that we model the waiting time, measured continuously in hours, from now until the next earthquake (of any magnitude) occurs in southern CA as a continuous random variable X with pdf

$$f_X(x) = 2e^{-2x}, \quad x \geq 0$$

This is the pdf of the “Exponential(2)” distribution.

1. Sketch the pdf of X . What does this tell you about waiting times?

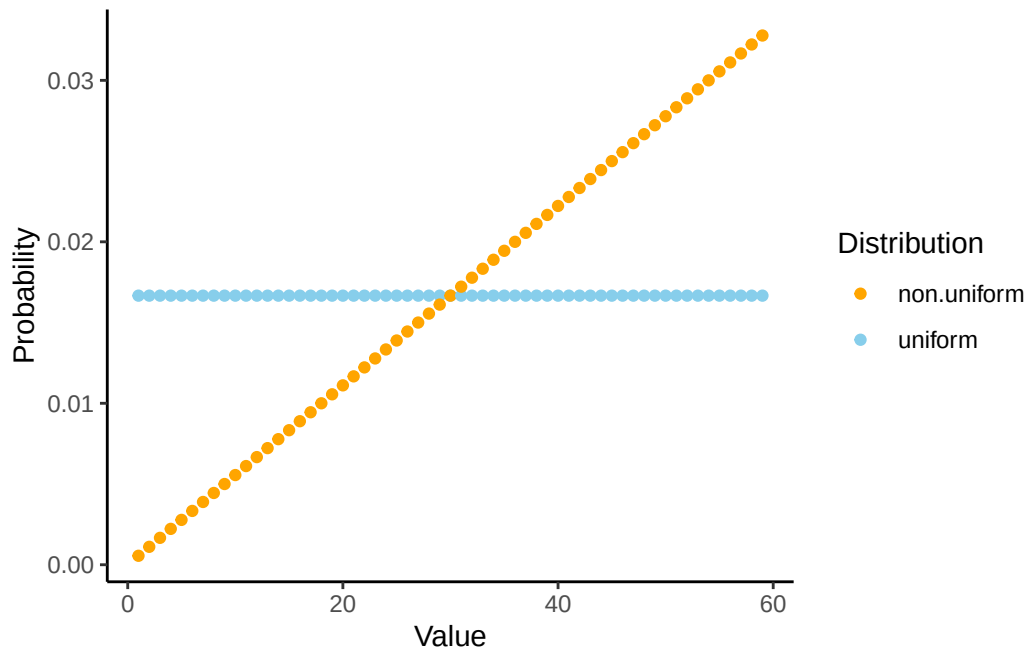


Figure 9.3: Probability of Regina arriving at each minute between noon (0) and 1:00PM (60), to the nearest minute, for the Uniform(0, 60) distribution (blue) and the distribution in Example 9.2 (orange).

2. Without doing any integration, approximate the probability that X rounded to the nearest minute is 0.5 hours.

3. Without doing any integration determine how much more likely that X rounded to the nearest minute is to be 0.5 than 1.0.

4. Compute and interpret $P(X > 0.25)$.

5. Compute and interpret $P(X \leq 3)$.

- A continuous random variable X has an **Exponential distribution** with *rate* parameter $\lambda > 0$ if its pdf is

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

- If X has an $\text{Exponential}(\lambda)$ distribution then

$$P(X > x) = e^{-\lambda x}, \quad x \geq 0$$

- Exponential distributions are often used to model the *waiting time* in a random process until some event occurs.
 - λ is the average *rate* at which events occur over time (e.g., 2 per hour)
 - $1/\lambda$ is the mean time between events (e.g., 1/2 hour)

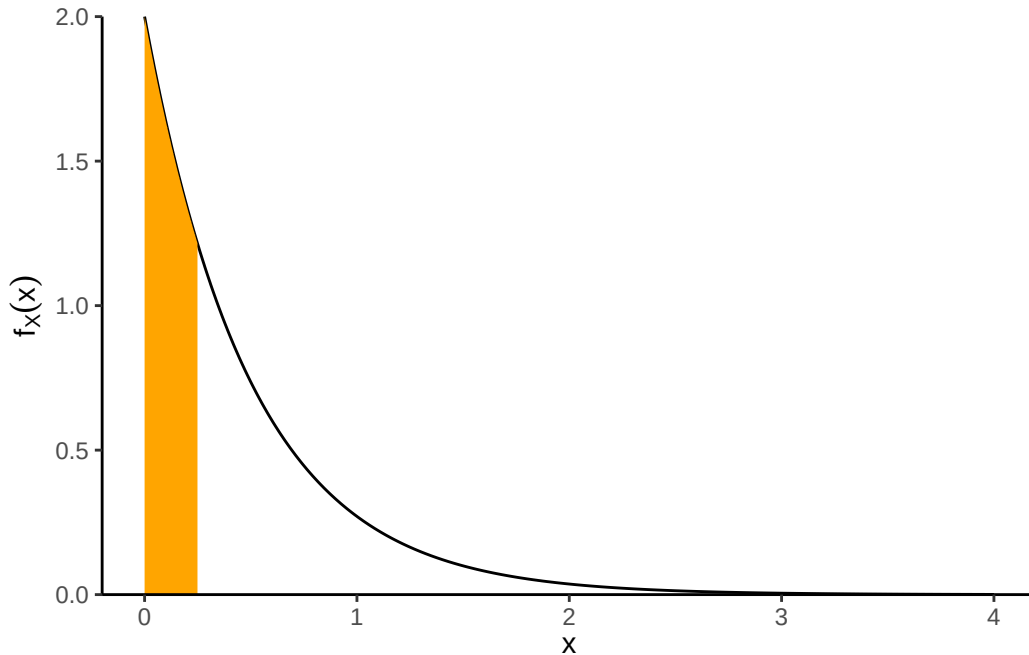


Figure 9.4: Probability density function of X with an Exponential distribution with rate 2. The shaded area represents $P(X < 0.25)$.

9.4 Expected Values

- The **expected value** (a.k.a. *expectation* a.k.a. *mean*), of a random variable X is a number denoted $E(X)$ representing the probability-weighted average value of X .
- The expected value of a continuous random variable with pdf f_X is defined as

$$E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$$

- Replace the generic bounds $(-\infty, \infty)$ with the possible values of the random variable
- Note well that $E(X)$ represents *a single number*.
- The expected value is the “balance point” (center of gravity) of a distribution.
- The expected value of a random variable X is defined by the probability-weighted average according to the underlying probability measure. But the expected value can also be interpreted as the **long-run average value**, and so can be approximated via simulation.
- Read the symbol $E(\cdot)$ as
 - Simulate lots of values of what’s inside $(\cdot)$
 - Compute the average. This is a “usual” average; just sum all the simulated values and divide by the number of simulated values.

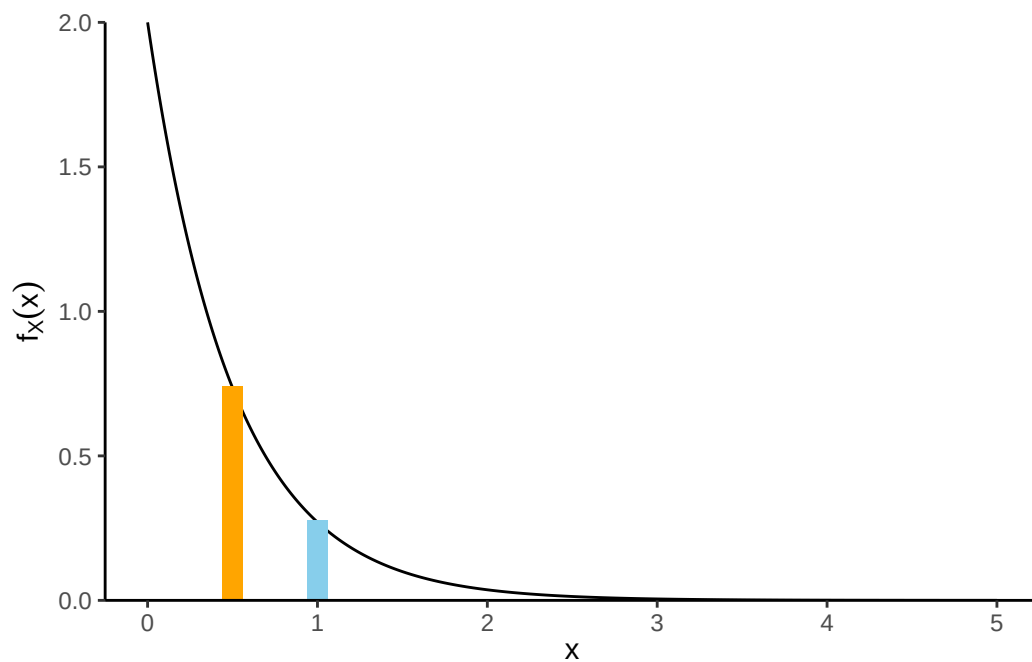


Figure 9.5: Illustration that the probability that X is close to x can be approximated by the area of a rectangle with height equal to the density at x , $f_X(x)$. The density height at $x = 0.5$ is roughly 2.7 times greater than the density height at $x = 1$ so the probability that X is close to 0.5 is roughly 2.7 times greater than the probability that X is close to 1.

Example 9.4. Continuing Example 9.3, where X , the waiting time (hours) from now until the next earthquake (of any magnitude) occurs in southern CA, has an Exponential distribution with rate 2.

1. Use simulation to approximate the long run average value of X .

2. Set up the integral to compute $E(X)$.

3. Interpret $E(X)$.

4. Compute $P(X = E(X))$.

5. Compute $P(X \leq E(X))$. (Is it equal to 50%?)

- If X has an $\text{Exponential}(\lambda)$ distribution then

$$E(X) = \frac{1}{\lambda}$$

- For example, if events happen at rate $\lambda = 2$ per hour, then the average waiting time between events is $1/\lambda = 1/2$ hour.

9.5 Law of the Unconscious Statistician

- The “**law of the unconscious statistician**” (**LOTUS**) says that the expected value of a transformed random variable can be found without finding the distribution of the transformed random variable, simply by applying the probability weights of the original random variable to the transformed values.

$$\text{Discrete } X \text{ with pmf } p_X: \quad E[g(X)] = \sum_x g(x)p_X(x) \quad (9.9)$$

$$\text{Continuous } X \text{ with pdf } f_X: \quad E[g(X)] = \int_{-\infty}^{\infty} g(x)f_X(x)dx \quad (9.10)$$

- LOTUS says we don’t have to first find the distribution of $Y = g(X)$ to find $E[g(X)]$; rather, we just simply apply the transformation g to each possible value x of X and then apply the corresponding weight for x to $g(x)$.
- Whether in the short run or the long run, in general

$$\text{Average of } g(X) \neq g(\text{Average of } X) \quad (9.11)$$

- In terms of expected values, in general

$$E(g(X)) \neq g(E(X)) \quad (9.12)$$

The left side $E(g(X))$ represents first transforming the X values and then averaging the transformed values. The right side $g(E(X))$ represents first averaging the X values and then plugging the average (a single number) into the transformation formula.

Example 9.5. Continuing Example 9.3, where X , the waiting time (hours) from now until the next earthquake (of any magnitude) occurs in southern CA, has an Exponential distribution with rate 2.

1. Use simulation to approximate the variance of X .
2. Set up the integral to compute $E(X^2)$.

3. Compute $\text{Var}(X)$. (The previous part shows $E(X^2) = 0.5$)

4. Compute $\text{SD}(X)$.

- The **variance** of a random variable X is

$$\text{Var}(X) = E\left((X - E(X))^2\right) \quad (9.13)$$

$$= E(X^2) - (E(X))^2 \quad (9.14)$$

- The **standard deviation** of a random variable is

$$\text{SD}(X) = \sqrt{\text{Var}(X)}$$

- Variance is the long run average squared deviation from the mean.
- Standard deviation measures, roughly, the long run average distance from the mean. The measurement units of the standard deviation are the same as for the random variable itself.
- The definition $E((X - E(X))^2)$ represents the concept of variance. However, variance is usually computed using the following equivalent but slightly simpler formula.

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

- That is, variance is the expected value of the square of X minus the square of the expected value of X .
- Variance has many nice theoretical properties. Whenever you need to compute a standard deviation, first find the variance and then take the square root at the end.
- If X has an $\text{Exponential}(\lambda)$ distribution then

$$\text{Var}(X) = \frac{1}{\lambda^2}$$

10 Percentiles

Example 10.1. Maggie and Seamus are babies who have just turned one. At their one-year visits to their pediatrician:

- Maggie is 76cm tall and in the 75th percentile of height for girls.
- Seamus is 72cm tall and in the 10th percentile of height for boys.

Explain what these percentiles mean.

- A distribution is characterized by its percentiles.
- Roughly, the value x is the p th **percentile** (a.k.a. quantile) of a distribution of a random variable X if p percent of values of the variable are less than or equal to x : $P(X \leq x) = p$.
- Any distribution can be represented by a spinner. A spinner basically describes a distribution by specifying all the percentiles. For example,
 - The 25th percentile goes 25% of the way around the axis (at “3 o’clock”)
 - The 50th percentile goes 50% of the way around the axis (at “6 o’clock”)
 - The 75th percentile goes 75% of the way around the axis (at “9 o’clock”)

10.1 Cumulative Distribution Functions

- Roughly, the value x is the p th percentile of a distribution of a random variable X if p percent of values of the variable are less than or equal to x : $P(X \leq x) = p$.
- The *cumulative distribution function (cdf)* of a random variable fills in the blank for any given x : x is the (blank) percentile. That is, for an input x , the cdf outputs $P(X \leq x)$.
- The **cumulative distribution function (cdf)** of a random variable X is the function, F_X defined by $F_X(x) = P(X \leq x)$. A cdf is defined for all real numbers x regardless of whether x is a possible value of X .

Example 10.2. Continuing Example 9.2 where Regina's arrival time X (minutes after noon) has pdf

$$f_X(x) = (2/3600)x, \quad 0 < x < 60.$$

Let F_X be the cdf of X .

1. Compute and interpret $F_X(15)$.
2. Compute and interpret $F_X(45)$.
3. Compute and interpret $F_X(45) - F_X(15)$.
4. Compute $F_X(-0.2)$
5. Compute $F_X(70)$
6. Find the cdf of X .

Example 10.3. Let X be the number of heads in 3 flips of a fair coin.

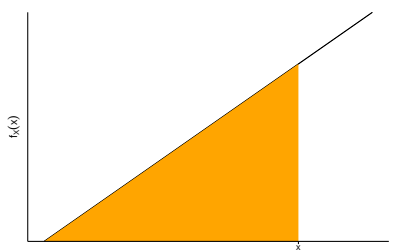


Figure 10.1: PDF

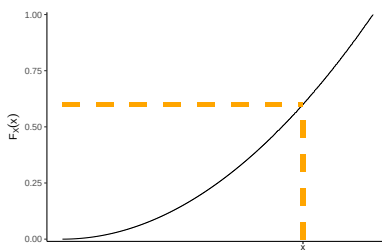


Figure 10.2: CDF

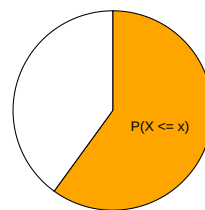


Figure 10.3: Spinner

1. Find the pmf of X and sketch a plot of it.

2. Find the cdf of X and sketch a plot of it.

3. Let Y be the number of tails in 3 flips of a fair coin. Find the cdf of Y .

- A cdf is defined for all values of x , regardless if x is a possible value of the RV.
- A cdf is a non-decreasing function: if $x_1 \leq x_2$ then $F_X(x_1) \leq F_X(x_2)$.
- A cdf approaches 0 as the input approaches $-\infty$: $\lim_{x \rightarrow -\infty} F_X(x) = 0$
- A cdf approaches 1 as the input approaches ∞ : $\lim_{x \rightarrow \infty} F_X(x) = 1$
- The cdf of a *discrete* random variable is a step function.
 - The steps occur at the possible values of the random variable.
 - The height of a particular step corresponds to the probability of that value, given by the pmf.
- The cdf of a *continuous* random variable is a continuous function.
 - The cdf of a *continuous* random variable is obtained by integrating the pdf, so

- The pdf of a *continuous* random variable is obtained by differentiating the cdf

$$F'_X = f_X \quad \text{if } X \text{ is continuous}$$

- For any random variable X with cdf F_X

$$F_X(b) - F_X(a) = P(a < X \leq b)$$

Whether the inequalities in the above event are strict ($<$) or not ($\leq$) matters for discrete random variables, but not for continuous.

- Random variables X and Y **have the same distribution** if their cdfs are the same, that is, if $F_X(u) = F_Y(u)$ for all u .
- That is, two random variables have the same distribution if all the percentiles are the same.

10.2 Quantile functions

- The *quantile function* (essentially the inverse cdf) fills in the following blank for a given $p \in [0, 1]$: the 100 p th percentile is (blank).

Example 10.4. Continuing Example 9.2 where Regina's arrival time X (minutes) had pdf

$$f_X(x) = (2/3600)x, \quad 0 < x < 60.$$

1. Find the median (i.e., 50th percentile) of X .
2. Find the 25th percentile of X .
3. Find the quantile function of X .

4. Construct a spinner for simulating values of X according to its distribution.

- For a *continuous* random variable X with cdf F_X , the **quantile function** Q_X is the inverse of the cdf, $Q_X(p) = F_X^{-1}(p)$.
- The *quantile function* (essentially the inverse cdf) fills in the following blank for a given $p \in [0, 1]$: the 100 p th percentile is (blank).
- For example, evaluating the quantile function at $p = 0.25$ outputs the 25th percentile.
- The quantile function can be used to create a spinner for a distribution. Basically, the values on the outside boundary of the spinner are scaled based on the quantile function (which is determined by the cdf). Intervals corresponding to regions of higher density (“more likely”) values are stretched out on the spinner boundary; intervals corresponding to regions of lower density (“less likely” values) are shrunk.

Spinner for pdf $f(x) = (2/3600)x$, $0 < x < 60$

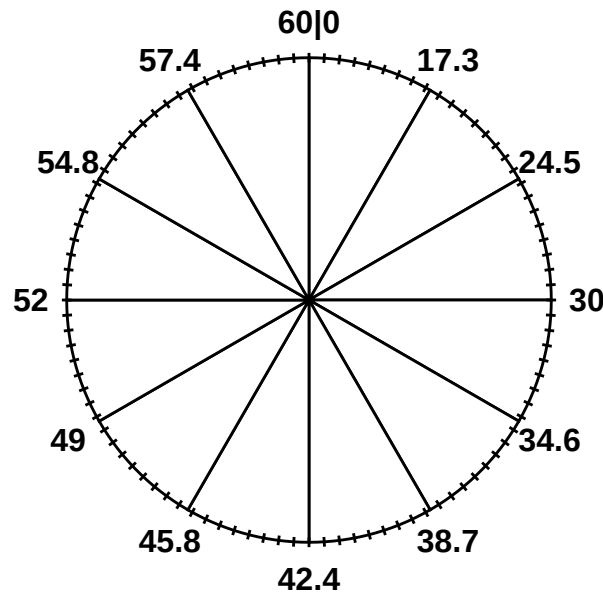


Figure 10.4: Spinner for X in Example 9.2. The axis is transformed according to the quantile function

- **Universality of the Uniform** (or “one spinner to rule them all”). Let F be a cdf and Q its corresponding quantile function. Let U have a Uniform(0, 1) distribution and define the random variable $X = Q(U)$. Then the cdf of X is F .

- Universality of the uniform might look complicated but all it basically says is that you can construct a spinner by putting the 25th percentile 25% of the way around, the 75th percentile 75% of the way around, etc.
- Actually, universality of the uniform says we don't have to create a new spinner. We can just spin the $\text{Uniform}(0, 1)$ spinner and transform each resulting value by plugging it into the quantile function.

11 Normal Distributions

- Normal distributions are probably the most important distributions in probability and statistics.
- A continuous random variable X has a **Normal** (a.k.a., Gaussian) distribution with mean $\mu \in (-\infty, \infty)$ and standard deviation $\sigma > 0$ if its pdf is

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right), \quad -\infty < x < \infty$$

- If X has a $\text{Normal}(\mu, \sigma)$ distribution then

$$\begin{aligned} E(X) &= \mu \\ \text{SD}(X) &= \sigma \end{aligned}$$

- A Normal density is a particular “bell-shaped” curve which is symmetric about its mean μ . The mean μ is a *location* parameter: μ indicates where the center and peak of the distribution is.
- The standard deviation σ is a *scale* parameter: σ indicates the distance from the mean to where the concavity of the density changes. That is, there are inflection points at $\mu \pm \sigma$.

Example 11.1. The pdfs in the plot below represent the distribution of hypothetical test scores in three classes. The test scores in each class follow a Normal distribution. Identify the mean and standard deviation for each class.

- Any Normal distribution follows the “empirical rule” which determines the percentiles that give a Normal distribution its particular bell shape.
- For a Normal distribution:
 - 38% of values are within 0.5 standard deviations of the mean
 - 68% of values are within 1 standard deviation of the mean
 - 87% of values are within 0.5 standard deviations of the mean
 - 95% of values are within 2 standard deviations of the mean
 - 99% of values are within 2.6 standard deviations of the mean
 - 99.7% of values are within 3 standard deviations of the mean
- The table below lists some percentiles of a Normal distribution.

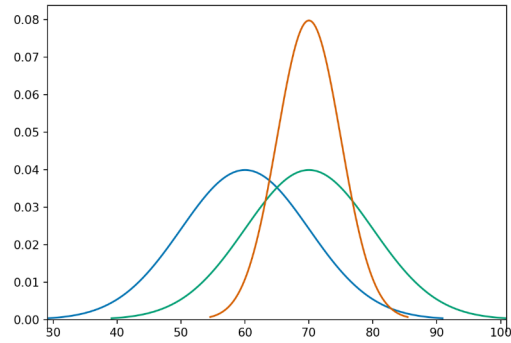


Figure 11.1

Table 11.1: Selected percentiles for a Normal distribution

Percentile	SDs away from the mean
0.1%	3.09 SDs below the mean
0.5%	2.58 SDs below the mean
1%	2.33 SDs below the mean
2.5%	1.96 SDs below the mean
10%	1.28 SDs below the mean
15.9%	1 SDs below the mean
25%	0.67 SDs below the mean
30.9%	0.5 SDs below the mean
50%	0 SDs above the mean
69.1%	0.5 SDs above the mean
75%	0.67 SDs above the mean
84.1%	1 SDs above the mean
90%	1.28 SDs above the mean
97.5%	1.96 SDs above the mean
99%	2.33 SDs above the mean
99.5%	2.58 SDs above the mean
99.9%	3.09 SDs above the mean

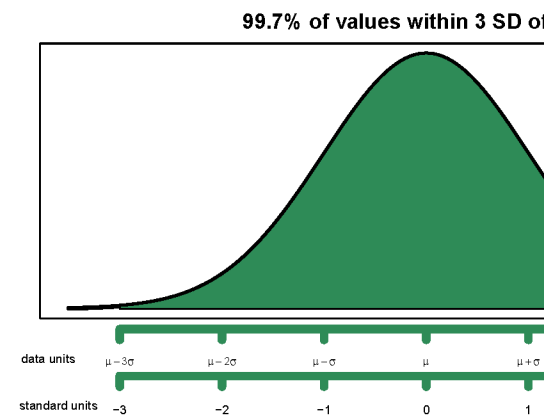
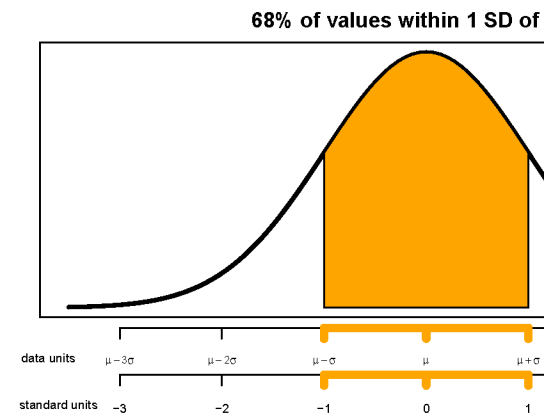
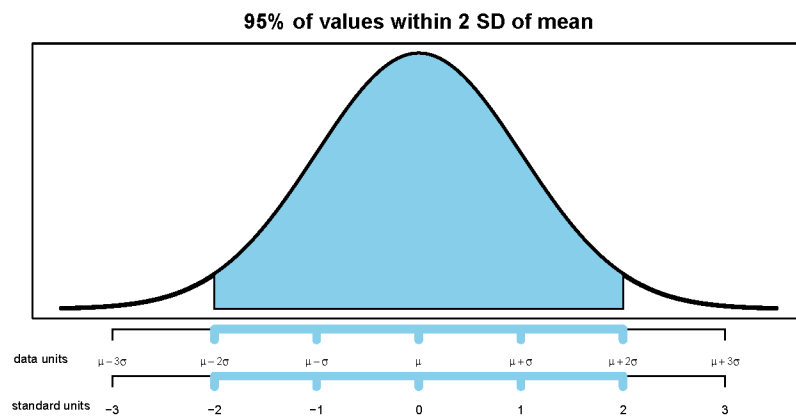
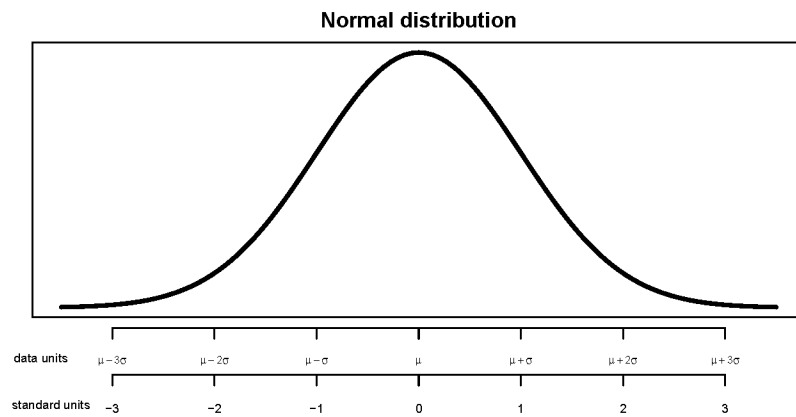


Figure 11.2

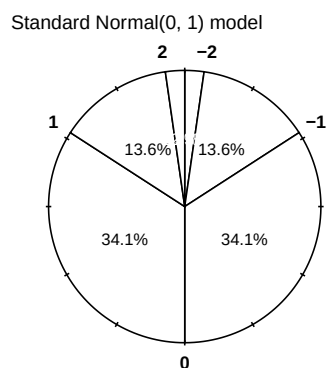
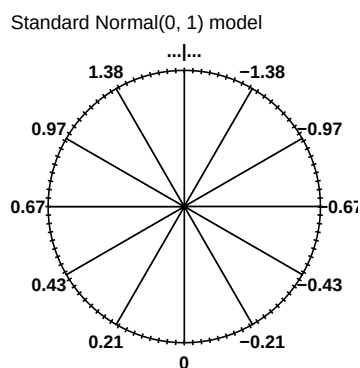


Figure 11.3: A standard Normal(0, 1) spinnerFigure 11.4: A standard Normal(0, 1) spinner.

The same spinner is displayed on both sides, with different features highlighted on the left and right. Only selected rounded values are displayed, but in the idealized model the spinner is infinitely precise so that any real number is a possible outcome. Notice that the values on the axis are *not* evenly spaced.

The same spinner is displayed on both sides, with different features highlighted on the left and right. Only selected rounded values are displayed, but in the idealized model the spinner is infinitely precise so that any real number is a possible outcome. Notice that the values on the axis are *not* evenly spaced.

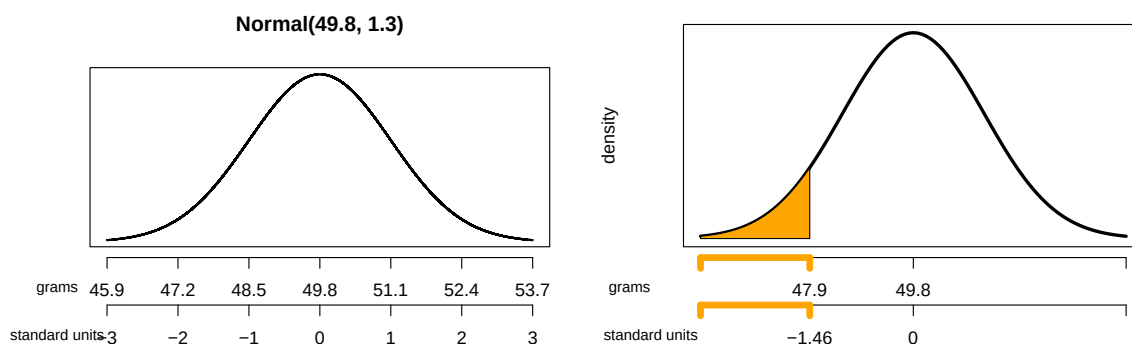
- The Normal(0, 1) distribution (mean 0, SD 1) is called the “Standard Normal” distribution
- Standardization:
 - If X has a Normal(μ , σ) distribution then $Z = \frac{X-\mu}{\sigma}$ has a Normal(0, 1) distribution
 - If Z has a Normal(0, 1) distribution then $X = \mu + \sigma Z$ has a Normal(μ , σ) distribution

Example 11.2. The wrapper of a package of candy lists a weight of 47.9 grams. Naturally, the weights of individual packages vary somewhat. Suppose package weights have an approximate Normal distribution with a mean of 49.8 grams and a standard deviation of 1.3 grams.

1. Sketch the distribution of package weights. Carefully label the variable axis. It is helpful to draw two axes: one in the measurement units of the variable, and one in standardized units.
2. Why wouldn't the company print the mean weight of 49.8 grams as the weight on the package?
3. Estimate the probability that a package weighs less than the printed weight of 47.9 grams.
4. Estimate the probability that a package weighs between 47.9 and 53.0 grams.
5. Suppose that the company only wants 1% of packages to be underweight. Find the weight that must be printed on the packages.

6. Find the 25th percentile (a.k.a., first (lower) quartile) of package weights.

7. Find the 75th percentile (a.k.a., third (upper) quartile) of package weights. How can you use the work you did in the previous part?



Example 11.3. Daily high temperatures (degrees Fahrenheit) in San Luis Obispo in August follow (approximately) A Normal distribution with a mean of 76.9 degrees F. The temperature exceeds 100 degrees Fahrenheit on about 1.5% of August days.

1. What is the standard deviation?

2. Suppose the mean increases by 2 degrees Fahrenheit. On what percentage of August days will the daily high temperature exceed 100 degrees Fahrenheit? (Assume the standard deviation does not change.)

3. A mean of 78.9 is 1.02 times greater than a mean of 76.9. By what (multiplicative) factor has the percentage of 100-degree days increased? What do you notice?

Example 11.4. In a large class, scores on midterm 1 follow (approximately) a $\text{Normal}(\mu_1, \sigma)$ distribution and scores on midterm 2 follow (approximately) a $\text{Normal}(\mu_2, \sigma)$ distribution. Note that the SD σ is the same on both exams. The 40th percentile of midterm 1 scores is equal to the 70th percentile of midterm 2 scores. Compute

$$\frac{\mu_1 - \mu_2}{\sigma}$$

(This is one statistical measure of *effect size*.)

Example 11.5. A bank uses an applicant's score on some criteria to decide whether or not to approve their loan application. Based on past history, the bank has determined that

- Scores for those who repay the loan follow a Normal distribution with mean 60 and SD 8
- Scores for those who do NOT repay the loan follow a Normal distribution with mean 40 and SD 12

When someone applies for a loan the bank does not know whether the applicant will eventually repay the loan. How can the bank use the applicant's score to decide whether or not to approve the loan?

1. Draw sketches of these two normal curves on the same axis.
2. Suggest a decision rule, based on an applicant's score, for deciding whether or not to approve the applicant's loan.
3. Describe the two kinds of classification errors that could be made in this situation.
4. Determine the probability that we incorrectly reject the application of someone who would repay. Shade in the corresponding region under the Normal curve, and interpret this probability.
5. Determine the probability that we incorrectly approve the application of someone who would NOT repay. Shade in the corresponding region under the Normal curve, and interpret this probability.
6. In which direction — smaller or larger — would you need to change the decision rule's cutoff value in order to decrease the probability that an applicant who would repay the loan is incorrectly rejected?
7. Would the probability of the other kind of error — incorrectly approving a loan for an

applicant who would not repay it — increase or decrease with this new cutoff value?

8. Determine the cutoff value needed to decrease the probability that an applicant who would repay the loan is incorrectly rejected to 0.05.
 9. Determine the other error probability with this new cut-off rule.
 10. Now repeat the two previous parts with the goal of decreasing to 0.05 the probability of incorrectly approving an applicant who would not repay the loan.
 11. If you consider the two kinds of errors to be equally serious, how might you decide which of the three decision rules considered so far is the best?
 12. Which error do you think the bank would consider more serious? In which direction would that move the threshold?
- Making decisions/predictions based on data often involves trade-offs.
 - Decreasing the probability of one kind of error often comes at the expense of increasing the probability of another kind of error.

12 Joint Distributions

Example 12.1. Roll a fair four-sided die twice. Let X be the sum of the two dice, and let Y be the larger of the two rolls (or the common value if both rolls are the same). Recall Table 5.1.

1. Compute and interpret $p_{X,Y}(5,3) = P(X = 5, Y = 3)$.
2. Construct a “flat” table displaying the distribution of (X, Y) pairs, with one pair in each row.
3. Construct a two-way displaying the joint distribution on X and Y .
4. Sketch a plot depicting the joint distribution of X and Y .
5. Starting with the two-way table, how could you obtain $P(X = 5)$?

Table 12.1: Flat table representing the joint distribution of the sum (X) and larger (Y) of two rolls of a four-sided die.

(x, y)	$P(X = x, Y = y)$
(2, 1)	0.0625
(3, 2)	0.1250
(4, 2)	0.0625
(4, 3)	0.1250
(5, 3)	0.1250
(5, 4)	0.1250
(6, 3)	0.0625
(6, 4)	0.1250
(7, 4)	0.1250
(8, 4)	0.0625

6. Starting with the two-way table, how could you obtain the marginal distribution of X ? of Y ?

7. Starting with the marginal distribution of X and the marginal distribution of Y , could you necessarily construct the two-way table of the joint distribution? Explain.

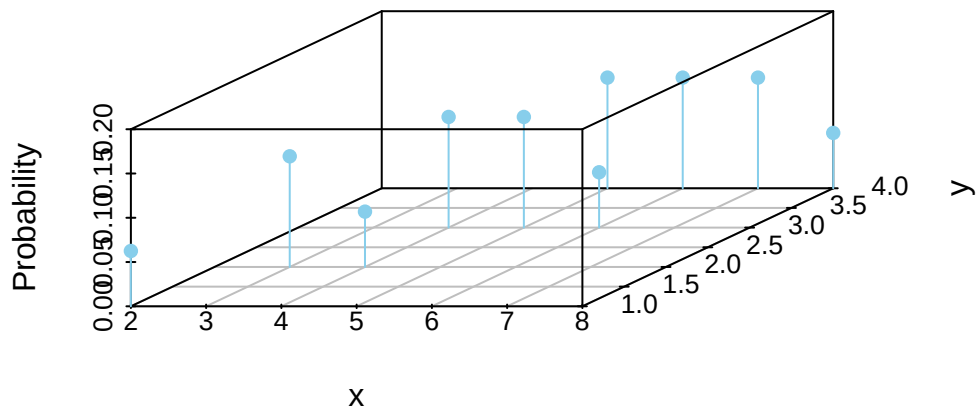
- The *joint distribution* of random variables X and Y is a probability distribution on (x, y) *pairs*, and describes how the values of X and Y vary together or jointly.
- Marginal distributions can be obtained from a joint distribution by “stacking”/“collapsing”/“aggregating” out the other variable.
- **In general, marginal distributions alone are not enough to determine a joint distribution.** (The exception is when random variables are *independent*.)

Table 12.2: Two-way table representing the joint distribution of the sum (X) and larger (Y) of two rolls of a four-sided die.

$x \setminus y$	1	2	3	4
2	1/16	0	0	0

3	0	2/16	0	0
4	0	1/16	2/16	0
5	0	0	2/16	2/16
6	0	0	1/16	2/16
7	0	0	0	2/16
8	0	0	0	1/16

3D Impulse Plot



Example 12.2. Continuing the dice rolling example, construct a spinner representing the joint distribution of X and Y .

- The **joint probability mass function (pmf)** of two *discrete* random variables (X, Y) defined on a probability space is the function defined by

$$p_{X,Y}(x, y) = P(X = x, Y = y) \quad \text{for all } x, y$$

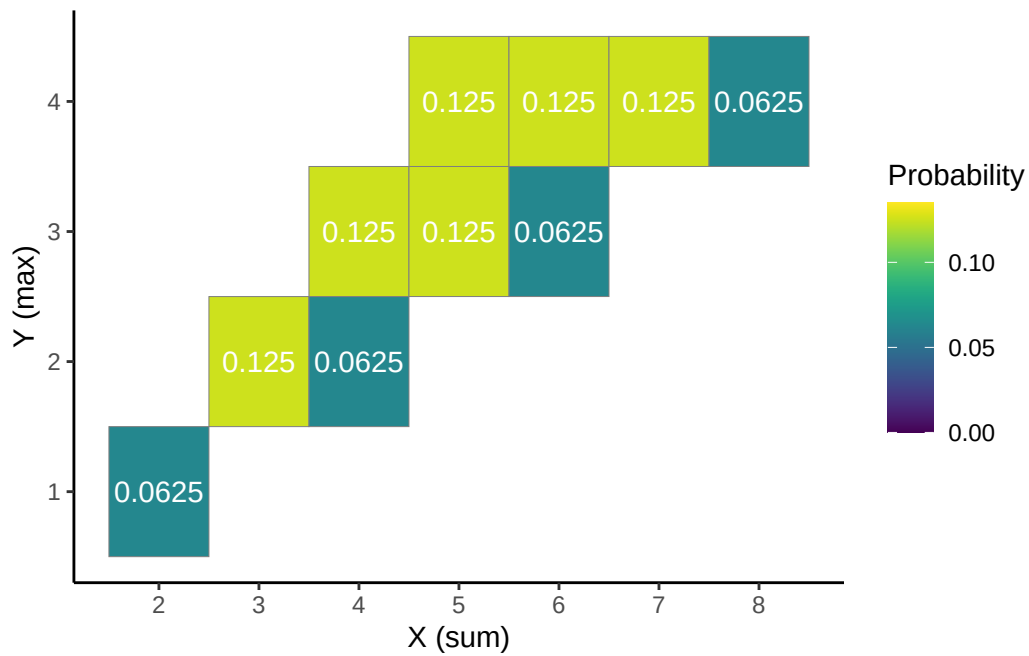


Figure 12.1: Tile plot representing the joint distribution of the sum (X) and larger (Y) of two rolls of a four-sided die.

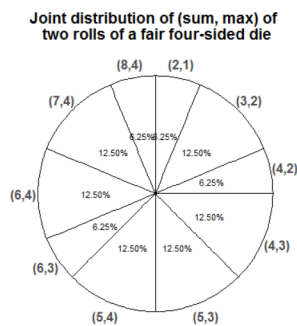


Figure 12.2: Spinner representing the joint distribution of the sum (X) and larger (Y) of two rolls of a fair four-sided die.

- Remember to specify the possible (x, y) pairs when defining a joint pmf.

Example 12.3. Let X be the number of home runs hit by the home team, and Y the number of home runs hit by the away team in a randomly selected Major League Baseball game. Suppose that X and Y have joint pmf

$$p_{X,Y}(x, y) = \begin{cases} e^{-2.3} \frac{1.2^x 1.1^y}{x!y!}, & x = 0, 1, 2, \dots; y = 0, 1, 2, \dots, \\ 0, & \text{otherwise.} \end{cases}$$

1. Compute and interpret the probability that the home teams hits 2 home runs and the away team hits 1 home run.
2. Construct a two-way table representation of the joint pmf (you can use software or a spreadsheet).
3. Compute and interpret the probability that each team hits at most 3 home runs.
4. Compute and interpret the probability that both teams combine to hit a total of 3 home runs.
5. Compute and interpret the probability that the home team and the away team hit the same number of home runs.

6. Compute and interpret the probability that the home team hits 2 home runs.

7. Compute and interpret the probability that the away team hits 1 home run.

- Recall that we can obtain marginal distributions from a joint distribution.
- *Marginal pmfs* are determined by the joint pmf via the law of total probability.
- If we imagine a plot with blocks whose heights represent the joint probabilities, the marginal probability of a particular value of one variable can be obtained by “stacking” all the blocks corresponding to that value.

$$p_X(x) = \sum_y p_{X,Y}(x, y) \quad \text{a function of } x \text{ only}$$
$$p_Y(y) = \sum_x p_{X,Y}(x, y) \quad \text{a function of } y \text{ only}$$

12.1 Joint probability density functions

- The joint distribution of two continuous random variables can be specified by a joint pdf, a *surface* specifying the density of (x, y) pairs.
- The probability that the (X, Y) pair of random variables lies in some region is the *volume* under the joint pdf surface over the region.
- The **joint probability density function (pdf)** of two *continuous* random variables (X, Y) defined on a probability space is the function $f_{X,Y}$ which satisfies, for any region

S

$$P[(X, Y) \in S] = \iint_S f_{X,Y}(x, y) dx dy$$

- A joint pdf is a surface with height $f_{X,Y}(x, y)$ at (x, y) .
- The probability that the (X, Y) pair of random variables lies in the region A is the *volume* under the pdf surface over the region A
- The height of the density surface at a particular (x, y) pair is related to the probability that (X, Y) takes a value “close to” (x, y) :

$$P(x - \epsilon/2 < X < x + \epsilon/2, y - \epsilon/2 < Y < y + \epsilon/2) = \epsilon^2 f_{X,Y}(x, y) \quad \text{for small } \epsilon$$

- The joint distribution is a distribution on (X, Y) pairs. A mathematical expression of a joint distribution is a function of both values of X and values of Y . Pay special attention to the possible values; the possible values of one variable might be restricted by the value of the other.
- The marginal distribution of Y is a distribution on Y values only, regardless of the value of X . A mathematical expression of a marginal distribution will have only values of the single variable in it; for example, an expression for the marginal distribution of Y will only have y in it (no x , not even in the possible values).

12.2 Independence of random variables

- Two random variables X and Y defined on a probability space with probability measure P are **independent** if $P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$ for all x, y . That is, two random variables are independent if their joint cdf is the product of their marginal cdfs.
- Random variables X and Y are independent if and only if the joint distribution factors into the product of the marginal distributions. The definition is in terms of cdfs, but analogous statements are true for pmfs and pdfs.

Discrete RVs X and Y are independent

$$\iff p_{X,Y}(x, y) = p_X(x)p_Y(y) \quad \text{for all } x, y$$

Continuous RVs X and Y are independent

$$\iff f_{X,Y}(x, y) = f_X(x)f_Y(y) \quad \text{for all } x, y$$

- Random variables X and Y are independent if and only if their joint distribution can be factored into the product of a function of values of X alone and a function of values of Y alone. That is, X and Y are independent if and only if there exist functions g and h for which

$$f_{X,Y}(x, y) \propto g(x)h(y) \quad \text{for all } x, y$$

- X and Y are independent if and only if the joint distribution factors into a product of the marginal distributions. The above result says that you can determine if that's true without first finding the marginal distributions.

Example 12.4. Let X be the number of home runs hit by the home team, and Y the number of home runs hit by the away team in a randomly selected Major League Baseball game. Suppose that X and Y are independent, X has a Poisson(1.2) distribution and Y has a Poisson(1.1) distribution.

1. Find the joint pmf of X and Y .
2. How could you have from the setup of Example 12.3 that X and Y were independent?

13 Covariance and Correlation

Quantities like expected value and variance summarize characteristics of the *marginal* distribution of a single random variable. When there are multiple random variables their *joint* distribution is of interest. Covariance and correlation summarize a characteristic of the joint distribution of two random variables, namely, the degree to which they “co-deviate from the their respective means”.

13.1 Covariance

- The **covariance** between two random variables X and Y is

$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - E[X])(Y - E[Y])] \\ &= E(XY) - E(X)E(Y)\end{aligned}$$

- Covariance is the long run average of the products of the paired deviations from the mean
- Covariance is the expected value of the product minus the product of expected values.

Example 13.1. Roll a fair four-sided die twice and let X be the sum of the two rolls, and Y the larger of the two rolls.

1. Specify the joint pmf of X and Y . (These first few parts are review of things we have covered before.)
2. Find the marginal pmf of X and $E(X)$.
3. Find the marginal pmf of Y and $E(Y)$.

4. Find $E(XY)$. Is it equal to $E(X)E(Y)$?

5. Find $\text{Cov}(X, Y)$. Why is the covariance positive?

6. Describe how you would use simulation to approximate $\text{Cov}(X, Y)$.

- $\text{Cov}(X, Y) > 0$ (positive association): above average values of X tend to be associated with above average values of Y
- $\text{Cov}(X, Y) < 0$ (negative association): above average values of X tend to be associated with below average values of Y
- $\text{Cov}(X, Y) = 0$ indicates that the random variables are **uncorrelated**: there is no overall positive or negative association. But be careful: if X and Y are uncorrelated there can still be a relationship between X and Y ; there is just no overall positive or negative association.. We will see examples that demonstrate that being uncorrelated does not necessarily imply that random variables are independent.

Example 13.2. Continuing Example 13.1. Let X be the sum of the two rolls, Y the larger of the two rolls, W the number of rolls equal to 4, and Z the number of rolls equal to 1.

1. Is $\text{Cov}(X, W)$ positive, negative, or zero? Why?

2. Is $\text{Cov}(X, Z)$ positive, negative, or zero? Why?

3. Let $V = W + Z$. Is $\text{Cov}(X, V)$ positive, negative, or zero? Why?

4. Is $\text{Cov}(W, Z)$ positive, negative, or zero? Why?

Example 13.3. What is another name for $\text{Cov}(X, X)$?

13.2 Correlation

Example 13.4. The covariance between height in inches and weight in pounds for football players is 96.

1. What are the measurement units of the covariance?
2. Suppose height were measured in feet instead of inches. Would the shape of the joint distribution change? Would the strength of the association between height and weight change? Would the value of covariance change?

- The **correlation (coefficient)** between random variables X and Y is

$$\begin{aligned}\text{Corr}(X, Y) &= \text{Cov}\left(\frac{X - E(X)}{\text{SD}(X)}, \frac{Y - E(Y)}{\text{SD}(Y)}\right) \\ &= \frac{\text{Cov}(X, Y)}{\text{SD}(X)\text{SD}(Y)}\end{aligned}$$

- The correlation for two random variables is the covariance between the corresponding standardized random variables. Therefore, correlation is a *standardized* measure of the association between two random variables.
- A correlation coefficient has no units and is measured on a universal scale. Regardless of the original measurement units of the random variables X and Y

$$-1 \leq \text{Corr}(X, Y) \leq 1$$

- $\text{Corr}(X, Y) = 1$ if and only if $Y = aX + b$ for some $a > 0$
- $\text{Corr}(X, Y) = -1$ if and only if $Y = aX + b$ for some $a < 0$
- Therefore, correlation is a standardized measure of the strength of the *linear* association between two random variables.
- Covariance is the correlation times the product of the standard deviations.

$$\text{Cov}(X, Y) = \text{Corr}(X, Y)\text{SD}(X)\text{SD}(Y)$$

Example 13.5. The covariance between height in inches and weight in pounds for football players is 96. Heights have mean 74 and SD 3 inches. Weights have mean 250 pounds and SD 45 pounds. Find the correlation between weight and height.

Example 13.6. Consider the probability space corresponding to two rolls of a fair four-sided die. Let X be the sum of the two rolls, Y the larger of the two rolls. Compute $\text{Corr}(X, Y)$.

Example 13.7. Try the [Guessing Correlations applet!](#)

Example 13.8. Sketch a scatterplot where X and Y have a correlation of 0, but they are not independent.

14 Conditional Distributions

- The *joint distribution* of random variables X and Y is a probability distribution on (x, y) pairs, and describes how the values of X and Y vary together or jointly.
- We can also study *conditional distributions* of random variables given the values of some random variables. How does the distribution of Y change for different values of X (and vice versa)?

Example 14.1. Roll a fair four-sided die twice. Let X be the sum of the two rolls, and let Y be the larger of the two rolls (or the common value if a tie). We have previously found the joint and marginal distributions of X and Y , displayed in the two-way table below.

$p_{X,Y}(x, y)$					
$x \setminus y$	1	2	3	4	$p_X(x)$
2	1/16	0	0	0	1/16
3	0	2/16	0	0	2/16
4	0	1/16	2/16	0	3/16
5	0	0	2/16	2/16	4/16
6	0	0	1/16	2/16	3/16
7	0	0	0	2/16	2/16
8	0	0	0	1/16	1/16
$p_Y(y)$	1/16	3/16	5/16	7/16	

1. Compute $p_{X|Y}(6|4) = P(X = 6|Y = 4)$.
2. Construct a table, plot, and spinner to represent the conditional distribution of X given $Y = 4$.

3. Construct a table, plot, and spinner to represent the conditional distribution of X given $Y = 3$.
4. Construct a table, plot, and spinner to represent the conditional distribution of X given $Y = 2$.
5. Construct a table, plot, and spinner to represent the conditional distribution of X given $Y = 1$.
6. Compute $p_{Y|X}(4|6) = P(Y = 4|X = 6)$.
7. Construct a table, plot, and spinner to represent the distribution of Y given $X = 6$.
8. Construct a table, plot, and spinner to represent the distribution of Y given $X = 5$.
9. Construct a table, plot, and spinner to represent the distribution of Y given $X = 4$.

- The **conditional distribution of Y given $X = x$** is the distribution of Y values over only those outcomes for which $X = x$. It is a distribution on values of Y only; treat x as a fixed constant when conditioning on the event $\{X = x\}$.
- Conditional distributions can be obtained from a joint distribution by *slicing and renormalizing*. The conditional distribution of Y given $X = x$, where x represents a particular number, can be thought of as:
 - the slice of the joint distribution corresponding to $X = x$, a distribution on values of Y alone with $X = x$ fixed
 - renormalized so that the slice accounts for 100% of the probability over the values of Y
- The *shape* of the conditional distribution of Y given $X =$ is determined by the shape of the slice of the joint distribution over values of Y for the fixed x .
- For each fixed x , the conditional distribution of Y given $X = x$ is a different distribution on values of the random variable Y . There is not one “conditional distribution of Y given X ”, but rather a *family of conditional distributions of Y* given different values of X .
- Each conditional distribution is a distribution, so we can summarize its characteristics like mean and standard deviation. The conditional mean and standard deviation of Y given $X = x$ represent, respectively, the long run average and variability of values of Y over only (X, Y) pairs with $X = x$.
- Since each value of x typically corresponds to a different conditional distribution of Y given $X = x$, the conditional mean and standard deviation will typically be functions of x .

Example 14.2. We have already discussed two ways for simulating an (X, Y) pair in the dice rolling example: simulate a pair of rolls and measure X (sum) and Y (max), or spin the joint distribution spinner for (X, Y) once.

1. Now describe another way for simulating an (X, Y) pair using the spinners in Example 14.1. (Hint: you’ll need one more spinner in addition to the four from the previous example.)
2. Describe in detail how you can simulate (X, Y) pairs and use the results to approximate $P(X = 6|Y = 4)$.

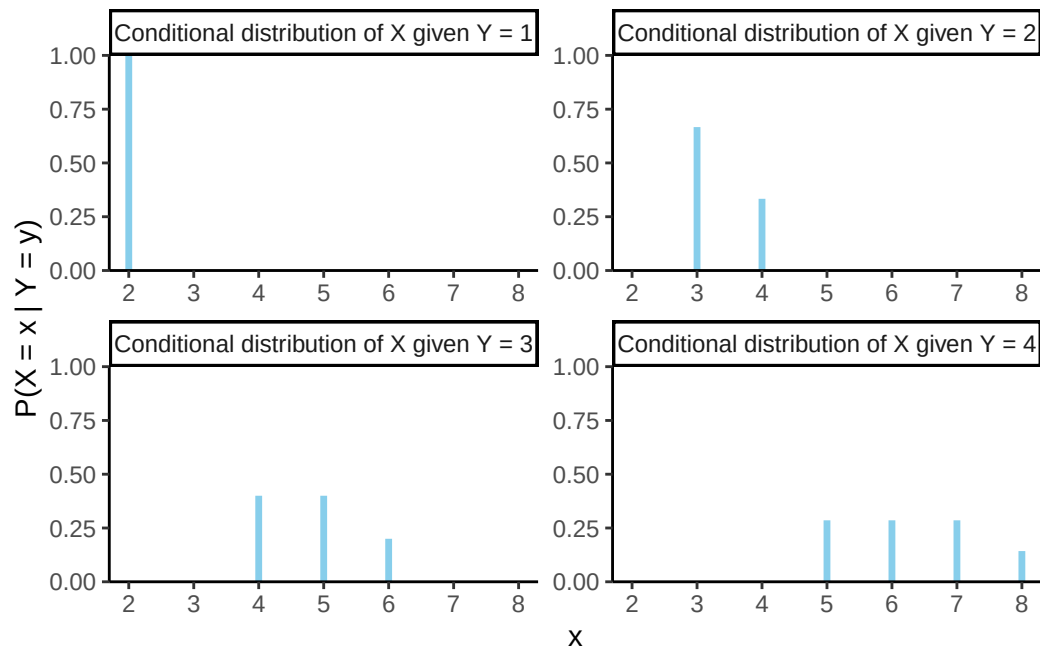
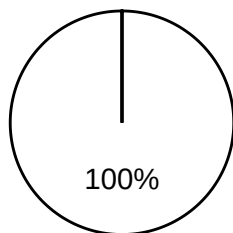
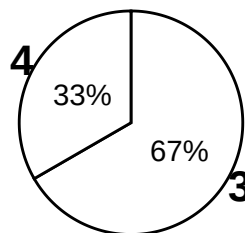


Figure 14.1: Impulse plots representing the family of conditional distributions of X given Y for the dice rolling example. Each plot represents a conditional distribution of X given $Y = y$ for a particular value of $y = 1, 2, 3, 4$.

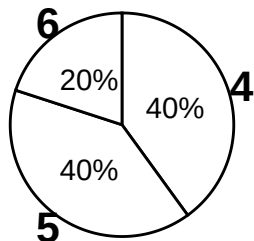
Distribution of X given $Y = 1$



Distribution of X given $Y =$



Distribution of X given $Y = 3$



Distribution of X given $Y =$

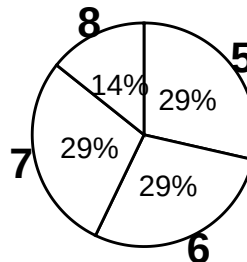


Figure 14.2: Spinners representing the family of conditional distributions of X given Y in the dice rolling example. Each spinner represents a conditional distribution of X given $Y = y$ for a particular value of $y = 1, 2, 3, 4$.

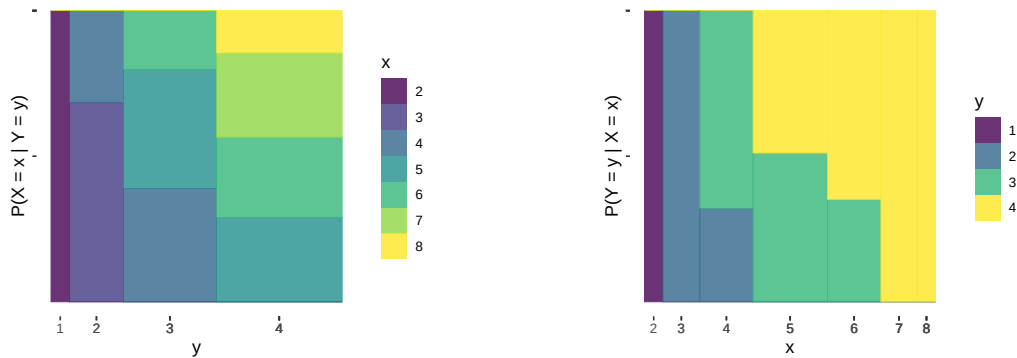
3. Describe in detail how you can simulate (X, Y) pairs and use the results to approximate the conditional distribution of X given $Y = 4$.

4. Describe in detail how you can simulate values from the conditional distribution of X given $Y = 4$ without simulating (X, Y) pairs.

- Rather than directly simulating from a joint distribution, we can simulate an (X, Y) pair in two stages:
 - Simulate a value of X from its marginal distribution. Call the simulated value x .
 - Given x , simulate a value of Y from the conditional distribution of Y given $X = x$. There will be a different distribution (spinner) for each possible value of x .
- This “marginal then conditional” process is essentially implementing the multiplication rule

$$\text{joint} = \text{conditional} \times \text{marginal}$$

- In many problems a joint distribution is naturally described by specifying the marginal distribution of X and the family of conditional distributions of Y given values of X
- Be sure to distinguish between joint, conditional, and marginal distributions.
- The joint distribution of X and Y is a distribution on (X, Y) pairs. A mathematical expression of a joint distribution is a function of both values of X and values of Y .
- The conditional distribution of Y given $X = x$ is a distribution on Y values (among (X, Y) pairs with a fixed value of $X = x$). A mathematical expression of a conditional distribution will involve both x and y , but x is treated like a fixed constant and y is treated as the variable. Note: the possible values of Y might depend on the value of x .
- The marginal distribution of Y is a distribution on Y values only, regardless of the value of X . A mathematical expression of a marginal distribution will have only values of the single variable in it; for example, an expression for the marginal distribution of Y will only have y in it (no x , not even in the possible values).



(a) Mosaic plot of conditional distributions of X given values of Y (a) Mosaic plot of conditional distributions of Y given values of X

- **Be careful** when conditioning with continuous random variables. Remember that the probability that a continuous random variable is equal to a particular value is 0; that is, for continuous X , $P(X = x) = 0$. - Mathematically, when we condition on $\{X = x\}$ we are really conditioning on $\{|X - x| < \epsilon\}$ — the event that the random variable X is within ϵ of the value x — and seeing what happens in the idealized limit when $\epsilon \rightarrow 0$.
- Practically, ϵ represents our “close enough” degree of precision, e.g., $\epsilon = 0.01$ if “within 0.01” is close enough.
- When conditioning on a continuous random variable X in a simulation, *never* condition on $\{X = x\}$; rather, condition on $\{|X - x| < \epsilon\}$ where ϵ represents the suitable degree of precision.

14.1 Conditional Expected Value

Example 14.3. Roll a fair four-sided die twice. Let X be the sum of the two rolls, and let Y be the larger of the two rolls (or the common value if a tie).

$p_{X,Y}(x,y)$					
$x \setminus y$	1	2	3	4	$p_X(x)$
2	1/16	0	0	0	1/16
3	0	2/16	0	0	2/16
4	0	1/16	2/16	0	3/16
5	0	0	2/16	2/16	4/16
6	0	0	1/16	2/16	3/16
7	0	0	0	2/16	2/16
8	0	0	0	1/16	1/16

$p_{X,Y}(x,y)$				
$p_Y(y)$	1/16	3/16	5/16	7/16

1. Compute and interpret $E(Y)$. How could you find a simulation-based approximation?
2. We have seen that the long run average value of Y is 3.125. Would you expect the conditional long run average value of Y given $X = 8$ to be greater than, less than, or equal to 3.125? Explain without doing any calculations. What about given $X = 3$?
3. How could you use simulation to approximate the conditional long run average value of Y given $X = 6$?
4. Compute and interpret $E(Y|X = 6)$.
5. Find $E(Y|X = x)$ for each possible value of x of X .
6. Compute and interpret $E(X|Y = 4)$. How could you find a simulation-based approximation?
7. Find $E(X|Y = y)$ for each possible value y of Y .

- The **conditional expected value** (a.k.a. *conditional expectation* a.k.a. *conditional mean*), of a random variable Y given the event $\{X = x\}$, defined on a probability space with measure P , is a *number* denoted $E(Y|X = x)$ representing the probability-weighted average value of Y , where the weights are determined by the conditional distribution of Y given $X = x$.

Discrete X, Y with conditional pmf $p_{Y|X}$:
$$E(Y|X = x) = \sum_y y p_{Y|X}(y|x)$$

- Remember, when conditioning on $X = x$, x is treated as a fixed constant. The conditional expected value $E(Y|X = x)$ is a *number* representing the mean of the conditional distribution of Y given $X = x$.
- The conditional expected value $E(Y|X = x)$ is the long run average value of Y over only those outcomes for which $X = x$.
- To approximate $E(Y|X = x)$, simulate many (X, Y) pairs, discard the pairs for which $X \neq x$, and average the Y values for the pairs that remain.

15 Joint Normal Distributions

- Jointly continuous random variables X and Y have a **Bivariate Normal** distribution with parameters $\mu_X, \mu_Y, \sigma_X > 0, \sigma_Y > 0$, and $-1 < \rho < 1$ if the joint pdf is {

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) \right]\right)$$

- }
If the pair (X, Y) has a $\text{BivariateNormal}(\mu_X, \mu_Y, \sigma_X, \sigma_Y, \rho)$ distribution

$$\begin{aligned} E(X) &= \mu_X \\ E(Y) &= \mu_Y \\ SD(X) &= \sigma_X \\ SD(Y) &= \sigma_Y \\ \text{Corr}(X, Y) &= \rho \end{aligned}$$

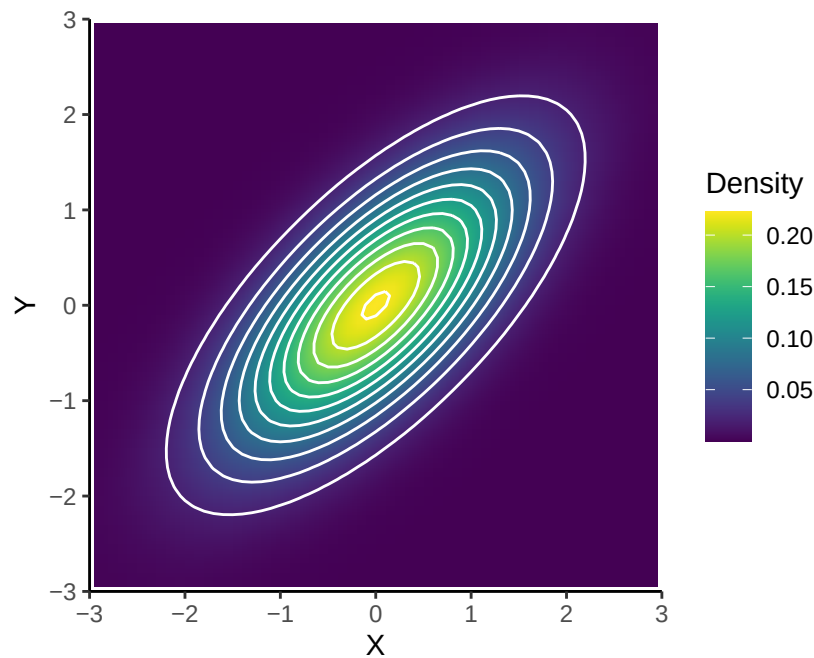


Figure 15.1: Bivariate Normal distribution with means (0, 0), standard deviations (1, 1) and correlation 0.7.

- A Bivariate Normal Density has elliptical contours. For each height $c > 0$ the set $\{(x, y) : f_{X,Y}(x, y) = c\}$ is an ellipse. The density decreases as (x, y) moves away from (μ_X, μ_Y) , most steeply along the minor axis of the ellipse, and least steeply along the major of the ellipse.
- A scatterplot of (x, y) pairs generated from a Bivariate Normal distribution will have a rough linear association and the cloud of points will resemble an ellipse.
- If X and Y have a Bivariate Normal distribution, then the marginal distributions are also Normal: X has a $\text{Normal}(\mu_X, \sigma_X)$ distribution and Y has a $\text{Normal}(\mu_Y, \sigma_Y)$.
- If X and Y have a Bivariate Normal distribution and $\text{Corr}(X, Y) = 0$ then X and Y are independent. (Remember, in general it is possible to have situations where the correlation is 0 but the random variables are not independent.)
- X and Y have a Bivariate Normal distribution if and only if every linear combination of X and Y has a Normal distribution. That is, X and Y have a Bivariate Normal distribution if and only if $aX + bY + c$ has a Normal distribution for all a, b, c .

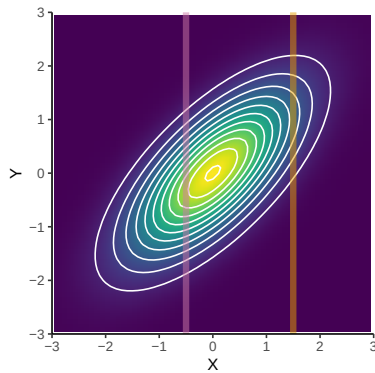


Figure 15.2: Bivariate Normal distribution with two conditional *slices* highlighted.

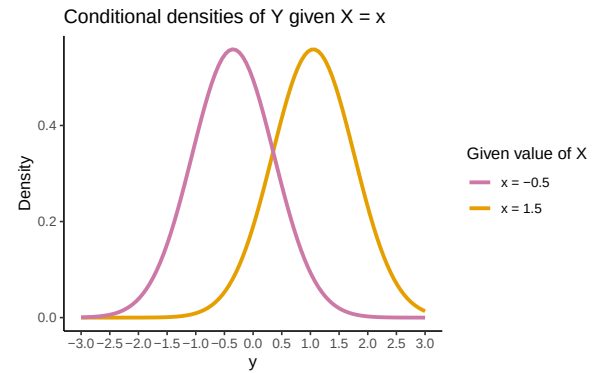


Figure 15.3: Two conditional Normal distributions.

- If X and Y have a Bivariate Normal distribution then any conditional distribution is Normal. The conditional distribution of Y given $X = x$ is

$$N\left(\mu_Y + \frac{\rho\sigma_Y}{\sigma_X}(x - \mu_X), \sigma_Y\sqrt{1 - \rho^2}\right)$$

- The conditional expected value of Y given $X = x$ is a *linear* function of x , called the *regression line of Y on X*:

$$E(Y|X = x) = \mu_Y + \rho\sigma_Y \left(\frac{x - \mu_X}{\sigma_X} \right)$$

- The regression line passes through the point of means (μ_X, μ_Y) and has slope

$$\frac{\rho\sigma_Y}{\sigma_X}$$

- The regression line estimates that if the given x value is z SDs above the mean of X , then the corresponding Y values will be, on average, ρz SDs away from the mean of Y

$$\frac{E(Y|X = x) - \mu_Y}{\sigma_Y} = \rho \left(\frac{x - \mu_X}{\sigma_X} \right)$$

- Since $|\rho| \leq 1$, for a given x value the corresponding Y values will be, on average, relatively closer to the mean of Y than the given x value is to the mean of X . This is known as *regression to the mean*.
- For Bivariate Normal distributions, the conditional variance of Y given $X = x$ does not depend on x :

$$SD(Y|X = x) = \sigma_Y \sqrt{1 - \rho^2}$$

Example 15.1. Suppose that SAT Math (M) and Reading (R) scores of CalPoly students have a Bivariate Normal distribution. Math scores have mean 640 and SD 80, Reading scores have mean 610 and SD 70, and the correlation between scores is 0.7.

1. Identify the distribution of Math scores. Find the probability that a student has a Math score above 700.
2. Compute and interpret $E(M|R = 700)$.
3. Compute and interpret $SD(M|R = 700)$.
4. Identify the conditional distribution of Math scores given the Reading score is 700. Find the probability that a student has a higher Math than Reading score if the student scores 700 on Reading.

5. Compute and interpret $E(M|R = 550)$.
 6. Compute and interpret $SD(M|R = 550)$.
 7. Identify the conditional distribution of Math scores given the Reading score is 550. Find the probability that a student has a higher Math than Reading score if the student scores 550 on Reading.
 8. Describe how you could simulate a single (M, R) pair.
 9. Find and interpret $E(M|R)$.
-
- If the pair (X, Y) has a joint Normal distribution then each of X and Y has a Normal distribution.
 - But the converse is not true. That is, if each of X and Y has a Normal distribution, it is not necessarily true that the pair (X, Y) has a joint Normal distribution
 - However, if X and Y are *independent* and each of X and Y has a Normal distribution, then the pair (X, Y) has a joint Normal distribution. (But joint Normal is much more general than two independent Normal random variables.)

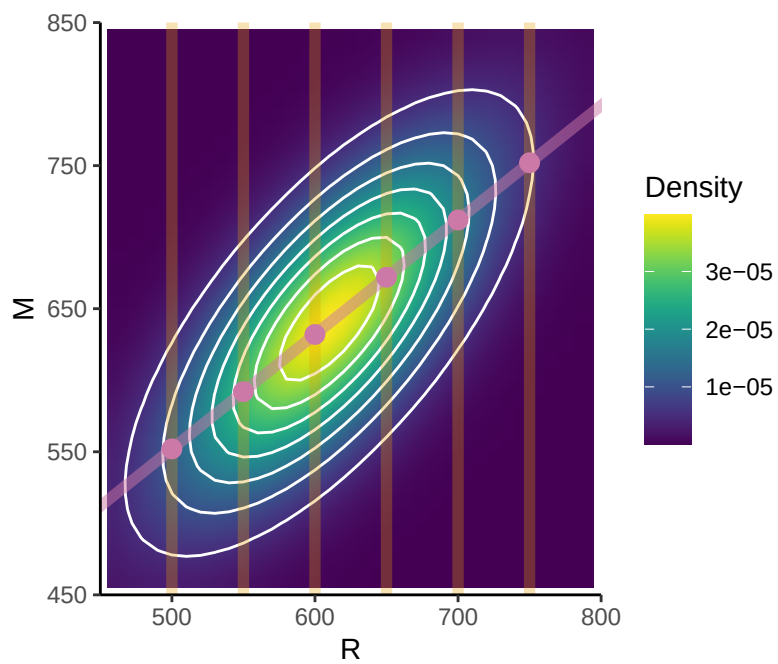


Figure 15.4: A Bivariate Normal distribution with some conditional distributions and conditional expected values highlighted.

16 Expected Values of Linear Combinations of Random Variables

16.1 Linear rescaling

If X is a random variable and a, b are non-random constants then

$$\begin{aligned}E(aX + b) &= aE(X) + b \\SD(aX + b) &= |a|SD(X) \\Var(aX + b) &= a^2Var(X)\end{aligned}$$

16.2 Linearity of expected value

Example 16.1. Refer to the tables and plots in [Example 5.29 here](#). Each scenario contains SAT Math (X) and Reading (Y) scores for 10 hypothetical students, along with the total score ($T = X + Y$) and the difference between the Math and Reading scores ($D = X - Y$, negative values indicate lower Math than Reading scores). Note that the 10 X values are the same in each scenario, and the 10 Y values are the same in each scenario, but the (X, Y) values are paired in different ways: the correlation is 0.78 in scenario 1, -0.04 (essentially 0) in scenario 2, and -0.94 in scenario 3.

1. What is the mean of $T = X + Y$ in each scenario? How does it relate to the means of X and Y ? Does the correlation affect the mean of $T = X + Y$?
2. What is the mean of $D = X - Y$ in each scenario? How does it relate to the means of X and Y ? Does the correlation affect the mean of $D = X - Y$?

- **Linearity of expected value.** For *any* two random variables X and Y ,

$$E(X + Y) = E(X) + E(Y)$$

- That is, the expected value of the sum is the sum of expected values, regardless of how the random variables are related.
- Therefore, you only need to know the marginal distributions of X and Y to find the expected value of their sum. (But keep in mind that the *distribution* of $X + Y$ will depend on the joint distribution of X and Y .)
- Whether in the short run or the long run,

$$\text{Average of } X + Y = \text{Average of } X + \text{Average of } Y$$

regardless of the joint distribution of X and Y .

- A **linear combination** of two random variables X and Y is of the form $aX + bY$ where a and b are non-random constants. Combining properties of linear rescaling with linearity of expected value yields the expected value of a linear combination.

$$E(aX + bY) = aE(X) + bE(Y)$$

- Linearity of expected value extends naturally to more than two random variables.

16.3 Variance of linear combinations of random variables

Example 16.2. Recall Example [16.1](#).

1. In which of the three scenarios is $\text{Var}(X + Y)$ the largest? Can you explain why?
2. In which of the three scenarios is $\text{Var}(X + Y)$ the smallest? Can you explain why?
3. In which scenario is $\text{Var}(X + Y)$ roughly equal to the sum of $\text{Var}(X)$ and $\text{Var}(Y)$?
4. In which of the three scenarios is $\text{Var}(X - Y)$ the largest? Can you explain why?

5. In which of the three scenarios is $\text{Var}(X - Y)$ the smallest? Can you explain why?
6. In which scenario is $\text{Var}(X - Y)$ roughly equal to the *sum* of $\text{Var}(X)$ and $\text{Var}(Y)$?

- **Variance of sums and differences of random variables.**

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X, Y)$$

Example 16.3. Assume that SAT Math (X) and Reading (Y) scores follow a Bivariate Normal distribution, Math scores have mean 527 and standard deviation 107, and Reading scores have mean 533 and standard deviation 100. Compute $\text{Var}(X + Y)$ and $\text{SD}(X + Y)$ for each of the following correlations.

1. $\text{Corr}(X, Y) = 0.77$

2. $\text{Corr}(X, Y) = 0.40$

3. $\text{Corr}(X, Y) = 0$

4. $\text{Corr}(X, Y) = -0.77$

Example 16.4. Continuing the previous example. Compute $\text{Var}(X - Y)$ and $\text{SD}(X - Y)$ for each of the following correlations.

1. $\text{Corr}(X, Y) = 0.77$

2. $\text{Corr}(X, Y) = 0.40$

3. $\text{Corr}(X, Y) = 0$

4. $\text{Corr}(X, Y) = -0.77$

- The variance of the sum is the sum of the variances if and only if X and Y are uncorrelated.

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) \quad \text{if } X, Y \text{ are uncorrelated}$$

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) \quad \text{if } X, Y \text{ are uncorrelated}$$

- The variance of the difference of uncorrelated random variables is the *sum* of the variances
- If a, b, c are non-random constants and X and Y are random variables then

$$\text{Var}(aX + bY + c) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X, Y)$$

Example 16.5. Suppose that SAT Math (M) and Reading (R) scores of CalPoly students have a Bivariate Normal distribution. Math scores have mean 640 and SD 80, Reading scores have mean 610 and SD 70, and the correlation between scores is 0.7.

1. Find the probability that a student has a total score above 1500.
 2. Find the probability that a student has a higher Math than Reading score.
-
- X and Y have a Bivariate Normal distribution if and only if every linear combination of X and Y has a Normal distribution. That is, X and Y have a Bivariate Normal distribution if and only if $aX + bY + c$ has a Normal distribution for all a, b, c .
 - In particular, if X and Y are independent and each has a Normal distribution then $aX + bY + c$ has a Normal distribution.

17 From Probability to Statistics

Example 17.1. In each of the following scenarios, one of (i) and (ii) is a *probability* and one is a *statistics* question. Classify each question as “probability” or “statistics”. Then discuss some general features of the probability questions, and some general features of the statistics questions

1. [U.S. Satisfaction](#)

- Assume that 30% of Americans are satisfied with the way things are going in the U.S. If a random sample of 1000 Americans is selected, what are the chances that more than 330 Americans in the sample are satisfied with the way things are going in the U.S?
- In a random sample of 1000 Americans, 330 Americans are satisfied with the way things are going in the U.S. What percent of all Americans are satisfied with the way things are going in the U.S?

2. Great white shark length

- Assume that lengths of female great white sharks follow a Normal distribution with mean 16 ft and standard deviation 3 ft. If a random sample of 40 female great white sharks is selected, what are the chances that the average length in the sample is between 16.5 and 17.5 ft?
- In a sample of 40 female great white sharks, the sample mean length is 17.0 ft. What is the average length of female great white sharks?

3. In a [clinical trial](#), 200 subjects are randomly assigned to receive either an experimental vaccine or no treatment (e.g., a placebo), and then each subject is tested to see if they developed immunity.

- If the vaccine is in reality not effective, what are the chances that the proportion of subjects who develop immunity is twice as large for those who received the vaccine than for those who did not?
- Among the 100 subjects who received the vaccine, 20 developed immunity; among the 100 subjects who did not receive the vaccine, 10 developed immunity. Is there evidence that the vaccine is effective?

4. Snap streaks

- Assume that lengths of streaks on Snapchat follow an Exponential distribution, with mean 75 for teen users and mean 40 for adult users. A random sample of 200 Snapchat users is selected, 100 teens and 100 adults. What are the chances that the sample mean streak length for teens is within 20 of the sample mean streak length for adults?
 - In a random sample of 100 teen Snapchat users, the sample mean streak length is 80. In a random sample of 100 adult Snapchat users, the sample mean streak length is 35. In general, how much longer do streak lengths for teen Snapchat users tend to be than for adult Snapchat users?
-
- **Probability:** Assume a model for a random (uncertain) process and evaluate probabilities of potential outcomes or events — what might the data look like?
 - In probability, a model is assumed and parameters are assumed to be known, but the data is unknown.
 - In probability, we typically call the model a “distribution” and the data “random variables”.
 - **Statistical inference:** Observe sample data and make conclusions about the process that generated the data
 - In statistics, the data is observed but parameters are unknown.
 - In statistics, we typically call the model the “population” and the data the “sample”.
 - *Statistical inference* involves using sample data to make conclusions about the population.
 - **Point estimation:** Estimate an unknown population parameter with a single number based on sample data
 - **Interval estimation:** Estimate an unknown population parameter with a plausible range of values based on sample data

- **Hypothesis testing:** Assess the evidence that the sample data provides regarding a particular claim about unknown population parameters.
- The *population distribution* is the distribution of individual values of the variable of interest. A *parameter*, generically denoted θ , is a number that summarizes the population distribution.
- The **population distribution**, denoted p_θ or $p_\theta(x)$, is a probability model for the distribution of values of a variable¹ X over all *individuals* in the population.
 - If the variable X is discrete, then p_θ represents a probability mass function (pmf): $p_\theta(x) = P(X = x)$
 - If the variable X is continuous, then p_θ represents a probability density function (pdf): $p_\theta(x) \approx \frac{1}{\epsilon}P(x - \epsilon/2 < X < x + \epsilon/2)$
- A **parameter**, generically denoted θ , is a characteristic of the population distribution, often computed as an expected value of some function of X .
 - In some cases θ represents a vector of multiple parameters.
 - For example, if p_θ represents a Normal distribution, then θ might represent the pair $\theta \equiv (\mu, \sigma)$ consisting of the population mean and the population standard deviation.
- A parameter is a *fixed number* but its value is *unknown*².
- Many statistical studies involve *random sampling* from a population.
- A **(simple) random sample** of size n is a collection of random variables $X_1, \dots, X_n$ that are *independent* and *identically distributed* (i.i.d.)
 - Identically distributed — each individual X_i has the same marginal distribution — the population distribution — since they are sampled from the same population:

$$X_i \sim p_\theta \quad \text{for all } i = 1, \dots, n$$

- Independent — the values are sampled independently, so the joint distribution³ is the product of marginal distributions:

$$p_\theta(x_1, \dots, x_n) = p_\theta(x_1)p_\theta(x_2) \cdots p_\theta(x_n) \quad \text{for all } x_1, \dots, x_n$$

¹Source: <http://www.nejm.org/doi/full/10.1056/NEJMoa013259>

²There are several ways to do this. The simplest is a *Bonferroni* adjustment, which splits the error rate evenly across all intervals. For example, for simultaneous 95% confidence in 10 intervals (i.e. a total error rate of 0.05), use the t^* factor for 99.5% confidence ($0.995 = 1 - 0.05/10$), $t^* = 2.8$, instead of 2 when computing the intervals. But the Tukey method is more widely used when comparing means.

³If the group sizes are not the same, you would compute a weighted average.

- A **statistic** is a characteristic of the *sample* which can be computed from the data. More precisely, a statistic is a function of $X_1, \dots, X_n$, but not of θ .
 - That is, a statistic is itself a *random variable*, and therefore has its own probability distribution, which describes how values of the statistic would vary from sample-to-sample over many (hypothetical) samples.
 - The probability distribution of a statistic is called a **sampling distribution**.
 - The *distribution* of a statistic will depend on θ , but the *value* of a statistic will not.
- A statistic $\hat{\theta} \equiv T(X_1, \dots, X_n)$ used to estimate a parameter θ is called a **(point) estimator**.
 - Naturally, we hope that there is some correspondence between our estimator $\hat{\theta}$ and the parameter it is estimating θ — for example, estimating the population mean with the sample mean — but the definition does not require this.
 - When the numerical value of a statistic is calculated from observed sample data, its value is called a *(point) estimate* of the parameter.
 - The estimator $\hat{\theta}$ is a RV whose value changes from sample to sample; an estimate is the observed value computed based on the results of a single sample.

Example 17.2. Jerry has just opened a car dealership in a new location. He assumes that the number of cars he'll sell in a day has a $\text{Poisson}(\mu)$ distribution. But he doesn't know the value of μ , so he plans to use data from his first few days of business to estimate μ . Let $X_1, \dots, X_n$ represent the number of cars sold for a sample of n days.

1. What plays the role of the population distribution? What is the parameter? How do you interpret the parameter in this context?
2. Suggest one estimator of μ . (Hint: what does μ represent for a Poisson distribution?)

3. Suppose he collects data on $n = 3$ days and observes $x_1 = 3$ cars sold on day 1, $x_2 = 0$ cars sold on day 2, and $x_3 = 2$ cars sold on day 3. Compute the value of your estimator from the previous part for this sample. That is, what is your estimate of μ based on this sample?

4. Suggest another estimator of μ . (Hint: what else does μ represent for a Poisson distribution?)

5. For a $\text{Poisson}(\mu)$ distribution μ is both the population mean and the population variance. So one reasonable estimator of μ is the sample mean $\bar{X}$. Another reasonable estimator is the sample variance. But there are two commonly used formulas for the sample variance, each of which provides an estimator of μ

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Compute the value of each of these estimators for the (3, 0, 2) sample. That is, what is your estimate of μ based on this sample if you use each of these estimators?

6. At his previous dealership at a different location, Jerry sold 2.3 cars per day on average. Consider the estimator of μ : $\frac{n}{n+100} \bar{X} + \frac{100}{n+100} (2.3)$. Explain in words what this estimator represents. In particular, what happens as n increases? Then compute the value of this estimator for the (3, 0, 2) sample.

7. According to the $\text{Poisson}(\mu)$ distribution, the probability of selling 0 cars in a day is

$p_0 = e^{-\mu}$. Rearranging, $\mu = -\log p_0$. Therefore, if $\hat{p}_0$ is the sample proportion of days with 0 cars sold, then another estimator of μ is $-\log \hat{p}_0$. Compute the value of this estimator for the (3, 0, 2) sample.

8. We have seen a few different, seemingly reasonable estimators of μ , and each produced a different estimate of μ based on the same (3, 0, 2) sample data. Which of these numbers is the best *estimate* of μ ? That is, which of these estimates is closest to μ ?

9. How might you investigate which of the *estimators* corresponds to the best estimation *procedure*?

- Because the true value of the parameter θ is unknown, we never know if the sample data provides a reasonable estimate of θ
- Rather, we evaluate the estimation *procedure* for potential values of θ : How does the estimation procedure perform over many possible samples given potential values of θ ?
- Some questions of interest when estimating a parameter θ
 - How do we find an estimator of θ ?
 - What are properties of “good” estimators?
 - How can we tell if one estimator is “better” than another?
 - Can we find a “best” estimator of θ ? (Short answer: no)
 - How do we determine the margin of error for an estimator?

18 Maximum Likelihood Estimation: Principles

Example 18.1. Suppose we are willing to assume a $\text{Poisson}(\mu)$ model for the number of cars sold in a day at a particular car dealership. But we don't know the value of μ , so we'll collect data and use it to estimate μ . Suppose first that we observe just a single day; let X be the number of cars sold on this day. (We'll look at a sample of n days soon.)

1. Does X take values on a discrete or continuous scale? What about μ ?
2. *First some probability questions.* Assume $\mu = 2.3$. Find the pmf of X and sketch a plot of it; what is this a function of? Find and interpret the probability that in a single day there are 3 cars sold.
3. *Now back to some statistics questions.* Now remember that in reality μ is unknown. Suppose that $x = 3$ home runs are observed in a single day. Now write the pmf plugging in everything we know; what is this a function of? Is this function a pmf?
4. Compute and interpret the probability that in a single day there are 3 cars sold if $\mu = 2.3$.

5. Compute and interpret the probability that in a single day there are 3 cars sold if $\mu = 1$.
6. Compute and interpret the probability that in a single day there are 3 cars sold if $\mu = 5$.
7. Compute and interpret the probability that in a single day there are 3 cars sold if $\mu = 3.5$.
8. Suppose that $x = 3$ cars are sold in a single day. Based just on this day, which value — 1, 2.3, 3.5, or 5 — would you choose as your estimate for μ ? Why?
9. We obviously have more choices for our estimate of μ than just 1, 2.3, 3.5, or 5. Describe in principle the process you would follow to find the estimate of μ based on a single day with 3 cars sold.
10. Suppose that we observe $x = 3$ cars sold in a single day. Plot the likelihood function; be sure to label your axes. Based on your plot, what would be your estimate of μ ?
11. Suppose that we observe $x = 3$ cars sold in a single day. Plot the log of the likelihood function. Compare the plots of the likelihood function and the log-likelihood; what do

you notice about where the maximum value occurs?

12. Now consider a general value of x cars sold in a single day. Carefully write the likelihood function, and the log-likelihood function. What are these functions of? How would you use them to determine your estimate of μ given x cars sold in a single day?

- Consider a single observation X from a population p_θ with parameter θ : Let $X \sim p_\theta$. The **likelihood function** given $X = x$ is defined as

$$L(\theta) \equiv L_x(\theta) = p_\theta(x)$$

- That is, the likelihood function is the probability (or density for continuous data X) of observing the given sample data x viewed as a *function of the unknown parameter(s) θ* . The observed value of x (data) is treated as a fixed constant.
- The *maximum likelihood estimate* of θ is the value of θ for which the observed data has the highest likelihood of occurring.
- The **maximum likelihood estimator (MLE)** of θ is the RV $\hat{\theta}$ which maximizes the likelihood, i.e. the value $\hat{\theta}$ which satisfies

$$L(\hat{\theta}) \geq L(\theta) \text{ for all } \theta.$$

- Remember that an estimator $\hat{\theta}$ is a function of X , the observed data.
- It is often convenient to consider the **log-likelihood function** of θ

$$\ell(\theta) \equiv \ell_x(\theta) = \log L_x(\theta)$$

- Because log is an increasing function it follows that $\hat{\theta}$ is the MLE of θ if and only if

$$\ell(\hat{\theta}) \geq \ell(\theta) \text{ for all } \theta.$$

Example 18.2. Suppose we are willing to assume a $\text{Poisson}(\mu)$ model for the number of cars sold in a day at a particular car dealership. But we don't know the value of μ , so we'll collect data and use it to estimate μ . But now we estimate μ based on a sample of n days. Let $X_1, \dots, X_n$ be the number of cars sold in a random sample of n days. For example, suppose $n = 3$ and we observe $x_1 = 3, x_2 = 0, x_3 = 2$.

1. *First some probability questions.* Assume $\mu = 2.3$ and $n = 3$. Find the joint pmf of (X_1, X_2, X_3) ; what is this a function of? Find and interpret the probability of observing the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$.
2. *Now back to some statistics questions.* Now remember that in reality μ is unknown. Suppose that the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$ is observed. Now write the joint pmf plugging in everything we know; what is this a function of? Is this function a pmf?
3. Compute the probability of observing the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$ if $\mu = 2.3$.
4. Compute the probability of observing the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$ if $\mu = 1$.
5. Compute the probability of observing the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$ if $\mu = 5$.
6. Compute the probability of observing the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$ if $\mu = 3.5$.
7. Suppose the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$ is observed. Which value — 1, 2.3, 3.5, or 5 — would you choose as your estimate for μ ? Why?

8. We obviously have more choices for our estimate of μ than just 1, 2.3, 3.5, or 5. Describe in principle the process you would follow to find the estimate of μ based on observing the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$.

 9. Suppose that we observe the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$. Plot the likelihood function; be sure to label your axes. Based on your plot, what would be your estimate of μ ?

 10. Suppose that we observe the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$. Plot the log of the likelihood function. Compare the plots of the likelihood function and the log-likelihood; what do you notice about where the maximum value occurs?

 11. Now consider a general sample of size n . Carefully write the likelihood function, and the log-likelihood function. What are these functions of? How would you use them to determine your estimate of μ given x cars sold in a single day?
-
- Consider a random sample of n values from a population p_θ with parameter θ : Let $X_1, \dots, X_n$ be i.i.d. $\sim p_\theta$.
 - The **likelihood function** of θ , given $X_1 = x_1, \dots, X_n = x_n$, is

$$L(\theta) \equiv L_{x_1, \dots, x_n}(\theta) = \prod_{i=1}^n p_\theta(x_i) = p_\theta(x_1) \times \dots \times p_\theta(x_n)$$

- That is, the likelihood function is the joint density¹ of the sample $X_1, \dots, X_n$, evaluated at the observed data values $x_1, \dots, x_n$, viewed as *a function of the (unknown) parameter(s) θ* .
- Note that $x_1, \dots, x_n$ represent distinct values (like $x_1 = 3, x_2 = 0, x_3 = 2$ in the example).
- The MLE of θ is the value of θ for which the observed sample data has the highest likelihood of occurring.
- The **maximum likelihood estimator (MLE)** of θ is the RV $\hat{\theta}$ which maximizes the likelihood, i.e. the value $\hat{\theta}$ which satisfies

$$L(\hat{\theta}) \geq L(\theta) \text{ for all } \theta.$$

- Remember that an estimator $\hat{\theta}$ is a function of $X_1, \dots, X_n$.
- The **log-likelihood function** of θ is

$$\begin{aligned} \ell(\theta) &\equiv \ell_{x_1, \dots, x_n}(\theta) = \log L(\theta) \\ &= \sum_{i=1}^n \log p_{\theta}(x_i) = \log p_{\theta}(x_1) + \dots + \log p_{\theta}(x_n) \end{aligned}$$

- Because log is an increasing function it follows that $\hat{\theta}$ is the MLE of θ if and only if

$$\ell(\hat{\theta}) \geq \ell(\theta) \text{ for all } \theta.$$

Example 18.3. Suppose instead of observing the number of cars sold on individual days, you only knew that in a sample of 3 days there were 5 cars sold in total.

1. *Probability question.* If $X_1, \dots, X_n$ are i.i.d. $\text{Poisson}(\mu)$, what is the distribution of $X_1 + \dots + X_n$?
2. Write the likelihood function for a sample of 3 days in which a total of 5 cars were sold. How does this compare to the likelihood function based on the sample $(3, 0, 2)$?
3. Find the MLE of μ based on a sample of 3 days in which a total of 5 cars are sold.

¹Source: <http://www.nejm.org/doi/full/10.1056/NEJMoa013259>

- In many situations, the full sample data is not necessary to find the MLE; rather, a few descriptive statistics are often sufficient to evaluate the likelihood and determine the MLE
- Poisson(μ): the MLE of μ is the sample mean $\bar{X}$
- Binomial(n, p), with n known: the MLE of p is the sample proportion $\hat{p} = X/n$, where X is the number of “successes” in the sample
- Exponential(λ): the MLE of λ is the sample rate $1/\bar{X}$; the usual situation is where the X ’s measure times between “events” and so $\bar{X}$ is the sample mean time between “events”.
- Normal(μ, σ), with both μ and σ unknown: the MLEs of μ , σ^2 , and σ , are, respectively

$$\hat{\mu} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$\hat{\sigma} = \sqrt{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}$$

Example 18.4. Assume a Poisson(μ) model for the number of cars sold in a day at a particular car dealership. In 3 days, 5 cars are sold.

1. Consider $\sqrt{\mu}$. Is $\sqrt{\mu}$ a parameter or a statistic? How would you interpret $\sqrt{\mu}$ in this context?
2. What do you think the MLE of $\sqrt{\mu}$ is?
3. Consider $e^{-\mu}$. Is $e^{-\mu}$ a parameter or a statistic? How would you interpret $e^{-\mu}$ in this context?

4. What do you think the MLE of $e^{-\mu}$ is?
5. Consider $\mu e^{-\mu}$. Is $\mu e^{-\mu}$ a parameter or a statistic? How would you interpret $\mu e^{-\mu}$ in this context?
6. What do you think the MLE of $\mu e^{-\mu}$ is?
7. Consider $\mu^2 e^{-\mu}/2$. Is $\mu^2 e^{-\mu}/2$ a parameter or a statistic? How would you interpret $\mu^2 e^{-\mu}/2$ in this context?
8. What do you think the MLE of $\mu^2 e^{-\mu}/2$ is?
9. We started by assuming a Poisson model, but how do we know that is a reasonable assumption? Suppose we observe the number of cars sold for a random sample of days. Suggest you might use the sample data to investigate whether a Poisson model is appropriate.

- A parameter is any characteristic of a population distribution
- For many populations, there are a few “main” parameters (like μ for $\text{Poisson}(\mu)$, or (μ, σ) for $\text{Normal}(\mu, \sigma)$), but any characteristic of a population is parameter and can be estimated
- **Invariance property of MLEs.** If $\hat{\theta}$ is the MLE of θ , then $g(\hat{\theta})$ is the MLE of $g(\theta)$.
- That is, the “MLE of a function is the function of the MLE”
- The invariance principle says that if you find the MLE of the “main parameters” then you automatically get the MLE of all the parameters that are based on it

Example 18.5. Suppose that within a certain population, credit scores follow a $\text{Normal}(\mu, \sigma)$ distribution. In a random sample of 5 individuals from this population, the credit scores are 600, 630, 680, 700, 770.

1. Compute the MLE of μ .
2. Compute the MLE of σ .
3. Explain in words what it means for these parameters to be the MLE.
4. Compute the MLE of the 16th percentile of credit scores for this population
5. Compute the MLE of the 90th percentile of credit scores for this population

6. Compute of the MLE of the population proportion of individuals with a credit score above 750.

- Maximum likelihood estimation procedures are widely applicable and maximum likelihood estimators have many nice theoretical properties.
- However, there are some drawbacks:
 - May require numerical maximization, which can be computationally intensive (especially if many parameters)
 - May not be reliable estimators in small samples. (Many of the nice theoretical properties of MLEs are true for large samples.)

19 Maximum Likelihood Estimation: Calculus

Example 19.1. We wish to estimate the parameter μ for a $\text{Poisson}(\mu)$ distribution based on a single observed value X .

1. Suppose that $x = 3$. Carefully write the likelihood function.
2. Suppose that $x = 3$. Carefully write the log-likelihood function.
3. Use calculus to find the MLE of μ given $x = 3$. Compare to what we found previously using graphical methods.
4. Now consider a general x . Carefully write the likelihood function.
5. Now consider a general x . Carefully write the log-likelihood function.

6. Use calculus to find the MLE of μ given x . Compare to what we found previously using graphical methods.

Example 19.2. We wish to estimate the parameter μ for a $\text{Poisson}(\mu)$ distribution based on a random sample of n values $X_1 \dots, X_n$.

1. Suppose that $n = 3$ and the sample is $(3, 0, 2)$. Carefully write the likelihood function.
2. Suppose that $n = 3$ and the sample is $(3, 0, 2)$. Carefully write the log-likelihood function.
3. Use calculus to find the MLE of μ given $n = 3$ and the sample $(3, 0, 2)$. Compare to what we found previously using graphical methods.
4. Now consider a general sample of size n . Carefully write the likelihood function.
5. Now consider a general sample of size n . Carefully write the log-likelihood function.

6. Use calculus to find the MLE of μ given a sample of size n . Compare to what we found previously using graphical methods.

20 Bias of Estimators

- MLE is one procedure for finding an estimator of a parameter θ . But when several estimators of θ are available, how do we decide which is “better”?
- The value of a parameter θ is unknown, so it is impossible to determine if any single estimate of θ is good or bad.
- An estimator is a random variable that returns different estimates of θ for different samples. So even if an estimator produces “good” estimates of θ for some samples, it might produce “bad” estimates of θ for others.
- Therefore, we can never determine for any particular sample if an estimator produces a good estimate of the true unknown value of θ .
- Rather, we evaluate the estimation *procedure*: Over many samples, does an estimator tend to produce reasonable estimates of θ for a variety of potential values of θ ?
- Remember: a statistic/estimator is a random variable whose sampling distribution describes how values of the estimator vary from sample-to-sample over many (hypothetical) samples.
 - We can estimate the degree of this variability by *simulating many hypothetical samples* from an assumed population distribution and computing the value of the statistic for each sample.
 - However, *in practice usually only a single sample is selected* and a single value of the statistic is observed, and we don’t know what the population distribution is.

Example 20.1. Consider a large population in which 20% of individuals have an annual income (\$ thousands) of 200, 40% of have an income of 70, and 40% have an income of 10. For this population, the population mean is

$$\mu = 10(0.4) + 70(0.4) + 200(0.2) = 72$$

Also, the population variance is

$$[10^2(0.4) + 70^2(0.4) + 200^2(0.2)] - 72^2 = 4816$$

and the the population SD is $\sigma = 69.4$.

Suppose our goal is to estimate σ based on a sample of size n . We’ll start by estimating the population variance σ^2 . In this example, we know the population distribution and we know the population variance $\sigma^2 = 4816$, but in practice the population distribution and the population variance would be unknown.

One estimator of the population variance σ^2 is

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

1. For $n = 2$, describe in detail how you would use simulation to approximate the distribution of $\hat{\sigma}^2$, and its expected value.
2. For $n = 2$ determine the distribution of $\hat{\sigma}^2$. (This is not a simulation; use ideas from the first half of the course.)
3. Find and interpret $P(\hat{\sigma}^2 = 0)$.
4. Compute and interpret $E(\hat{\sigma}^2)$. Compare $E(\hat{\sigma}^2)$ to σ^2 ; what does this say?

- $\hat{\theta}$ is an **unbiased** estimator of θ if

$$E(\hat{\theta}) = \theta \quad \text{for all potential values of } \theta$$

- Roughly speaking, an unbiased estimator does not tend to systematically overestimate or underestimate the parameter of interest, regardless of the true value of the parameter.
- The **bias (function)** of an estimator is the difference between its expected value (over many hypothetical samples) and the true (but unknown) value of the parameter

$$\text{bias}_{\theta}(\hat{\theta}) = E(\hat{\theta} - \theta) = E(\hat{\theta}) - \theta, \quad \text{a function of } \theta$$

Example 20.2. Continuing Example 20.1. Another commonly used definition of the sample variance is

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

1. For $n = 2$, determine the distribution of S^2 .
2. Compute and interpret $E(S^2)$. Compare $E(S^2)$ to σ^2 ; what does this say?
3. The *sample standard deviation* is commonly defined as $S = \sqrt{S^2}$, that is,

$$S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$$

For $n = 2$, determine the distribution of S .

4. Compute and interpret $E(S)$. Compare $E(S)$ to σ ; what does this say?

- For a random sample $X_1, \dots, X_n$, the **sample variance** is commonly defined as

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

- The **sample standard deviation** is $S = \sqrt{S^2}$.

$$S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$$

- The sample variance S^2 is an *unbiased* estimator of the population variance σ^2

$$E(S^2) = \sigma^2$$

- However, the sample SD S is a *biased* estimator of the population SD σ

$$E(S) < \sigma$$

- Though the bias decreases as the sample size increases

Example 20.3. Let $X_1, \dots, X_n$ be an (i.i.d.) random sample from a population with population mean μ . Let $\bar{X} = (X_1 + \dots + X_n)/n$ be the sample mean. It can be shown that

$$E(\bar{X}) = \mu$$

There are three means above; explain what all the means mean, and what the equation says.

- The **sample mean** is

$$\bar{X}_n = \frac{X_1 + \dots + X_n}{n}$$

- The sample mean $\bar{X}$ is an *unbiased* estimator of the population mean μ .

$$E(\bar{X}_n) = \mu$$

- Over many random samples, sample means do not systematically overestimate or underestimate the population mean.

Example 20.4. Recall that in the car dealership problem we were trying to estimate μ for a Poisson(μ) distribution. We considered a few different estimators of μ . Determine which of the following estimators are unbiased estimators of μ based on a random sample $X_1, \dots, X_n$ of size n from a Poisson(μ) distribution. If the estimator is not unbiased, identify its bias function.

1. $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$.

2. $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$

3. $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$. (Hint: $\hat{\sigma}^2 = \frac{n-1}{n} S^2$.)

4. $\frac{n}{n+100} \bar{X} + \frac{100}{n+100} (2.3)$. (Recall this estimator was like a weighted average of the mean from the old dealership and the sample mean for the new dealership.)

5. 2.3. (This estimator ignores the sample data from the new dealership entirely and just uses the mean from the old dealership as the estimator.)

Example 20.5. Continuing the Poisson(μ) problem. Suppose $n = 3$. Imagine we had not already derived the bias function of the estimator $\frac{n}{n+100} \bar{X} + \frac{100}{n+100} (2.3)$

1. Describe in detail how you would use simulation to approximate the bias of $\frac{n}{n+100} \bar{X} + \frac{100}{n+100} (2.3)$ when $\mu = 2.3$.

2. Describe in detail how you would use simulation to approximate the bias *function* of $\frac{n}{n+100} \bar{X} + \frac{100}{n+100} (2.3)$.

Example 20.6. Continuing the Poisson(μ) problem. Let $\theta = e^{-\mu}$. We know $\bar{X}$ is an unbiased estimator of μ . We also know that $e^{-\bar{X}}$ is the MLE of $e^{-\mu}$. Is $\hat{\theta} = e^{-\bar{X}}$ an unbiased estimator of $\theta = e^{-\mu}$?

- In general, if $\hat{\theta}$ is an unbiased estimator of θ , $g(\hat{\theta})$ is usually NOT an unbiased estimator of $g(\theta)$ (unless g is linear).
 - Because in general $E(g(\hat{\theta})) \neq g(E(\hat{\theta}))$.

Example 20.7. Continuing the Poisson(μ) problem. Consider the estimator $\hat{\sigma}^2$. Recall that the bias function is $\text{bias}_{\mu}(\hat{\sigma}^2) = -\frac{\mu}{n}$. What happens to the bias function as $n \rightarrow \infty$. What does this mean?

- An estimator $\hat{\theta}_n$ is an **asymptotically unbiased** estimator of θ if $E(\hat{\theta}_n) \rightarrow \theta$ as $n \rightarrow \infty$ for all θ , that is, if $\text{bias}_{\theta}(\hat{\theta}_n) \rightarrow 0$ as $n \rightarrow \infty$ for all θ .
- For an asymptotically unbiased estimator, if the sample size is large $E(\hat{\theta}) \approx \theta$, regardless of the true value of θ . That is, if the sample size is large the estimator is “nearly unbiased”.
- MLEs are often biased, but in general, MLEs are asymptotically unbiased

21 Variance of Estimators

- The bias function of an estimator measures the estimator's tendency to overestimate or underestimate the parameter, as a function of potential values of the parameter.
- But bias is only one consideration. Bias only considers the average value of the estimator over many samples, which is just one feature of the estimator's sampling distribution.

Example 21.1. Consider a large population in which 20% of individuals have an annual income (\$ thousands) of 200, 40% of have an income of 70, and 40% have an income of 10. For this population, the population mean is

$$\mu = 10(0.4) + 70(0.4) + 200(0.2) = 72$$

Also, the population variance is

$$[10^2(0.4) + 70^2(0.4) + 200^2(0.2)] - 72^2 = 4816$$

and the the population SD is $\sigma = 69.4$.

Suppose our goal is to estimate μ based on a sample of size n . In this example, we know the population distribution and we know the population mean $\mu = 72$, but in practice the population distribution and the population mean would be unknown.

One estimator of the population mean μ is

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

First suppose $n = 2$.

1. Determine the distribution of $\bar{X}$.

2. Find and interpret $P(\bar{X} \geq 105)$.

3. Compute $E(\bar{X})$. How does it relate to μ ? Why?

4. Compute and interpret $\text{Var}(\bar{X})$. How does it relate to the population variance σ^2 ?

5. Compute and interpret $\text{SD}(\bar{X})$. How does it relate to the population standard deviation σ ?

 - A **(simple) random sample** of size n is a collection of random variables $X_1, \dots, X_n$ that are *independent* and *identically distributed* (i.i.d.)
 - A **statistic** is a characteristic of the *sample* which can be computed from the data. More precisely, a statistic is a function of $X_1, \dots, X_n$, but not of θ .
 - That is, a statistic is itself a *random variable*, and therefore has its own probability distribution, which describes how values of the statistic would vary from sample-to-sample over many (hypothetical) samples.
 - The probability distribution of a statistic is called a **sampling distribution**.
 - Statistics exhibit **sample-to-sample variability**: the value of a statistic varies from sample to sample.
 - We can estimate the degree of this variability by *simulating many hypothetical samples* and computing the value of the statistic for each sample.
 - The resulting standard deviation, called the **standard error**, measures the sample-to-sample variability of the statistic over many (hypothetical) samples of the same size.

- However, *in practice usually only a single sample is selected* and a single value of the statistic is observed.
- For many statistics (e.g., means and proportions) — but not all — statistics from larger random samples vary less, from sample to sample, than statistics from smaller random samples.

Example 21.2. Continuing Example 21.1, now consider $n = 8$.

1. How could you use simulation to approximate the distribution of $\bar{X}$, and its expected value and SD?
 2. Find and interpret $E(\bar{X})$ without first finding the distribution of $\bar{X}$, and compare to the simulation results.
 3. Find and interpret $SD(\bar{X})$ without first finding the distribution of $\bar{X}$, and compare to the simulation results. What does this measure variability of? Compare to $n = 2$.
 4. Use simulation to find and interpret $P(\bar{X} \geq 105)$. Compare to $n = 2$.
- For an i.i.d. random sample $X_1, \dots, X_n$, the **sample mean** is the *statistic*

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i = \frac{X_1 + \dots + X_n}{n}$$

- The sample mean $\bar{X}$ is an *unbiased* estimator of the population mean μ :

$$E(\bar{X}_n) = \mu$$

- Over many random samples, sample means do not consistently overestimate or consistently underestimate the population mean
- Sample-to-sample variability of sample means decreases as the sample size increases

$$\text{SD}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}$$

$$\text{sample-to-sample SD of sample means} = \frac{\text{unit-to-unit SD of values of the variable}}{\sqrt{\text{sample size}}}$$

- The more variable the individual values of the variable are (i.e. the larger the population SD of the variable, σ), the more variable sample means will be (i.e. the larger than the SE).
- Sample means are less variable than individual values of the variable
- Over many random samples, sample means from larger random samples vary less, from sample to sample, than sample means from smaller random samples.
- The SD of the sample means decreases as the sample size increases, but this reduction in SD follows a “square root rule”. For example, if the sample size is increased by a factor of 4, then the SD is reduced by a factor of 2 (not 4).

Example 21.3. Consider again estimating μ for a Poisson(μ) distribution based on a random sample $X_1, \dots, X_n$ of size n . We have seen that both $\bar{X}$ and S^2 are unbiased estimators of μ . If we want to choose between these two estimators, how do we decide?

1. Assume $n = 3$ and $\mu = 2.3$. Describe in full detail how you could conduct a simulation to approximate the sample-to-sample distribution of $\bar{X}$ and its expected value and standard deviation. Then conduct the simulation and record the results. What does the standard deviation measure?
2. Assume $n = 3$ and $\mu = 2.3$. Describe in full detail how you could conduct a simulation to approximate the sample-to-sample distribution of S^2 and its expected value and standard deviation. Then conduct the simulation and record the results. What does the standard deviation measure?

3. Compare the simulation results for $\bar{X}$ and S^2 when $n = 3$ and $\mu = 2.3$. Based on the simulation results, which estimator of μ is preferred when $\mu = 2.3$ — $\bar{X}$ or S^2 ? Why? But then explain why this information isn't very helpful.
4. Assuming $n = 3$ and $\mu = 2.3$, compute $E(\bar{X})$ and $SD(\bar{X})$, and compare to the simulation results.
5. For a general n and $\mu > 0$, find expressions for $E(\bar{X})$ and $SD(\bar{X})$.
6. It can be shown that

$$\text{Var}(S^2) = \frac{2\mu^2}{n-1} + \frac{\mu}{n}$$

Sketch a plot of the variance functions of both $\bar{X}$ and S^2 as a function of μ . Regardless of the value of μ , which estimator has smaller variance? Which of these two unbiased estimators of μ , $\bar{X}$ or S^2 , is preferred?

- When choosing between two *unbiased* estimators, the one with smaller variance is generally preferred.
- Remember that the variance of an estimator will be a function of the unknown parameter θ , so we need to consider the variance *function* for various potential values of θ .

22 Mean Square Error of an Estimator

- MLE is one procedure for finding an estimator of a parameter θ . But when several estimators of θ are available, how do we decide which is “better”?
- The value of a parameter θ is unknown, so it is impossible to determine if any single estimate of θ is good or bad.
- An estimator is a random variable that returns different estimates of θ for different samples. So even if an estimator produces “good” estimates of θ for some samples, it might produce “bad” estimates of θ for others.
- Therefore, we can never determine for any particular sample if an estimator produces a good estimate of the true unknown value of θ .
- Rather, we evaluate the estimation *procedure*: Over many samples, does an estimator tend to produce reasonable estimates of θ for a variety of potential values of θ ?
- Remember: a statistic/estimator is a random variable whose sampling distribution distribution describes how values of the estimator vary from sample-to-sample over many (hypothetical) samples.
 - We can estimate the degree of this variability by *simulating many hypothetical samples* from an assumed population distribution and computing the value of the statistic for each sample.
 - However, *in practice usually only a single sample is selected* and a single value of the statistic is observed, and we don’t know what the population distribution is.

Example 22.1. Continuing the situation of estimating μ for a Poisson(μ) distribution based on a random sample $X_1, \dots, X_n$ of size n . Consider the estimators

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$
$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \left(\frac{n-1}{n}\right) S^2$$

We have seen that S^2 is an unbiased estimator of μ but $\hat{\sigma}^2$ is not. Does that mean we should prefer S^2 over $\hat{\sigma}^2$?

1. Assume $n = 3$ and $\mu = 2.3$. Use simulation to approximate the distribution of $\hat{\sigma}^2$ and its expected value and standard deviation.

2. Compare the simulation results to those for S^2 (from a previous handout.) For which estimator is the bias smaller? For which estimator is the variance smaller? Is there a clear preference between the two estimators when $\mu = 2.3$?

- Estimators can be evaluated based on both bias and variability.
- Mean square error is a combined measure of both an estimator's bias and its variability.
- The **mean square error (MSE)** of an estimator $\hat{\theta}$ of a parameter θ is

$$\text{MSE}_{\theta}(\hat{\theta}) = \text{E} \left((\hat{\theta} - \theta)^2 \right), \quad \text{a function of } \theta$$

- Mean square error measures, on average, how far the estimator deviates from the parameter it is estimating.
- The MSE of an estimator is a function of the parameter θ . It can be shown that¹

$$\begin{aligned} \text{MSE}_{\theta}(\hat{\theta}) &= \text{Var}(\hat{\theta}) + (\text{E}(\hat{\theta}) - \theta)^2 \\ &= \text{Var}(\hat{\theta}) + (\text{bias}_{\theta}(\hat{\theta}))^2 \end{aligned}$$

- There can be many “reasonable” estimators of a parameter. Given two estimators, it's often the situation that neither one has smaller MSE for all potential values of θ . Choosing an estimator often involves a tradeoff between bias and variability.
3. Use the simulation results to compute the MSE of S^2 and $\hat{\sigma}^2$ when $\mu = 2.3$. Determine which estimator has smaller MSE when $\mu = 2.3$, but then explain why this information is not very useful.

¹Source: <http://www.nejm.org/doi/full/10.1056/NEJMoa013259>

4. Recall

$$\text{Var}(S^2) = \frac{2\mu^2}{n-1} + \frac{\mu}{n}$$

Find and plot the MSE functions of S^2 and $\hat{\sigma}^2$. Which estimator is preferred? Discuss.

Example 22.2. Continuing estimating μ for a $\text{Poisson}(\mu)$ distribution. Consider $\bar{X}$ and the constant estimator 2.3.

1. Find the MSE of the constant estimator 2.3.
2. Find the MSE of $\bar{X}$.
3. Suppose $n = 3$. Plot the MSE functions of the two estimators. Does either estimator have a better MSE? Explain.
4. Donny Don't says: "neither estimator has smaller MSE for all values of μ . So we'll use the constant estimator 2.3 when μ is near 2.3 (between 1.57 and 3.36 if $n = 3$) and we'll use $\bar{X}$ for other values of μ ". Do you agree with Donny's strategy? Do you see any problems with it?
5. What happens to the MSEs as n increases? In particular, what happens to the range of values of μ for which the constant estimator 2.3 has smaller MSE than $\bar{X}$?

- Given a parameter θ , it is never possible to find a single estimator that has the smallest MSE for all potential values of θ .
- Choosing an estimator often involves a tradeoff between bias and variability.
- MLEs generally have good properties. For many situations, roughly,
 - The MLE is asymptotically unbiased; that is, the bias is small when the sample size is large
 - The variance of an MLE is about as small as it can be (for an unbiased estimator)
- But there are many reasonable estimation procedures besides MLE (e.g., Bayes estimators).
- Bias in estimation is not necessarily bad. Being willing to accept a little bit of bias can often lead to a beneficial reduction in variability.
 - On the other hand, try to minimize bias in *data collection*: how the sample is selected, how the variables are measured, etc
 - A lot of the theory assumes we have a perfect random sample, which is never true in practice

23 Central Limit Theorem

- A **statistic** is a characteristic of the *sample* which can be computed from the data.
 - That is, a statistic is itself a *random variable*, and therefore has its own probability distribution, which describes how values of the statistic would vary from sample-to-sample over many (hypothetical) samples.
 - The probability distribution of a statistic is called a **sampling distribution**.
- Statistics exhibit **sample-to-sample variability**: the value of a statistic varies from sample to sample.
 - We can estimate the degree of this variability by *simulating many hypothetical samples* and computing the value of the statistic for each sample.
 - The resulting standard deviation, called the **standard error**, measures the sample-to-sample variability of the statistic over many (hypothetical) samples of the same size.
 - However, *in practice usually only a single sample is selected* and a single value of the statistic is observed.
- For many statistics (e.g., proportions, means, differences in means) — but not all —
 - Statistics from larger random samples vary less, from sample to sample, than statistics from smaller random samples.
 - The pattern of sample-to-sample variability of the statistic follows, approximately, a Normal distribution

Example 23.1. In a previous example, we consider a discrete population with a population mean of $\mu = 72$ and a population standard deviation of $\sigma = 69.4$. Now we'll consider several other populations with a population mean of 72 and a population standard deviation of 69.4. We'll use an [applet](#) to simulate the sampling distribution of the sample mean $\bar{X}$ for samples of size n from several different populations. Enter 72 in the box for population mean, and 69.4 in the box for population SD.

1. Regardless of the shape of the population, what will $E(\bar{X})$ be? How we do interpret this?

2. Regardless of the shape of the population, what will $SD(\bar{X})$ be? How do we interpret this?

3. Choose a super-small sample size, like $n = 2$. Choose a population shape, and then run the simulation to generate many values of the sample mean. Then repeat for different population shapes. When the sample size is small, does the *shape* of the sample-to-sample distribution of sample means (plot on right) depend on the shape of the population (plot on the left)?

4. Now increase the sample size, and repeat for the several populations. What happens to the *shape* of the sample-to-sample distribution of sample means as the sample size increases?

5. Now choose a not-super-small sample size, like $n = 500$. Choose a population, and simulate *a single sample*. Look at the simulated sample (the middle plot), then repeat to simulate a few samples. Then repeat for the different populations. Does the *distribution of individual values within the sample* depend on the shape of the population?

6. When the sample size is large, does the *sample-to-sample distribution of sample means* depend on the shape of the population?

- For a representative random sample, the *distribution of individual values within the observed sample* should resemble the population distribution
 - Regardless of the sample size
- For many statistics (e.g., proportions, means, differences of means) — but not all — if the sample size is large enough the pattern of *sample-to-sample variability of values of the statistic* follows a Normal distribution
 - Regardless of the shape of the population distribution of individual values of the variable.
- The **Central Limit Theorem (CLT)** says that if n is large enough,

$\bar{X}_n$ has an approximate $N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$ distribution.

- The CLT says that if the sample size is large enough, the sample-to-sample distribution of sample means is approximately Normal, *regardless of the shape of the population distribution*.
- The above CLT is for a single population mean, but similar results apply for other statistics like sample proportions, differences in two sample means, differences in two sample proportions.

Example 23.2. Recall the population where every individual has an income of 10, 70, or 200 with probability 0.4, 0.4, 0.2. Now suppose that every individual has an income of 10, 70, 200, or 2000 with probability 0.4, 0.4, 0.19, 0.01. Simulate sample means for many samples of size $n = 30$ from this population. Is the distribution of sample means approximately Normal? What if $n = 100$?

- How large a sample size is required for normality of the sampling distribution of sample means does depend on the shape of the population distribution of individual values.
 - If the population distribution is Normal, then the sample-to-sample distribution of sample means is Normal for any sample size.
 - If the population distribution is “close to Normal” — e.g. symmetric, light tails — then smaller samples sizes are sufficient for the sample-to-sample distribution of sample means to be approximately Normal.
 - If the population distribution is “far from Normal” — e.g. severe skewness, heavy tails, extreme outliers — then larger sample sizes are required for the sample-to-sample distribution of sample means to be approximately Normal.

- You should certainly be aware that some population distributions require a large sample size for the CLT to kick in. However, in many situations the sample-to-sample distributions of sample means (or other statistics) is approximately Normal even for small or moderate sample sizes.

24 Confidence Intervals

- The estimators we have considered so far are *point* estimators; we estimate the population parameter θ , an unknown number, by the observed sample statistic $\hat{\theta}$, a number which can be computed based on sample data $X_1, \dots, X_n$
- For a particular sample a particular numerical value of $\hat{\theta}$ is observed, but there are many possible randomly selected samples and many possible values of the random variable $\hat{\theta}$.
- While for any particular sample we hope the observed value of $\hat{\theta}$ is close to the true value of θ , there will be some error in estimation due to natural sample-to-sample variability.
- Because of this natural sample-to-sample variability, it is usually better to provide a range of values when estimating a parameter
- In determining a plausible range of values, there is a trade-off between
 - Accuracy: the wider the interval the more confident we are that it actually contains the true parameter value
 - Precision: but the wider the interval the less precise of an estimate it provides
- The usual approach is to specify a desired degree of accuracy. Specifically, interval estimate procedures are constructed so that they will be accurate “most of the time”.
- A *confidence interval* estimates a population parameter based on sample data with
 - a plausible range of values and
 - a *confidence level* that the estimated range contains the true parameter value.
 - * By far, the most commonly used confidence level is 95%.
 - * But see Example 24.5 and related material about multiple confidence intervals
- A $1 - \alpha$ **confidence interval** for θ is an *interval estimator* of θ based on the sample $X_1, \dots, X_n$ of the form $[L(X_1, \dots, X_n), U(X_1, \dots, X_n)]$ which satisfies

$$P(L(X_1, \dots, X_n) \leq \theta \leq U(X_1, \dots, X_n)) = 1 - \alpha$$

- $L(X_1, \dots, X_n)$ and $U(X_1, \dots, X_n)$ represent, respectively, the lower and upper endpoints of the interval. L and U are *random variables* that vary from sample-to-sample.
- Typical values of α include 0.05 and 0.01. We express $1 - \alpha$ as a percentage. For example, $\alpha = 0.05$ corresponds to a 95% confidence interval.
- Often $L = \hat{\theta} - M$ and $U = \hat{\theta} + M$, where the **margin of error** M — where M can be a RV — accounts for both the sample-to-sample variability of $\hat{\theta}$ and the confidence level.
 - The margin of error is often of the form: multiple of standard error

- The multiple is determined by the confidence level and the distribution of the point estimator $\hat{\theta}$ (e.g., 2 for 95% confidence when the estimator follows a Normal distribution.)
 - * The larger the confidence level, the larger the margin of error — and the wider the range of plausible values — required to achieve that level of confidence.
- The **standard error** is the estimated sample-to-sample SD of the estimator, $SE(\hat{\theta}) = \widehat{SD}(\hat{\theta})$.
 - * The less variable the statistic (from sample to sample) the smaller the margin of error and the narrower the interval.
 - * In general, the larger the sample size, the smaller the SE of the statistic, the smaller the margin of error, and the narrower the interval.

Example 24.1. Consider the general problem of estimating a population mean μ based on a random sample of size n . A natural point estimator of μ is the sample mean $\bar{X}$. The central limit theorem says that, if n is large enough, $\bar{X}$ has an approximate Normal($\mu, \sigma/\sqrt{n}$) distribution, where σ is the population standard deviation. We'll use this theory to derive a confidence interval for μ .

1. Fill in the blank: for 95% of samples, the sample mean $\bar{X}$ is within [blank] of the population mean μ .

2. Fill in the blank: for 95% of samples, the interval with endpoints $\bar{X} \pm$ [blank] will contain the population mean μ .

3. The previous part suggests a formula for a CI for μ . However, this is not a practically useful formula; explain why. Then suggest how we can fix it.

4. Everything else being equal, how does the sample size affect the confidence interval and its margin of error: the larger the sample size...? Does this make sense?

5. Everything else being equal, how does the confidence level (e.g. 95%) affect the confidence interval and its margin of error: the larger the confidence level...? Does this make sense?

- A one-sample t confidence interval for a population mean μ has endpoints

$$\bar{X} \pm t^* \frac{S}{\sqrt{n}}$$

where S is the sample standard deviation $S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$ and t^* , the multiple corresponding to the confidence level, is from the t distribution with $n - 1$ degrees of freedom.

- For all practical purposes you can treat t distributions like Normal distributions and use the usual empirical rule
 - Technically, t^* is a little bigger than z^* to account for the additional sample-to-sample variability in S .
 - But in most situations encountered in practice, t^* and z^* are essentially the same.
- Assumptions:
 - Simple random sample from a large population, that is, independent observations
 - No bias in data collection
 - Population mean is the parameter you want to estimate

Confidence level	90%	95%	99%	99.7%	99.99%
Multiple ($z^* \approx t^*$)	1.7	2.0	2.6	3.0	4.0

Example 24.2. A frequently used example data set is the [Ames housing data set](#) which consists of residential properties sold in Ames, Iowa from 2006 to 2010. For more information about the variables in this data set, refer to the [data documentation](#).

Suppose we wish to use this data set to estimate μ , the *population mean sale price of single family homes*.

In the sample, there are 2425 single family homes with a sample mean sale price (\$K) of 185 and sample standard deviation 83.

1. Compute a 95% confidence interval.
2. Write a clearly worded sentence containing the conclusion of the confidence interval in context.
3. To what extent can you generalize your results? That is, what is the relevant “population”? What are some issues to consider?
4. Assuming that this is a random sample from the appropriate population, does the 95% confidence interval you computed in the previous part provide an accurate estimate of the corresponding parameter? That is, is μ contained within the CI? Explain.
5. Assuming that this is a random sample from the appropriate population, are we reasonably confident that μ is greater than 180K? Explain.
6. What if we wanted 95% confidence, but we wanted the margin of error to be no bigger than 1K. How could we achieve this?

- Because the value of the parameter is *unknown*, we never know for sure whether any one confidence interval contains the parameter or not.
- But the procedure guarantees that in the long run — that is, over many unbiased random samples — “most” confidence intervals will contain the parameter, regardless of what the true value of the parameter is. “Most” is determined by the level of confidence.
- Any value within the confidence interval is a plausible value of the parameter
- Values outside of the confidence interval are not plausible values of the parameter
- The CI could be wrong; we are confident that it is correct, but we’ll never know for sure
- There are two sources of differences between a statistic and a corresponding parameter
 - systematic errors due to **bias**
 - “chance errors” due to **sampling variability**
- While there is no bias inherent in simple random sampling from a population, bias can be present in other ways, including undercoverage, nonresponse, wording of questions, etc.
- Bias concerns the *center* of the sampling distribution of a statistic. A biased sampling procedure produces values which consistently overestimate or consistently underestimate the respective parameter.
- The inference procedures — confidence intervals and hypothesis tests — we will encounter assume that the data are collected in an unbiased way from a random sample.
- If there is bias present in data collection then the observed statistic (i.e. the center of the confidence interval) is likely to be a poor estimate of the parameter.
- The margin of error in a confidence interval only accounts for sampling variability, NOT bias.
- To see if bias is present — in which case the confidence interval cannot be trusted — ask: how was the data collected?

Example 24.3. For each of the following statements, identify if it is a valid interpretation of the confidence interval from Example 24.2. If not, explain why.

1. We estimate that 95% of single family homes in the population have sale price (\$K) between 181 and 188.
2. We estimate with 95% confidence that the sample mean sale price (\$K) of single family homes is between 181 and 188.

3. If many samples of this size were selected about 95% would produce a CI of (181, 188).

 4. In about 95% of samples of this size, the population mean sale price of single family homes will be between 181 and 188.

 5. If many hypothetical unbiased random samples of this size were selected, and a 95% confidence interval computed based on each sample, about 95% of these confidence intervals would contain the population mean sale price.
-
- A CI is a conclusion about a population parameter — a number that *summarizes* the population (e.g. the population mean) — and not about individual values of a variable.
 - The conclusions from inference regard the *population* from which the random sample was selected; there is nothing to conclude for the sample.
 - No one particular sample (e.g. the one observed) or CI is necessarily good or representative of the many other possible samples or CIs.
 - The randomness is in the *sampling procedure*, which determines L and U , and not the parameter. The parameter θ is a fixed number. After selecting the sample, there is no randomness left. Either the CI contains the parameter or not; you just don't know which it is.
 - We are confident in our observed estimate because we are confident in our estimation *procedure*
 - By considering what would happen over many hypothetical random samples we are able to determine the necessary margin of error so that our CIs would contain the parameter in 95% of samples.
 - Because the value of the parameter is unknown, we never know for sure whether

any one confidence interval contains the parameter or not.

- But the procedure guarantees that in the long run — that is, over many unbiased random samples — “most” confidence intervals will contain the parameter, regardless of what the true value of the parameter is. “Most” is determined by the level of confidence.

Example 24.4. Continuing Example 24.2.

1. Consider the distribution of sale price. What shape do you expect this distribution to have? Why?
 2. A student says, “The distribution of sale price is clearly skewed to the right. Since it’s not Normal the confidence interval we computed, and the resulting conclusion about the population mean, is not valid.” Do you agree with this critique? Explain.
 3. Considering that the distribution of sale price is skewed to the right, suggest a valid criticism of the confidence interval we computed. Hint: what parameter is it estimating?
- The t interval for the population *mean* is valid as long as the sample size is large enough, regardless of the shape of the population distribution.
 - However, if the population distribution is highly skewed, then the population *median* might be a more appropriate parameter to consider as a measure of center.
 - Which is more appropriate, the population mean or the population median, depends on the purpose for the inference.
 - ALWAYS PLOT YOUR DATA, and consider the distribution of the variable. If the values in a representative sample are skewed or include large outliers, the same is also plausibly true for the population it’s supposed to represent.

Example 24.5. Suppose we wish to estimate population mean GPA of students at each of the 23 CSU campuses. For each campus, you select a random sample of students, collect the

GPA's of the students, and then use the sample data to compute a 95% confidence interval. That is you compute 23 separate 95% confidence intervals, based on 23 independent random samples, for each of 23 population means.

1. First just consider Cal Poly. What is the probability that your confidence interval based on the Cal Poly sample contains the true population mean GPA at Cal Poly?
 2. What is the probability that all 23 of your 95% confidence intervals contain their respective population mean?
 3. How confident are you that all 23 of your 95% confidence intervals contain their respective population mean?
- When computing multiple confidence intervals, the multiple (z^* or t^*) of each confidence interval should be increased to account for the **issue of multiple comparisons**.
 - For example, with a multiple of 2 in a *single* confidence interval, we are 95% confident that the interval contains the corresponding parameter.
 - However, if we use a multiple of 2 for each of 10 confidence intervals, we are only 60% confident¹ that *all* of the 10 95% confidence intervals contain their respective parameters.
 - A **Bonferroni** adjustment, splits the error rate evenly across all intervals. For example, for simultaneous 95% confidence in 10 intervals (i.e. a total error rate of 0.05), use the z^*/t^* multiple for 99.5% confidence when constructing each interval.
 - A Bonferroni adjustment guarantees **simultaneous confidence**
 - But it is conservative, and tends to produce wider intervals than necessary

¹If the group sizes are not the same, you would compute a weighted average.

- There are other methods; for example, the *Tukey-Kramer method* is often used for *all pairwise comparisons of means* (e.g., in ANOVA)
4. Suppose we want simultaneous 95% confidence that all 23 CIs contain their respective mean. Find the z^* multiple corresponding to a Bonferroni adjustment. How many times wider will each CI be than the unadjusted intervals?
 5. How confident are you that all 23 of your adjusted confidence intervals contain their respective population mean?

Example 24.6. Continuing with the Ames housing data set. Now suppose we want to compare sale price of single family homes with other homes (including condos, townhouses, etc). In particular, suppose we wish to estimate $\mu_S - \mu_N$, the difference in population mean sale price between single family homes and non-single family homes.

1. Suggest a point estimator, compute its variance, and suggest a formula for a CI for $\mu_S - \mu_N$.

2. The table below summarizes the sample data. Compute a 95% confidence interval.

	count	mean	std	min	25%	50%	75%	max
SingleFamily								
False	505.0	161.511398	60.394023	55.000	120.0	147.4	190.0	392.5
True	2425.0	184.812041	82.821802	12.789	130.0	165.0	220.0	755.0

3. Write a clearly worded sentence interpreting the confidence interval from the previous part in context.

4. Are we reasonably confident that single family homes tend to have higher sale prices than other homes? Explain.

- For many statistics (e.g. means, differences in means, proportions, differences in proportions) the sample-to-sample pattern of variability of values of the statistic over many unbiased random samples has an approximate Normal shape centered at the true value of the population parameter

– In such situations confidence intervals take the form

statistic $\pm$ multiple of its SE

- If two samples are selected *independently* from each of two populations with parameters θ_1 and θ_2 , then an approximate $1 - \alpha$ CI for $\theta_1 - \theta_2$ has endpoints

$$\hat{\theta}_1 - \hat{\theta}_2 \pm t^* \text{SE}(\hat{\theta}_1 - \hat{\theta}_2)$$

where

$$\text{SE}(\hat{\theta}_1 - \hat{\theta}_2) = \sqrt{\text{SE}(\hat{\theta}_1)^2 + \text{SE}(\hat{\theta}_2)^2}$$

- The above assumes that $\hat{\theta}_1$ and $\hat{\theta}_2$ each have an approximate Normal distribution, and so, since they are independent, $\hat{\theta}_1 - \hat{\theta}_2$ also has an approximate Normal distribution.
- A difference provides an absolute comparison between groups.
 - Ratios can be used to provide relative comparisons; however, the CI formulas are more complicated, but we'll see another approach soon

25 Bootstrap Confidence Intervals

The value of any statistic varies from sample to sample. Statistical inference procedures such as confidence intervals are based on sampling distributions which describe how the values of a statistic vary from sample to sample over many possible samples. In particular, the standard error of a statistic measures its degree of sample-to-sample variability.

If a population distribution is assumed, then we can approximate the sampling distribution of a statistic and characteristics like SE via simulation. But what if the population distribution is completely unknown? For some statistics — most notably means, proportions and certain transformations of them — the Central Limit Theorems and its relatives (e.g., Delta Method) guarantee that the sampling distribution will be approximately Normal for large samples, regardless of the population distribution. However, not every statistic has an approximate Normal distribution, so we need other methods for approximating sampling distributions. But how do we estimate the sampling distribution of a statistic based only on sample data? Bootstrapping provides a simple but extremely versatile and powerful method for doing so.

Example 25.1. For purposes of illustration, we'll start with a small data set. The table below contains data on all the quidditch matches (for which the final score was discernible) in the *Harry Potter* books¹. The sample mean is $\bar{x} = 340$ and the sample SD is $\hat{\sigma} = 134.8$ points². Say we want a confidence interval for μ , the population mean quidditch score, based on this sample.

200	230	250	260	380	470	590
-----	-----	-----	-----	-----	-----	-----

1. Warmup exercise: If we assume that quidditch scores follow a $N(345, 140)$ distribution, explain how you would use simulation to approximate the sampling distribution of the sample mean. Note: we really want to approximate the sampling distribution without assuming anything about the population distribution; this is just to remind you of how we have done things so far. Also note: the values 345 and 140 are just made up; we're assuming three things: (1) scores follow a Normal distribution, (2) population mean is 345, (3) population SD is 140.

¹Source: <http://www.nejm.org/doi/full/10.1056/NEJMoa013259>

²There are several ways to do this. The simplest is a *Bonferroni* adjustment, which splits the error rate evenly across all intervals. For example, for simultaneous 95% confidence in 10 intervals (i.e. a total error rate of 0.05), use the t^* factor for 99.5% confidence ($0.995 = 1 - 0.05/10$), $t^* = 2.8$, instead of 2 when computing the intervals. But the Tukey method is more widely used when comparing means.

2. Warmup exercise: How could you estimate the SE of the sample mean based on the above simulation? Compare to the theoretical formula.
3. Now suppose we make no assumptions about the population distribution. How could you estimate the population distribution based only on the observed sample data? Construct a spinner corresponding to this distribution.
4. Continuing the previous part, explain how you would use simulation to approximate the sampling distribution of the sample mean *without knowing the population distribution*.
5. Suppose that instead of using a spinner, you wrote each observed value on a piece of paper. How would you use the pieces of paper to simulate a sample?
6. Conduct by hand a few repetitions of the simulation from the previous part.
7. Use software to perform the simulation. How would you use the simulation results to approximate the SE? Compare to the usual formula; hint: what plays the role of

“population” SD in this simulation?

8. How could you find a confidence interval for the population mean?

- The sampling distribution of a statistic can be approximated by simulating many samples *from the population* and computing the value of the statistic for each simulated sample.
- The idea behind **bootstrapping** is that if the sample data is representative of the population, then we can approximate the process of selecting a sample from the population by *treating the observed sample as if it were the population and selecting a sample — called a resample or bootstrap sample — from the observed sample.*
- A **bootstrap sample** *resamples* the data from the *observed sample*, drawing the same number of observations, but *with replacement*.
- The **bootstrap distribution** of a statistic is obtained by simulating many bootstrap samples and computing the value of the statistic for each sample
 - Simulate a (re)samples of size n with replacement from the observed sample
 - Compute the value of the statistic for the simulated bootstrap sample
 - Repeat many times, recording the value of the statistic for each repetition
- The **bootstrap standard error** of a statistic is the standard deviation of the bootstrap distribution of the statistic.
- The bootstrap distribution of a statistic approximates some, but not all, features of the sampling distribution of the statistic.
 - The shape of the bootstrap distribution usually approximates the shape of the sampling distribution
 - The variability of the bootstrap distribution usually approximates the variability of the sampling distribution. That is, the bootstrap SE usually approximates the true SE.
 - The center of the bootstrap distribution does NOT accurately approximate the center of the sampling distribution
 - The bootstrap estimate of bias usually approximates the bias of the statistic.
- There are many methods for constructing bootstrap confidence intervals for parameters; we’ll only study a few.

- **Bootstrap Normal interval:** If the bootstrap distribution for the corresponding statistic is approximately Normal, then an approximate 95% confidence interval has endpoints

$$\text{observed statistic} \pm 2 \times \text{bootstrap SE}$$

- **Bootstrap percentile interval:** The 95% bootstrap percentile interval is the interval with endpoints equal to the 2.5th and 97.5th percentiles of the bootstrap distribution.
- One of the main advantages of bootstrapping is that it can be used to approximate the SE of *any statistic*.

Example 25.2. Continuing the previous exercise, say we want to estimate the population *median* quidditch score.

1. Describe in detail how would use simulation to approximate the bootstrap distribution. Then conduct the simulation. What are the possible values in the bootstrap distribution? Why?
2. Which bootstrap confidence interval would be more appropriate, Normal or percentile? Find and interpret the appropriate confidence interval.

Example 25.3. Using the Ames housing data set, suppose we wish to estimate the population *median* sale price of single family homes.

1. Describe in detail how you would use bootstrapping to find a confidence interval for the population median.

2. Conduct the simulation and find and interpret the confidence interval.

Example 25.4. Continuing with the Ames housing data set. Now suppose we want to compare sale price of single family homes with other homes.

1. Describe in detail how you would use bootstrapping to find a confidence interval for the *difference in population means*.
2. Conduct the simulation and find and interpret the confidence interval.
3. Describe in detail how you would use bootstrapping to find a confidence interval for the *ratio of population medians*.
4. Conduct the simulation and find and interpret the confidence interval.

Example 25.5. Continuing with the Ames housing data. Suppose we want to estimate the *correlation* between sale price and square footage (“Gr Liv Area”).

1. Describe in detail how you would use bootstrapping to find a confidence interval for the *population correlation coefficient* between sale price and square footage for single family

homes.

2. Conduct the simulation and find and interpret the confidence interval.
 3. Describe in detail how you would use bootstrapping to find a confidence interval for the *difference in population correlation coefficients* between sale price and square footage for single family homes and other homes.
 4. Conduct the simulation and find and interpret the confidence interval.
- The accuracy of bootstrap procedures is primarily influenced by the size of the original sample.
 - Bootstrapping methods do not overcome the weaknesses of small samples.

Example 25.6. Consider again the quidditch data

200 230 250 260 380 470 590

1. What is the center of the *sampling distribution* of the sample mean? Why? Given a single observed sample, what is the center of the *bootstrap distribution* of the sample mean?

2. Suppose we want to estimate the population SD? Is the sample SD an unbiased estimator of the population SD? Does this cause any issues for CIs?

3. How could we use bootstrapping to estimate the bias of the sample SD (as an estimator of the population SD)?

- The bootstrap distribution of a statistic approximates many, but not all, features of the sampling distribution of the statistic.
 - In particular, the center of the bootstrap distribution will generally not be the same as the center of the sampling distribution.
 - This can lead to coverage problems (e.g., less than the stated 95%) for bootstrap percentile intervals, especially for statistics which are biased estimators of their parameters.
- Bootstrap percentile intervals do not fully account for differences between the observed statistic and the true parameter
- Remember, there are two sources of differences between a statistic and a corresponding parameter
 - systematic errors due to *estimation bias*³
 - “chance errors” due to natural *sample-to-sample variability*
- The usual confidence procedures assume that there is no bias in estimation. The margin of error of a confidence interval typically only accounts for natural sample-to-sample variability. So if there is bias in estimation, the actual coverage probability of the confidence interval procedure might be less than the nominal confidence level.
- Recall that the bias of an estimator $\hat{\theta}$ of a parameter θ is $E(\hat{\theta}) - \theta$
 - That is, the bias of an estimator is the difference between the center of its sampling distribution and the true (but unknown) value of the parameter
 - If a statistic is *unbiased*, the center of its sampling distribution should be equal to the true (but unknown) value of the parameter.
- Bootstrapping treats the observed sample data as the population for the purposes of simulation. Therefore, bootstrapping treats the observed value of the statistic as the true value of the parameter for the purposes of simulation.

³If the group sizes are not the same, you would compute a weighted average.

- If a statistic is an unbiased estimator of a parameter, the center of its bootstrap distribution should be equal to *the observed value of the statistic*.
- The difference between the center of the bootstrap distribution of the statistic and the observed value of the statistic provides a bootstrap estimate of the statistic's bias
- A *bootstrap pivotal confidence interval* essentially uses the distribution of $\hat{\theta} - \theta$, rather than just $\hat{\theta}$, to construct a confidence interval that more fully accounts for both bias and variability.
- The 95% **bootstrap pivotal confidence interval** is the interval with endpoints
lower endpoint = $2 \times \text{observed statistic} - 97.5\text{th percentile of bootstrap distribution}$
upper endpoint = $2 \times \text{observed statistic} - 2.5\text{th percentile of bootstrap distribution}$

4. Find a 95% bootstrap pivotal interval for the population mean. Compare to the percentile interval.
5. Find a 95% bootstrap pivotal interval for the population SD. Compare to the percentile interval.

26 Null Hypothesis Testing

- Statisticians view conjectures about a population as competing *hypotheses*. These hypotheses usually constitute claims about the values of certain parameters.
- A statistical **(null) hypothesis test** consists of using data to make a decision regarding two explanations (hypotheses).
- The **null hypothesis**, denoted H_0 , represents the “status quo/no difference/no effect/no association” hypothesis.
 - If the null hypothesis is true then any observed difference/effect/association is just due to “random chance” resulting from natural sampling variability
- The **alternative hypothesis**, denoted H_a (or H_1), typically corresponds to what the researchers are hoping to gather evidence to support, and thus is a statement in line with the research conjecture
- A statistical hypothesis test assesses if the data provide evidence to rule out the “random chance” explanation (which corresponds to the null hypothesis) in favor of the explanation provided by the research conjecture (alternative hypothesis).
- The null hypothesis is often of the form

$$H_0 : \text{parameter} = \text{hypothesized value}$$

where the hypothesized value is a specific number determined by the research question (often 0 for “no difference”).

- The alternative hypotheses is often of one of the following forms

Case 1a: $H_a : \text{parameter} > \text{hypothesized value}$

Case 1b: $H_a : \text{parameter} < \text{hypothesized value}$

Case 2: $H_a : \text{parameter} \neq \text{hypothesized value}$ (“two-sided”)

Which of the three cases above is appropriate is determined by the research conjecture. When in doubt, use a two-sided ($\neq$) alternative.

Example 26.1. Did the New England Patriots deliberately deflate footballs in an NFL playoff game against the Indianapolis Colts on January 18, 2015? The following table¹ summarizes the decrease in air pressure (psi), from pre-game to halftime, for the footballs from each team that were checked at halftime by officials.

¹Source: <http://www.nejm.org/doi/full/10.1056/NEJMoA013259>

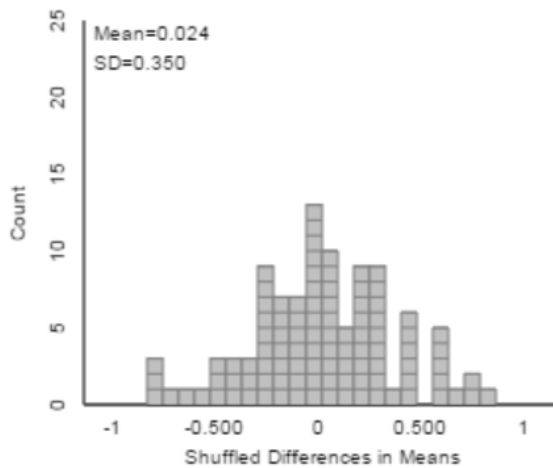
Deflation														
Team	(psi)											Size	Mean	SD
Patriots	1.00	1.65	1.35	1.80	1.40	0.90	0.65	1.40	1.55	2.00	1.60	11	1.391	0.402
Colts	0.30	0.25	0.50	0.45								4	0.375	0.119

1. Compute the observed difference in sample means.
2. If there were no difference between the Patriots and the Colts footballs, what would you expect the difference in sample means to be? Is the fact that we observed another value necessarily evidence that the Patriots footballs tended to deflate more than the Colts?
3. Suppose we want to assess if it is plausible that the Patriots didn't interfere with the footballs, and the observed difference in sample means is just due to chance variability. Specify how the "null distribution" of the statistic could be simulated with cards. Note: this is NOT bootstrapping; how is it different?
4. Use cards to perform one repetition of the above simulation, record the results here, and compute the appropriate difference in means.

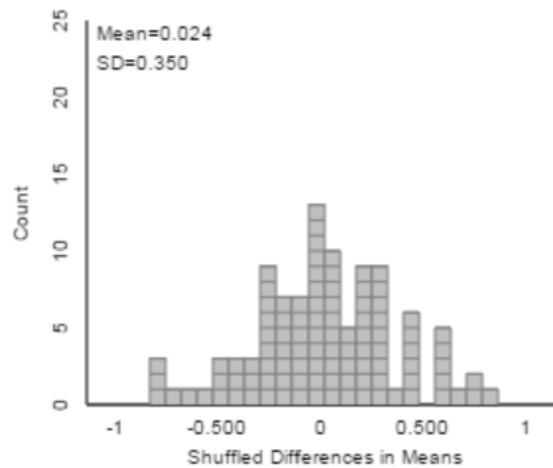
Deflation														
Team	(psi)											Size	Mean	SD
Patriots												11		
Colts			X	X	X	X	X	X				4		

5. The following plot displays results from an [applet](#) used to run 10000 repetitions of the simulation (the plot on the left displays the results from just the first 100 repetitions). Estimate the probability of observing a difference in sample means as extreme as the observed difference just by chance if there were no real difference between the Patriots and Colts footballs.

Total Shuffles = 100



Total Shuffles = 100



Count Samples

(a) First 100 repetitions

Count Samples

(b) 10000 repetitions

Figure 26.1: Permutation simulation of null distribution in Example 26.1.

6. Is it plausible that the observed difference in means is just due to random chance? To support your answer, interpret the probability from the previous part in context.

- To assess the plausibility of the “random chance” explanation for the observed result — corresponding to the null hypothesis being true — we need to know the pattern of sampling variability of the statistic that would be expected just due to random chance if the null hypothesis were true. This is called the **null distribution** of the statistic.
- If the actual observed value of the statistic is not consistent with this pattern, then we have convincing evidence that the results did not occur by random chance alone, and we have evidence to reject the null hypothesis
- The (simulated) null distribution provides a picture of what typical values of the statistic would be expected just by random chance alone if the null hypothesis were true.
- The question is then: how surprising is the result we actually observed from the perspective of the null distribution? If the result would be very surprising, then “random chance” is not a plausible explanation for the observed result.

- This gives us evidence to rule out a “random chance” explanation like in favor of a “research conjecture” explanation.
- In other words, the more surprising the observed result would be just due to random chance, the stronger the evidence to **reject the null hypothesis** in favor of the alternative hypothesis.
- We quantify “how surprising” with a p-value.
 - The **p-value** is the probability of observing a value at least as extreme as the value actually observed, when the null hypothesis is true.
 - The p-value can be estimated by the proportion (out of the total number of repetitions) of values in the simulated null distribution that are *at least as extreme* as the actual observed value of the statistic.
 - * Which tail(s) constitutes “at least as extreme” is determined by the form of the *alternative hypothesis*
- The smaller the p-value — that is, the closer the p-value is to 0 — the less plausible it is that the observed result occurred just by random chance alone.
- Thus, **the smaller the p-value the stronger the evidence to reject the null hypothesis in favor of the alternative hypothesis.**
- If the observed results provide strong evidence that the data did not arise by random chance alone — that is, when the p-value is small — then there is evidence of a **statistical difference**.
- How small is a small p-value? You should think of a sliding scale: the closer the p-value gets to 0 the stronger the evidence to reject the null hypothesis in favor of the alternative. Some guidelines are below, but these should *not* be followed too strictly (especially when performing many hypothesis tests simultaneously):
- Do NOT rely strictly on arbitrary thresholds like $p\text{-value} < 0.05$.

p-value	Evidence to reject the null hypothesis?
Above 0.10	Little to no evidence
Between 0.01 and 0.10	Weak evidence
Between 0.001 and 0.01	Moderate evidence
Between 0.0001 and 0.001	Strong evidence
Less than 0.0001	Very strong evidence

Example 26.2. Carrying a heavy backpack can cause chronic shoulder, neck, and back pain. A [common recommendation](#) is that a daily backpack should not weigh more than 10 percent of the wearer’s body weight. Do Cal Poly students follow this recommendation? That is, do the backpacks of Cal Poly students weigh, on average, less than 10% of their bodyweight? In a random sample of 100 Cal Poly students the sample mean backpack weight as percent of bodyweight is 7.71 percent with sample standard deviation 0.366 percent.

1. Identify the main parameter of interest with both words and an appropriate symbol.
2. Translate the research question into a hypothesis testing problem by specifying, with both words and appropriate symbols, the null hypothesis and the alternative hypothesis.
3. Identify the null distribution of sample means. (You might need to make some assumptions.)
4. Compute the p-value.
5. Interpret the p-value in context.
6. Does the p-value provide evidence that backpacks of CalPoly students weigh, on average, less than 10% of their bodyweight? Explain.
7. Does the p-value provide evidence that backpacks of CalPoly students weigh, on average, *much* less than 10% of their bodyweight? Explain.

8. How could we address the question in the previous part?

9. Do Cal Poly students follow the 10% recommendation? Is the analysis above the only way to address this question? Can you suggest another?

One-sample t test

- Situation: question concerns a single numerical variable of interest
- Parameter: μ is the population mean (i.e. the mean value of the variable over all units in the population)
- Goal: assess the plausibility of a claim about μ regarding a hypothesized value μ_0
- The null hypothesis is $H_0 : \mu = \mu_0$
- The alternative hypothesis is one of $H_a : \mu > \mu_0$, $H_a : \mu < \mu_0$, or $H_a : \mu \neq \mu_0$
 - When in doubt, use a two-sided alternative and a two-tailed p-value.
- Statistics: $\bar{x}$ is the sample mean; s is the sample standard deviation
- Test statistic: the standardized value of $\bar{x}$, where
 - the mean is computed assuming the null hypothesis, $H_0 : \mu = \mu_0$, is true, and
 - the unknown SD ($\sigma/\sqrt{n}$) is estimated with the standard error $s/\sqrt{n}$
$$t = \frac{\text{observed} - \text{hypothesized}}{\text{SE}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$
- The p-value is the probability of observing a value at least as extreme as t , in the tail direction indicated by the alternative hypothesis, for the t -distribution with $n-1$ degrees of freedom.
 - In almost all situations encountered in practice you can use the usual empirical rule for Normal distributions
- Assumptions:
 - Simple random sample from a large population, that is, independent observations

- No bias in data collection
- Population mean is the parameter you want to estimate

Example 26.3. Continuing with the Ames housing data set. Now suppose we want to compare sale prices of single family homes with other homes (including condos, townhouses, etc). In particular, do single family homes tend to have higher sale prices on average than other homes?

The table below summarizes the sample data.

	count	mean	std	min	25%	50%	75%	max
SingleFamily								
False	505.0	161.511398	60.394023	55.000	120.0	147.4	190.0	392.5
True	2425.0	184.812041	82.821802	12.789	130.0	165.0	220.0	755.0

1. Explain how you could use simulation to approximate the p-value.
2. Use the sample data to conduct by hand an appropriate hypothesis test.
3. Write a clearly worded sentence interpreting the p-value in context.
4. Does the p-value provide evidence that single family homes tend to have higher sale prices on average than other homes? Explain.
5. Does the p-value provide evidence that single family homes tend to have *much* higher sale prices on average than other homes? Explain.

6. How could we address the question in the previous part?

Two-sample t test for a difference in population (or treatment) means.

- Situation: binary explanatory variable (population/treatment 1, 2) and numerical response variable
- Parameter: $\mu_1 - \mu_2$ is the difference in population means (treatment means)
- Goal: assess the strength of the evidence against the “no difference” or “no treatment effect” hypothesis
- The null hypothesis is $H_0 : \mu_1 - \mu_2 = 0$
- The alternative hypothesis is one of $H_a : \mu_1 - \mu_2 > 0$, $H_a : \mu_1 - \mu_2 < 0$, or $H_a : \mu_1 - \mu_2 \neq 0$
 - When in doubt, use a two-sided alternative and a two-tailed p-value.
- Statistic: $\bar{x}_1 - \bar{x}_2$ is the difference in sample means for samples (groups) of size n_1, n_2
- The SE of the difference in sample means is

$$SE(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{SE^2(\bar{x}_1) + SE^2(\bar{x}_2)}$$

- Test statistic:

$$t = \frac{\text{observed} - \text{hypothesized}}{SE} = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$
- The p-value is the probability of observing a value at least as extreme as t , in the tail direction indicated by the alternative hypothesis, for the t -distribution with (complicated) degrees of freedom.
 - In almost all situations encountered in practice you can use the usual empirical rule for Normal distributions
- Assumptions:
 - Either
 - * The data are obtained through two unbiased and independent random samples from large populations

- Or from one unbiased random sample which is classified according to two variables, and the two groups can be considered independent of each other.
- * The data are obtained from an experiment in which subjects were randomly assigned to two treatments.
 - No bias in data collection
 - Difference in population means is the parameter you want to estimate
- The t procedures for population means are valid as long as the sample size is large enough, regardless of the shape of the population distribution.
- **However, if the population distributions are highly skewed, then the population *medians* might be more appropriate parameters to consider as measures of center.**
- Which is more appropriate, the population means or the population medians, depends on the purpose for the inference.
- Also remember that sample means and SDs, and hence t procedures, are sensitive to outliers and extreme values.

Example 26.4. In Example 26.2 and Example 26.3 we both performed a null hypothesis test and computed a confidence interval. Discuss the advantages and disadvantages of each approach.

- Confidence intervals and hypothesis tests are commonly used in conjunction, but they address different questions
 - Hypothesis test: How strong is the evidence that there is a difference (or association or effect)?
 - Confidence interval: How large is the difference (or association or effect)?
- A confidence interval
 - provides a range of plausible values
 - but the cut-off between plausible and implausible is arbitrary (e.g. 95% confidence corresponds to $\alpha = 0.05$)
 - * But be especially careful about interpreting confidence when conducting multiple confidence intervals.
- A null hypothesis test
 - Considers the plausibility of only a single hypothesized value

- But provides a precise measure — the p-value — of the plausibility of the particular hypothesized value in light of the observed data
 - * But be especially careful about interpreting p-values when conducting multiple hypothesis tests.

27 Some issues with hypothesis testing

- Remember: the smaller the p-value, the stronger the evidence to reject the null hypothesis.
 - But how small is small enough?
- A “significance level”, usually denoted α , is a pre-specified cut-off for how small the p-value must be in order to reject the null hypothesis. The most common value is $\alpha = 0.05$.
 - HOWEVER, in general, do NOT rely too strictly on a significance level α .
 - In fact, for years the [American Statistical Association](#) has said to stop using values like 0.05 to make binary reject/not reject decisions.
- If the p-value is small the result is sometimes referred to as “statistically significant”.
 - HOWEVER, we will see that this is a poor choice of words.
 - In fact, for years the [American Statistical Association](#) has said to stop using the term “statistically significant”. Quote: “‘statistically significant’—don’t say it and don’t use it.”
- Reporting the actual p-value and interpreting it as a sliding scale of weak/moderate/strong evidence to reject the null hypothesis is much more informative than just reporting “reject H_0 at level $\alpha = 0.05$ ” or “p-value < 0.05 ”.
- Remember that hypothesis testing and p-values are just small parts of a much bigger picture. A hypothesis test provides some information, but there are many important questions that hypothesis testing and p-values do not have the ability to address.

Example 27.1. A 2007 paper¹ suggests that in coin-tossing there is a particular “dynamical bias” that causes a coin to be slightly more likely to land the same way up as it started. That is, the paper claims that a coin is more likely to land facing Heads up if it starts facing Heads up than if it starts facing Tails up; likewise if it starts facing Tails up. Is this really true? Two students at Berkeley investigated this phenomenon by tossing a coin many, many times.

1. Define the parameter p in words and state the null and alternative hypotheses using symbols.

¹Source: <http://www.nejm.org/doi/full/10.1056/NEJMoa013259>

2. Suppose that out of 10000 flips, 5083 landed the same way they started. Based on this data, can you conclude that a coin is more likely to land the same way it started using a strict “significance level” of $\alpha = 0.05$? Compute the p-value and state your conclusion at level $\alpha = 0.05$ in context. Note: One goal of the example is to show you one reason why using a strict α is a bad idea.
3. Repeat the previous part assuming 5082 out of 10000 flips landed the same way they started.
4. Compare the two previous parts. Give one reason why you should be wary of using a “significance level” (like 0.05) too strictly.
5. The students actually tossed the coin 40,000 times!². Of the 40,000 flips, 20,245 landed the same way they started. Compute the p-value and state a conclusion in context. (Do NOT use a strict α level.) Which result—20245 out of 40000 or 5083 out of 10000—gives stronger evidence to reject the null hypothesis?
6. Would you say the results of the real study, in the previous part, are newsworthy? Or important? Or “significant”? Using this problem as an example, explain why “statistically significant” is a poor choice of words. In other words, explain why “statistical significance” is not the same as practical importance.

²If the group sizes are not the same, you would compute a weighted average.

- In general, do NOT rely too strictly on a significance level α .
- Reporting the actual p-value and interpreting it as strong/moderate/weak evidence to reject the null hypothesis is more informative than just reporting “reject H_0 at level $\alpha = 0.05$ ” or “p-value < 0.05 ”.
- Beware the meaning of “significant”.
- A “statistically significant” result is not necessarily practically important or meaningful. Rather, it is just is more extreme than what would be expected to occur just due to chance if the null hypothesis were true, but not necessarily large in practical terms.
- Especially when the sample size is large, even a small observed difference can result in a small p-value.
- On the other hand, an observed result might have practical significance even when the p-value is not small.
- A confidence interval estimates the size of the difference, which helps in judging practical significance. It is also important to consider *effect size*.
- Do NOT use the terminology “statistically significant”.

Example 27.2. (For this exercise, you’ll need to believe it’s possible for a person to have ESP.) “Ganzfeld studies” test psychic ability (like ESP). Ganzfeld studies involve two people, a “sender” and a “receiver”, who are placed in separate acoustically-shielded rooms. The sender looks at a “target” image on a television screen (which may be a static photograph or a short movie segment playing repeatedly) and attempts to *transmit with their mind* information about the target to the receiver. The receiver is then shown four possible choices of targets, one of which is the correct target and the other three are “decoys.” The receiver must choose the one transmitted by the sender³.

1. We will now conduct some Ganzfeld tests to see if anyone in the class has ESP.
 - Find a partner. Identify one person as the sender and one as the receiver.
 - The sender rolls a four-sided die, looks at it (without letting the receiver see), and then tries to transmit the number rolled to the mind of the receiver.
 - After receiving the mental image, the receiver identifies the number. Record whether the answer is correct or not.
 - Repeat the above process for a total of 9 trials with the same receiver. The receiver should record the number of trials (out of 9) they got correct.
 - Then switch sender/receiver roles and do it again. Each receiver should record the number of trials (out of 9) that they got correct.

³This number assumes independence of confidence intervals — $0.95^{10} \approx 0.60$ — but similar ideas apply more generally.

2. Use your results to conduct a hypothesis test to see if you have evidence that you did better than just guessing (and so you have some kind of ESP!)

3. Suppose we had set a cut-off of 0.05. Based on this strict cutoff, do you have enough evidence to reject the null hypothesis? That is, based on this cutoff, is your p-value small enough to conclude that you have ESP? (Hint: if you got at least 5 out of 9 correct your answer should be “yes”.)

4. Consider a student who rejects the null hypothesis. Could their conclusion be wrong? How?

5. Consider a student who fails to reject the null hypothesis. Could their conclusion be wrong? How? (Again, you’ll need to believe it’s possible for a person to have ESP.)

6. Using the cutoff of 0.05, did any receiver in our class (of about 30 students) discover evidence that they have ESP? Should we be surprised that this happened?

- The null hypothesis H_0 is “innocent until ‘proven’ guilty”. That is, the null hypothesis is not rejected unless there is convincing evidence against it.
- When making a conclusion — reject H_0 , or fail to reject H_0 — there are two errors that could be made:
 - **Type I error:** Reject H_0 when H_0 is actually true (unknownst to the researcher)
 - * “False alarm” or “false discovery”
 - * Analogous to sending an innocent person to jail.
 - **Type II error:** Fail to reject H_0 when H_0 is actually false (unknownst to the researcher)
 - * “Missed opportunity”
 - * Analogous to letting a guilty person go free.
- The probability of making a Type I error can be managed by setting a “significance” level α .
- If decisions are made according to the rule “reject the null hypothesis only when p-value $< \alpha$ ” then the probability of making a Type I error is (at most) α .
- But there is a tradeoff: decreasing the probability of making a Type I error will increase probability of making a Type II error.
- The **power** of a test is the probability of correctly rejecting the null hypothesis when the the null hypothesis is false.
 - The power is the probability of making a “true discovery”.

		What is true (unknownst to the jury)	
		Null hypothesis is true (defendant is innocent)	Null hypothesis is false (defendant is guilty)
What jury decides (based on evidence)	Reject null hypothesis (Jury finds defendant guilty)	Type I error (false alarm)	Correct Decision
	Do not reject null hypothesis (Jury finds defendant not guilty)	Correct Decision	Type II error (missed opportunity)

- Hypothesis tests are appropriate when there is a clearly defined research question and conjecture (i.e. alternative hypothesis), and sample data is collected for this purpose.
- If many hypothesis tests are performed, eventually you will reject a null hypothesis — just by chance — even if all the null hypotheses are really true.
 - For example, in a set of 35 tests all conducted using strict level $\alpha = 0.05$, there is about an 84 percent chance of rejecting at least one of the null hypotheses when all 35 null hypotheses are true.

- Beware of **p-hacking**, which involves repeating hypothesis tests until a small p-value is found
- The issue of multiple testing/p-hacking is another reason why you should not rely too strictly on a “significance” level α .
- Do NOT use a strict level α to make a strict reject H_0 /fail to reject H_0 conclusion based on observed sample data about a particular hypothesized value or research question
 - Rather, report the p-value and assess strength of evidence in favor of rejecting the null hypothesis on a sliding scale.
 - But remember the p-value doesn’t tell you anything about how big or meaningful the difference is—but rather just if it’s not due to random chance—especially if the sample size is large
 - Be especially careful about using α or interpreting p-values when conducting multiple hypothesis tests on the same data. Beware p-hacking.
- Do USE “significance” levels
 - For research planning purposes — to manage probability of Type I error, to investigate power, and to determine an appropriate sample size — *before* collecting data
 - When determining a *range* of plausible values for the parameter, e.g., to determine the cutoff between plausible/not plausible in a confidence interval.
 - * A 95% confidence is essentially the range of hypothesized values that would not be rejected at level 0.05 based on the observed data.
- Remember that hypothesis testing and p-values are just small parts of a much bigger picture. A hypothesis test provides some information, but there are many important questions that hypothesis testing and p-values do not have the ability to address.

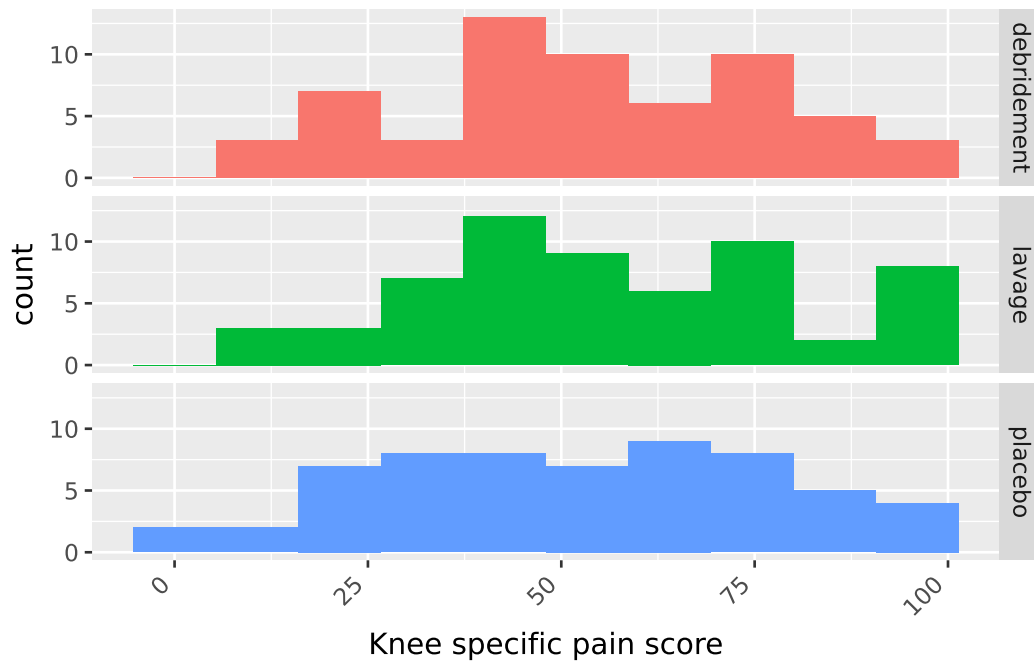
28 Comparing multiple means: ANOVA F procedures

Example 28.1. When medical therapy fails to relieve the pain of osteoarthritis of the knee, arthroscopic lavage or debridement is often recommended. More than 650,000 such procedures are performed each year at a cost of roughly \$5,000 each. A recent double blind experiment investigated the effectiveness of these procedures¹. 180 subjects with osteoarthritis of the knee were randomly assigned to receive one of three treatments: arthroscopic lavage (which involved flushing the knee joint with fluid), arthroscopic debridement (which involved both lavage and shaving of cartilage and removal of debris), and placebo surgery (which involved incisions in the knee, but no insertion of instruments). Various outcomes were measured; the summaries below summarize scores two years after the procedure on a knee-specific pain scale (0-100, with higher scores indicating more severe pain). Does any procedure tend to result in better pain scores on average?

1. State the null and alternative hypotheses (in words is fine).
2. The summarizes the sample data. Do you think there will be any evidence to reject the null hypothesis?
3. Suggest a statistic which could be used to perform a hypothesis test.

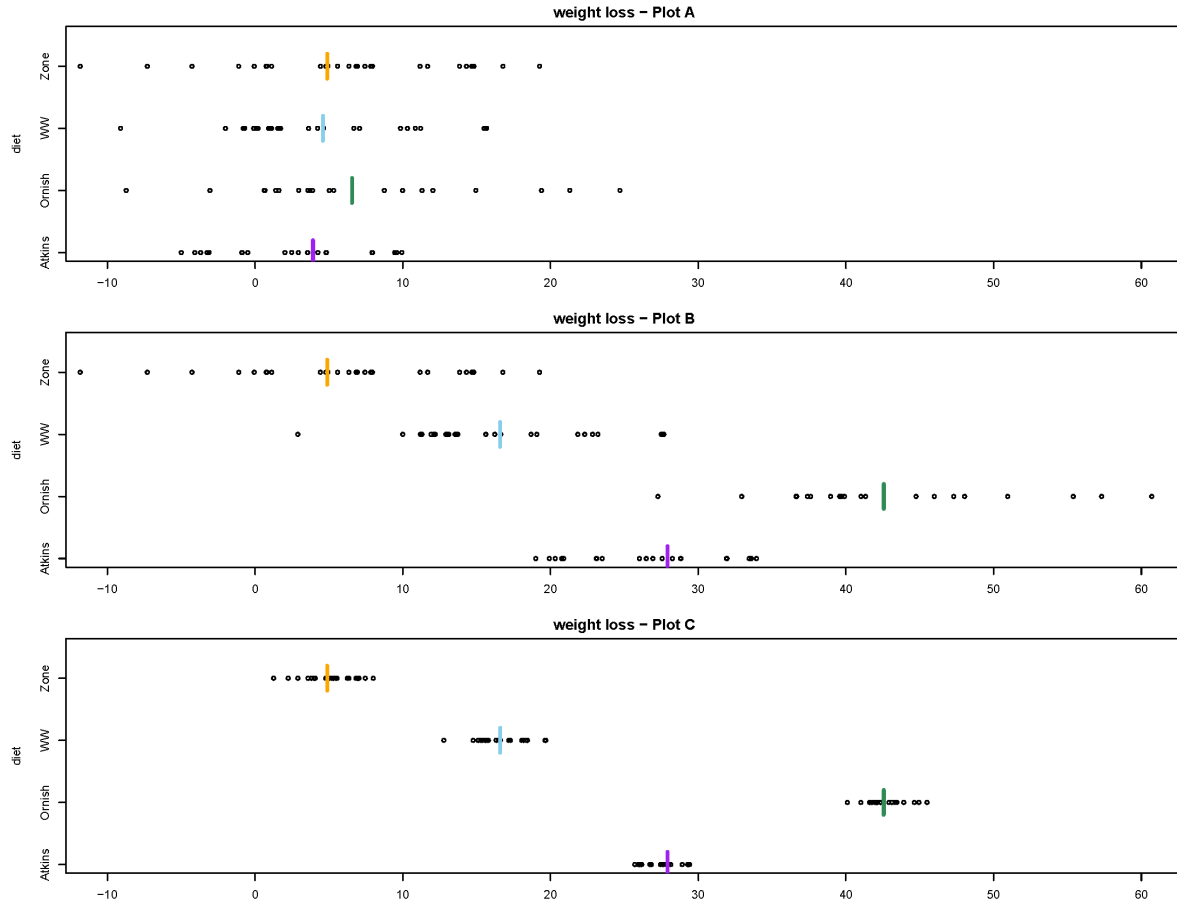
¹Source: <http://www.nejm.org/doi/full/10.1056/NEJMoa013259>

	count	mean	std	min	25%	50%	75%	max
treatment								
debridement	60.0	53.983333	23.304536	10.0	40.75	52.0	76.25	97.0
lavage	60.0	56.733333	24.150564	8.0	41.00	53.5	73.50	99.0
placebo	60.0	52.450000	25.079146	3.0	31.00	54.0	70.25	97.0



Example 28.2. Consider the following plots, which are all on the same scale. (This example just illustrates some general ideas so the context isn't important, but the data relate to a study on different types of diet plans (Atkins, etc) and weight loss.) Plot A displays the study data. Plots B and C show hypothetical data sets with the same group sizes as in Plot A. Group means are indicated by the colored vertical lines.

- The SD for each group in Plot B equals the SD for the corresponding group in Plot A.
- The mean for each group in Plot C equals the mean for the corresponding group in Plot B.



- Rank the plots in order from weakest to strongest evidence to reject the null hypothesis of no difference in means between treatments. That is, rank the plots in order from largest to smallest p-value.
- Compare plots A and B.
 - The variability *between* groups in Plot B is (**choose one:** $>$, $<$, $=$) the variability *between* groups in Plot A.
 - The variability *within* groups in Plot B is (**choose one:** $>$, $<$, $=$) the variability *within* groups in Plot A.
- Compare plots B and C.
 - The variability *between* groups in Plot C is (**choose one:** $>$, $<$, $=$) the variability *between* groups in Plot B.
 - The variability *within* groups in Plot C is (**choose one:** $>$, $<$, $=$) the variability *within* groups in Plot B.

- Rank the plots in order from smallest to largest F statistic.

- Example 28.2 illustrates the idea behind the F statistic: we reject the null hypothesis of no difference in means between groups/populations/treatments if the observed variability *between* groups is large relative to the observed variability *within* groups.
- Roughly, we reject the null hypothesis if the ratio

$$\frac{\text{variability between groups}}{\text{variability within groups}}$$

is large enough. This idea is known as **analysis of variance (ANOVA)**.

- The name ANOVA refers to *what* we are doing, and not *why* we are doing it.
- Remember, we are analyzing the variance to make a conclusion about the *means*
- The F statistic quantifies the above ratio. The larger the value of the F statistic the stronger the evidence to reject the null hypothesis (of no difference) in favor of the alternative hypothesis (of some difference). That is, the larger the F statistic the smaller the p-value.

Example 28.3. Continuing Example 28.1. The computation of the F statistic is full of technical details. We will only do a rough hand computation which illustrates the main ideas, but we'll usually rely on software to compute it. The computation is somewhat simplified when the sample size is the same for each explanatory variable group.

- What statistic can be used to measure the variability *within* the Debridement group? Lavage? Placebo? How could you combine these numbers to to measure, in a single number, *variability within groups*?
- If the null hypothesis is true, what would you expect about the group means? What statistic can be used to measure the variability *between* groups?

3. Regardless of whether or not the null hypothesis is true, which of the values from the previous parts would you expect to be larger? Why?
4. So far, we have used the group means and group SDs in our calculations. What other feature of the groups do you think will influence the results? How?
5. Compute the F statistic.
6. What value would you expect F to be if the null hypothesis is true? Specify in detail how you could use simulation to approximate the null distribution of the F statistic.
7. Figure 28.1 displays results from an applet used to run 10000 repetitions of the simulation. How could you use the simulation results to approximate the p-value? Could you use the empirical rule for Normal distributions?
8. State a conclusion in context. Are the results “significant”?
9. If we were to compute a series of pairwise confidence intervals for the difference in group means — Debridement — Lavage, Debridement — Placebo, Lavage — Placebo — what value would each of the confidence intervals contain? Why?

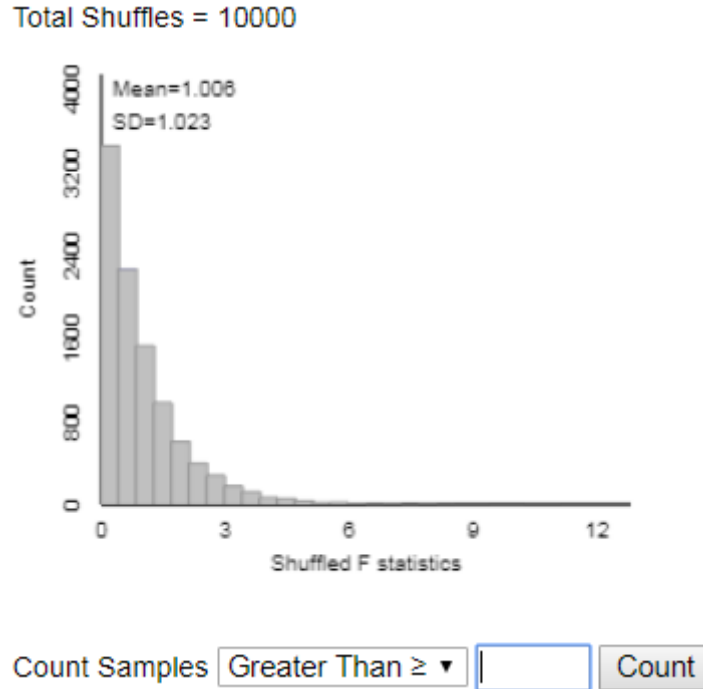


Figure 28.1: Permutation null distribution of F statistic for Example 28.1.

Hypothesis test for comparing multiple means: “One-way ANOVA F test”

- Situation: N observations of numerical response variable; categorical explanatory variable with g groups
- Parameters: μ_j 's are the population/process/treatment response means; σ is the common group SD of responses
- Goal: test for any differences in means
- The null hypothesis is $H_0 : \mu_1 = \dots = \mu_g$ (“no difference” or “no treatment effect”)
- The alternative hypothesis is not H_0 ; that is, there is some difference in means between groups, $H_1 : \mu_j \neq \mu_k$ for some $j \neq k$
- Test statistic: one-way ANOVA F statistic

$$F = \frac{\sum_{j=1}^g n_j (\bar{x}_j - \bar{x})^2 / (g - 1)}{\sum_{j=1}^g (n_j - 1) s_j^2 / (N - g)}$$

- The p-value is the probability of observing a value at least as large as the observed F based on an F distribution with $g - 1$ degrees of freedom in the numerator and $N - g$ degrees of freedom in the denominator
- Assumptions:
 - Either:
 - * The data are obtained through multiple unbiased and independent random samples from large populations
 - Or from one unbiased random sample which can be classified into multiple groups, and the groups can be considered independent of each other.
 - * The data are obtained from an experiment in which subjects were to two treatments.
 - No bias in data collection
 - Comparing *means* is an appropriate analysis
 - Equal variance: True SD of responses is same, σ , for each population/process/treatment.
 - * Typically satisfied through random assignment.
 - * Might not be satisfied if the data are sampled from different populations/processes.
 - * One rule of thumb is that the largest observed sample SD should be no more than twice the smallest.
- Special case: when the group size n is the same for all g groups is

$$F = \frac{\text{group size} \times \text{variance of group means}}{\text{mean of group variances}}$$

$$= \frac{n \sum_{j=1}^g (\bar{x}_j - \bar{x})^2 / (g - 1)}{\sum_{j=1}^g s_j^2 / g}$$

- The calculations are traditionally summarized in an **ANOVA table**

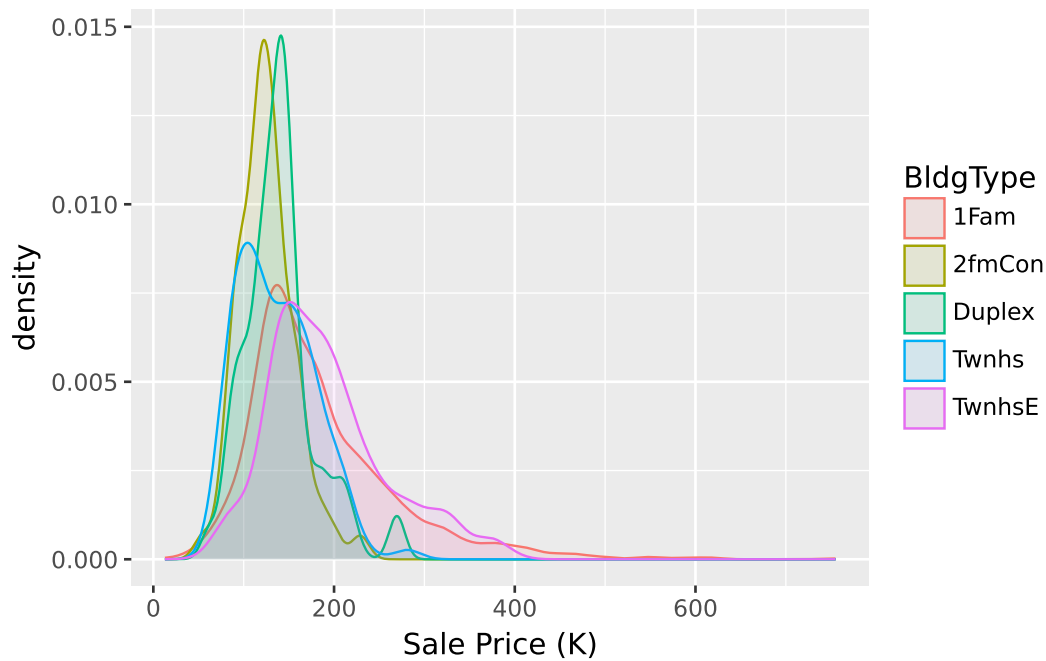
Source	df	SS (Sum of squares)	MS (Mean Square)	F
Between groups (a.k.a. “treatment”)	$g - 1$	$SS(\text{between}) = \sum_{j=1}^g n_j (\bar{x}_j - \bar{x})^2$	$\frac{SS(\text{between})}{g - 1}$	$\frac{MS(\text{between})}{MS(\text{within})}$
Within groups (a.k.a. “error”)	$N - g$	$SS(\text{within}) = \sum_{j=1}^g (n_j - 1) s_j^2$	$\frac{SS(\text{within})}{N - g}$	
Total	$N - 1$	$(SS(\text{between}) + SS(\text{within})) = \sum_{i=1}^N (x_i - \bar{x})^2$		

Table 28.4: ANOVA table for Example 28.4.

	sum_sq	df	F	PR(>F)
C(BldgType)	6.454111e+05	4.0	26.151358	2.475923e-21
Residual	1.804713e+07	2925.0	NaN	NaN

Example 28.4. Continuing with the Ames housing data set. Now suppose we want to compare sale price of homes by building type: single-family (1Fam), two-family (2fmCon), duplex, townhouse end unit (TwnhsE), townhouse inside unit (Twnhs).

	count	mean	std	min	25%	50%	75%	max
BldgType								
1Fam	2425.0	184.812041	82.821802	12.789	130.0000	165.000	220.000	755.00
2fmCon	62.0	125.581710	31.089240	55.000	106.5625	122.250	140.000	228.95
Duplex	109.0	139.808936	39.498974	61.500	118.8580	136.905	153.337	269.50
Twnhs	101.0	135.934059	41.938931	73.000	100.5000	130.000	170.000	280.75
TwnhsE	233.0	192.311914	66.191738	71.000	145.0000	180.000	222.000	392.50

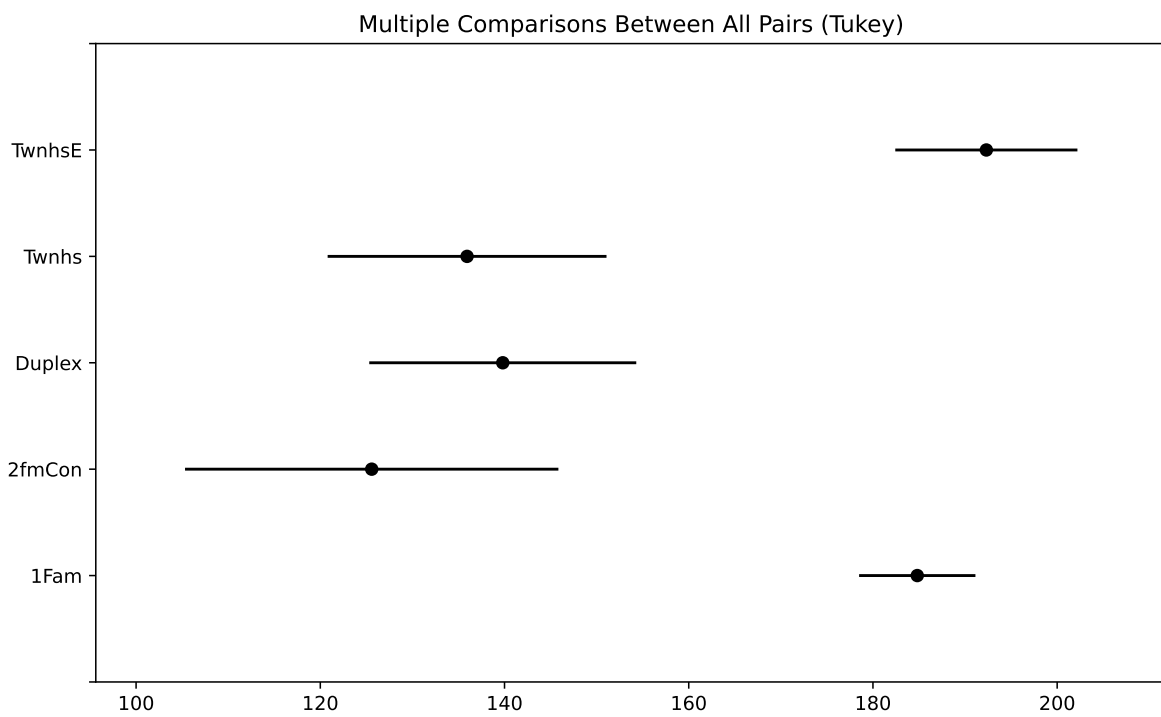
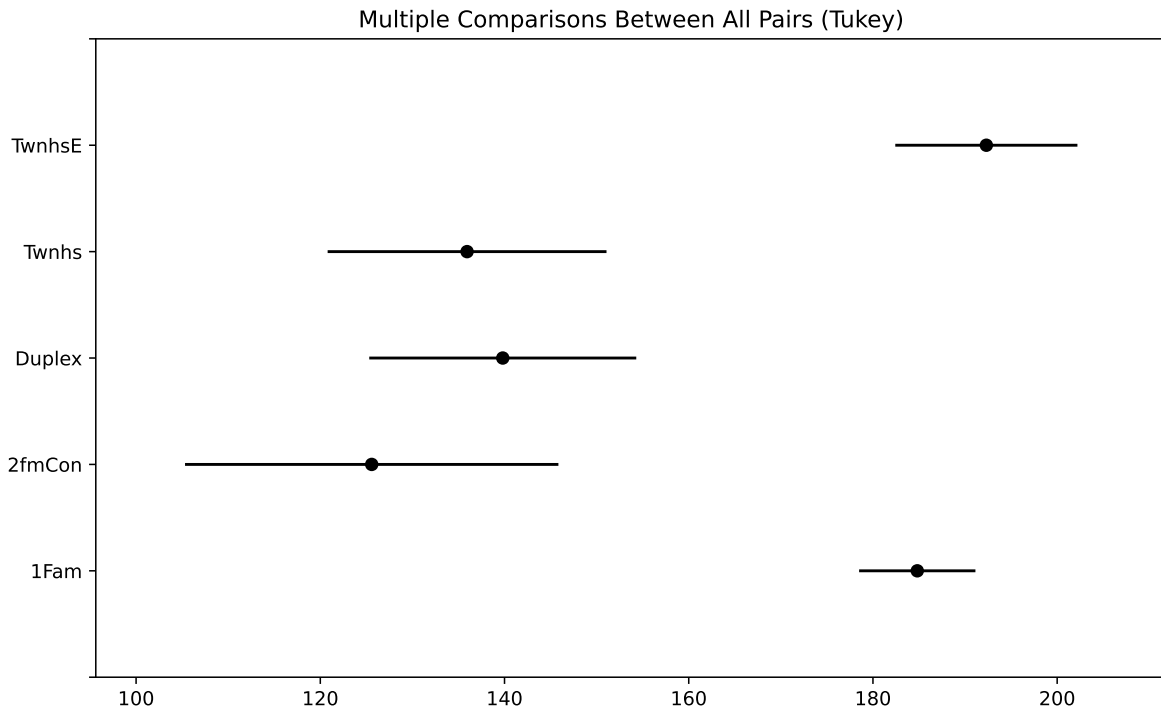


1. Table 28.4 is the ANOVA table. State the conclusion of the ANOVA F test in context.

Table 28.5: Tukey pairwise 95% confidence intervals for Example 28.4.

Multiple Comparison of Means - Tukey HSD, FWER=0.05						
group1	group2	meandiff	p-adj	lower	upper	reject
1Fam	2fmCon	-59.2303	0.0	-86.8048	-31.6559	True
1Fam	Duplex	-45.0031	0.0	-65.9952	-24.0111	True
1Fam	Twnhs	-48.878	0.0	-70.6511	-27.1049	True
1Fam	TwnhsE	7.4999	0.6328	-7.2051	22.2048	False
2fmCon	Duplex	14.2272	0.786	-19.8771	48.3316	False
2fmCon	Twnhs	10.3523	0.9255	-24.2382	44.9429	False
2fmCon	TwnhsE	66.7302	0.0	36.0924	97.368	True
Duplex	Twnhs	-3.8749	0.9965	-33.4861	25.7363	False
Duplex	TwnhsE	52.503	0.0	27.6234	77.3825	True
Twnhs	TwnhsE	56.3779	0.0	30.8358	81.9199	True

2. Consider the conclusion of the F test. What are some natural follow-up questions? How might you use the data to answer them?
3. Table 28.5 contains the results of Tukey pairwise 95% confidence intervals. Interpret these intervals in context.



- Rejecting the null hypothesis based on an F statistic only tells you that “the mean for

at least one of the treatments (populations) is not the same as the mean for at least one of the other treatments (populations)”.

- The natural follow-up questions are: which treatment (population) means differ? and by how much?
- One approach when you have rejected the null hypothesis is to calculate — using software — all pairwise confidence intervals. In particular, which intervals do not contain zero?
- When computing multiple confidence intervals, the multiple (z^* or t^*) should be increased² to account for the **issue of multiple comparisons** (recall Example 24.5 and related notes).
- **Tukey’s method** is a widely used method for constructing simultaneous confidence intervals for *all pairwise comparisons of means*. The confidence intervals computed using Tukey’s method already contain an adjustment to account for multiple comparisons.

²There are several ways to do this. The simplest is a *Bonferroni* adjustment, which splits the error rate evenly across all intervals. For example, for simultaneous 95% confidence in 10 intervals (i.e. a total error rate of 0.05), use the t^* factor for 99.5% confidence ($0.995 = 1 - 0.05/10$), $t^* = 2.8$, instead of 2 when computing the intervals. But the Tukey method is more widely used when comparing means.

Homework 1

1. In each of the following, write a clearly worded sentence interpreting the numerical value of the probability as a long run relative frequency in context. (Just take the numerical values as given for now. We'll see how to compute probabilities like these later.)
 - a. The probability of rolling doubles when you roll two fair six-sided dice is $1/6$.
 - b. The probability of rolling doubles on three consecutive rolls of two fair six-sided dice is 0.00463.
 - c. The probability that the sum of 100 rolls of a fair six-sided die is less than 370 is 12%.
 - d. Roll a fair six-sided die until you roll a 6 three times and then stop. The probability that you roll the die at least 10 times is 0.822.
2. Suppose your subjective probabilities for the 2025 World Series champion satisfy the following conditions.
 - The Mets and Yankees are equally likely to win
 - The Phillies are 1.5 times more likely than the Mets to win
 - The Dodgers are 2 times more likely than the Phillies to win
 - The winner is as likely to be among these four teams — Mets, Yankees, Phillies, Dodgers — as not.

Construct a table of your subjective probabilities like the one in Example 1.3.

3. Suppose that you have applied to two graduate schools, A and B. Your subjective probability of being accepted is 0.6 for school A and 0.7 for school B. Hint: the fact that these are subjective probabilities does not change the way you solve the problem. Remember, the mathematics of probability is the same regardless of whether the probabilities represent long run relative frequencies or subjective degrees of relative likelihood. Just take the 0.6 and 0.7 values as given and use them to solve the following parts.
 - a. Interpret the subjective probabilities: How many times more likely than not are you to be accepted at school A? How many times more likely are you to be accepted at school B than at school A?
 - b. What is the largest possible probability of being accepted by both schools? Under what scenario (however unrealistic) would this be true? Explain.

- c. What is the smallest possible probability of being accepted by both schools? Under what scenario (however unrealistic) would this be true? Explain.
 - d. Explain why your subjective probability of being accepted by both schools is not necessarily 0.42.
 - e. **For the remaining parts, suppose your subjective probability of being accepted at both schools is 0.55.** If you are accepted at school A, what is your probability of also being accepted at school B? For this and the remaining parts in addition to computing the probability represent it with proper notation.
 - f. If you are accepted at school A, what is your probability of not being accepted at school B?
 - g. If you are not accepted at school A, what is your probability of being accepted at school B?
 - h. If you are accepted at school B, what is your probability of also being accepted at school A?
 - i. If you are not accepted at school B, what is your probability of being accepted at school A?
4. For each of the following, create your own “which of (a) or (b) is larger?” example. You can use the ones from the handout as examples, but you should choose your own contexts. Identify the correct answer and write a sentence or two explaining the answer to a student who is confused. You can use illustrations in your explanations, but don’t do any calculations.
- a. Reverse the direction of conditioning (like the 6 foot tall and NBA example)
 - b. Compare conditional and unconditional probabilities (like man versus man who is greater than 6 feet tall)
 - c. Two “particulars” that people might think are not equally likely but are (like HH-HHH versus HTHTT).
 - d. Comparing “the particular” and “the general” (HTHTT versus 2 H in 5 flips).
 - e. Comparing the probability of happening once versus at least once (like you winning the lottery versus someone winning the lottery)
5. The “matching problem” involves n distincts objects labeled $1, \dots, n$ which are placed in n distinct boxes labeled $1, \dots, n$, with exactly one object placed in each box. Suppose the objects are placed in the boxes uniformly at random, so that any possible arrangement is equally likely. We’re interested in the number of matches (the number of objects for which their label matches the label of the box in which they are placed.) Let C be the event that at least one object is placed in the correct spot (at least one match).
- a. Describe in full detail how, in principle, you would physical objects (like cards) to simulate a single repetition of the placement of objects in boxes, and the number of matches.
 - b. Describe in full detail how, in principle, you could conduct a simulation and use the results to approximate $P(C)$.

- c. [This applet conducts a simulation of the matching problem](#) (in the context of babies in a hospital being mixed up and returned to parents uniformly at random.) In the applet, “Number of babies” represents n and “Number of trials” represents the number of simulated repetitions. Starting with $n = 4$, try different values for the number of repetitions and hit the “Randomize” button to run the simulation. It helps to start with just a few repetitions so that you can see how the simulation is working, before jumping to thousands of repetitions. Use the simulation results to approximate $P(C)$ when $n = 4$.
 - d. How do you expect $P(C)$ to depend on n ? As n increases, do you expect $P(C)$ to increase, decrease, or stay about the same? Just think about it before proceeding; what does your intuition say?
 - e. Now try a few different values of n (say $n = 5$, $n = 10$, and $n = 20$, but you’re free to choose other values). For each value of n run the simulation and use the results to approximate $P(C)$. Record your results.
 - f. What do the simulation results suggest about the relationship between $P(C)$ and n ? (Remember that there is simulation margin of error, based on the number of repetitions. Two numerical approximations that are within the margin of error of each other are essentially the same approximation.)
6. Katniss throws a dart at a circular dartboard with radius 1 foot. Suppose that Katniss’s dart lands at a uniformly random location on the dartboard (and she never misses the dartboard).
- a. Compute the probability that Katniss’s dart lands within 1 inch of the center of the dartboard.
 - b. Compute the probability that Katniss’s dart lands more than 1 inch but less than 2 inches away from the center of the dartboard.

Homework 2

Note: you can typically use software to compute probabilities for named distributions. But to get some practice with formulas, you should compute probabilities by hand for this assignment.

Problem 1

Consider three tennis players Arya (“A”), Brienne (“B”), and Cersei (“C”). One of these players is better than the other two, who are equally good/bad. When the best player plays either of the others, she has a $2/3$ probability of winning the match. When the other two players play each other, each has a $1/2$ probability of winning the match. But you do not know which player is the best. Based on watching the players warm up, you start with subjective probabilities of 0.5 that A is the best, 0.35 that B is the best, and 0.15 that C is the best. A and B will play the first match.

1. Suppose that A beats B in the first match. Compute your posterior probability that each of A, B, C is best given that A beats B in the first match.
2. Compare the posterior probabilities from the previous part to the prior probabilities. Explain how your probabilities changed, and why that makes sense.
3. **Coding required.** Code and run a simulation to simulate (1) who is the best player according to your prior distribution, (2) who wins the first match given who is the best player; then use the results to approximate the probabilities from part a.
4. Suppose instead that B beats A in the first match. Compute your posterior probability that each of A, B, C is best given that B beats A in the first match.
5. Compare the posterior probabilities from the previous part to the prior probabilities. Explain how your probabilities changed, and why that makes sense.
6. Now suppose again that A beats B in the first match, and also that A beats C in the second match. Compute your posterior probability that each of A, B, C is best given the results of the first two matches. (Hint: use as the prior your posterior probabilities from the previous part.) Explain how your probabilities changed, and why that makes sense.

Problem 2

Solve Example [4.3](#).

Problem 3

Consider a “best-of-5” series of games between two teams: games are played until one of the teams has won 3 games (requiring at most 5 games total). Suppose one team, team A, is better than the other, having a 0.55 probability of winning any particular game. Assume the results of the games are independent (and ignore advantage, etc). Let X represent the number of games played in the series. (Hint: It’s helpful to first construct a two-way table of probabilities with the number of games played and which team wins, and then use it to answer the following questions. It will also help to list some outcomes, like AABA (team A wins game 1, 2, and 4, and B wins game 3).)

- Compute the probability that team A wins the series in 3 games.
- Compute the probability that the series ends in 3 games.
- Compute the probability that team A wins the series.
- Are the events “team A wins the series” and “the series ends in 3 games” independent? Explain by comparing relevant probabilities.
- Let X represent the number of games played in the series. Find the distribution of X .

Problem 4

Randomly select a county in the U.S. Let X be the leading digit in the county’s population. [For example](#), if the county’s population is 10,040,000 (Los Angeles County) then $X = 1$; if 3,170,000 (Orange County) then $X = 3$; if 283,000 (SLO County) then $X = 2$; if 30,600 (Lassen County) then $X = 3$. The possible values of X are 1, 2, ..., 9. You might think that X is equally likely to be any of its possible values. However, a more appropriate model is to assume that X has pmf

$$p_X(x) = \begin{cases} \log_{10}(1 + \frac{1}{x}), & x = 1, 2, \dots, 9, \\ 0, & \text{otherwise} \end{cases}$$

This distribution is known as [Benford’s law](#).

- Construct a table specifying the distribution of X , and the corresponding spinner.
- Find $P(X \geq 3)$

- c. **Coding required.** Code and run a simulation and use the results to approximate the distribution of X . (In R, there are built in functions to simulate Benford's law, but I want you to use the `sample` function. In R, `log` is natural log; `log10` is base-10 log.)

Problem 5

Maya is a basketball player who makes 40% of her three point field goal attempts. Suppose that at the end of every practice session, she attempts three pointers until she makes one and then stops. Let X be the total number of shots she attempts in a practice session. Assume shot attempts are independent, each with probability of 0.4 of being successful.

- What are the possible values that X can take? Is X discrete or continuous?
- Explain why X does *not* have a Binomial distribution.
- Describe in detail how you could, in principle, conduct a simulation using physical objects (coins, cards, dice, etc) and how you would use the results to approximate the distribution of X .
- Compute and interpret $P(X = 1)$.
- Compute and interpret $P(X = 2)$.
- Compute and interpret $P(X = 3)$.
- Find the probability mass function of X . Be sure to specify the possible values.
- Construct a table, plot, and spinner corresponding to the distribution of X .
- Compute $P(X > 5)$ without summing. (Hint: what needs to be true about the first 5 attempts for $X > 5$?)

Problem 6

Suppose the number of earthquakes per hour, for a certain range of magnitudes in a certain region, follows a Poisson distribution with parameter 0.7.

- Compute and interpret the probability that there is at least one earthquake of this size in the region in any given hour.
- Compute and interpret the probability that there are exactly 3 earthquakes of this size in the region in any given hour.
- Interpret the value 0.7 in context.
- Construct a table, plot, and spinner corresponding to a $\text{Poisson}(0.7)$ distribution.

Problem 7

Suppose that a total of 350 students at a college are taking a particular statistics course. The college offers five sections of the course, each taught by a different instructor. The class sizes are shown in the following table.

Section	A	B	C	D	E
Number of students	35	35	35	35	210

We are interested in: What is the average class size?

- Suppose we randomly select one of the 5 *instructors*. Let X be the class size for the selected instructor. Specify the distribution of X . (A table is fine.)
- Compute and interpret $E(X)$.
- Compute and interpret $P(X = E(X))$.
- Suppose we randomly select one of the 350 *students*. Let Y be the class size for the selected student. Specify the distribution of Y . (A table is fine.)
- Compute and interpret $E(Y)$.
- Compute and interpret $P(Y = E(Y))$.
- Comment on how these two expected values compare, and explain why they differ as they do. Which average would you say is more relevant?

Homework 3

Problem 1

(This is an overly simplified example from actuarial science.) A life insurance company sells a term insurance policy to a 21 year old man (the “insured”) that pays \$100,000 to a designated beneficiary if the insured dies within the next 5 years, and pays nothing if the insured does not die before age 26. The probability that a randomly chosen 21-year-old man will die each year can be found in mortality tables. The company collects a premium of \$250 at the beginning of each year of the 5 years of the term, as long as the insured is alive, as payment for the insurance. The amount Y (in dollars) that the company earns on this policy is \$250 per year, less the \$100,000 that it pays out in the event the insured dies. The company doesn’t pay out anything if the insured does not die in the 5 year term.

Age at death	Earnings Y	Probability
21	−99,750	0.00183
22		0.00186
23		0.00189
24		0.00191
25		0.00193
26 or older		

- Find the distribution of Y by completing the above table. For example, the value -99,750 provides an example of how Y is calculated when the insured dies in the first year of the policy.
- Compute $E(Y)$.
- Write a sentence explaining in this context in what sense the number from the previous part is “expected”.
- Would you be willing to sell such an insurance policy to one of your 21 year old friends? Explain why this is good business for the insurance company, but not for you.
- Compute $\text{Var}(Y)$
- Compute $\text{SD}(Y)$.
- Compute the probability that Y is more than 1 standard deviation above its mean.

Problem 2

In a certain population, household income X (\$ thousands) follows the pdf

$$f_X(x) = cx^{-2.5}, \quad x \geq 30$$

for an appropriate constant c . Note that the measurement units are \$ thousands, so 1 represents 1 thousand dollars.

- What are the possible values of X ? Sketch the pdf of X .
- Find the value of c .
- Find the probability that a household has an income over \$200K.
- Find the probability that a household has an income exactly equal to \$200K.
- Without integrating, find the probability that a household has an income, rounded to the nearest thousand dollars, of \$200K.
- Without integrating, determine how many times more likely it is for a household to have an income, rounded to the nearest thousand dollars, of \$100K than \$200K.

Problem 3

In the meeting time problem, assume that Regina's R and Cady's Y arrival times (measured in fractions of the hour after noon, e.g. 0.25 represents 12:15), each follow a Uniform(0, 1) distribution, independently of each other. Let $W = |R - Y|$ be the amount of time (hours) the first person to arrive waits for the second person to arrive. It can be shown that W has pdf

$$f_W(w) = 2(1 - w), \quad 0 < w < 1.$$

- Sketch a plot of the pdf. Describe roughly what this means in context.
- Coding required.** Code and run a simulation to approximate the distribution of W . Does the histogram reasonably match up with the pdf? Then use the simulation results to approximate $P(W < 0.25)$.
- Use the pdf to compute $P(W < 0.25)$ and interpret the value. Set up an integral, but sketch a picture and use geometry to compute.

Problem 4

(Continued.) In the Regina/Cady problem, let $W = |R - Y|$ be the amount of time the first person to arrive has to wait for the second person. Recall that W is a continuous random variable with pdf

$$f_W(w) = 2(1 - w), \quad 0 < w < 1.$$

- a. **Coding required.** Use your simulation results from before to approximate $E(W)$, $E(W^2)$, $\text{Var}(W)$, and $\text{SD}(W)$.
- b. Compute and interpret $E(W)$.
- c. Compute $E(W^2)$.
- d. Compute $\text{Var}(W)$.
- e. Compute $\text{SD}(W)$.
- f. Explain in words in detail how you could use the $\text{Uniform}(0, 1)$ spinner and simulation to approximate $\text{Var}(W)$.
- g. Compute the probability that W is more than 1 SD away from its mean.

Problem 5

Katniss throws a dart at a circular dartboard with radius 1 foot. Suppose that Katniss's dart lands at a uniformly random location on the dartboard (and she never misses the dartboard).

- a. Compute the probability that Katniss's dart lands within 1 inch of the center of the dartboard.
- b. Compute the probability that Katniss's dart lands more than 1 inch but less than 2 inches away from the center of the dartboard.

Homework 4

Problem 1

In a certain region, times (minutes) between occurrences of earthquakes (of any magnitude) have a distribution with percentiles displayed in the table below.

- Construct a spinner corresponding to this distribution.
- What percent of times are between 26.8 and 110.0 minutes?
- Let F be the cdf. Evaluate and interpret $F(42.8)$.
- Let Q be the quantile function. Evaluate and interpret $Q(0.9)$.
- Sketch (by hand) a histogram of this distribution.

Problem 2

(Continued from previous HW.) In the Regina/Cady problem, let $W = |R - Y|$ be the amount of time the first person to arrive has to wait for the second person. Recall that W is a continuous random variable with pdf

$$f_W(w) = 2(1 - w), \quad 0 < w < 1.$$

- Let F be the cdf. Evaluate and interpret $F(0.25)$.
- Find an expression for the cdf of W . Set up an integral, but sketch a picture and use geometry to compute.
- Find and interpret the 25th percentile of W . (You can do the next part first if you want, but it might help to

Percentile	Value (minutes)
10th	12.6
20th	26.8
30th	42.8
40th	61.3
50th	83.2
60th	110.0
70th	144.5
80th	193.1
90th	276.3

start with a specific number like in this part.) c. Find the quantile function of W . d. Sketch a spinner corresponding to the distribution of W . Label the 25th, 50th, and 75th percentiles. e. **Coding required.** Use your simulation results from before to approximate the 25th, 50th, and 75th percentiles.

Problem 3

Solve Example 11.3 from the Normal distribution handout (the problem on daily high temperatures in SLO.)

Problem 4

Solve Example 11.4 from the Normal distribution handout (the problem on “effect size”).

Problem 5

The latest series of collectible Lego Minifigures contains 3 different Minifigure prizes (labeled 1, 2, 3). Each package contains a single unknown prize. Suppose we only buy 3 packages and we consider as our sample space outcome the results of just these 3 packages (prize in package 1, prize in package 2, prize in package 3). For example, 323 (or (3, 2, 3)) represents prize 3 in the first package, prize 2 in the second package, prize 3 in the third package. Let X be the number of distinct prizes obtained in these 3 packages. Let Y be the number of these 3 packages that contain prize 1. Suppose that each package is equally likely to contain any of the 3 prizes, regardless of the contents of other packages. There are 27 possible, equally likely outcomes

- Evaluate X and Y for each of the outcomes.
- Construct a two-way table representing the *joint distribution* of X and Y .
- Sketch a plot representing the joint distribution of X and Y .
- Construct a spinner corresponding to the joint distribution of X and Y .
- Describe two ways to simulate an (X, Y) pair.
- Identify the *marginal* distribution of X , and construct a corresponding spinner.
- Identify the *marginal* distribution of Y , and construct a corresponding spinner.
- Describe in detail in words how you could conduct a simulation and use the results to approximate. (This part is asking you to describe the process in words; not to write code.)
 - $P(X = 3)$
 - $P(X = 2, Y = 1)$

box1	box2	box3	X	Y
1	1	1		
2	1	1		
3	1	1		
1	2	1		
2	2	1		
3	2	1		
1	3	1		
2	3	1		
3	3	1		
1	1	2		
2	1	2		
3	1	2		
1	2	2		
2	2	2		
3	2	2		
1	3	2		
2	3	2		
3	3	2		
1	1	3		
2	1	3		
3	1	3		
1	2	3		
2	2	3		
3	2	3		
1	3	3		
2	3	3		
3	3	3		

- iii. $E(X)$
- iv. $E(Y)$
- v. $E(XY)$
- vi. $SD(X)$
- vii. $SD(Y)$
- 1. $Cov(X, Y)$
- i. $Corr(X, Y)$
- i. **Coding required.** Code and run the simulation from the previous part and use the simulation results to approximate each of
 - i. $P(X = 3)$
 - ii. $P(X = 2, Y = 1)$
 - iii. $E(X)$
 - iv. $E(Y)$
 - v. $E(XY)$
 - vi. $SD(X)$
 - vii. $SD(Y)$
 - 1. $Cov(X, Y)$
 - i. $Corr(X, Y)$
- j. Compute
 - i. $P(X = 3)$
 - ii. $P(X = 2, Y = 1)$
 - iii. $E(X)$ iii. $E(Y)$
 - iv. $E(XY)$
 - v. $SD(X)$ iii. $SD(Y)$
 - 1. $Cov(X, Y)$
 - i. $Corr(X, Y)$

Homework 5

Problem 1

(Continuing from a previous HW) The latest series of collectible Lego Minifigures contains 3 different Minifigure prizes (labeled 1, 2, 3). Each package contains a single unknown prize. Suppose we only buy 3 packages and we consider as our sample space outcome the results of just these 3 packages (prize in package 1, prize in package 2, prize in package 3). For example, 323 (or (3, 2, 3)) represents prize 3 in the first package, prize 2 in the second package, prize 3 in the third package. Let X be the number of distinct prizes obtained in these 3 packages. Let Y be the number of these 3 packages that contain prize 1. Suppose that each package is equally likely to contain any of the 3 prizes, regardless of the contents of other packages; let P denote the corresponding probability measure.

- a. Find the conditional distribution of Y given $X = x$ for each possible value of x of X .
- b. Compute and interpret $E(Y|X = x)$ for each possible value of x of X .
- c. Find the conditional distribution of X given $Y = y$ for each possible value of y of Y .
- d. Compute and interpret $E(X|Y = y)$ for each possible value of y of Y .
- e. Explain how you could use spinners to implement the “marginal then conditional” method to simulate an (X, Y) pair.
- f. Suppose you have simulated many (X, Y) pairs. Explain how you could use the simulation results to approximate:
 - i. $P(X = 1|Y = 0)$
 - ii. the conditional distribution of X given $Y = 0$
 - iii. $E(X|Y = 0)$

Problem 2

Xavier and Yolanda are playing roulette. They both place bets on red on the same 3 spins of the roulette wheel before Xavier has to leave. (Remember, the probability that any bet on red on a single spin wins is $18/38$.) After Xavier leaves, Yolanda places bets on red on 2 more spins of the wheel. Let X be the number of bets that Xavier wins and let Y be the number that Yolanda wins.

- Identify by name the marginal distribution of X . Be sure to specify the values of any relevant parameters. Compute $E(X)$.
- Identify by name the marginal distribution of Y . Be sure to specify the values of any relevant parameters. Compute $E(Y)$.
- The joint distribution of X and Y is represented in the table below. Explain why $p_{X,Y}(1, 4) = 0$ and $p_{X,Y}(2, 1) = 0$.
- Compute $p_{X,Y}(2, 3)$. (Yes, the table tells you it's 0.1766, but you have to show how this number can be computed based on the assumptions of the problem.)
- Are X and Y independent? Justify your answer with an appropriate calculation.
- Compute and interpret $P(X + Y) = 4$.
- Make a table representing the marginal distribution of $X + Y$.
- Use the table from the previous part to compute $E(X + Y)$, and verify that it is equal to $E(X) + E(Y)$.
- Without doing any computation, determine if $\text{Var}(X + Y)$ will be greater than, less than, or equal to $\text{Var}(X) + \text{Var}(Y)$. Explain your reasoning.

x, y	0	1	2	3	4	5
0	0.0404	0.0727	0.0327	0	0	0
1	0	0.1090	0.1963	0.0883	0	0
2	0	0	0.0981	0.1766	0.0795	0
3	0	0	0	0.0294	0.0530	0.0238

Problem 3

Percent returns for assets X , Y , and Z follow a joint distribution with

- Mean 10 and standard deviation 15 for asset X ,
- Mean 5 and standard deviation 3 for asset Y ,
- Correlation of -0.6 between asset X and asset Y
- Asset Z yields a constant return of 1 percent.

An investment portfolio has 60% of its funds in asset X , 30% in asset Y , and 10% in asset Z .

- Let R be the portfolio return. Express R in terms of X, Y, Z .
- Compute $E(R)$.
- Compute $\text{Cov}(X, Y)$.
- Compute $\text{Cov}(X, Z)$.
- Compute $\text{SD}(R)$.

Problem 4

Devi and Paxton are meeting. Arrival times are measured in minutes after noon, with negative times representing arrivals before noon. Devi's arrival time follows a Normal distribution with mean 20 and SD 15 minutes, and Paxton's arrival time follows a Normal distribution with mean 25 and SD 10 minutes, independently of each other.

For each of the following, find the appropriate standardized value and make a ballpark estimate of the probability. Then use software to compute the probability.

- a. Compute the probability that Devi arrives before noon.
- b. Compute the probability that Devi arrives first.
- c. Compute the probability that the first person to arrive has to wait more than 15 minutes for the second person to arrive.
- d. **Coding required.** Code and run a simulation and use the results to approximate the probability from the previous part (just part c). You should first simulate (Devi, Paxton) arrival time pairs, and go from there.

Problem 5

(Continued.) Devi and Paxton are meeting. Arrival times are measured in minutes after noon, with negative times representing arrivals before noon. Devi's arrival time follows a Normal distribution with mean 20 and SD 15 minutes, and Paxton's arrival time follows a Normal distribution with mean 25 and SD 10 minutes.

For each of the following, find the appropriate standardized value and make a ballpark estimate of the probability. Then use software to compute the probability.

1. Assume the pairs of arrival times follow a Bivariate Normal distribution with correlation 0.8
 - a. Compute the probability that Devi arrives first given that Paxton arrives at 12:10.
 - b. Compute the probability that the first person to arrive has to wait more than 15 minutes for the second person to arrive.
 - c. **This part is optional.** Code and run a simulation and use the results to approximate the probability from the previous part (just part b). You should first simulate (Devi, Paxton) arrival time pairs, and go from there.
2. Assume the pairs of arrival times follow a Bivariate Normal distribution with correlation -0.7
 - a. Compute the probability that Devi arrives first given that Paxton arrives at 12:10.
 - b. Compute the probability that the first person to arrive has to wait more than 15 minutes for the second person to arrive.

- c. **This part is optional.** Code and run a simulation and use the results to approximate the probability from the previous part (just part b). You should first simulate (Devi, Paxton) arrival time pairs, and go from there.

Homework 6

Problem 1

You have a brand new coin and you want to estimate p , the probability that it lands on heads. You flip the coin n times and observe X heads.

- Suggest an estimator of p ; it should be an expression involving X and n . Describe in words what your estimator is. (Hint: what is the most obvious estimator you can think of?)
- Suppose you observe $x = 6$ heads in $n = 10$ flips. Use your estimator to compute an estimate of p based on this sample.
- Consider the estimator (called the “plus four” estimator)

$$\frac{X + 2}{n + 4} = \left(\frac{4}{n + 4} \right) (0.5) + \left(\frac{n}{n + 4} \right) \left(\frac{X}{n} \right)$$

Explain in words what this estimator does. (The two expressions in the equation above are just different ways of expressing the estimator.)

- Use the estimator from the previous part and compute an estimate of p based on a sample with 6 heads in 10 flips.
- If you observe 6 heads in 10 flips, which of your two estimates is a better estimate of p ? Explain.
- Compute and interpret the probability of observing 6 heads in 10 flips if $p = 0.6$.
- Compute and interpret the probability of observing 6 heads in 10 flips if $p = 0.571$.
- Compare the values from the two previous parts. If you observed 6 heads in 10 flips, which of the values — 0.6 or 0.571 — would you choose as your estimate of p ? Explain your reasoning.
- Suppose you observe 6 heads in 10 flips. Carefully write the likelihood function.
- Continuing the previous part, use any software of your choice and plot the likelihood function. Based on your plot, what value appears to be the maximum likelihood estimate of p based on a sample with 6 heads in 10 flips? (You can do this just by zooming in on the plot.)
- Continuing the previous part, carefully write the log-likelihood function.
 - Use calculus to find the maximum likelihood estimate of p based on observing 6 heads in 10 flips.
- Now suppose you observe x heads in 10 flips. Carefully write the likelihood function.

- n. Continuing the previous part, carefully write the log-likelihood function.
- o. Use calculus to find a general expression for the maximum likelihood estimate of p based on observing x heads in 10 flips. (Your answer will be an expression involving x .)

Problem 2

Suppose that times (minutes) between earthquakes (of any magnitude in a certain region) follow an Exponential distribution with rate parameter λ . Let $X_1, \dots, X_n$ be a random sample of times (minutes) between earthquakes for n earthquakes. It can be shown that the MLE of λ is $1/\bar{X}$.

1. Explain intuitively why $1/\bar{X}$ is a reasonable estimator of λ . Hint: it's not just because it's the MLE; think of properties of Exponential distributions.
2. Suppose the observed sample is: 20, 37, 13, 10, 25 minutes. Carefully write the likelihood function. (For a random sample from a continuous distribution, to find the likelihood function you plug each observed value into the pdf and then form the product.)
3. Use any software you want to carefully plot the likelihood function from the previous part. Then use your plot to verify that the MLE of λ based on this sample is $1/\bar{x} = 5/105 = 0.0476$. (You can just zoom in on the plot.)
4. Write a clearly worded sentence reporting in context your estimate of λ from the previous part.

Homework 7

Problem 1

Assume a $\text{Poisson}(\mu)$ model for the number of home runs hit (in total by both teams) in a MLB game. Let $X_1, \dots, X_n$ be a random sample of home run counts for n games.

Suppose we want to estimate $\theta = \mu e^{-\mu}$, the probability that any single game has exactly 1 HR (for $\text{Poisson}(\mu)$, $P(X = 1) = e^{-\mu} \mu^1 / 1! = \mu e^{-\mu}$). Consider two estimators of θ :

- $\hat{\theta} = \bar{X} e^{-\bar{X}}$
 - $\hat{p}$ = sample proportion of 1s = $\frac{\text{number of games in the sample with 1 HR}}{\text{sample size}}$
- a. Compute the value of $\hat{\theta}$ based on the sample (3, 0, 1, 4, 0). Write a clearly worded sentence reporting in context this estimate of θ .
 - b. Compute the value of $\hat{p}$ based on the sample (3, 0, 1, 4, 0). Write a clearly worded sentence reporting in context your estimate of θ .
 - c. Which of these two estimators is the MLE of θ in this situation? Explain, without doing any calculations.
 - d. It can be shown that $\hat{p}$ is an unbiased estimator of θ . Explain in words what this means.
 - e. Is $\hat{\theta}$ an unbiased estimator of θ ? Explain. (You don't have to derive anything; just apply a general principle.)
 - f. Suppose $\mu = 2.3$ and $n = 5$. Explain in full detail how you would use simulation to approximate the bias of $\hat{\theta}$ in this case.
 - g. **Coding required.** Conduct the simulation from the previous part and approximate bias of $\hat{\theta}$ when $\mu = 2.3$ and $n = 5$.
 - h. Explain in full detail how you would use simulation to approximate the bias *function* of $\hat{\theta}$ when $n = 5$.
 - i. **Coding required.** Conduct the simulation from the previous part and plot the approximate bias function when $n = 5$. For what values of μ does $\hat{\theta}$ tend to overestimate μ ? Underestimate? For what values of μ is the bias the worst?

Problem 2

Continuing Problem 1.

- a. It can be shown that $\text{Var}(\hat{p}) = \frac{\theta(1-\theta)}{n}$. Compute $\text{Var}(\hat{p})$ when $\mu = 2.3$ and $n = 5$. Then write a clearly worded sentence interpreting this value.
- b. Suppose $\mu = 2.3$ and $n = 5$. Explain in full detail how you would use simulation to approximate the variance of $\hat{\theta}$.
- c. **Coding required.** Conduct the simulation from the previous part and approximate the variance of $\hat{\theta}$ when $\mu = 2.3$ and $n = 5$. Then write a clearly worded sentence interpreting this value.
- d. Which estimator has smaller variance when $\mu = 2.3$ (and $n = 5$)? Answer, but then explain why this information alone is not really useful.
- e. Explain in full detail how you would use simulation to approximate the variance function of $\hat{\theta}$ (if $n = 5$).
- f. **Coding required.** Conduct the simulation from the previous part and plot the approximate variance function. Compare to the variance function of $\hat{p}$. Based on variability alone, which estimator is preferred?

Problem 3

Continuing Problems 1 and 2

- a. Compute $\text{MSE}(\hat{p})$ when $\mu = 2.3$ and $n = 5$. (You can do the next part first if you want, but it helps to work with specific numbers first.)
- b. Derive the MSE function of $\hat{p}$. (Hint: use facts from previous parts.)
- c. Suppose $\mu = 2.3$ (and $n = 5$). Explain in full detail how you would use simulation to approximate the MSE of $\hat{\theta}$.
- d. **Coding required.** Conduct the simulation from the previous part and approximate the MSE of $\hat{\theta}$ when $\mu = 2.3$ (and $n = 5$).
- e. Which estimator has smaller MSE when $\mu = 2.3$ (and $n = 5$)? Answer, but then explain why this information alone is not really useful.
- f. Explain in full detail how you would use simulation to approximate the MSE *function* of $\hat{\theta}$ (if $n = 5$).
- g. **Coding required.** Conduct the simulation from the previous part and plot the approximate MSE function. Compare to the MSE function of $\hat{p}$.
- h. Compare the MSEs of the two estimators for $n = 5$ and a few other values of n . Is there a clear preference between these two estimators? Discuss.

Homework 8

Problem 1

In roulette, a bet on a single number has a $1/38$ probability of success and pays 35-to-1. That is, if you bet 1 dollar, your net winnings are -1 with probability $37/38$ and +35 with probability $1/38$. Consider betting on a single number on each of n spins of a roulette wheel. Let $\bar{X}_n$ be your average net winnings per bet.

For each of the values $n = 10$, $n = 100$, $n = 1000$:

1. Compute $E(\bar{X}_n)$
2. Compute $SD(\bar{X}_n)$
3. **Coding required.** Run a simulation to determine if the distribution of $\bar{X}_n$ is approximately Normal
4. **Coding required.** Use simulation to approximate $P(\bar{X}_n > 0)$, the probability that you come out ahead after n bets
5. If $n = 1000$ use the Central Limit Theorem to approximate $P(\bar{X}_{1000} > 0)$, the probability that you come out ahead after 1000 bets.
6. The casino wants to determine how many bets on a single number are needed before they have (at least) a 99% probability of making a profit. (Remember, the casino profits if you lose; that is, if $\bar{X}_n < 0$.) Use the Central Limit Theorem to determine the minimum number of bets (keeping in mind that n must be an integer). You can assume that whatever n is, it's large for the CLT to kick in.

Problem 2

The standard measurement of the alcohol content of drinks is alcohol by volume (ABV), which is given as the volume of ethanol as a percent of the total volume of the drink. In a sample of 67 brands of beer, the mean ABV is 4.61 percent and the standard deviation of ABV is 0.754 percent.

- First, to get some practice, answer the following questions by computing by hand using only the information provided here.
- **Coding required.** Then, use Python to produce the summaries and analysis needed to answer the questions. Be sure to also include appropriate plots of the [beer data](#).

1. Compute a 95% confidence interval for the appropriate population mean.
2. Write a clearly worded sentence reporting your confidence interval in context.
3. Is 4.5% a plausible value of the parameter? Explain briefly.
4. One of the brands of beer in the sample is O'Doul's, a non-alcoholic beer. The ABV for O'Doul's is 0.4% (it has a bit of alcohol.) Suppose O'Doul's is removed from the data set. Compute the sample mean ABV of the remaining 66 brands.
5. The sample SD of ABV of the remaining 66 brands is 0.55 percent. Explain intuitively why this value is smaller than the sample SD of all 67 brands.
6. Compute the 95% confidence interval based on the sample with O'Doul's removed. Compare to the original interval, both in terms of center of the CI and its width. Explain briefly.
7. Based on the interval based on the sample with O'Doul's removed, is 4.5% a plausible value of the parameter? Explain briefly.
8. Which of the analyses is more appropriate: with or without O'Doul's? Explain your reasoning.

Problem 3

Do non-smokers have better lungs than smokers? One measure of lung function is forced expiratory volume (FEV), the amount of air (in liters) an individual can exhale in the first second of forceful breath. Larger values of FEV are indicative of better functioning lungs. In a study, FEV was measured for a group of 654 subjects, along with whether they smoked and other variables like age (years). Note: I'm purposely not yet giving you much information about how the sample was collected.

1. **Coding required.** Use Python to summarize the [FEV data](#). Summarize the distribution of FEV both overall and separately for smokers and non-smokers.
2. Suppose you want to fill in the blanks in the following sentence: 95% of FEV values in the population are between [blank] and [blank]. Assuming this sample is representative, provide reasonable values that fill in the blanks, and explain your reasoning. Hint: it is better to give a correct rough estimate than a precise but incorrect answer, so think before trying to compute anything. (Since you don't have much information about the sample, you don't have to clarify yet what "population" is appropriate.)
3. Using appropriate summary statistics, compute by hand a 95% confidence for the appropriate difference in population means.
4. **Coding required.** Use Python to compute the confidence interval and compare to what you computed by hand.
5. Write a clearly worded sentence interpreting the confidence interval in context.
6. Assuming this is a representative sample, are we reasonably confident that one group (smokers or non-smokers) tends to have better FEV? Why? Which group?
7. **Coding required.** You should have observed something surprising in the previous parts. Investigate some of the other variables in the data set and produce a few appropriate

plots or summaries that provide an explanation for the surprising result. Write a clearly worded sentence or two summarizing your explanation.

Problem 4

(I had much longer instructions written, but I think they weren't helpful. If you have questions about how the applet is working, don't hesitate to ask.)

You are going to use a [simulation applet](#) that randomly generates confidence intervals for a population mean to help you understand some ideas. The applet calculates a confidence interval for each set of randomly generated data.

- In the box for “Statistic” select means “Means”.
- In the box next to “Method” select “t”.
- Start with “Number of intervals” equal to 1 at first to see what is happening, but then change it to lots.

Experiment with different simulations; be sure to change

- Distribution
- Population Mean
- Population SD
- Sample size
- Confidence level

Then write a paragraph or two or some bullet points of your main observations. Based on this applet, what can you say about how confidence intervals work? How does changing each of the inputs affect the confidence intervals and the Results? Include a few screenshots from the applet to support your conclusions.

Homework 9

Problem 1

The following table displays the time (measured continuously in minutes) until the first goal was scored in each of 20 professional hockey games. The sample mean is 12.1 minutes, the sample median is 8.9 minutes, and the sample SD is 13.1 minutes.

4.4	14.9	2.8	1.4	1.1	8.1	11.7	3.9	2.4	15.7
8.8	0.6	5.1	10.4	9.1	13.7	27.8	9.0	46.0	44.7

1. You wish to estimate the population *median* time until the first goal is scored. Describe in detail in words how you could use the sample data and simulation to find an appropriate 95% bootstrap percentile confidence interval.
2. **Coding required.** Code and run the simulation from the previous part using the [hockey data](#). Summarize the results; provide a histogram of the bootstrap distribution, the value of the bootstrap SE, and a 95% bootstrap percentile confidence interval.
3. The following summarizes the bootstrap distribution of the sample median based on the sample data and an appropriate simulation.

Min	2.5th percentile	Median	97.5th percentile	Max	Mean	SD
1.25	4.15	8.90	12.70	19.75	8.52	2.10

Using the above information, compute the endpoints of each of the following bootstrap 95% confidence intervals for the population median.

- a. Normal interval
 - b. Percentile interval
4. Write a clearly worded sentence reporting the bootstrap percentile confidence interval from the previous part in context.

Problem 2

Two different machines (A and B) that fill packages of a certain candy are calibrated to a weight of 50 grams. Naturally, the weights of individual packages vary somewhat in the production process, but too much variation is undesirable. A small sample of packages is taken from each machine; the weights (grams) are in the following table.

Machine						Mean	SD
A	47.1	48.7	50.1	50.2	50.5	49.3	1.4
B	48.6	50.5	50.6	51.4	52.0	50.6	1.3

1. Describe in detail in words how you could use the sample data and simulation to find a 95% bootstrap percentile confidence interval for the *ratio of the variances* of packages weights for the two machines.
2. **Coding required.** Write code to implement the procedure from the previous part, and run the simulation to find a 95% bootstrap percentile confidence interval. Include your code and output.
3. Write a clearly worded sentence reporting the confidence interval from the previous part in context.

Problem 3

Continuing Problem 2.

Describe in detail how you could use simulation to approximate the p-value of the permutation test which uses the ratio of variances as the test statistic.

Problem 4

Scores on a certain standardized test (e.g. GRE) are approximately normally distributed with mean 525 and standard deviation 100 (assume that the scale is 200-800). An agency claims that students who take their test preparation class have a higher mean score than the national average. To test the validity of the claim we consider

$$H_0 : \mu = 525 \quad H_a : \mu > 525,$$

where μ is the mean test score of all students who take the preparation class. Suppose that we take a simple random sample of n students who have taken the class, and let $\bar{y}$ be the mean test score for the sample. You can assume that $\sigma = 100$ and just use the usual empirical rule (that is, don't worry about t distributions.)

1. Compute the p -value of the test if $n = 100$ and $\bar{y} = 541.4$.
2. Compute the p -value of the test if $n = 100$ and $\bar{y} = 541.5$.
3. Compute the p -value of the test if $n = 100,000$ and $\bar{y} = 526$.
4. Write a short paragraph explaining why (1) we should not rely on cutoffs like $\alpha = 0.05$ strictly; (2) why the term statistically “significant” is a poor choice of words which should not be used. Use the results from the previous parts and this context to support your explanation.

Problem 5

Psychologists have shown that we are often able to “chunk” information which allows us to remember more information (typically about seven chunks). A study was performed in previous statistics classes to investigate this idea. Students were given 20 seconds to memorize a sequence of 30 letters. After 20 seconds, everyone was asked to write down, in order, as many of the letters as they could. Every student saw the same sequence of 20 letters, JFKCIAF-BIUSASATGPABFFLOLNBACPR, but students were randomly assigned to see the letters presented (chunked) in one of two different ways

- a “meaningful” grouping: JFK-CIA-FBI-USA-SAT-GPA-BFF-LOL-NBA-CPR
- a “not meaningful” grouping: JFKC-IAF-BIU-SASA-TGP-ABF-FLO-LN-BAC-PR

Each person’s score was the number of correct letters in a row before the first mistake. (For example, JFKCIABIUSAT yields a score of 6 because of missing the F.) We want to know whether those given the meaningful “JFK” sequence (with the letters grouped into familiar acronyms) would tend to remember more letters in sequence, on average, than those given the not meaningful “JFKC” sequence.

1. **Coding required.** Use Python to summarize the [letters data](#).
2. State the null and alternative hypothesis in words and symbols
3. Explain in detail how, in principle, you would use index cards to conduct an appropriate simulation and use the simulation results to compute the p -value.
4. Compute by hand the t -statistic and the p -value.
5. **Coding required.** Use Python to conduct the hypothesis test, and compare the results to what you computed by hand.
6. Write a clearly worded sentence reporting the conclusion of the hypothesis test in context.
7. Compute by hand an appropriate 95% confidence interval.
8. **Coding required.** Use Python to compute the confidence interval, and compare the results to what you computed by hand.
9. Write a clearly worded sentence reporting the conclusion of the confidence interval in context.

Problem 6

The data in this exercise comes from the study: [Singh R, Meier T, Kuplicki R, Savitz J, et al., “Relationship of Collegiate Football Experience and Concussion With Hippocampal Volume and Cognitive Outcome,” JAMA, 311\(18\), 2014.](#)

The study included 3 groups, with 25 cases in each group. The control group consisted of healthy individuals with no history of brain trauma who were comparable to the other groups in age, sex, and education. The second group consisted of NCAA Division 1 college football players with no history of concussion, while the third group consisted of NCAA Division 1 college football players with a history of concussion. High resolution MRI was used to collect brain hippocampus volume (microliters).

1. **Coding required.** Use Python to summarize the [brain data](#).
2. Compute by hand the ANOVA F statistic.
3. Describe in full detail how (in principle) you could use index cards to simulate the null distribution of the ANOVA F statistic and how you would use the simulation results to find the p-value. (You don’t have to compute the p-value; just explain how you could find it after you performed the simulation.)
4. Optional: use this [applet to perform the permutation test and approximate the p-value](#).
5. **Coding required.** Use Python to conduct the hypothesis test, and compare the results to what you computed by hand.
6. Write a clearly worded sentence containing the conclusion of the hypothesis test in the context of the problem.
7. **Coding required.** Use Python to compute Tukey pairwise 95% confidence intervals.
8. Which of the confidence intervals contain 0, and which do not? Explain what this means.
9. Write a clearly worded sentence interpreting each of the three confidence intervals in context.

Homework 1 Solutions

Problem 1

In each of the following, write a clearly worded sentence interpreting the numerical value of the probability as a long run relative frequency in context. (Just take the numerical values as given for now. We'll see how to compute probabilities like these later.)

- a. The probability of rolling doubles when you roll two fair six-sided dice is $1/6$.
- b. The probability of rolling doubles on three consecutive rolls of two fair six-sided dice is 0.00463.
- c. The probability that the sum of 100 rolls of a fair six-sided die is less than 370 is 12%.
- d. Roll a fair six-sided die until you roll a 6 three times and then stop. The probability that you roll the die at least 10 times is 0.822.

Solution

- a. In about 16.7% sets of two rolls both rolls in the set are 6.
- b. In about 0.463% of sets of three rolls all three rolls are 6. (If the sets consisted of more rolls, say 10 rolls, the probability of at least 3 consecutive 6s somewhere in the set would be greater.)
- c. In about 12% of sets of 100 rolls the sum of the 100 rolls is less than 370.
- d. In about 82.2% of sets of 9 rolls there are at most 2 6s.

Problem 2

Suppose your subjective probabilities for the 2022 World Series champion satisfy the following conditions.

- The White Sox and Brewers are equally likely to win
- The Astros are 1.5 times more likely than the White Sox to win
- The Dodgers are 2 times more likely than the Astros to win

- The winner is as likely to be among these four teams — Dodgers, Astros, White Sox, Brewers — as not

Construct a table of your subjective probabilities like the one in Example 1.3.

Solution

See [Example 1.5 in my probability notes](#).

Problem 3

Suppose that you have applied to two graduate schools, A and B. Your subjective probability of being accepted is 0.6 for school A and 0.7 for school B. Hint: the fact that these are subjective probabilities does not change the way you solve the problem. Remember, the mathematics of probability is the same regardless of whether the probabilities represent long run relative frequencies or subjective degrees of relative likelihood. Just take the 0.6 and 0.7 values as given and use them to solve the following parts.

- Interpret the subjective probabilities: How many times more likely than not are you to be accepted at school A? How many times more likely are you to be accepted at school B than at school A?
- What is the largest possible probability of being accepted by both schools? Under what scenario (however unrealistic) would this be true? Explain.
- What is the smallest possible probability of being accepted by both schools? Under what scenario (however unrealistic) would this be true? Explain.
- Explain why your subjective probability of being accepted by both schools is not necessarily 0.42.
- For the remaining parts, suppose your subjective probability of being accepted at both schools is 0.55.** If you are accepted at school A, what is your probability of also being accepted at school B? For this and the remaining parts in addition to computing the probability represent it with proper notation.
- If you are accepted at school A, what is your probability of not being accepted at school B?
- If you are not accepted at school A, what is your probability of being accepted at school B?
- If you are accepted at school B, what is your probability of also being accepted at school A?
- If you are not accepted at school B, what is your probability of being accepted at school A?

Solution

- a. The probability of being accepted at both schools can be no more than the probability of being accepted at either school, 0.6 and 0.7. So we try 0.6 and see if we get valid probabilities.

	Accepted by A	Not accepted by A	Total
Accepted by B	0.6	0.1	0.7
Not accepted by B	0	0.3	0.3
Total	0.6	0.4	1.00

All the probabilities above are valid, so the largest the probability of being accepted at both schools can be is 0.6. This occurs if getting into school A guarantees that you will also get into school B.

- b. Trying a subjective probability of being accepted at both schools leads to a negative probability of not being accepted at either school. So try a probability of 0 for the probability of not being accepted at either school.

	Accepted by A	Not accepted by A	Total
Accepted by B	0.3	0.4	0.7
Not accepted by B	0.3	0	0.3
Total	0.6	0.4	1.00

So the smallest the probability of being accepted at both schools can be is 0.3. This only occurs if you're guaranteed of getting into at least one of the two schools.

- c. Your chances of getting into one school might change depending if you've gotten into the other. For example, your chances of getting to Cal Poly are probably higher if you know you've already gotten into Stanford.
- d. For the remaining parts, suppose your subjective probability of being accepted at both schools is 0.55. If you are accepted at school A, what is your probability of also being accepted at school B?

	Accepted by A	Not accepted by A	Total
Accepted by B	0.55	0.15	0.70
Not accepted by B	0.05	0.25	0.30
Total	0.60	0.40	1.00

If you are accepted at school A, then your probability of also being accepted at school B is $P(B|A) = 0.55/0.60 = 0.917$. (91.7% of students like you who get into school A also get into school B.)

- e. If you are accepted at school A, what is your probability of not being accepted at school B?

If you are accepted at school A, then your probability of not being accepted at school B is $P(B^c|A) = 0.05/0.60 = 0.083$. (8.3% of students like you who get into school A do not get into school B.)

- f. If you are not accepted at school A, what is your probability of being accepted at school B?

If you are not accepted at school A, then your probability of being accepted at school B is $P(B|A^c) = 0.15/0.40 = 0.375$. (37.5% of students like you who do not get into school A do get into school B.)

- g. If you are accepted at school B, what is your probability of also being accepted at school A?

If you are accepted at school B, then your probability of being accepted at school A is $P(A|B) = 0.55/0.7 = 0.786$. (78.6% of students like you who get into school B also get into school A.)

- h. If you are not accepted at school B, what is your probability of being accepted at school A?

If you are not accepted at school B, then your probability of being accepted at school A is $P(A|B^c) = 0.05/0.30 = 0.167$. (16.7% of students like you who do not get into school B do get into school A.)

Problem 4

For each of the following, create your own “which of (a) or (b) is larger?” example. You can use the ones from the handout as examples, but you should choose your own contexts. Identify the correct answer and write a sentence or two explaining the answer to a student who is confused. You can use illustrations in your explanations, but don’t do any calculations.

- Reverse the direction of conditioning (like the 6 foot tall and NBA example)
- Compare conditional and unconditional probabilities (like man versus man who is greater than 6 feet tall)
- Two “particulars” that people might think are not equally likely but are (like HHHHHH versus HTHTTT).
- Comparing “the particular” and “the general” (HTHTTT versus 2 H in 5 flips).

- e. Comparing the probability of happening once versus at least once (like you winning the lottery versus someone winning the lottery)

Solution

Results vary. See [Section 1.5 of my probability notes](#) for discussion.

Problem 5

The “matching problem” involves n distinct objects labeled $1, \dots, n$ which are placed in n distinct boxes labeled $1, \dots, n$, with exactly one object placed in each box. Suppose the objects are placed in the boxes uniformly at random, so that any possible arrangement is equally likely. We’re interested in the number of matches (the number of objects for which their label matches the label of the box in which they are placed.) Let C be the event that at least one object is placed in the correct spot (at least one match).

- a. Describe in full detail how, in principle, you would use physical objects (like cards) to simulate a single repetition of the placement of objects in boxes, and the number of matches.
- b. Describe in full detail how, in principle, you could conduct a simulation and use the results to approximate $P(C)$.
- c. [This applet conducts a simulation of the matching problem](#) (in the context of babies in a hospital being mixed up and returned to parents uniformly at random.) In the applet, “Number of babies” represents n and “Number of trials” represents the number of simulated repetitions. Starting with $n = 4$, try different values for the number of repetitions and hit the “Randomize” button to run the simulation. It helps to start with just a few repetitions so that you can see how the simulation is working, before jumping to thousands of repetitions. Use the simulation results to approximate $P(C)$ when $n = 4$.
- d. How do you expect $P(C)$ to depend on n ? As n increases, do you expect $P(C)$ to increase, decrease, or stay about the same? Just think about it before proceeding; what does your intuition say?
- e. Now try a few different values of n (say $n = 5$, $n = 10$, and $n = 20$, but you’re free to choose other values). For each value of n run the simulation and use the results to approximate $P(C)$. Record your results.
- f. What do the simulation results suggest about the relationship between $P(C)$ and n ? (Remember that there is simulation margin of error, based on the number of repetitions. Two numerical approximations that are within the margin of error of each other are essentially the same approximation.)

Solution

- a. Use a box with 4 tickets, labeled 1, 2, 3, 4. Shuffle the tickets and draw all 4 *without* replacement and record the tickets drawn *in order*. Let X be the number of tickets that match their spot in order. (For example, if the tickets are drawn in the order 2431 then the realized value of X is 1 since only ticket 3 matches its spot in the order.) Since $C = \{X \geq 1\}$, event C occurs if $X \geq 1$ and does not occur if $X = 0$. We could record the realization of event C as “True” or “False”.
- b. Repeat the previous part many times. Count the number of repetitions on which C occurs and divide by the total number of repetitions to approximate $P(C)$.
- c. See the applet.
- d. You might think the probability might decrease with n , since the probability that any particular object goes in its correct spot decreases with n . On the other hand, as n increases, the more chances there are to get at least one object in the correct spot. These considerations move in opposite directions; what happens as n increases?
- e. When $n = 4$, $P(X = 0)$ is approximately 0.375; when $n = 5$, $P(X = 0)$ is approximately 0.367; but for $n \geq 6$, $P(X = 0)$ is approximately 0.368 for any value of n .
- f. The probability of no matches essentially does not depend on n ! We’ll investigate a little more in class.

Problem 6

Katniss throws a dart at a circular dartboard with radius 1 foot. Suppose that Katniss’s dart lands at a uniformly random location on the dartboard (and she never misses the dartboard).

- a. Compute the probability that Katniss’s dart lands within 1 inch of the center of the dartboard.
- b. Compute the probability that Katniss’s dart lands more than 1 inch but less than 2 inches away from the center of the dartboard.

Solution

See [Example 3.30 in my probability notes](#)

Homework 2 Solutions

Note: you can typically use software to compute probabilities for named distributions. But to get some practice with formulas, you should compute probabilities by hand for this assignment.

Problem 1

Consider three tennis players Arya (“A”), Brienne (“B”), and Cersei (“C”). One of these players is better than the other two, who are equally good/bad. When the best player plays either of the others, she has a $2/3$ probability of winning the match. When the other two players play each other, each has a $1/2$ probability of winning the match. But you do not know which player is the best. Based on watching the players warm up, you start with subjective probabilities of 0.5 that A is the best, 0.35 that B is the best, and 0.15 that C is the best. A and B will play the first match.

1. Suppose that A beats B in the first match. Compute your posterior probability that each of A, B, C is best given that A beats B in the first match.
2. Compare the posterior probabilities from the previous part to the prior probabilities. Explain how your probabilities changed, and why that makes sense.
3. **Coding required.** Code and run a simulation to simulate (1) who is the best player according to your prior distribution, (2) who wins the first match given who is the best player; then use the results to approximate the probabilities from part a.
4. Suppose instead that B beats A in the first match. Compute your posterior probability that each of A, B, C is best given that B beats A in the first match.
5. Compare the posterior probabilities from the previous part to the prior probabilities. Explain how your probabilities changed, and why that makes sense.
6. Now suppose again that A beats B in the first match, and also that A beats C in the second match. Compute your posterior probability that each of A, B, C is best given the results of the first two matches. (Hint: use as the prior your posterior probabilities from the previous part.) Explain how your probabilities changed, and why that makes sense.

best_player	prior	likelihood_A_beats_B	product	posterior
A	0.50	0.6667	0.3333	0.6349
B	0.35	0.3333	0.1167	0.2222
C	0.15	0.5000	0.0750	0.1429
Total	1.00	1.5000	0.5250	1.0000

Solution

a) Hypotheses are which player is best (A, B, C). Evidence is that A beats B. The likelihood is the probability that A beats B given each of the best players.

- If A is best, probability A beats B is $2/3$.
- If B is best, probability A beats B is $1/3$.
- If C is best, probability A beats B is $1/2$.

b) A's probability of being the best increased, which makes sense because A won the match. B's probability of being the best decreased considerably, which makes sense because B lost the match. C's probability of being the best decreased slightly, despite C not being involved in the match.

c) Simulation below.

```
N_rep = 10000

# initialize a data.frame for storing results
best_winner = data.frame(best = character(),
                          winner = character())

for (i in 1:N_rep) {
  best = sample(c("A", "B", "C"), size = 1, prob = c(0.5, 0.35, 0.15))
  if (best == "A") {
    winner = sample(c("A", "B"), size = 1, prob = c(2 / 3, 1 / 3))
  } else if (best == "B"){
    winner = sample(c("A", "B"), size = 1, prob = c(1 / 3, 2 / 3))
  } else {
    winner = sample(c("A", "B"), size = 1, prob = c(1 / 2, 1 / 2))}
  best_winner[i, ] = c(best, winner)
}

# first few rows; each row is a (best player, first match winner) pair
head(best_winner)
```

	best	winner
1	B	B
2	C	A
3	A	A
4	B	B
5	B	A
6	A	A

```
# tabulate results with proportions
table(best_winner) / N_rep
```

	winner	
best	A	B
A	0.3298	0.1637
B	0.1126	0.2440
C	0.0753	0.0746

```
# Approximate P(A is best | A wins first match)
nrow(best_winner[(best_winner$best == "A") & (best_winner$winner == "A"), ]) / nrow(best_winner)
```

```
[1] 0.6370485
```

```
# Approximate P(B is best | A wins first match)
nrow(best_winner[(best_winner$best == "B") & (best_winner$winner == "A"), ]) / nrow(best_winner)
```

```
[1] 0.2175005
```

```
# Approximate P(C is best | A wins first match)
nrow(best_winner[(best_winner$best == "C") & (best_winner$winner == "A"), ]) / nrow(best_winner)
```

```
[1] 0.145451
```

d) Hypotheses are which player is best (A, B, C). Evidence is that B beats A. The likelihood is the probability that B beats A given each of the best players.

- If A is best, probability B beats A is $1/3$.
- If B is best, probability B beats A is $2/3$.
- If C is best, probability B beats A is $1/2$.

best_player	prior	likelihood_B_beats_A	product	posterior
A	0.50	0.3333	0.1667	0.3509
B	0.35	0.6667	0.2333	0.4912
C	0.15	0.5000	0.0750	0.1579
Total	1.00	1.5000	0.4750	1.0000

best_player	prior	likelihood_A_beats_C	product	posterior
A	0.6349	0.6667	0.4233	0.7273
B	0.2222	0.5000	0.1111	0.1909
C	0.1429	0.3333	0.0476	0.0818
Total	1.0000	1.5000	0.5820	1.0000

- e) A's probability of being the best decreased, which makes sense because A lost the match. B's probability of being the best increased, which makes sense because B won the match. C's probability of being the best changed slightly, despite C not being involved in the match.
- f) The prior is the posterior from the first part. Evidence is that A beats C. The likelihood is the probability that A beats C given each of the best players.
- If A is best, probability A beats C is $2/3$.
 - If B is best, probability A beats C is $1/2$.
 - If C is best, probability A beats C is $1/3$.

By winning both matches, A's probability of being the best has increased considerably. By losing their only matches, B's and C's probabilities of being the best have decreased considerably.

Problem 2

Solve Example 4.3.

Solution

See [Example 3.16 in my probability notes](#)

Problem 3

Consider a “best-of-5” series of games between two teams: games are played until one of the teams has won 3 games (requiring at most 5 games total). Suppose one team, team A, is better than the other, having a 0.55 probability of winning any particular game. Assume the results of the games are independent (and ignore advantage, etc). Let X represent the number of games played in the series. (Hint: It’s helpful to first construct a two-way table of probabilities with the number of games played and which team wins, and then use it to answer the following questions. It will also help to list some outcomes, like AABA (team A wins game 1, 2, and 4, and B wins game 3).)

- Compute the probability that team A wins the series in 3 games.
- Compute the probability that the series ends in 3 games.
- Compute the probability that team A wins the series.
- Are the events “team A wins the series” and “the series ends in 3 games” independent? Explain by comparing relevant probabilities.
- Let X represent the number of games played in the series. Find the distribution of X .

Solution

- It is helpful to construct a two-way table to answer the following questions.

	x			
	3	4	5	Total
A wins	$0.55^3 = 0.166$	$3(0.55^3)(0.45) = 0.22553$	$6(0.55^3)(0.45)^2 = 0.202$	0.593
A does not win	$0.45^3 = 0.091$	$3(0.45^3)(0.55) = 0.15053$	$6(0.45^3)(0.55)^2 = 0.165$	0.407
Total	0.258	0.375	0.368	1

There is only one outcome for which A wins in 3 games: AAA. There are three outcomes in which A wins in 4 games: AABA, ABAA, BAAA. (Not AAAB because then the series would be over in 3 games.) Since the games are independent, an outcome like AABA has probability $(0.55^3)(0.45)$, so the probability that A wins in 4 games is $3(0.55^3)(0.45)$. There are six outcomes in which A wins in 5 games: AABBA, ABABA, BAABA, ABBA, BABAA, BBAAA. (Not outcomes like AAABB or AABAB because then the series would be over in 3 or 4 games.) Since the games are independent, an outcome like AABBA has probability $(0.55^3)(0.45)^2$, so the probability that A wins in 5 games is $6(0.55^3)(0.45)$. You can fill in the rest of the table similarly.

The probability that team A wins the series in 3 games is $P(X = 3, A) = 0.55^3 = 0.166$.

- Either the stronger team wins 3 in a row or loses 3 in a row. The probability is $0.55^3 + (1 - 0.55)^3 = 0.2575$.

- c. The probability that team A wins the series is the sum of first row: $P(A) = 0.593$.
- d. No. $P(X = 3, A) = 0.55^3 = 0.166 \neq 0.152 = (0.2575)(0.593) = P(X = 3)P(A)$. Alternatively, $P(X = 3|A) = 0.166/0.593 = 0.2799 \neq 0.2575 = P(X = 3)$. Given that A wins the series the series it more likely to end in 3 games than when B wins the series.
- e. Total row above. X can take values 3, 4, or 5. Consider first the ways in which the stronger team wins in 4 games: AABA, ABAA, BAAA. (For example AABA, means the stronger team wins game 1, 2, and 4, and the weaker team wins game 3). Each of these outcomes has probability $0.55^3(0.45)$ so the probability that team A wins in 4 games is $3(0.55)^3(0.45)$. Similarly, the probability that team B wins in 4 games is $3(0.45)^3(0.55)$. So

$$P(X = 4) = 3(0.55)^3(0.45) + 3(0.45)^3(0.55) = 0.3750$$

You could find $P(X = 5)$ in a similar way, or just use the fact that the probabilities have to add up to 1. The distribution of X is given by the following table.

x	$P(X = x)$
3	0.2575
4	0.3750
5	0.3675

Problem 4

Randomly select a county in the U.S. Let X be the leading digit in the county's population. [For example](#), if the county's population is 10,040,000 (Los Angeles County) then $X = 1$; if 3,170,000 (Orange County) then $X = 3$; if 283,000 (SLO County) then $X = 2$; if 30,600 (Lassen County) then $X = 3$. The possible values of X are $1, 2, \dots, 9$. You might think that X is equally likely to be any of its possible values. However, a more appropriate model is to assume that X has pmf

$$p_X(x) = \begin{cases} \log_{10}(1 + \frac{1}{x}), & x = 1, 2, \dots, 9, \\ 0, & \text{otherwise} \end{cases}$$

This distribution is known as [Benford's law](#).

- Construct a table specifying the distribution of X , and the corresponding spinner.
- Find $P(X \geq 3)$

- c. **Coding required.** Code and run a simulation and use the results to approximate the distribution of X . (In R, there are built in functions to simulate Benford's law, but I want you to use the `sample` function. In R, `log` is natural log; `log10` is base-10 log.)

Solution

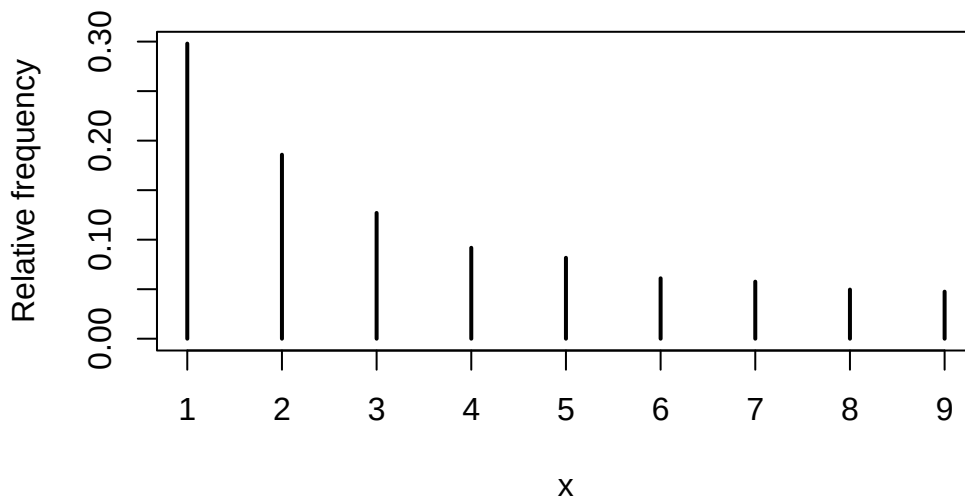
See [Example 4.6 of my probability notes](#)

```
possible_x = 1:9

p_x = log10(1 + 1 / possible_x)

x = sample(possible_x, size = 10000, prob = p_x, replace = TRUE)

plot(table(x) / 10000,
      ylab = "Relative frequency")
```



Problem 5

Maya is a basketball player who makes 40% of her three point field goal attempts. Suppose that at the end of every practice session, she attempts three pointers until she makes one and

then stops. Let X be the total number of shots she attempts in a practice session. Assume shot attempts are independent, each with probability of 0.4 of being successful.

- What are the possible values that X can take? Is X discrete or continuous?
- Explain why X does *not* have a Binomial distribution.
- Describe in detail how you could, in principle, conduct a simulation using physical objects (coins, cards, dice, etc) and how you would use the results to approximate the distribution of X .
- Compute and interpret $P(X = 1)$.
- Compute and interpret $P(X = 2)$.
- Compute and interpret $P(X = 3)$.
- Find the probability mass function of X . Be sure to specify the possible values.
- Construct a table, plot, and spinner corresponding to the distribution of X .
- Compute $P(X > 5)$ without summing. (Hint: what needs to be true about the first 5 attempts for $X > 5$?)

Solution

- X can take values 1, 2, 3, Even though it is unlikely that X is very large, there is no fixed upper bound. Even though X can take infinitely many values, X is a discrete random variables because it takes *countably* many possible values.
- X does not have a Binomial distribution since the number of trials is not fixed.
- Construct a spinner that lands on “success” with probability 0.4. Spin the spinner until it lands on success and then stop and count the number of spins. This is one repetition, resulting in one value of X . Repeat many times to simulate many values of X . Summarize the simulated values to approximate the distribution. For example, counts the number of repetitions resulting in $X = 1$ and divide by the total number of repetitions to approximate $P(X = 1)$.
- $X = 1$ only if she makes her first attempt, so $P(X = 1) = 0.4$. If Maya does this every practice, then in about 40% of practices she will make her first three pointer on her first attempt.
- $X = 2$ only if she misses her first attempt and makes her second attempt, so since the attempts are independent $P(X = 2) = (0.6)(0.4) = 0.24$. If Maya does this every practice, then in about 24% of practices she will make her first three pointer on her second attempt.
- In order for X to be 3, Maya must miss her first two attempts and make her third. Since the attempts are independent $P(X = 3) = (1 - 0.4)^2(0.4) = 0.144$. If Maya does this every practice, then in about 14.4% of practices she will make her first three pointer on her third attempt.

x	p(x)
1	0.4000
2	0.2400
3	0.1440
4	0.0864
5	0.0518
6	0.0311
7	0.0187
8	0.0112
9	0.0067

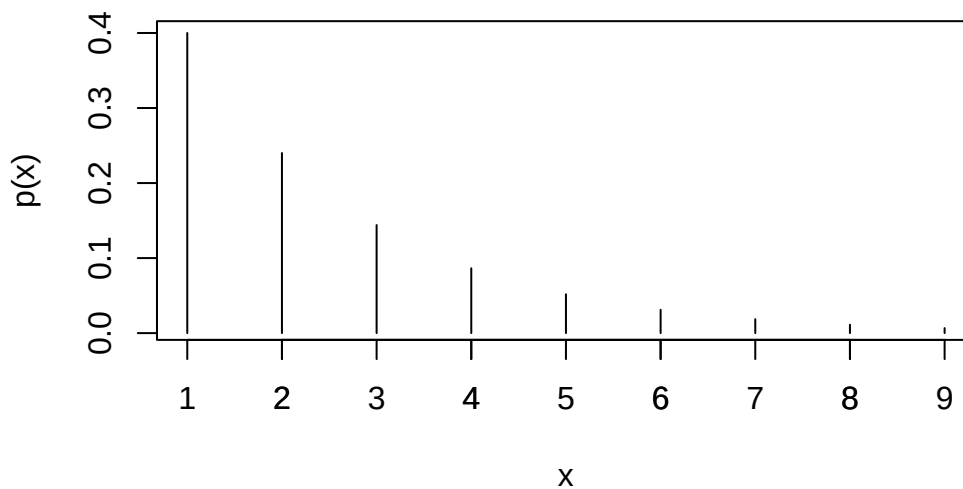
- g. In order for X to take value x , the first success must occur on attempt x , so the first $x - 1$ attempts must be failures.

$$p_X(x) = (1 - 0.4)^{x-1}(0.4), \quad x = 1, 2, 3, \dots$$

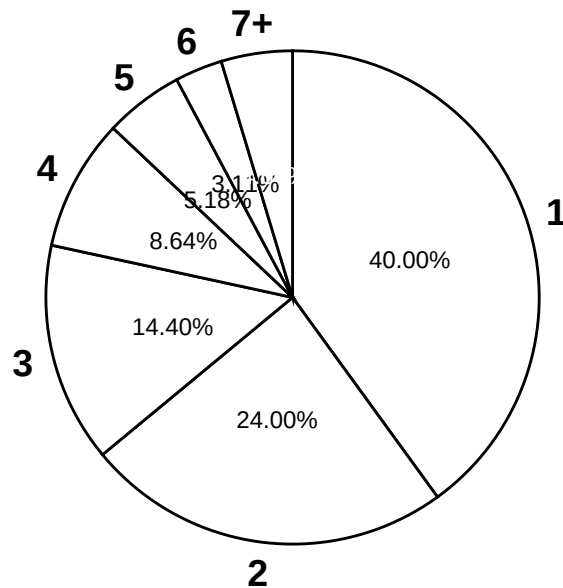
- h. See below. This distribution is called the Geometric distribution with parameter 0.4.
- i. The key is to realize that Maya requires more than 5 attempts to obtain her first success if and only if the first 5 attempts are failures. Therefore,

$$P(X > 5) = (1 - 0.4)^5 = 0.078$$

Geometric(0.4) pmf



Geometric(0.4) distribution



Problem 6

Suppose the number of earthquakes per hour, for a certain range of magnitudes in a certain region, follows a Poisson distribution with parameter 0.7.

- Compute and interpret the probability that there is at least one earthquake of this size in the region in any given hour.
- Compute and interpret the probability that there are exactly 3 earthquakes of this size in the region in any given hour.
- Interpret the value 0.7 in context.
- Construct a table, plot, and spinner corresponding to a Poisson(0.7) distribution.

Solution

Let X be the number of earthquakes in the region in a randomly selected hour. The pmf is

$$p_X(x) = e^{-0.7} \frac{0.7^x}{x!}, \quad x = 0, 1, 2, \dots$$

- $P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-0.7} \frac{0.7^0}{0!} = 1 - e^{-0.7} = 1 - 0.497 = 0.503$. In about 50% of hour long intervals there is at least one quake.

x	p(x)
0	0.4966
1	0.3476
2	0.1217
3	0.0284
4	0.0050
5	0.0007

- b. $P(X = 3) = e^{-0.7} \frac{0.7^3}{3!} = 0.028$. In about 3% of hour long intervals there are exactly 3 quakes.
- c. Over many hour-long time intervals, there are 0.7 earthquakes per hour on average.
- d. See below for table and spinner

```
x = 0:5

mu = 0.7

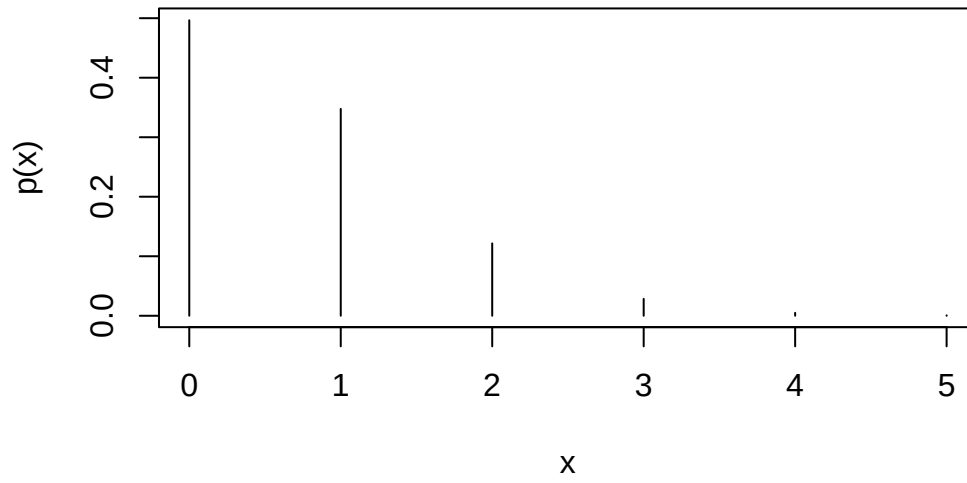
px = dpois(x, mu)

df = data.frame(x, px)

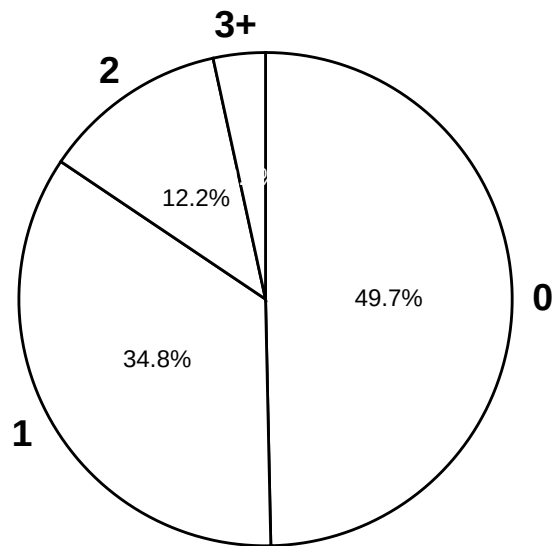
df |>
  kbl(col.names = c("x", "p(x)"),
      digits = 4) |>
  kable_styling()
```

```
plot(x, px, type = "h", xlab = "x", ylab = "p(x)", main = "Poisson(0.7) pmf")
```

Poisson(0.7) pmf



Poisson(0.7) distribution



Problem 7

Suppose that a total of 350 students at a college are taking a particular statistics course. The college offers five sections of the course, each taught by a different instructor. The class sizes are shown in the following table.

Section	A	B	C	D	E
Number of students	35	35	35	35	210

We are interested in: What is the average class size?

- Suppose we randomly select one of the 5 *instructors*. Let X be the class size for the selected instructor. Specify the distribution of X . (A table is fine.)
- Compute and interpret $E(X)$.
- Compute and interpret $P(X = E(X))$.
- Suppose we randomly select one of the 350 *students*. Let Y be the class size for the selected student. Specify the distribution of Y . (A table is fine.)
- Compute and interpret $E(Y)$.
- Compute and interpret $P(Y = E(Y))$.
- Comment on how these two expected values compare, and explain why they differ as they do. Which average would you say is more relevant?

Solution

- X takes the value 210 with probability $1/5$ and 35 with probability $4/5$.
- $E(X) = 210(1/5) + 35(4/5) = 70$ is the average number of students per class from the instructors' perspective. If we randomly select an instructor, record the number of students in the instructor's class, and repeat, the long run average number of students will be 70.
- $P(X = E(X)) = P(X = 70) = 0$. No instructor has a class whose size is equal to the average class size (from the instructor's perspective).
- Y takes the value 210 with probability $210/350$ and 35 with probability $140/350$.
- $E(Y) = 210(210/350) + 35(140/350) = 140$ is the average number of students per class from the students' perspective. If we randomly select a student, record the number of students in the selected student's class, and repeat, the long run average number of students will be 140.
- $P(Y = E(Y)) = P(Y = 140) = 0$. No student is in a class whose size is equal to the average class size (from the students' perspective).
- Colleges usually report average class size from the instructor/class perspective, which in this case would be 70 students. From the students' perspective, the average class size is not even close to 70! In fact, it's twice that size. Some students (140 of them, which

is 40% of the total of 350 students) have the benefit of a small class size of 35. But most students (210 of them, which is 60% of the students) are stuck in a large class of 210 students. In other words, most students would be pretty seriously misled if they chose this college based on the advertised average class size of 35 students per class. From the students' perspective, it seems that 140 is the more relevant average to report. However, neither average really adequately represents what is happening here: there is wide variability in class sizes between large and small.

Homework 3 Solutions

Problem 1

(This is an overly simplified example from actuarial science.) A life insurance company sells a term insurance policy to a 21 year old man (the “insured”) that pays \$100,000 to a designated beneficiary if the insured dies within the next 5 years, and pays nothing if the insured does not die before age 26. The probability that a randomly chosen 21-year-old man will die each year can be found in mortality tables. The company collects a premium of \$250 at the beginning of each year of the 5 years of the term, as long as the insured is alive, as payment for the insurance. The amount Y (in dollars) that the company earns on this policy is \$250 per year, less the \$100,000 that it pays out in the event the insured dies. The company doesn’t pay out anything if the insured does not die in the 5 year term.

Age at death	Earnings Y	Probability
21	99,750	0.00183
22		0.00186
23		0.00189
24		0.00191
25		0.00193
26 or older		

- Find the distribution of Y by completing the above table. For example, the value -99,750 provides an example of how Y is calculated when the insured dies in the first year of the policy.
- Compute $E(Y)$.
- Write a sentence explaining in this context in what sense the number from the previous part is “expected”.
- Would you be willing to sell such an insurance policy to one of your 21 year old friends? Explain why this is good business for the insurance company, but not for you.
- Compute $\text{Var}(Y)$
- Compute $\text{SD}(Y)$.
- Compute the probability that Y is more than 1 standard deviation above its mean.

Solution

Age at death	Earnings Y	Probability
21	99,750	0.00183
22	-99,500	0.00186
23	-99,250	0.00189
24	-99,000	0.00191
25	-98,750	0.00193
26 or older	1,250	0.99058

- See above. The earnings are computed according to the contract; the insurance company only needs to pay out the 100000 if the insured dies before age 26. Since all the probabilities must add up to 1, the probability in the blank is $1 - (0.00183 + 0.00186 + 0.00189 + 0.00191 + 0.00193) = 0.99058$
- $E(Y) = (-99750)(0.00183) + \dots + (-98750)(0.00193) + (1250)(0.99058) = 303.23$
- If the insurance company sells a large number of these policies to 21 year-old men then the insurance company will expect to make, on average, a profit of \$303.36 per policy. For example, if they sell 1 million policies, their total profit would be expected to be close to \$303 million.
- If you sell one policy, you are either going to make a profit of \$1250, or you will pay out about \$100,000. The chances are high you will make a profit of \$1250, but there is a risk that you will have to pay out around \$100,000. This risk is probably too great to take on considering the relatively small (in comparison) reward.

On the other hand, the insurance company sells *many* policies. They operate in the long run. Over many policies they make a profit, on average, of about \$300 per policy. For example, for a million policies this equates to a profit of \$300 million. There would be some variability around this, but in the long run the average will win out. The company could be sure that they would only pay out the \$100,000 benefit for less than 1% of policyholders, and they will be earning \$1,250 on many other policies, more than enough to offset the relatively few \$100,000 payouts there are.

- $E(Y^2) = (-99750)^2(0.00183) + \dots + (-98750)^2(0.00193) + (1250)^2(0.99058) = 94328849$. Then $\text{Var}(Y) = E(Y^2) - (E(Y))^2 = 94328849 - (303)^2 = 94236827$. This is a giant number because it's measured in dollars².
- $\text{SD}(Y) = \sqrt{\text{Var}(Y)} = 9707$ dollars.
0. One SD above the mean here is $303 + 9707 = 10000$ and $P(Y > 10000) = 0$ since it's not possible to have a value of this Y which is more than 1 SD above its mean.

Problem 2

In a certain population, household income X (\$ thousands) follows the pdf

$$f_X(x) = cx^{-2.5}, \quad x \geq 30$$

for an appropriate constant c . Note that the measurement units are \$ thousands, so 1 represents 1 thousand dollars.

- What are the possible values of X ? Sketch the pdf of X .
- Find the value of c .
- Find the probability that a household has an income over \$200K.
- Find the probability that a household has an income exactly equal to \$200K.
- Without integrating, find the probability that a household has an income, rounded to the nearest thousand dollars, of \$200K.
- Without integrating, determine how many times more likely it is for a household to have an income, rounded to the nearest thousand dollars, of \$100K than \$200K.

Solution

- The smallest possible value of X is 30 (thousand dollars). The pdf has a peak at 30 and decreases as x increases. The density is 0 for $x < 30$. We can sketch the general shape of $f(x)$ without knowing the value of c ; c just tells us how to label the density axis so the total area under the curve is 1.
- c is such that the total area under the curve is 1

$$1 = \int_{30}^{\infty} cx^{-2.5} dx = c30^{-1.5}/1.5$$

Solve for $c = (1.5)30^{1.5} = 246.5$. That is

$$f_X(x) = (1.5)(30^{1.5})x^{-2.5}, \quad x \geq 30$$

- The probability that a household has an income over \$200K is

$$P(X > 200) = \int_{200}^{\infty} (1.5)30^{1.5}x^{-2.5} dx = (30/200)^{1.5} = 0.058$$

About 5.8% percent of households in this area have an income over 200 thousand dollars.

- X is continuous so $P(X = 200) = 0$
- In the measure units of X , 1 thousand dollars is 1 unit long, a small range in the scale of the variable. Can integrate but since it's a small range of values in the scale of the variable, can approximate

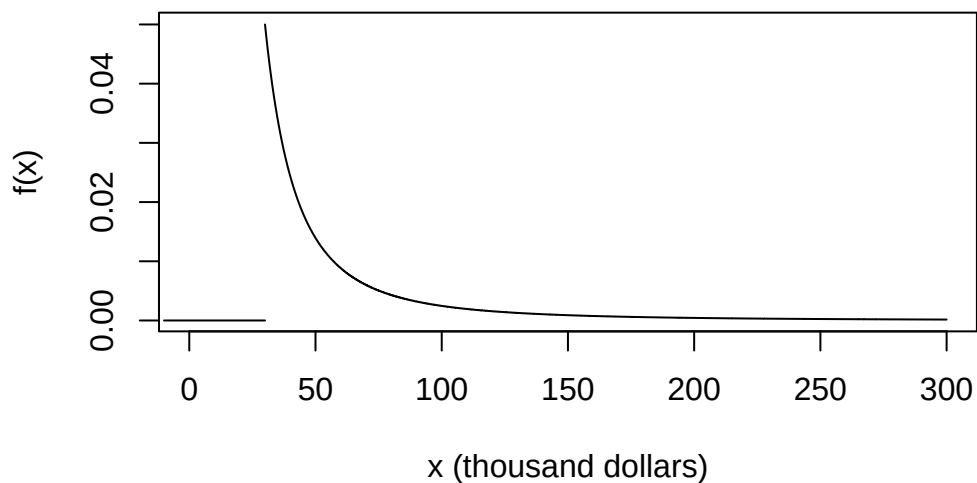
$$P(200 - 0.5 < X < 200 + 0.5) \approx f_X(200)(1) = (1.5)30^{1.5}200^{-2.5}(1) = 0.000436$$

About 0.04% of households in this area have an income, to the nearest thousand dollars, of 200 thousand dollars.

f. Ratio of the densities; note we don't need to know the value of c to solve this part.

$$\frac{f_X(100)}{f_X(200)} = \frac{100^{-2.5}}{200^{-2.5}} = 5.66$$

In this area, a household is 5.66 times more likely to have an income, rounded to the nearest thousand dollars, of \$100K than \$200K.



Problem 3

In the meeting time problem, assume that Regina's R and Cady's Y arrival times (measured in fractions of the hour after noon, e.g. 0.25 represents 12:15), each follow a Uniform(0, 1) distribution, independently of each other. Let $W = |R - Y|$ be the amount of time (hours) the first person to arrive waits for the second person to arrive. It can be shown that W has pdf

$$f_W(w) = 2(1 - w), \quad 0 < w < 1.$$

- Sketch a plot of the pdf. Describe roughly what this means in context.
- Coding required.** Code and run a simulation to approximate the distribution of W . Does the histogram reasonably match up with the pdf? Then use the simulation results to approximate $P(W < 0.25)$.
- Use the pdf to compute $P(W < 0.25)$ and interpret the value. Set up an integral, but sketch a picture and use geometry to compute.

Solution

- See plot below. The density is highest at $w = 0$ and decreases linearly to 0 at $w = 1$. Waiting time is most likely to be close to 0 minutes, and likelihood of waiting close to w hours decreases as w increases.
- See simulation results below. pdf seems reasonable.
- Integrate the pdf over $(0, 0.25)$. This gives an area of a trapezoid; the complementary area is a triangle with a base of $(1 - 0.25)$, a height of $2(1 - 0.25)$, and an area of $(1/2)(1 - 0.25)(2(1 - 0.25)) = (1 - 0.25)^2$

$$P(W < 0.25) = \int_0^{0.25} 2(1 - w)dw = -(1 - w)^2 \Big|_{w=0}^{w=0.25} = 1 - (1 - 0.25)^2 = 0.4375$$

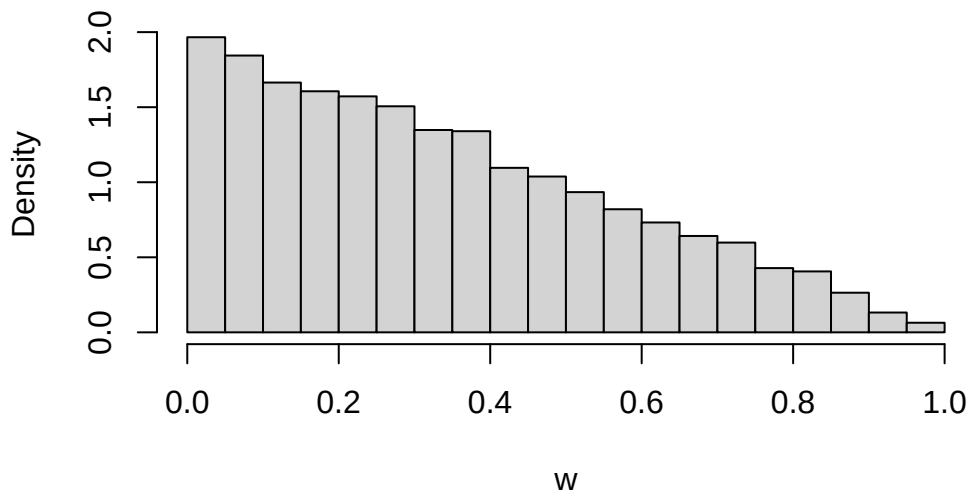
If they meet over many days, on any 43.75% of days the first person will have to wait less than 15 minutes for the second person.

```
N_rep = 10000

x = runif(N_rep, 0, 1)
y = runif(N_rep, 0, 1)

w = abs(x - y)

hist(w,
     freq = FALSE,
     main = "")
```



```
sum(w < 0.25) / N_rep
```

```
[1] 0.4326
```

Problem 4

(Continued.) In the Regina/Cady problem, let $W = |R - Y|$ be the amount of time the first person to arrive has to wait for the second person. Recall that W is a continuous random variable with pdf

$$f_W(w) = 2(1 - w), \quad 0 < w < 1.$$

- Coding required.** Use your simulation results from before to approximate $E(W)$, $E(W^2)$, $\text{Var}(W)$, and $\text{SD}(W)$.
- Compute and interpret $E(W)$.
- Compute $E(W^2)$.
- Compute $\text{Var}(W)$.
- Compute $\text{SD}(W)$.
- Explain in words in detail how you could use the $\text{Uniform}(0, 1)$ spinner and simulation to approximate $\text{Var}(W)$.
- Compute the probability that W is more than 1 SD away from its mean.

Solution

```
mean(w)
```

```
[1] 0.3351759
```

```
mean(w ^ 2)
```

```
[1] 0.168477
```

```
var(w)
```

```
[1] 0.0561398
```

```
sd(w)
```

```
[1] 0.2369384
```

- See above.
- Use the definition of EV for continuous RV

$$E(W) = \int_0^1 w f_W(w) dw = \int_0^1 w (2(1-w)) dw = 1/3$$

If they were to meet like this over many days, the long run average waiting time will be 1/3 hour (20 minutes).

- Use LOTUS

$$E(W^2) = \int_0^1 w^2 f_W(w) dw = \int_0^1 w^2 (2(1-w)) dw = 1/6$$

- We did all the work in the previous parts. $\text{Var}(W) = E(W^2) - E(W)^2 = 1/6 - (1/3)^2 = 1/18$ square-hours.
- $\text{SD}(W) = 0.236$ hours (about 14 minutes)
- Explain how you could use the Uniform(0, 1) spinner and simulation to approximate $\text{Var}(W)$.
 - Spin the Uniform(0, 1) spinner twice to get R, Y and set $W = |R - Y|$.
 - Repeat many times to simulate many values of W .
 - Average the values of W .

- For each simulated value of W , subtract the average of W and square to get the squared deviation
 - Average the squared deviations to approximate $\text{Var}(W)$.
- g. $P(|W - E(W)| > \text{SD}(W)) = P(W > 1/3 + 0.236) + P(W < 1/3 - 0.236)$. In this case you can find the corresponding probability/area in this case with geometry. Also, integrating gives

$$\int_0^{1/3-0.236} 2(1-w)dw + \int_{1/3+0.236}^1 2(1-w)dw = 0.37$$

Homework 4 Solutions

Problem 1

In a certain region, times (minutes) between occurrences of earthquakes (of any magnitude) have a distribution with percentiles displayed in the table below.

- Construct a spinner corresponding to this distribution.
- What percent of times are between 26.8 and 110.0 minutes?
- Let F be the cdf. Evaluate and interpret $F(42.8)$.
- Let Q be the quantile function. Evaluate and interpret $Q(0.9)$.
- Sketch (by hand) a histogram of this distribution.

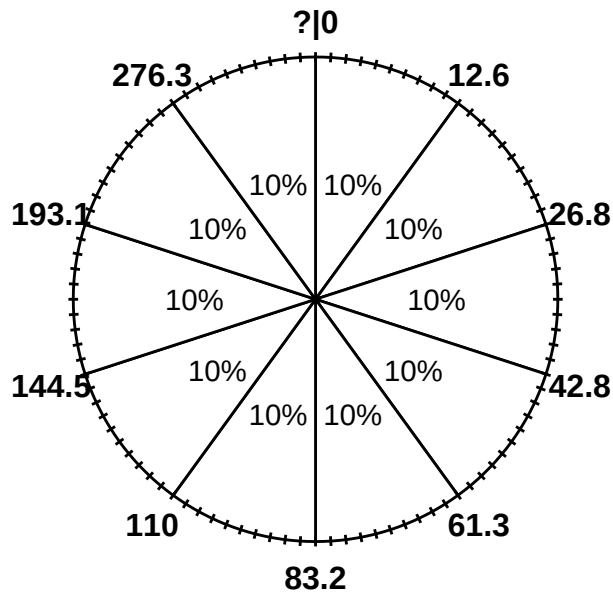
Solution

- See below. The 50th percentile goes at “6 o’clock” (50% of the way around), etc.
- 60% are below 110.0 and 20% are below 26.8, so 40% are between 26.8 and 110.0 minutes.
- $F(42.8) = P(X \leq 42.8) = 0.3$, since 42.8 minutes is the 30th percentile.
- $Q(0.9) = 276.3$, since 276.3 minutes is the 90th percentile.
- We always end with a histogram with even bins, but sometimes it helps to start sketching a histogram with uneven bins. We know that 10% of the values are between 0 and 12.6, 10% are between 12.6 and 26.8, 10% are between 26.8 and 42.8, etc. So if we split the bins of the histogram based on these percentiles, each bin would correspond to a relative frequency of 10%. HOWEVER, AREA in a histogram represents relative frequency. So

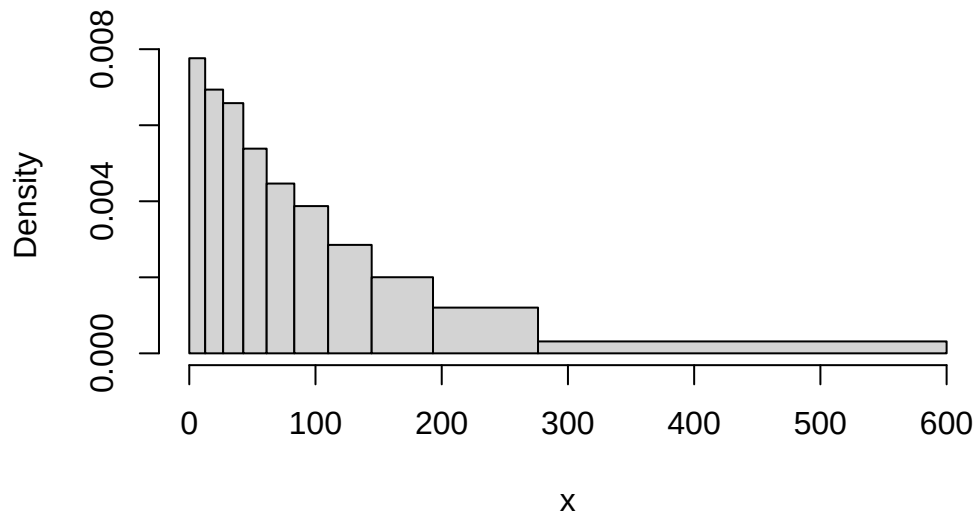
Percentile	Value (minutes)
10th	12.6
20th	26.8
30th	42.8
40th	61.3
50th	83.2
60th	110.0
70th	144.5
80th	193.1
90th	276.3

each of the bins would have the same area, but they are unequally spaced, so they would not have the same height. For example, the first bin has area of 10% over a base of (0, 12.6), while the 80-90 percentile bin has area 10% spread over the interval (193.1, 276.3), so the bar for (193.1, 276.3) will have a smaller height than the bar for (0, 12.6). Sketching a histogram with these unevenly spaced bins first gives an idea of the shape; then you can sketch a histogram with a similar shape but equally spaced bins. We can see that density is highest near 0 and decreases as x increases; notice the stretching and shrinking on the spinner. (The highest percentile is the 90th, so we don't know what the upper bound is that almost all values fall below.)

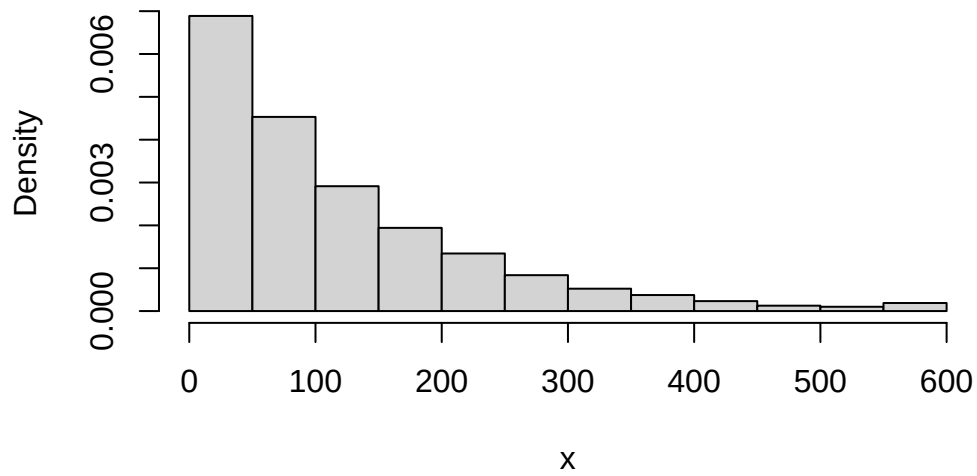
Marginal distribution of time (minutes)



Histogram – uneven bins



Histogram – even bins



Problem 2

(Continued.) In the Regina/Cady problem, let $W = |R - Y|$ be the amount of time the first person to arrive has to wait for the second person. Recall that W is a continuous random variable with pdf

$$f_W(w) = 2(1 - w), \quad 0 < w < 1.$$

- Let F be the cdf. Evaluate and interpret $F(0.25)$.
- Find an expression for the cdf of W . Set up an integral, but sketch a picture and use geometry to compute.
- Find and interpret the 25th percentile of W . (You can do the next part first if you want, but it might help to start with a specific number like in this part.)
- Find the quantile function of W .
- Sketch a spinner corresponding to the distribution of W . Label the 25th, 50th, and 75th percentiles.
- Coding required.** Use your simulation results from before to approximate the 25th, 50th, and 75th percentiles.

Solutions

- $F(0.25) = P(W \leq 0.25)$, which we computed on a previous HW to be $1 - (1 - 0.25)^2 = 0.4375$. On about 44% of days Regina arrives within the first 15 minutes (0.25 hours).
- Similar to computing $F_W(0.25) = P(W \leq 0.25)$ from an earlier problem, but replace 0.25 with a generic w (and remember to change the dummy variable of integration). For $0 < w < 1$,

$$F_W(w) = P(W < w) = \int_0^w 2(1 - u)du = -(1 - u)^2 \Big|_{u=0}^{u=w} = 1 - (1 - w)^2$$

Technically the cdf is defined for any value of w , so it's really

$$F_W(w) = \begin{cases} 0, & w \leq 0 \\ 1 - (1 - w)^2, & 0 < w < 1 \\ 1, & w \geq 1 \end{cases}$$

- The 25th percentile satisfies

$$0.25 = P(W \leq w) = F_W(w) = 1 - (1 - w)^2$$

Solve to get $w = 1 - \sqrt{1 - 0.25} = 0.134$. The waiting time is at most 0.134 hours (about 8 minutes) on 25% of days.

d. To find the p percentile q , use the same process as in the previous part.

$$p = P(W \leq w) = F_W(w) = 1 - (1 - w)^2$$

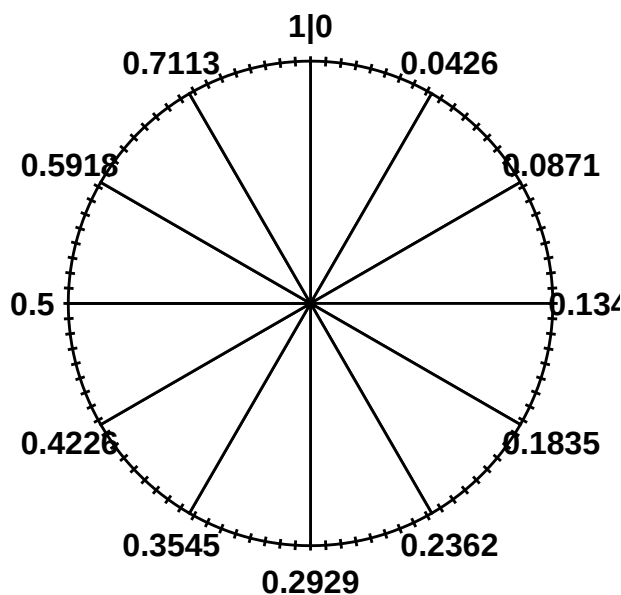
Solve to get $w = 1 - \sqrt{1 - p}$. So the quantile function is $Q_W(p) = 1 - \sqrt{1 - p}$, $0 \leq p \leq 1$.

e. See spinner below. We can use the quantile function to label the spinner.

- The 25th percentile of $Q_W(0.25) = 1 - \sqrt{1 - 0.25} = 0.134$ hours (8 minutes) would go at the 3 o'clock point.
- The 50th percentile of $Q_W(0.5) = 1 - \sqrt{1 - 0.5} = 0.293$ hours (17.6 minutes) would go at the 6 o'clock point.
- The 75th percentile of $Q_W(0.75) = 1 - \sqrt{1 - 0.75} = 0.5$ hours (30 minutes) would go at the 9 o'clock point.
- Notice that the values around the spinner are unequally spaced. The density of W is higher near 0 so intervals near 0 are stretched out; the density of W is lower near 1 so intervals near 1 are shrunk.

f. See simulation results below. Use the `quantile` function.

W spinner



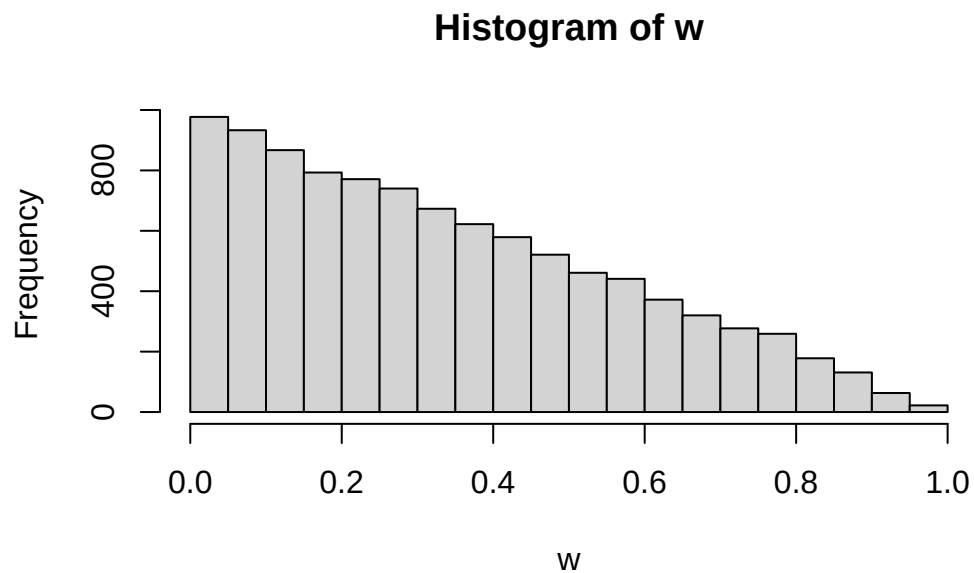
```
N_rep = 10000
```

```
x = runif(N_rep, 0, 1)
```

```
y = runif(N_rep, 0, 1)
```

```
w = abs(x - y)
```

```
hist(w)
```



```
quantile(w, 0.25)
```

```
25%  
0.1352974
```

```
quantile(w, c(0.25, 0.5, 0.75))
```

```
25%      50%      75%  
0.1352974 0.2941531 0.5021829
```

Problem 3

Solve Example 11.3 from the Normal distribution handout (the problem on daily high temperatures in SLO.)

Solution

See [Example 7.12 here](#)

Problem 4

Solve Example 11.4 from the Normal distribution handout (the problem on “effect size”).

Solution

See [Example 7.13 here](#)

Problem 5

The latest series of collectible Lego Minifigures contains 3 different Minifigure prizes (labeled 1, 2, 3). Each package contains a single unknown prize. Suppose we only buy 3 packages and we consider as our sample space outcome the results of just these 3 packages (prize in package 1, prize in package 2, prize in package 3). For example, 323 (or (3, 2, 3)) represents prize 3 in the first package, prize 2 in the second package, prize 3 in the third package. Let X be the number of distinct prizes obtained in these 3 packages. Let Y be the number of these 3 packages that contain prize 1. Suppose that each package is equally likely to contain any of the 3 prizes, regardless of the contents of other packages. There are 27 possible, equally likely outcomes

New names:

```
* `` -> `...4`  
* `` -> `...5`
```

- Evaluate X and Y for each of the outcomes.
- Construct a two-way table representing the *joint distribution* of X and Y .
- Sketch a plot representing the joint distribution of X and Y .
- Construct a spinner corresponding to the joint distribution of X and Y .
- Describe two ways to simulate an (X, Y) pair.
- Identify the *marginal* distribution of X , and construct a corresponding spinner.
- Identify the *marginal* distribution of Y , and construct a corresponding spinner.
- Describe in detail in words how you could conduct a simulation and use the results to approximate. (This part is asking you to describe the process in words; not to write code.)

box1	box2	box3	X	Y
1	1	1	1	3
2	1	1	2	2
3	1	1	2	2
1	2	1	2	2
2	2	1	2	1
3	2	1	3	1
1	3	1	2	2
2	3	1	3	1
3	3	1	2	1
1	1	2	2	2
2	1	2	2	1
3	1	2	3	1
1	2	2	2	1
2	2	2	1	0
3	2	2	2	0
1	3	2	3	1
2	3	2	2	0
3	3	2	2	0
1	1	3	2	2
2	1	3	3	1
3	1	3	2	1
1	2	3	3	1
2	2	3	2	0
3	2	3	2	0
1	3	3	2	1
2	3	3	2	0
3	3	3	1	0

- i. $P(X = 3)$
- ii. $P(X = 2, Y = 1)$ iii. $E(X)$ iii. $E(Y)$
- iii. $E(XY)$
- iv. $SD(X)$ iii. $SD(Y)$
- v. $Cov(X, Y)$
- vi. $Corr(X, Y)$

i. **Coding required.** Code and run the simulation from the previous part and use the simulation results to approximate each of

- i. $P(X = 3)$
- ii. $P(X = 2, Y = 1)$ iii. $E(X)$ iii. $E(Y)$
- iii. $E(XY)$
- iv. $SD(X)$ iii. $SD(Y)$
- v. $Cov(X, Y)$
- vi. $Corr(X, Y)$

j. Compute

- i. $P(X = 3)$
- ii. $P(X = 2, Y = 1)$ iii. $E(X)$ iii. $E(Y)$
- iii. $E(XY)$
- iv. $SD(X)$ iii. $SD(Y)$
- v. $Cov(X, Y)$
- vi. $Corr(X, Y)$

Solution

These don't quite match the assignment, but the ideas are similar.

- a. One dimension will have possible values of x , the other possible values of y . The interior cells should contain the probability of each (x, y) pair.

	y				
x	0	1	2	3	Total
1	2/27	0	0	1/27	3/27
2	6/27	6/27	6/27	0	18/27
3	0	6/27	0	0	6/27
Total	8/27	12/27	6/27	1/27	1

- b. Make a tile plot with color or shading representing probability; see below.
- c. The spinner is below; it returns the 6 possible (x, y) pairs with the proper probabilities.

box1	box2	box3	X	Y
1	1	1	1	3
2	1	1	2	2
3	1	1	2	2
1	2	1	2	2
2	2	1	2	1
3	2	1	3	1
1	3	1	2	2
2	3	1	3	1
3	3	1	2	1
1	1	2	2	2
2	1	2	2	1
3	1	2	3	1
1	2	2	2	1
2	2	2	1	0
3	2	2	2	0
1	3	2	3	1
2	3	2	2	0
3	3	2	2	0
1	1	3	2	2
2	1	3	3	1
3	1	3	2	1
1	2	3	3	1
2	2	3	2	0
3	2	3	2	0
1	3	3	2	1
2	3	3	2	0
3	3	3	1	0

- d. Method 1: Simulate from the probability space: Label 3 cards: 1, 2, 3. Shuffle the cards and deal one, then replace the card and repeat, dealing a total of 3 cards, e.g., 212. Measure X and Y for the outcome.

Method 2: Simulate from the distribution. Spin the spinner from the previous part once to generate an (X, Y) pair.

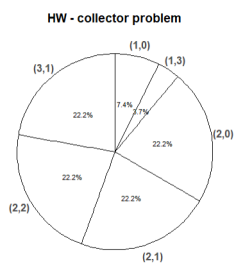
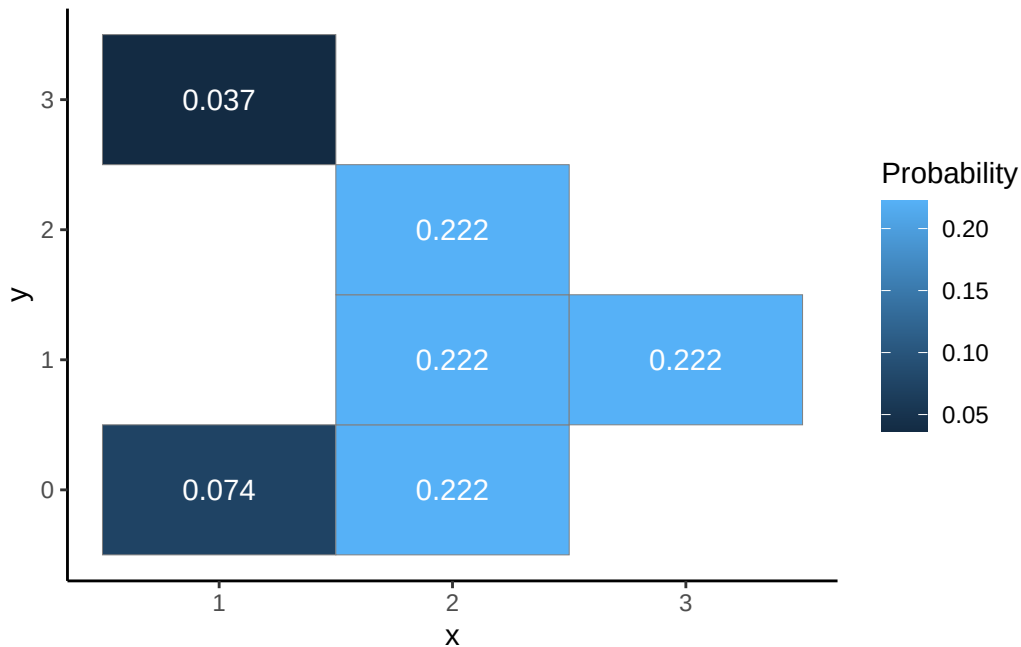
- e. The possible values of X are in the leftmost column and the probabilities are in the Total column. See previous HW solutions for plot and spinner.
- f. The possible values of Y are in the top row (0, 1, 2, 3) and the probabilities are in the Total row. See previous HW solutions for plot and spinner.

- g. Method 1: Simulate from the probspace: Draw three “boxes” and observe the prize sequence, and measure X .

Method 2: Simulate from marginal dist: Spin the X spinner once and observe the value.

Method 3: Simulate from joint dist: Spin the joint distribution spinner once to obtain an (X, Y) pair, and ignore Y . (This is the one we didn’t see on a previous HW.)

- h. Describe in detail how you could conduct a simulation to and use the results to approximate
- $P(X = 3)$. Simulate many values of X , count the number of simulated values equal to 3, and divide by the number of simulated values (i.e., find the simulated relative frequency of 3)
 - $P(X = 2, Y = 1)$. Simulate many (X, Y) *pairs*, count the number of simulated pairs in which $X = 2$ and $Y = 1$, and divide by the number of simulated values (i.e., find the simulated relative frequency of the event $\{X = 2, Y = 1\}$)
 - The long run average value of X . Simulate many values of X and compute their average (i.e., sum the simulated values of X and divide by the number of simulated values.)
 - The long run average value of XY . Simulate many (X, Y) *pairs*, find the product XY for each pair, and compute the average value of the products (i.e., sum the simulated values of XY and divide by the number of simulated pairs.)



There are many ways of doing the simulation. I'll use a for loop

```
# Number of repetitions
N_rep = 10000

# Initialize place holders for X and Y
# Each of X and Y will eventually contain 10000 values
# but initially they are empty
X = vector(length = N_rep)
Y = vector(length = N_rep)

for (i in 1:N_rep) {
  # within the loop a single repetition is performed
```

```

# simulate the 3 prizes
outcome = sample(1:3, size = 3, replace = TRUE)

# find the number of distinct prizes
x = length(unique(outcome))

# count the number of prize 1s
y = sum(outcome == 1)

# record the values in X and Y
X[i] = x
Y[i] = y
}

# After the loop runs each of X and Y will have 10000 values

# X counts
table(X)

```

```

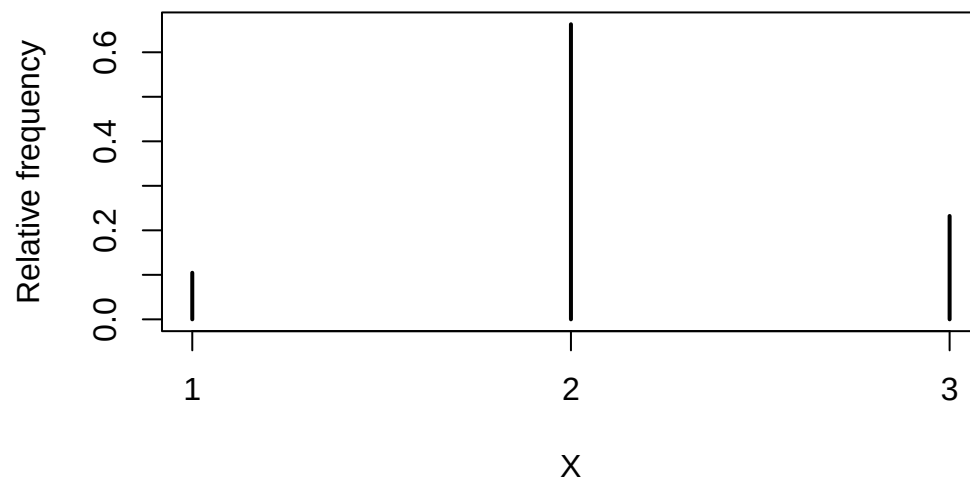
X
  1    2    3
1047 6630 2323

```

```

# Rough plot of approximate distribution of X
plot(table(X) / N_rep,
     xlab = "X",
     ylab = "Relative frequency")

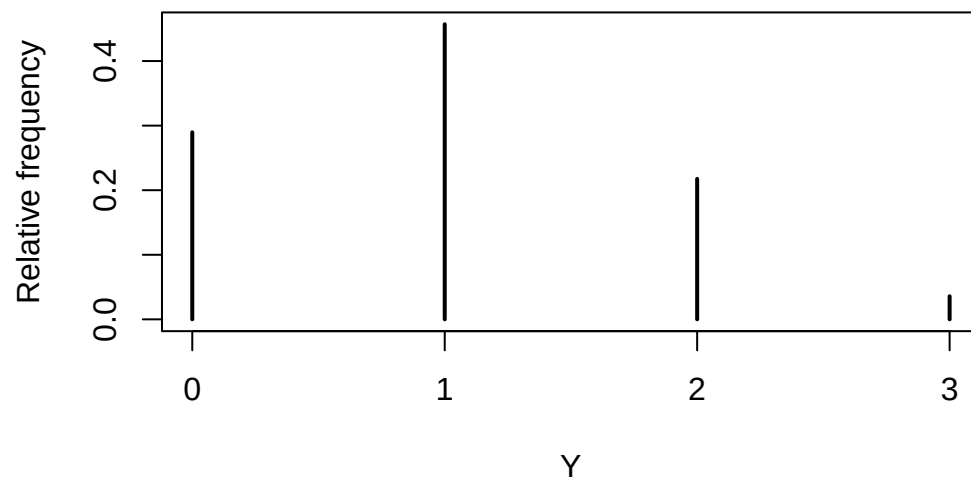
```



```
# Y counts  
table(Y)
```

```
Y  
  0   1   2   3  
2897 4570 2175 358
```

```
# Rough plot of approximate distribution of Y  
plot(table(Y) / N_rep,  
      xlab = "Y",  
      ylab = "Relative frequency")
```



```
# (X, Y) counts
table(X, Y)
```

	Y			
X	0	1	2	3
1	689	0	0	358
2	2208	2247	2175	0
3	0	2323	0	0

```
# Approximate E(X)
mean(X)
```

```
[1] 2.1276
```

```
# Approximate E(Y)
mean(Y)
```

```
[1] 0.9994
```



```
# Approximate SD(X)
sd(X)
```

```
[1] 0.5663482
```

```
# Approximate SD(Y)
sd(Y)
```

```
[1] 0.8065139
```

```
# Approximate E(XY)
mean(X * Y)
```

```
[1] 2.1237
```

```
# Approximate Cov(X, Y)
cov(X, Y)
```

```
[1] -0.002623702
```

```
# Approximate Corr(X, Y)
cor(X, Y)
```

```
[1] -0.005744063
```

```
# Approximate P(X = 3)
sum(X == 3) / N_rep
```

```
[1] 0.2323
```

```
# Approximate P(X = 2, Y = 1)
sum((X == 2) * (Y == 1)) / N_rep
```

```
[1] 0.2247
```

Homework 5 solutions

Problem 1

(Continuing from a previous HW) The latest series of collectible Lego Minifigures contains 3 different Minifigure prizes (labeled 1, 2, 3). Each package contains a single unknown prize. Suppose we only buy 3 packages and we consider as our sample space outcome the results of just these 3 packages (prize in package 1, prize in package 2, prize in package 3). For example, 323 (or (3, 2, 3)) represents prize 3 in the first package, prize 2 in the second package, prize 3 in the third package. Let X be the number of distinct prizes obtained in these 3 packages. Let Y be the number of these 3 packages that contain prize 1. Suppose that each package is equally likely to contain any of the 3 prizes, regardless of the contents of other packages; let P denote the corresponding probability measure.

- Find the conditional distribution of Y given $X = x$ for each possible value of x of X .
- Compute and interpret $E(Y|X = x)$ for each possible value of x of X .
- Find the conditional distribution of X given $Y = y$ for each possible value of y of Y .
- Compute and interpret $E(X|Y = y)$ for each possible value of y of Y .
- Explain how you could use spinners to implement the “marginal then conditional” method to simulate an (X, Y) pair.
- Suppose you have simulated many (X, Y) pairs. Explain how you could use the simulation results to approximate:
 - $P(X = 1|Y = 0)$
 - the conditional distribution of X given $Y = 0$
 - $E(X = 1|Y = 0)$

Solution

- Each row of the table below represents a different conditional distribution of Y given $X = x$. For example, the row for $x = 1$ represents the conditional distribution of Y given $X = 1$: Given $X = 1$, Y is 0 with probability $2/3$ and 3 with probability $1/3$.

	y				
	0	1	2	3	Total
x					

1	2/3	0	0	1/3	1
2	1/3	1/3	1/3	0	1
3	0	1	0	0	1

- b. Take expected values according to each conditional distribution. In general, $E(Y|X = x)$ depends on x , but in this case $E(Y|X = x) = 1$ for all values of x .; regardless of how many distinct prizes people obtain in their, the average number of prize 1s obtains is 1.

$$\begin{array}{ll}
 x & E(Y|X = x) \\
 1 & 0(2/3) + 3(1/3) = 1 \\
 2 & 0(1/3) + 1(1/3) + 2(1/3) = 1 \\
 3 & 1(1) = 1
 \end{array}$$

- c. Each column of the table below represents a different conditional distribution of X given $Y = y$. For example, the column for $y = 1$ represents the conditional distribution of X given $Y = 1$: Given $Y = 1$, X is 1 with probability $1/4$ and 2 with probability $3/4$.

	y			
	0	1	2	3
x				
1	1/4	0	0	1
2	3/4	1/2	1	0
3	0	1/2	0	0
Total	1	1	1	1

- d. Take expected values according to each conditional distribution. For example, $E(X|Y = 0) = 1.75$; among the people who never get prize 1 in their 3 boxes, the average number of distinct prizes they obtain is 1.75.

$$\begin{array}{ll}
 y & E(X|Y = y) \\
 0 & 1(1/4) + 2(3/4) = 1.75 \\
 1 & 2(1/2) + 3(1/2) = 2.5 \\
 2 & 2(1) = 2 \\
 3 & 1(1) = 1
 \end{array}$$

- e. Simulate a value of X from its marginal distribution: 1, 2, 3 with probability $3/27$, $18/27$, $6/27$ (see previous HW for spinner). Given the value of x , simulate Y from the appropriate conditional distribution of Y given $X = x$:
- if $x = 1$, simulate Y from a spinner that returns values 0, 3 with probability $2/3$, $1/3$
 - if $x = 2$, simulate Y from a spinner that returns values 0, 1, 2 with probability $1/3$, $1/3$, $1/3$
 - if $x = 3$ then Y is 1 with probability 1.
 - (You could also simulate Y first and then given $Y = y$ simulate X from the appropriate conditional distribution of X given $Y = y$.)
- f. Simulate many (X, Y) pairs using any appropriate method.
- i. Discard all pairs for which Y is not 0, count the number of remaining pairs in which $X = 1$ and divide by the number of non-discarded pairs (all with $Y = 0$).
 - ii. Discard all pairs for which Y is not 0, and summarize the X values to find the conditional relative frequency of each possible X value as in the previous part (given $Y = 0$, the possible values of X are 1 and 2).
 - iii. Discard all pairs for which Y is not 0 and sum all the X values for the remaining pairs and divide by the number of non-discarded pairs to find the conditional average.
 - iv. (In all the above, be careful about the margin of error.)

Problem 2

Xavier and Yolanda are playing roulette. They both place bets on red on the same 3 spins of the roulette wheel before Xavier has to leave. (Remember, the probability that any bet on red on a single spin wins is $18/38$.) After Xavier leaves, Yolanda places bets on red on 2 more spins of the wheel. Let X be the number of bets that Xavier wins and let Y be the number that Yolanda wins.

- a. Identify by name the marginal distribution of X . Be sure to specify the values of any relevant parameters. Compute $E(X)$.
- b. Identify by name the marginal distribution of Y . Be sure to specify the values of any relevant parameters. Compute $E(Y)$.
- c. The joint distribution of X and Y is represented in the table below. Explain why $p_{X,Y}(1, 4) = 0$ and $p_{X,Y}(2, 1) = 0$.
- d. Compute $p_{X,Y}(2, 3)$. (Yes, the table tells you it's 0.1766, but you have to show how this number can be computed based on the assumptions of the problem.)
- e. Are X and Y independent? Justify your answer with an appropriate calculation.
- f. Compute and interpret $P(X + Y) = 4$.
- g. Make a table representing the marginal distribution of $X + Y$.

- h. Use the table from the previous part to compute $E(X + Y)$, and verify that it is equal to $E(X) + E(Y)$.
- i. Without doing any computation, determine if $\text{Var}(X + Y)$ will be greater than, less than, or equal to $\text{Var}(X) + \text{Var}(Y)$. Explain your reasoning.

x, y	0	1	2	3	4	5
0	0.0404	0.0727	0.0327	0	0	0
1	0	0.109	0.1963	0.0883	0	0
2	0	0	0.0981	0.1766	0.0795	0
3	0	0	0	0.0294	0.053	0.0238

Solution

- a. X has a Binomial distribution with parameters $n = 3$ and $p = 18/38$, so $E(X) = np = 3(18/38) = 1.42$.
- b. Y has a Binomial distribution with parameters $n = 5$ and $p = 18/38$, so $E(X) = np = 5(18/38) = 2.37$.
- c. In the first 3 bets Yolanda wins the same number of games as Xavier, and Yolanda only plays two more games. So Y can't be more than $X + 2$, and Y can't be less than X .
- d. To have $X = 2$ and $Y = 3$ we need to have exactly two wins in the first 3 bets (where both play) exactly 1 win in the last 2 bets (Yolanda only). The number of wins in the first 3 bets is Binomial(3, 18/38), the number of wins in the last two bets is Binomial(2, 18/38) and the first 3 bets are independent of the last two bets. So the probability is $\text{dbinom}(2, 3, 18 / 38) * \text{dbinom}(1, 2, 18 / 38) = 0.177$ or

$$p_{X,Y}(2, 3) = P(X = 2, Y = 3) = \left(\binom{3}{1} (18/38)^2 (20/38)^1 \right) \left(\binom{2}{1} (18/38)^1 (20/38)^1 \right) = 0.177$$

- e. No. X and Y are not independent. For example $P(X = 0) = 0.15$ but $P(X = 0|Y = 0) = 1$.
- f. Sum over the (X, Y) pairs for which $X + Y = 4$: $(0, 4), (1, 3), (2, 2), (3, 1), (4, 0)$. So $0 + 0.0883 + 0.0981 + 0 = 0.1864$. Suppose they do this every time they visit the casino. Over many visits to the casino, together they win a total of 4 bets in 18.6% of visits
- g. The possible values are $0, 1, \dots, 8$. Fill in the probabilities like in the previous part. (Alternative $X + Y$ has the same distribution as $2X + Z$ where X and Z are independent and Z (number of wins in last two bets) has a Binomial(2, 18/38) distribution).

Value	Probability
0	0.0404
1	0.0727
2	0.1417
3	0.1963
4	0.1864
5	0.1766
6	0.1089
7	0.0530
8	0.0238

- h. From the table $E(X+Y) = 0(0.0404) + 1(0.0727) + \dots + 8(0.0238) = 3.79$, and $E(X) + E(Y) = 3(18/38) + 5(18/38) = 3.79$. (These are exactly equal; any differences are just rounding errors)
- i. There is a positive association so $\text{Var}(X + Y)$ will be greater than $\text{Var}(X) + \text{Var}(Y)$.

Problem 3

Percent returns for assets X , Y , and Z follow a joint distribution with

- Mean 10 and standard deviation 15 for asset X ,
- Mean 5 and standard deviation 3 for asset Y ,
- Correlation of -0.6 between asset X and asset Y
- Asset Z yields a constant return of 1 percent.

An investment portfolio has 60% of its funds in asset X , 30% in asset Y , and 10% in asset Z .

- Let R be the portfolio return. Express R in terms of X, Y, Z .
- Compute $E(R)$.
- Compute $\text{Cov}(X, Y)$.
- Compute $\text{Cov}(X, Z)$.
- Compute $\text{SD}(R)$.

Solution

- $R = 0.6X + 0.3Y + 0.1Z$
- Linearity of expected value

$$E(R) = E(0.6X + 0.3Y + 0.1Z) = 0.6E(X) + 0.3E(Y) + 0.1E(Z) = 0.6(10) + 0.3(5) + 0.1(1) = 7.6$$

- c. $\text{Cov}(X, Y) = \text{Corr}(X, Y)\text{SD}(X)\text{SD}(Y) = (-0.6)(15)(3) = -27$
- d. Z is constant so $\text{Cov}(X, Z) = 0$
- e. Use properties of variance and covariance

$$\begin{aligned}
 \text{Var}(0.6X + 0.3Y + 0.1Z) &= \text{Var}(0.6X + 0.3Y) && Z \text{ is constant, no affect variability} \\
 &= \text{Var}(0.6X) + \text{Var}(0.3Y) + 2\text{Cov}(0.6X, 0.3Y) && \text{variance of sum} \\
 &= 0.6^2\text{Var}(X) + 0.3^2\text{Var}(Y) + 2(0.6)(0.3)\text{Cov}(X, Y) && \text{rules for variance} \\
 &= 0.6^2\text{SD}(X)^2 + 0.3^2\text{SD}(Y)^2 + 2(0.6)(0.3)\text{Corr}(X, Y)\text{SD}(X)\text{SD}(Y) && \text{correlation} \\
 &= 0.6^2(15^2) + 0.3^2(3)^2 + 2(0.6)(0.3)(-0.6)(15)(3) \\
 &= 72.09
 \end{aligned}$$

$$\text{So } \text{SD}(R) = \sqrt{\text{Var}(R)} = \sqrt{72} = 8.49.$$

Problem 4

Devi and Paxton are meeting. Arrival times are measured in minutes after noon, with negative times representing arrivals before noon. Devi's arrival time follows a Normal distribution with mean 20 and SD 15 minutes, and Paxton's arrival time follows a Normal distribution with mean 25 and SD 10 minutes, independently of each other.

For each of the following, find the appropriate standardized value and make a ballpark estimate of the probability. Then use software to compute the probability.

- a. Compute the probability that Devi arrives before noon.
- b. Compute the probability that Devi arrives first.
- c. Compute the probability that the first person to arrive has to wait more than 15 minutes for the second person to arrive.
- d. **Coding required.** Code and run a simulation and use the results to approximate the probability from the previous part (just part c). You should first simulate (Devi, Paxton) arrival time pairs, and go from there.

Solution

- a. If V is Devi's arrival time, $V \sim N(20, 15)$. We want $P(V < 0)$. The standardized value for 0 is $(0 - 20)/15 = -1.333$; that is, 0 is 1.333 SDs below the mean. From the empirical rule, 87% of values are within 1.5 SDs of the mean, so our estimated probability is more than 6.5%. For software, see below; $P(V < 0) = 0.09$.

- b. If X is Paxton's arrival time, we want $P(X > V) = P(X - V > 0)$. Since X and V are independent each with a Normal distribution, then $X - V$ has a Normal distribution. The mean is $20 - 15 = 5$, and since X and V are independent

$$\text{Var}(X - V) = \text{Var}(X) + \text{Var}(V) - 2\text{Cov}(X, V) = 10^2 + 15^2 - 0 = 325$$

So $\text{SD}(X - V) = \sqrt{325} = 18.0$. We want $P(X - V > 0)$. The standardized value for 0 is $(0 - 5)/18 = -0.277$; that is, 0 is 0.277 SDs below the mean. The probability will be more than 50%. For software, see below; $P(X - V > 0) = 0.61$.

- c. We want $P(|X - V| > 15) = P(X - V > 15) + P(X - V < -15)$. The standardized value for 15 is $(15 - 5)/18 = 0.556$. The standardized value for -15 is $(-15 - 5)/18 = -1.11$. The probability that a value is more than 1 SD below the mean is about 0.16; the probability that a value is more than 0.5 SDs above the mean is about 0.31. So the probability that we want is less than 0.47. It's $P(|X - V| > 15) = 0.42$.

```
pnorm((0 - 20) / 15)
```

```
[1] 0.09121122
```

```
pnorm(0, 20, 15)
```

```
[1] 0.09121122
```

```
1 - pnorm((0 - 5) / 18)
```

```
[1] 0.6094085
```

```
1 - pnorm(0, 5, 18)
```

```
[1] 0.6094085
```

```
1 - pnorm((15 - 5) / 18) + pnorm((-15 - 5) / 18)
```

```
[1] 0.4225176
```

```
1 - pnorm(15, 5, 18) + pnorm(-15, 5, 18)
```

```
[1] 0.4225176
```



```
N_rep = 10000
```

```
V = rnorm(N_rep, 20, 15)
```

```
X = rnorm(N_rep, 25, 10)
```

```
sum(abs(V - X) > 15) / N_rep
```

```
[1] 0.4184
```

```
cor(V, X)
```

```
[1] -0.0001885177
```

```
mean(V)
```

```
[1] 19.93839
```

```
sd(V)
```

```
[1] 14.79381
```

```
mean(X)
```

```
[1] 25.04524
```

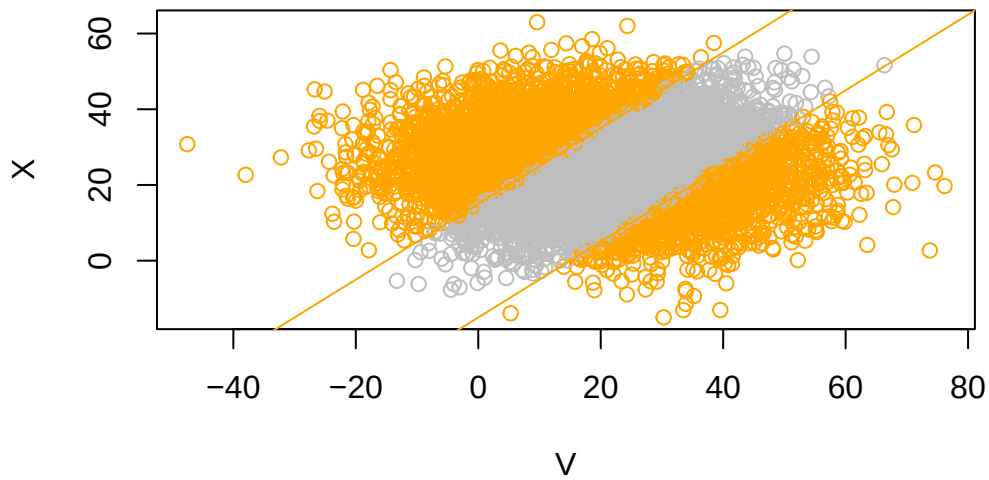
```
sd(X)
```

```
[1] 10.06369
```

```
plot(V, X, col = ifelse(abs(V - X) > 15, "orange", "gray"))
```

```
abline(a = 15, b = 1, col = "orange")
```

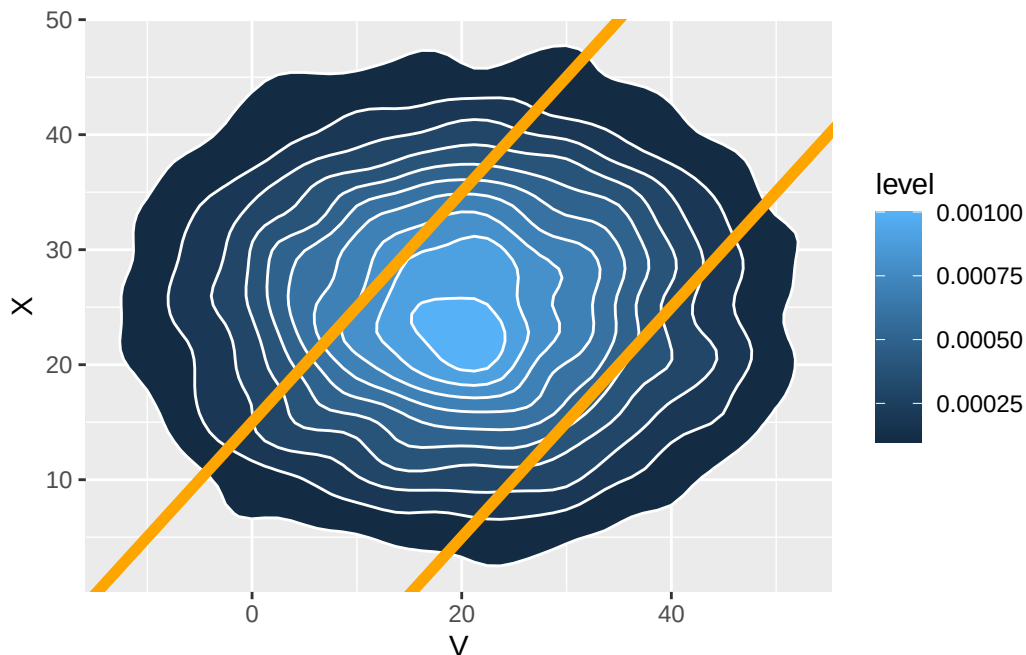
```
abline(a = -15, b = 1, col = "orange")
```



```
ggplot(data.frame(V, X), aes(x = V, y = X)) +  
  stat_density_2d(aes(fill = ..level..), geom = "polygon", colour="white") +  
  geom_abline(slope = 1, intercept = c(-15, 15), col = "orange", size = 2)
```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.

Warning: The dot-dot notation (`..level..`) was deprecated in ggplot2 3.4.0.
i Please use `after\_stat(level)` instead.



Problem 5 part 1

(Continued.) Devi and Paxton are meeting. Arrival times are measured in minutes after noon, with negative times representing arrivals before noon. Devi's arrival time follows a Normal distribution with mean 20 and SD 15 minutes, and Paxton's arrival time follows a Normal distribution with mean 25 and SD 10 minutes.

For each of the following, find the appropriate standardized value and make a ballpark estimate of the probability. Then use software to compute the probability.

Assume the pairs of arrival times follow a Bivariate Normal distribution with correlation 0.8

- Compute the probability that Devi arrives first given that Paxton arrives at 12:10.
- Compute the probability that the first person to arrive has to wait more than 15 minutes for the second person to arrive.
- Coding required.** Code and run a simulation and use the results to approximate the probability from the previous part (just part b). You should first simulate (Devi, Paxton) arrival time pairs, and go from there.

Solution

Let V be Devi's arrival time and let X be Paxton's arrival time.

- a. We want $P(V < X|X = 10) = P(V < 10|X = 10)$. Paxton's time is 1.5 SDs below the mean, $\frac{10-25}{10} = -1.5$. Devi's conditional mean arrival time is 1.2 SDs below the mean, $0.8(-1.5) = -1.2$, so Devi's conditional mean arrival time is $E(V|X = 10) = 20 - 1.2(15) = 2$. Devi's conditional SD is $SD(V|X = 10) = 15\sqrt{1 - 0.8^2} = 9$. So $(V|(X = 10)) \sim N(2, 9)$. $P(V < 10|X = 10) = P(Z < \frac{10-2}{9}) = P(Z < 0.889)$; ballpark estimate a little less than 84%. Software (below) gives 0.81.
- b. We want $P(|X - V| > 15) = P(X - V > 15) + P(X - V < -15)$. Since X and V have a Bivariate Normal distribution, then $X - V$ has a Normal distribution. The mean is $20 - 15 = 5$, and

$$\text{Var}(X - V) = \text{Var}(X) + \text{Var}(V) - 2\text{Cov}(X, V) = 10^2 + 15^2 - 2(0.8)(10)(15) = 85$$

So $SD(X - V) = \sqrt{85} = 9.2$. The standardized value for 15 is $(15 - 5)/9.2 = 1.08$. The standardized value for -15 is $(-15 - 5)/9.2 = -2.17$. The probability that a value is more than 2 SDs below the mean is about 0.025; the probability that a value is more than 1 SD above the mean is about 0.16. So the probability that we want is less than 0.185. It's $P(|X - V| > 15) = 0.15$.

```
pnorm((10 - 2) / 9)
```

```
[1] 0.8129686
```

```
pnorm(10, 2, 9)
```

```
[1] 0.8129686
```

```
1 - pnorm((15 - 5) / 9.2) + pnorm((-15 - 5) / 9.2)
```

```
[1] 0.1533838
```

```
1 - pnorm(15, 5, 9.2) + pnorm(-15, 5, 9.2)
```

```
[1] 0.1533838
```

```
N_rep = 10000
```

```
V = rnorm(N_rep, 20, 15)
```

```
X = rnorm(N_rep, 25 + 10 * 0.8 * (V - 20) / 15, 10 * sqrt(1 - 0.8 ^ 2))
```

```
sum(abs(V - X) > 15) / N_rep
```

```
[1] 0.1498
```

```
cor(V, X)
```

```
[1] 0.8044512
```

```
mean(V)
```

```
[1] 20.09649
```

```
sd(V)
```

```
[1] 15.01435
```

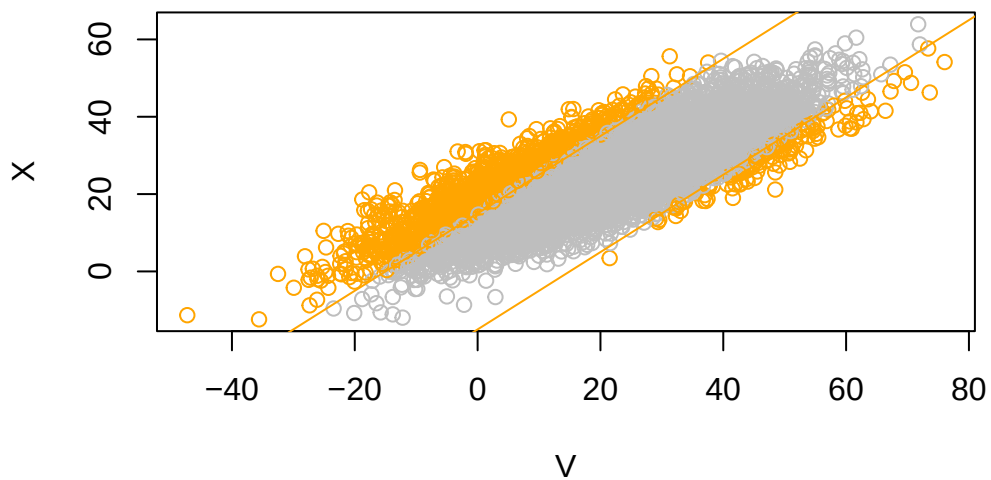
```
mean(X)
```

```
[1] 25.02947
```

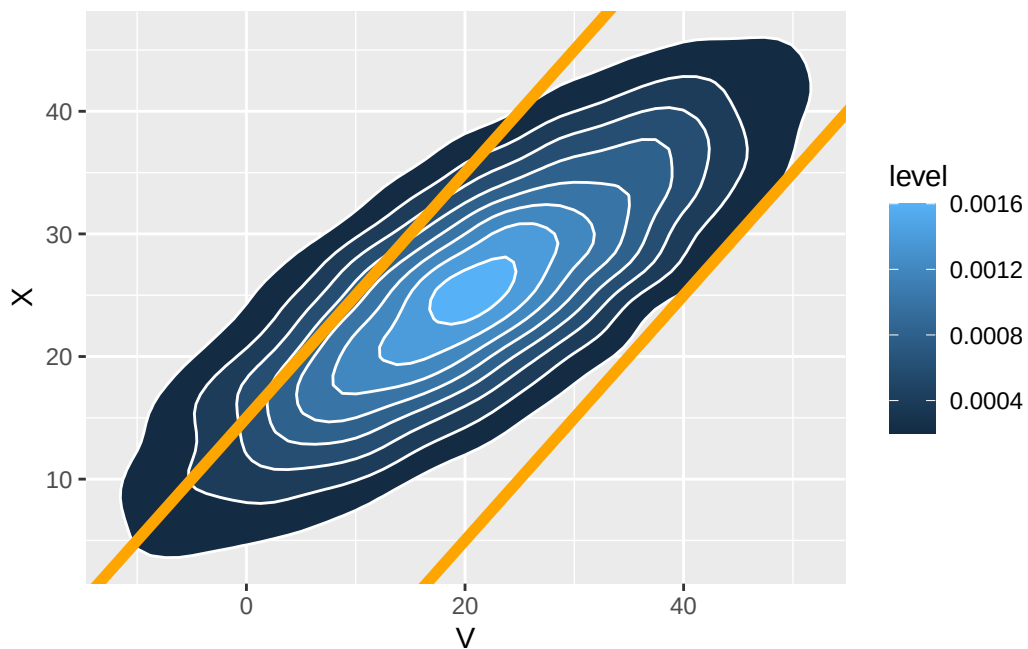
```
sd(X)
```

```
[1] 10.01573
```

```
plot(V, X, col = ifelse(abs(V - X) > 15, "orange", "gray"))  
abline(a = 15, b = 1, col = "orange")  
abline(a = -15, b = 1, col = "orange")
```



```
ggplot(data.frame(V, X), aes(x = V, y = X)) +
  stat_density_2d(aes(fill = ..level..), geom = "polygon", colour="white") +
  geom_abline(slope = 1, intercept = c(-15, 15), col = "orange", size = 2)
```



Problem 5 part 2

(Continued.) Devi and Paxton are meeting. Arrival times are measured in minutes after noon, with negative times representing arrivals before noon. Devi's arrival time follows a Normal distribution with mean 20 and SD 15 minutes, and Paxton's arrival time follows a Normal distribution with mean 25 and SD 10 minutes.

For each of the following, find the appropriate standardized value and make a ballpark estimate of the probability. Then use software to compute the probability.

Assume the pairs of arrival times follow a Bivariate Normal distribution with correlation -0.7

- Compute the probability that Devi arrives first given that Paxton arrives at 12:10.
- Compute the probability that the first person to arrive has to wait more than 15 minutes for the second person to arrive.
- Coding required.** Code and run a simulation and use the results to approximate the probability from the previous part (just part b). You should first simulate (Devi, Paxton) arrival time pairs, and go from there.

Solution

- We want $P(V < X | X = 10) = P(V < 10 | X = 10)$. Paxton's time is 1.5 SDs below the mean, $\frac{10-25}{10} = -1.5$. Devi's conditional mean arrival time is 1.05 SDs *above* the mean, $(-0.7)(-1.5) = 1.05$, so Devi's conditional mean arrival time is $E(V|X = 10) = 20 + 1.05(15) = 35.75$. Devi's conditional SD is $SD(V|X = 10) = 15\sqrt{1 - 0.7^2} = 10.7$. So $(V|(X = 10)) \sim N(35.75, 10.7)$. $P(V < 10 | X = 10) = P(Z < \frac{10-35.75}{10.7}) = P(Z < -2.4)$; ballpark estimate is about 0.005. Software (below) gives 0.008.
- We want $P(|X - V| > 15) = P(X - V > 15) + P(X - V < -15)$. Since X and V have a Bivariate Normal distribution, then $X - V$ has a Normal distribution. The mean is $20 - 25 = -5$, and

$$\text{Var}(X - V) = \text{Var}(X) + \text{Var}(V) - 2\text{Cov}(X, V) = 10^2 + 15^2 - 2(-0.7)(10)(15) = 535$$

So $SD(X - V) = \sqrt{535} = 23.1$. The standardized value for 15 is $(15 - (-5))/23.1 = 0.43$. The standardized value for -15 is $(-15 - (-5))/23.1 = -0.86$. The probability that a value is more than 1 SD below the mean is about 0.16; the probability that a value is more than 0.5 SD above the mean is about 0.31. So the probability that we want is about 0.47. It's $P(|X - V| > 15) = 0.53$.

```
pnorm((10 - 35.75) / 10.7)
```

```
[1] 0.008052175
```

```
pnorm(10, 35.75, 10.7)
```

```
[1] 0.008052175
```

```
1 - pnorm((15 - 5) / 23.1) + pnorm((-15 - 5) / 23.1)
```

```
[1] 0.5258432
```

```
1 - pnorm(15, 5, 23.1) + pnorm(-15, 5, 23.1)
```

```
[1] 0.5258432
```

```
N_rep = 10000
```

```
V = rnorm(N_rep, 20, 15)
```

```
X = rnorm(N_rep, 25 + 10 * (-0.7) * (V - 20) / 15, 10 * sqrt(1 - 0.7 ^ 2))
```

```
sum(abs(V - X) > 15) / N_rep
```

```
[1] 0.5249
```

```
cor(V, X)
```

```
[1] -0.6996195
```

```
mean(V)
```

```
[1] 20.02093
```

```
sd(V)
```

```
[1] 15.11597
```

```
mean(X)
```

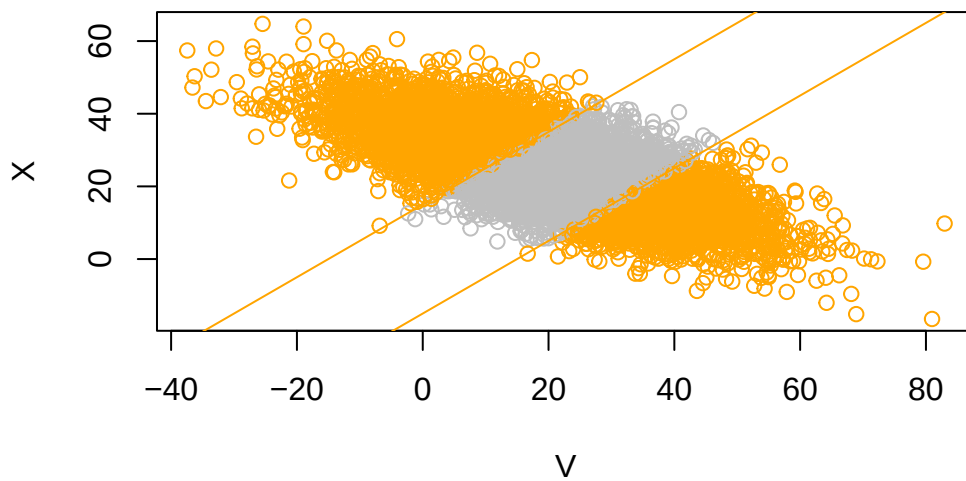
```
[1] 24.89673
```



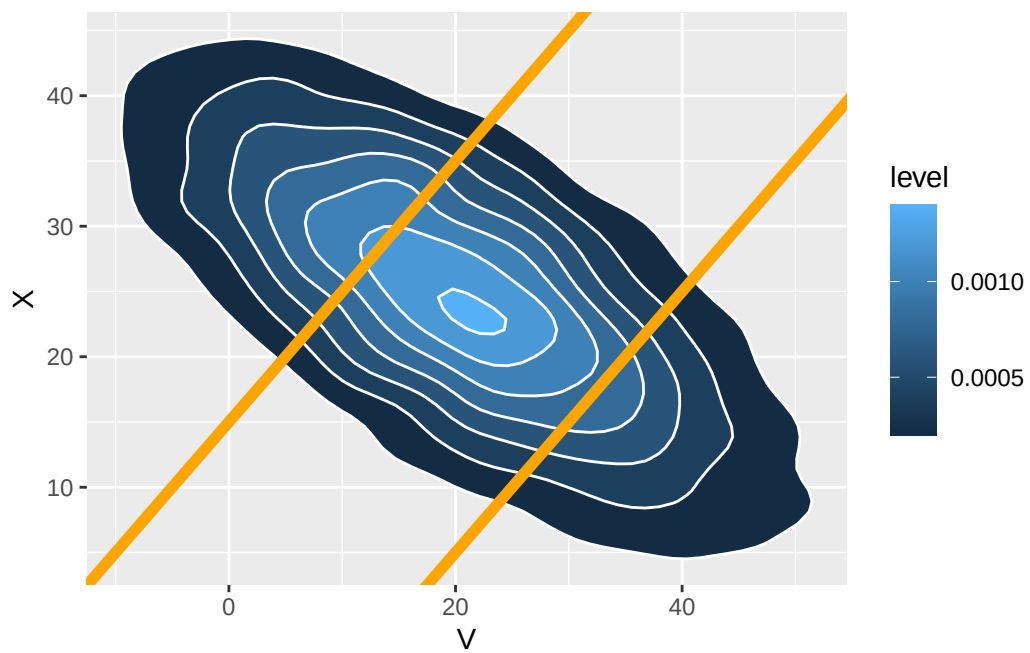
```
sd(X)
```

```
[1] 10.01412
```

```
plot(V, X, col = ifelse(abs(V - X) > 15, "orange", "gray"))  
abline(a = 15, b = 1, col = "orange")  
abline(a = -15, b = 1, col = "orange")
```



```
ggplot(data.frame(V, X), aes(x = V, y = X)) +  
  stat_density_2d(aes(fill = ..level..), geom = "polygon", colour="white") +  
  geom_abline(slope = 1, intercept = c(-15, 15), col = "orange", size = 2)
```



Homework 6 solutions

Problem 1

You have a brand new coin and you want to estimate p , the probability that it lands on heads. You flip the coin n times and observe X heads.

- Suggest an estimator of p ; it should be an expression involving X and n . Describe in words what your estimator is. (Hint: what is the most obvious estimator you can think of?)
- Suppose you observe $x = 6$ heads in $n = 10$ flips. Use your estimator to compute an estimate of p based on this sample.
- Consider the estimator (called the “plus four” estimator)

$$\frac{X+2}{n+4} = \left(\frac{4}{n+4}\right)(0.5) + \left(\frac{n}{n+4}\right)\left(\frac{X}{n}\right)$$

Explain in words what this estimator does. (The two expressions in the equation above are just different ways of expressing the estimator.)

- Use the estimator from the previous part and compute an estimate of p based on a sample with 6 heads in 10 flips.
- If you observe 6 heads in 10 flips, which of your two estimates is a better estimate of p ? Explain.
- Compute and interpret the probability of observing 6 heads in 10 flips if $p = 0.6$.
- Compute and interpret the probability of observing 6 heads in 10 flips if $p = 0.571$.
- Compare the values from the two previous parts. If you observed 6 heads in 10 flips, which of the values — 0.6 or 0.571 — would you choose as your estimate of p ? Explain your reasoning.
- Suppose you observe 6 heads in 10 flips. Carefully write the likelihood function.
- Continuing the previous part, use any software of your choice and plot the likelihood function. Based on your plot, what value appears to be the maximum likelihood estimate of p based on a sample with 6 heads in 10 flips? (You can do this just by zooming in on the plot.)
- Continuing the previous part, carefully write the log-likelihood function.
 - Use calculus to find the maximum likelihood estimate of p based on observing 6 heads in 10 flips.
- Now suppose you observe x heads in 10 flips. Carefully write the likelihood function.

- n. Continuing the previous part, carefully write the log-likelihood function.
- o. Use calculus to find a general expression for the maximum likelihood estimate of p based on observing x heads in 10 flips. (Your answer will be an expression involving x .)

Solutions

- a. Seems like the most natural estimator is X/n , the sample proportion of H.
- b. The observed sample proportion is $6/10 = 0.6$, so the estimator X/n would estimate p to be 0.6 for this sample.
- c. The “plus four” estimator basically appends 2 successes and 2 failures to the sample data and then computes the sample proportion based on the appended sample. (It is also a weighted average between the observe sample proportion X/n and 0.5.)
- d. Based on the 6/10 sample, the plus four estimator estimates p to be $(6 + 2)/(10 + 4) = 8/14 = 0.571 = (10/14)(0.6) + (4/14)(0.5) = 0.571$.
- e. We don’t know which of 0.6 and 0.571 is a better estimate because we don’t know what p is; that’s the reason why we’re trying to estimate it.
- f. $\text{dbinom}(6, 10, 0.6) = 0.251 = \binom{10}{6}(0.6)^6(1 - 0.6)^{10-6}$. If the probability that the coin lands on H is 0.6, then about 25.1 percent of sets of 10 flips would result in 6 H.
- g. $\text{dbinom}(6, 10, 0.571) = 0.186 = \binom{10}{6}(0.571)^6(1 - 0.571)^{10-6}$. If the probability that the coin lands on H is 0.571, then about 18.6 percent of sets of 10 flips would result in 6 H.
- h. The likelihood of observing 6 H in 10 flips is greater if $p = 0.6$ than if $p = 0.571$, so if we’re trying to maximum the likelihood, we could pick 0.6 as our estimate of p based on the 6/10 sample.
- i. When $n = 10$, X has a Binomial(10, p) distribution. The pmf is $\binom{10}{x}p^x(1 - p)^{10-x}$. Plug in the observed data $x = 6$ and view the likelihood as a function of p

$$L(p) = \binom{10}{6}p^6(1 - p)^{10-6}, \quad 0 < p < 1$$

- j. See below. It seems like the overall maximum occurs at a value of 0.6.
- k. Take logs

$$\ell(p) = \log L(p) = \log \left(\binom{10}{6}p^6(1 - p)^{10-6} \right) = 6 \log p + (10 - 6) \log(1 - p) + \log \left(\binom{10}{6} \right)$$

- l. Differentiate the log-likelihood with respect to p . (Don’t forget the chain rule for $\log(1 - p)$; the derivate of $1 - p$ on the inside is -1 .)

$$\ell'(p) = \frac{6}{p} + \frac{10 - 6}{1 - p}(-1) + 0$$

Set $\ell'(\hat{p}) = 0$ and solve to get $\hat{p} = 0.6$. So the MLE of p based on a sample of size 10 with 6 successes is $6/10 = 0.6$, the observed sample proportion of success.

m. Replace 6 with a general x

$$L(p) = \binom{10}{x} p^x (1-p)^{10-x}, \quad 0 < p < 1$$

n. Continuing the previous part, carefully write the log-likelihood function.

$$\ell(p) = \log L(p) = \log \left(\binom{10}{x} p^x (1-p)^{10-x} \right) = x \log p + (10-x) \log(1-p) + \log \binom{10}{x}$$

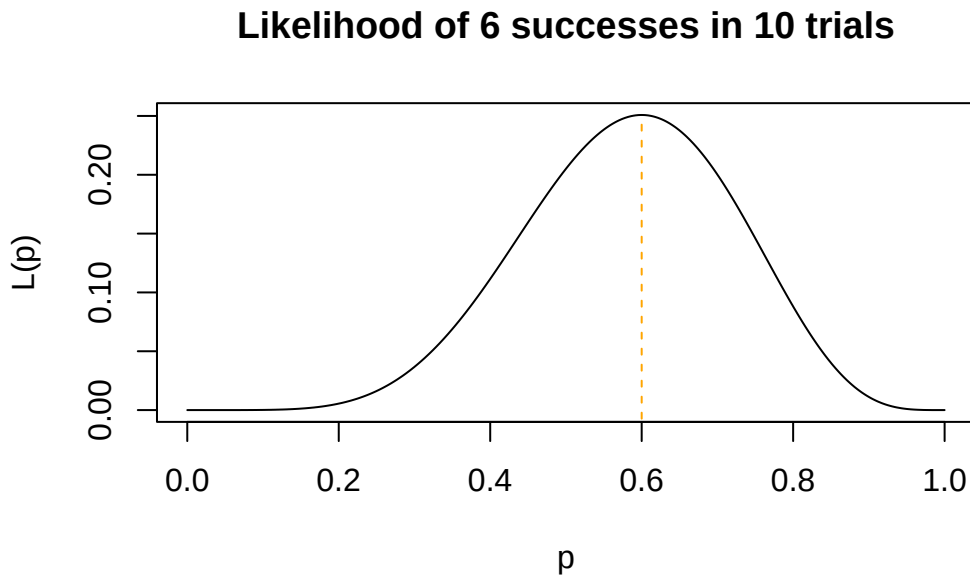
o. Differentiate the log-likelihood with respect to p . x is treated like a constant just like 6 above. (Don't forget the chain rule for $\log(1-p)$; the derivative of $1-p$ on the inside is -1 .)

$$\ell'(p) = \frac{x}{p} + \frac{10-x}{1-p}(-1) + 0$$

Set $\ell'(\hat{p}) = 0$ and solve to get $\hat{p} = x/10$. So the MLE of p based on a sample of size 10 with x successes is $x/10$, the observed sample proportion of success.

```
p = seq(0, 1, 0.001)

plot(p, dbinom(6, 10, p), type = "l",
     xlab = "p",
     ylab = "L(p)",
     main = "Likelihood of 6 successes in 10 trials")
segments(x0 = 0.6, x1 = 0.6, y0 = -1, y1 = dbinom(6, 10, 0.6), col = "orange", lty = 2)
```



Problem 2

Suppose that times (minutes) between earthquakes (of any magnitude in a certain region) follow an Exponential distribution with rate parameter λ . Let $X_1, \dots, X_n$ be a random sample of times (minutes) between earthquakes for n earthquakes. It can be shown that the MLE of λ is $1/\bar{X}$.

1. Explain intuitively why $1/\bar{X}$ is a reasonable estimator of λ . Hint: it's not just because it's the MLE; think of properties of Exponential distributions.
2. Suppose the observed sample is: 20, 37, 13, 10, 25 minutes. Carefully write the likelihood function. (For a random sample from a continuous distribution, to find the likelihood function you plug each observed value into the pdf and then form the product.)
3. Use any software you want to carefully plot the likelihood function from the previous part. Then use your plot to verify that the MLE of λ based on this sample is $1/\bar{x} = 5/105 = 0.0476$. (You can just zoom in on the plot.)
4. Write a clearly worded sentence reporting in context your estimate of λ from the previous part.

Solution

1. For Exponential, the population mean is the reciprocal of the rate, $1/\lambda$. It seems reasonable to estimate the Exponential population mean $1/\lambda$ with $\bar{X}$, so it seems reasonable to estimate the rate λ with $1/\bar{X}$.
2. Remember the Exponential pdf is $\lambda e^{-\lambda x}$. There will be a factor in the likelihood for each observed x values.

$$L(\lambda) = (\lambda e^{-20\lambda}) (\lambda e^{-37\lambda}) (\lambda e^{-13\lambda}) (\lambda e^{-10\lambda}) (\lambda e^{-25\lambda}) = \lambda^5 e^{-\lambda(20+37+13+10+25)}, \quad \lambda > 0$$

3. See below. It looks like the MLE of λ based on this sample is $1/\bar{x} = 5/105 = 0.0476$.
4. We estimate that earthquakes occur on average at rate 0.0476 earthquakes per minute (2.86 earthquakes per hour).

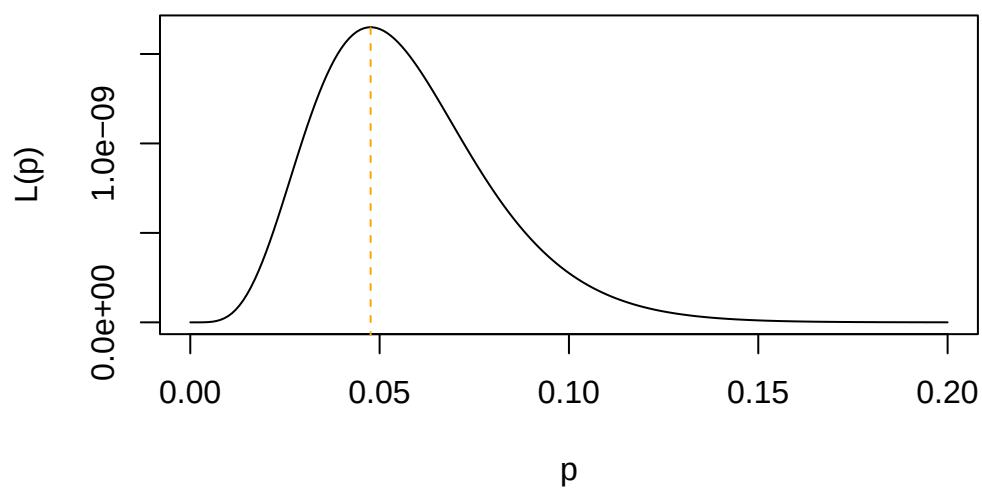
```
x = c(20, 37, 13, 10, 25)

xbar = mean(x)

lambda = seq(0, 0.2, 0.001)

plot(lambda, lambda ^ 5 * exp(-lambda * sum(x)), type = "l",
      xlab = "p",
      ylab = "L(p)",
      main = "Likelihood of a total time of 105 minutes for 5 quakes")
segments(x0 = 1 / xbar, x1 = 1 / xbar, y0 = -1, y1 = (1 / xbar) ^ 5 * exp(-(1 / xbar) * sum(x)))
```

Likelihood of a total time of 105 minutes for 5 quakes



Homework 7 solutions

Problem 1

Assume a $\text{Poisson}(\mu)$ model for the number of home runs hit (in total by both teams) in a MLB game. Let $X_1, \dots, X_n$ be a random sample of home run counts for n games.

Suppose we want to estimate $\theta = \mu e^{-\mu}$, the probability that any single game has exactly 1 HR. Consider two estimators of θ

- $\hat{\theta} = \bar{X}e^{-\bar{X}}$
 - $\hat{p}$ = sample proportion of 1s = $\frac{\text{number of games in the sample with 1 HR}}{\text{sample size}}$
- a. Compute the value of $\hat{\theta}$ based on the sample (3, 0, 1, 4, 0). Write a clearly worded sentence reporting in context this estimate of θ .
 - b. Compute the value of $\hat{p}$ based on the sample (3, 0, 1, 4, 0). Write a clearly worded sentence reporting in context your estimate of θ .
 - c. Which of these two estimators is the MLE of θ in this situation? Explain, without doing any calculations.
 - d. It can be shown that $\hat{p}$ is an unbiased estimator of θ . Explain what this means.
 - e. Is $\hat{\theta}$ an unbiased estimator of θ ? Explain. (You don't have to derive anything; just apply a general principle.)
 - f. Suppose $\mu = 2.3$ and $n = 5$. Explain in full detail how you would use simulation to approximate the bias of $\hat{\theta}$.
 - g. Conduct the simulation from the previous part and plot the approximate of $\hat{\theta}$ when $\mu = 2.3$ and $n = 5$.
 - h. Explain in full detail how you would use simulation to approximate the bias function of $\hat{\theta}$ when $n = 5$.
 - i. Conduct the simulation from the previous part and plot the approximate bias function when $n = 5$. For what values of μ does $\hat{\theta}$ tend to overestimate μ ? Underestimate? For what values of μ is the bias the worst?

Solution

- a. For the sample (3, 0, 1, 4, 0), the value of $\bar{X}$ is $8/5$ and the value of $\hat{\theta}$ is $(8/5)e^{-8/5} = 0.323$. Using this estimator, based on this sample we would estimate that 32.3 percent of baseball games have 1 home run.

- b. For the sample (3, 0, 1, 4, 0) there is 1 game with 1 HR so $\hat{p} = 1/5 = 0.2$. Using this estimator, based on this sample we would estimate that 20 percent of baseball games have 1 home run.
- c. We know that the MLE of μ for a Poisson distribution is $\bar{X}$, so by the invariance principle of MLEs, $\bar{X}e^{-\bar{X}}$ is the MLE of $\mu e^{-\mu}$.
- d. Over many samples, $\hat{p}$ does not consistent over or under estimate θ .
- e. No. $\bar{X}$ is an unbiased estimator of μ but nonlinear transformations do not preserve unbiasedness. $\hat{\theta}$ is not unbiased estimator of θ .

$$E(\hat{\theta}) = E(\bar{X}e^{-\bar{X}}) \neq E(\bar{X})e^{-E(\bar{X})} = \mu e^{-\mu} = \theta$$

- f. For $\mu = 2.3$
- Simulate a sample of size n from a Poisson(2.3) distribution, say (3, 0, 1, 4, 0) for $n = 5$.
 - For the simulated sample compute the sample mean $\bar{X}$ and the value of $\hat{\theta} = \bar{X}e^{-\bar{X}}$, say $8/5$ and $(8/5)e^{-8/5} = 0.323$.
 - Repeat many times, simulating many samples of size n and many values of $\hat{\theta}$ for the chosen μ .
 - Average the values of $\hat{\theta}$ and subtract $\theta = 2.3e^{-2.3}$ to approximate the bias when $\mu = 2.3$.
- g. See below.
- h. We carry out the process in the previous part for many values of μ . Choose a grid of μ values, say 0.01, 0.02, ..., 10
- Choose a value of μ from the grid, say 2.3
 - Simulate a sample of size n from a Poisson(μ) distribution, say (3, 0, 1, 4, 0) for $n = 5$.
 - For the simulated sample compute the sample mean $\bar{X}$ and the value of $\hat{\theta} = \bar{X}e^{-\bar{X}}$, say $8/5$ and $(8/5)e^{-8/5} = 0.323$.
 - Repeat many times, simulating many samples of size n and many values of $\hat{\theta}$ for the chosen μ .
 - Average the values of $\hat{\theta}$ and subtract $\theta = \mu e^{-\mu}$ to approximate the bias for the chosen μ .
 - Repeat the above process for all the values of μ in the grid and plot the approximate bias as a function of μ to approximate the bias function.
- i. See below.

Problem 2

Continuing Problem 2.

- It can be shown that $\text{Var}(\hat{p}) = \frac{\theta(1-\theta)}{n}$. Compute $\text{Var}(\hat{p})$ when $\mu = 2.3$ and $n = 5$. Then write a clearly worded sentence interpreting this value.
- Suppose $\mu = 2.3$ and $n = 5$. Explain in full detail how you would use simulation to approximate the variance of $\hat{\theta}$.
- Conduct the simulation from the previous part and approximate the variance of $\hat{\theta}$ when $\mu = 2.3$ and $n = 5$. Then write a clearly worded sentence interpreting this value.
- Which estimator has smaller variance when $\mu = 2.3$ (and $n = 5$)? Answer, but then explain why this information alone is not really useful.
- Explain in full detail how you would use simulation to approximate the variance function of $\hat{\theta}$ (if $n = 5$).
- Conduct the simulation from the previous part and plot the approximate variance function. Compare to the variance function of $\hat{p}$. Based on variability alone, which estimator is preferred?

Solution

- When $\mu = 2.3$, $\theta = 2.3e^{-2.3} = 0.231$, so if $n = 5$

$$\text{Var}_{\mu=2.3}(\hat{p}) = \frac{0.231(1-0.231)}{5} = 0.035$$

This number measures the sample-to-sample variance of sample proportions of games with 1 HR over many samples of 5 games, assuming that $\mu = 2.3$.

- For $\mu = 2.3$
 - Simulate a sample of size n from a Poisson(2.3) distribution, say (3, 0, 1, 4, 0) for $n = 5$.
 - For the simulated sample compute the sample mean $\bar{X}$ and the value of $\hat{\theta} = \bar{X}e^{-\bar{X}}$, say $8/5$ and $(8/5)e^{-8/5} = 0.323$.
 - Repeat many times, simulating many samples of size n and many values of $\hat{\theta}$ for the chosen μ .
 - Average the values of $\hat{\theta}$, for each value of $\hat{\theta}$ find its deviation from the average, and then average the squared deviations. This number approximates $\text{Var}(\hat{\theta})$ when $\mu = 2.3$
- See below. The value measures the sample-to-sample variance of values of $\hat{\theta}$ (estimates of the probability that a game has 1 HR), assuming that $\mu = 2.3$.
- $\hat{\theta}$ has smaller variance when $\mu = 2.3$, but that by itself isn't useful because we don't know what μ is.
- We carry out the process in the previous part for many values of μ . Choose a grid of μ values, say 0.01, 0.02, ..., 10
 - Choose a value of μ from the grid, say 2.3
 - Simulate a sample of size n from a Poisson(μ) distribution, say (3, 0, 1, 4, 0) for $n = 5$.

- For the simulated sample compute the sample mean $\bar{X}$ and the value of $\hat{\theta} = \bar{X}e^{-\bar{X}}$, say $8/5$ and $(8/5)e^{-8/5} = 0.323$.
 - Repeat many times, simulating many samples of size n and many values of $\hat{\theta}$ for the chosen μ .
 - Average the values of $\hat{\theta}$, for each value of $\hat{\theta}$ find its deviation from the average, and then average the squared deviations. This number approximates $\text{Var}(\hat{\theta})$ for the chosen μ .
 - Repeat the above process for all the values of μ in the grid and plot the approximate bias as a function of μ to approximate the bias function.
- f. See below. Seems like $\hat{\theta}$ has smaller variance regardless of μ .

Problem 3

Continuing Problems 2 and 3

- Compute $\text{MSE}(\hat{p})$ when $\mu = 2.3$ and $n = 5$. (You can do the next part first if you want, but it helps to work with specific numbers first.)
- Derive the MSE function of $\hat{p}$. (Hint: see facts from previous parts.)
- Suppose $\mu = 2.3$ (and $n = 5$). Explain in full detail how you would use simulation to approximate the MSE of $\hat{\theta}$.
- Conduct the simulation from the previous part and approximate the MSE of $\hat{\theta}$ when $\mu = 2.3$ (and $n = 5$).
- Which estimator has smaller MSE when $\mu = 2.3$ (and $n = 5$)? Answer, but then explain why this information alone is not really useful.
- Conduct the simulation from the previous part and plot the approximate MSE function. Compare to the MSE function of $\hat{p}$.
- Compare the MSEs of the two estimators for $n = 5$ and a few other values of n . Is there a clear preference between these two estimators? Discuss.

Solution

- Since $\hat{p}$ is unbiased, its MSE is just its variance.
- Since $\hat{p}$ is unbiased, its MSE function is just its variance function.
- For $\mu = 2.3$
 - Simulate a sample of size n from a Poisson(2.3) distribution, say (3, 0, 1, 4, 0) for $n = 5$.
 - For the simulated sample compute the sample mean $\bar{X}$ and the value of $\hat{\theta} = \bar{X}e^{-\bar{X}}$, say $8/5$ and $(8/5)e^{-8/5} = 0.323$.
 - Repeat many times, simulating many samples of size n and many values of $\hat{\theta}$ for the chosen μ .

- For each value of $\hat{\theta}$ subtract $\theta = 2.3e^{-2.3}$ and square the difference, then average these squared differences to approximate the MSE when $\mu = 2.3$.
- See below. The squared bias is fairly small compared to the variance, so the MSE is pretty close to the variance function.
 - $\hat{\theta}$ has smaller MSE when $\mu = 2.3$, but this isn't really useful because we don't know what μ is.
 - See below.
 - It seems that (at least for the values of n that we considered) $\hat{\theta}$ has smaller MSE than $\hat{p}$ for all μ . The sample proportion doesn't use the fact that the population distribution is Poisson, and it treats each value as either 1 or not. $\bar{X}e^{-\bar{X}}$ uses all the data and the fact that the population distribution is Poisson, so its better tuned to this problem.

Simulation

For mu = 2.3

```
n = 5

mu = 2.3

N_samples = 10000

theta_hats = vector(length = N_samples)

for (i in 1:N_samples) {
  # simulate a sample of size n from the Poisson(mu) population
  x = rpois(n, mu)

  # compute theta-hat
  theta_hats[i] = mean(x) * exp(-mean(x))
}

# Approximate the bias of theta-hat for this mu
mean(theta_hats) - mu * exp(-mu)
```

```
[1] 0.003998485
```

```
# Approximate the variance of theta-hat for this mu
var(theta_hats)
```

```
[1] 0.005706511
```

```
# Approximate the MSE of theta-hat for this mu
mean((theta_hats - mu * exp(-mu)) ^ 2)
```

```
[1] 0.005721928
```

For all mu

```
n = 5

# grid of mu values: 0.1, 0.2, 0.3, ..., 4.9, 5
mus = seq(0.1, 5, 0.1)

N_samples = 10000

# initialize vectors that will record bias, var, mse as functions of mu
bias_thetahat = vector()
var_thetahat = vector()
mse_thetahat = vector()

for (mu in mus) {

  # true value of theta for this mu
  theta = mu * exp(-mu)

  # for each mu simulate many samples and many values of sigma-hat^2
  theta_hats = vector(length = N_samples)

  for (i in 1:N_samples) {
    # simulate a sample of size n from the Poisson(mu) population
    x = rpois(n, mu)

    # compute theta-hat
    theta_hats[i] = mean(x) * exp(-mean(x))
```

```

}

# Approximate bias of theta-hat for this mu and add to the list
bias_thetahat = c(bias_thetahat, mean(theta_hats) - theta)

# Approximate variance of sigma-hat^2 for this mu and add to the list
var_thetahat = c(var_thetahat, var(theta_hats))

# Approximate MSE of sigma-hat^2 for this mu and add to the list
# Can just do bias^2 + var, but can also use definition of MSE
mse_thetahat = c(mse_thetahat, mean((theta_hats - theta) ^ 2))
}

```

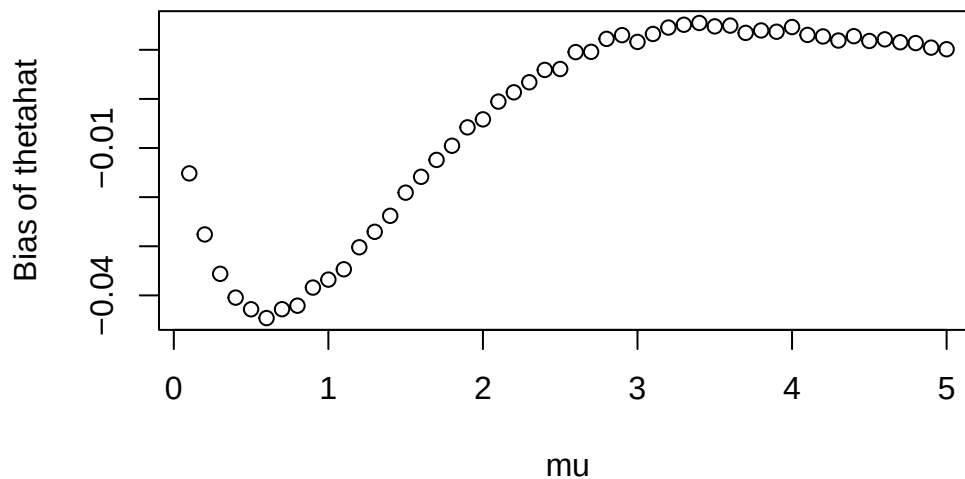
Bias function

Any “wiggles” are due to natural simulation margin of error (can make these go away by simulating even more samples)

```

plot(mu, bias_thetahat,
     xlab = "mu",
     ylab = "Bias of thetihat")

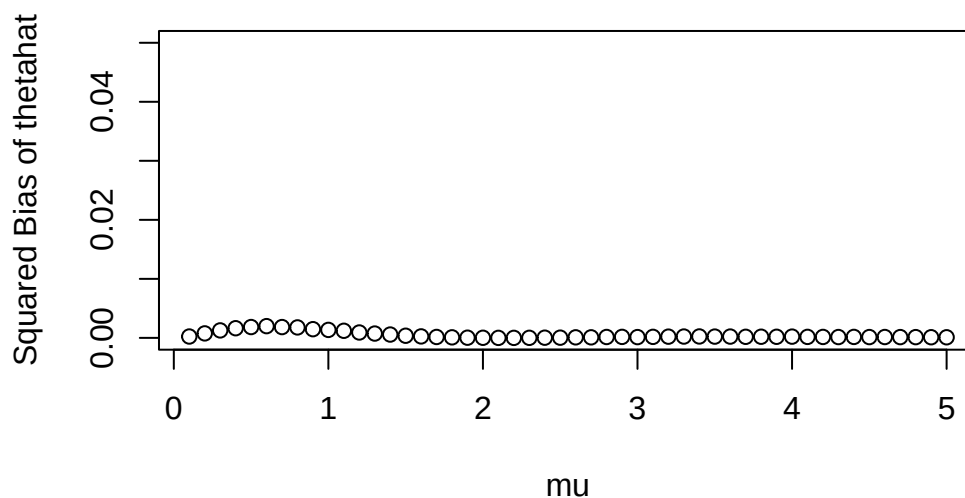
```



Squared bias function

Any “wiggles” are due to natural simulation margin of error (can make these go away by simulating even more samples)

```
plot(mus, bias_thetahat ^ 2,
     xlab = "mu",
     ylab = "Squared Bias of thetahat",
     ylim = c(0, 0.05)) # set common y-axis scale
```

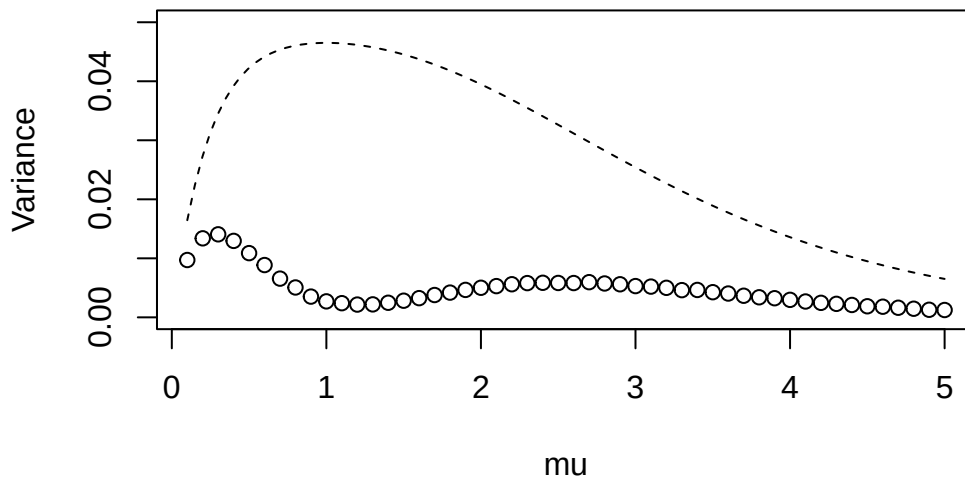


Variance function

Any “wiggles” are due to natural simulation margin of error (can make these go away by simulating even more samples)

```
plot(mus, var_thetahat,
     xlab = "mu",
     ylab = "Variance",
     ylim = c(0, 0.05), # set common y-axis scale
     main = "Variance functions of theta-hat (dots) and p-hat (dashed)")
curve(x * exp(-x) * (1 - x * exp(-x)) / n, add = TRUE,
     lty = 2)
```

Variance functions of theta-hat (dots) and p-hat (dashed)

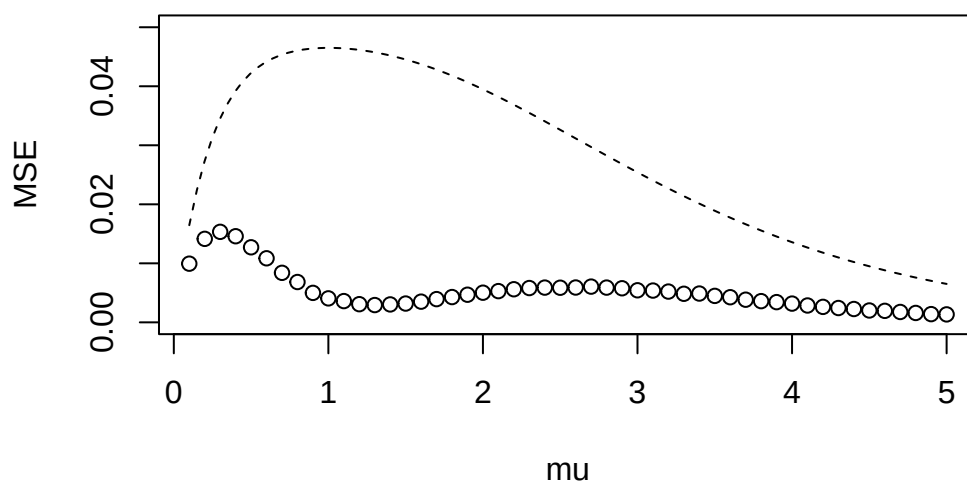


MSE function

Any “wiggles” are due to natural simulation margin of error (can make these go away by simulating even more samples)

```
plot(mus, mse_thetahat,  
     xlab = "mu",  
     ylab = "MSE",  
     ylim = c(0, 0.05), # set common y-axis scale  
     main = "MSE functions of theta-hat (dots) and p-hat (dashed)")  
curve(x * exp(-x) * (1 - x * exp(-x)) / n, add = TRUE,  
      lty = 2)
```


MSE functions of theta-hat (dots) and p-hat (dashed)



Different n

```
n = 50

# grid of mu values: 0.1, 0.2, 0.3, ..., 4.9, 5
mus = seq(0.1, 5, 0.1)

N_samples = 10000

# initialize vectors that will record bias, var, mse as functions of mu
bias_thetahat = vector()
var_thetahat = vector()
mse_thetahat = vector()

for (mu in mus) {

  # true value of theta for this mu
  theta = mu * exp(-mu)

  # for each mu simulate many samples and many values of sigma-hat^2
  theta_hats = vector(length = N_samples)
```

```

for (i in 1:N_samples) {
  # simulate a sample of size n from the Poisson(mu) population
  x = rpois(n, mu)

  # compute theta-hat
  theta_hats[i] = mean(x) * exp(-mean(x))
}

# Approximate bias of theta-hat for this mu and add to the list
bias_thetahat = c(bias_thetahat, mean(theta_hats) - theta)

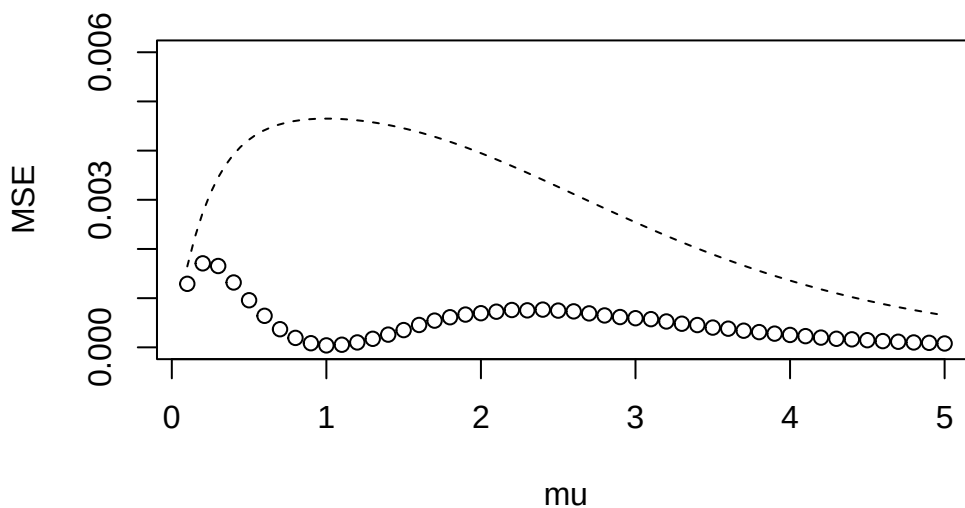
# Approximate variance of sigma-hat^2 for this mu and add to the list
var_thetahat = c(var_thetahat, var(theta_hats))

# Approximate MSE of sigma-hat^2 for this mu and add to the list
# Can just do bias^2 + var, but can also use definition of MSE
mse_thetahat = c(mse_thetahat, mean((theta_hats - theta) ^ 2))
}

plot(mus, mse_thetahat,
     xlab = "mu",
     ylab = "MSE",
     ylim = c(0, 0.006), # set common y-axis scale
     main = "n = 50:MSE functions of theta-hat (dots) and p-hat (dashed)")
curve(x * exp(-x) * (1 - x * exp(-x)) / n, add = TRUE,
      lty = 2)

```

n = 50: MSE functions of theta-hat (dots) and p-hat (dashed)



```
n = 20

# grid of mu values: 0.1, 0.2, 0.3, ..., 4.9, 5
mus = seq(0.1, 5, 0.1)

N_samples = 10000

# initialize vectors that will record bias, var, mse as functions of mu
bias_thetahat = vector()
var_thetahat = vector()
mse_thetahat = vector()

for (mu in mus) {

  # true value of theta for this mu
  theta = mu * exp(-mu)

  # for each mu simulate many samples and many values of sigma-hat^2
  theta_hats = vector(length = N_samples)

  for (i in 1:N_samples) {
    # simulate a sample of size n from the Poisson(mu) population
    x = rpois(n, mu)
```

```

    # compute theta-hat
    theta_hats[i] = mean(x) * exp(-mean(x))

}

# Approximate bias of theta-hat for this mu and add to the list
bias_thetahat = c(bias_thetahat, mean(theta_hats) - theta)

# Approximate variance of sigma-hat^2 for this mu and add to the list
var_thetahat = c(var_thetahat, var(theta_hats))

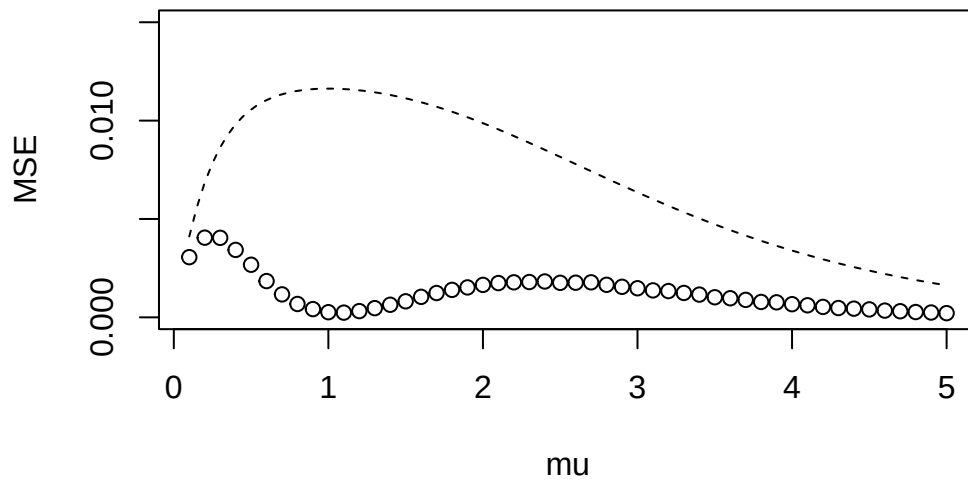
# Approximate MSE of sigma-hat^2 for this mu and add to the list
# Can just do bias^2 + var, but can also use definition of MSE
mse_thetahat = c(mse_thetahat, mean((theta_hats - theta) ^ 2))

}

plot(mus, mse_thetahat,
     xlab = "mu",
     ylab = "MSE",
     ylim = c(0, 0.015), # set common y-axis scale
     main = "n = 20:MSE functions of theta-hat (dots) and p-hat (dashed)")
curve(x * exp(-x) * (1 - x * exp(-x)) / n, add = TRUE,
      lty = 2)

```

n = 20: MSE functions of theta-hat (dots) and p-hat (dashed)



Homework 8 solutions

See code and results in [Colab notebook](#).

Problem 1

In roulette, a bet on a single number has a $1/38$ probability of success and pays 35-to-1. That is, if you bet 1 dollar, your net winnings are -1 with probability $37/38$ and +35 with probability $1/38$. Consider betting on a single number on each of n spins of a roulette wheel. Let $\bar{X}_n$ be your average net winnings per bet.

For each of the values $n = 10$, $n = 100$, $n = 1000$:

1. Compute $E(\bar{X}_n)$
2. Compute $SD(\bar{X}_n)$
3. **Coding required.** Run a simulation to determine if the distribution of $\bar{X}_n$ is approximately Normal
4. **Coding required.** Use simulation to approximate $P(\bar{X}_n > 0)$, the probability that you come out ahead after n bets
5. If $n = 1000$ use the Central Limit Theorem to approximate $P(\bar{X}_{1000} > 0)$, the probability that you come out ahead after 1000 bets.
6. The casino wants to determine how many bets on a single number are needed before they have (at least) a 99% probability of making a profit. (Remember, the casino profits if you lose; that is, if $\bar{X}_n < 0$.) Use the Central Limit Theorem to determine the minimum number of bets (keeping in mind that n must be an integer). You can assume that whatever n is, it's large for the CLT to kick in.

Solution

Let X be the winnings on a single bet. The population mean is

$$\mu = E(X) = (35)(1/38) + (-1)(37/38) = -2/38 \approx -0.05$$

The population variance is

$$\sigma^2 = \text{Var}(X) = E(X^2) - (E(X))^2 = (35)^2(1/38) + (-1)^2(37/38) - (-2/38)^2 = 33.2$$

The population SD is $\sigma = 5.76$ dollars.

1. $E(\bar{X}_n) = \mu = -2/38$ for any n
2. $SD(\bar{X}_n) = \frac{\sigma}{\sqrt{n}} = \frac{5.76}{\sqrt{n}}$, so `r = round(5.7626 / sqrt(c(10, 100, 1000)), 3)` for $n = 10, 100, 1000$. As the sample size increases the sample-to-sample variability of sample means decreases
3. See simulation results in Colab notebook. Distribution does not look anything like Normal for $n = 10$ or $n = 100$. Shape of sample-to-sample distribution of sample means looks approximately Normal for $n = 1000$ (but there is still discrete versus continuous issue).
4. See simulation results in the Colab notebook. Your expected winnings are negative so the game is biased against you. In the long run, you lose, and the casino wins. So you don't want to play in the long run. In the short run, there is a tradeoff: the more games you play the more chances you have to win at least once, and winning at least once can offset your losses. If you just play 10 games, the probability of winning at least once doesn't offset your losses as effectively as when you play 100 games. In the short run, your probability of coming out ahead increases as you play more games, but eventually it starts to decrease with n as you play more and more games. (There is some weird discreteness in terms of number of possibilities that happens so the pattern is a little more complicated.) However, remember that $E(\bar{X}_n)$ is negative for any n . For example, consider $n = 35$. Suppose that every day you go to the casino and play 35 games. Then on about 60% of days you'll end up ahead for that day. However, even though you end up ahead on more days than not, your winnings on the days you end up ahead are not enough to offset your losses on the other days. Over many days of 35 bets every day, you will lose about $35(2/38) = 1.84$ dollars on average per day.
5. Assuming $n = 1000$ is large enough to the CLT to kick, $\bar{X}_n$ has a Normal distribution with mean $-2/38$ and $SD\ 5.76/\sqrt{1000} = r = \text{round}(5.7626 / \text{sqrt}(1000), 3)$. A value of 0 is $(0 - (-2/38))/0.182 = 0.289$ SDs above the mean. From the empirical rule, the probability will be between 0.5 and 0.31. Software (below) shows it's 0.386.
6. Assuming n is large enough to the CLT to kick, $\bar{X}_n$ has a Normal distribution with mean $-2/38$ and $SD\ 5.76/\sqrt{n}$. We want to find n so that $P(\bar{X}_n < 0) = 0.99$. That is, we want 0 to be the 99th percentile. For a Normal distribution, the 99th percentile is 2.33 SDs above the mean. So we want to find n so that $0 = -2/38 + 2.33(5.76/\sqrt{n})$. Solving for n gives $n = (2.33(5.76)/(2/38))^2 = 64878$. (Roulette spins happen about once every minute at a casino. Even if the casino only has a single wheel with a single bet on each spin, the casino would easily clear more than the required number of bets in a day. Your bets are essentially free money for the casino.)

Problem 2

The standard measurement of the alcohol content of drinks is alcohol by volume (ABV), which is given as the volume of ethanol as a percent of the total volume of the drink. In a sample of 67 brands of beer, the mean ABV is 4.61 percent and the standard deviation of ABV is 0.754

percent.

- First, to get some practice, answer the following questions by computing by hand using only the information provided here.
 - **Coding required.** Then, use Python to produce the summaries and analysis needed to answer the questions. Be sure to also include appropriate plots of the [beer data](#).
1. Compute a 95% confidence interval for the appropriate population mean.
 2. Write a clearly worded sentence reporting your confidence interval in context.
 3. Is 4.5% a plausible value of the parameter? Explain briefly.
 4. One of the brands of beer in the sample is O'Doul's, a non-alcoholic beer. The ABV for O'Doul's is 0.4% (it has a bit of alcohol.) Suppose O'Doul's is removed from the data set. Compute the sample mean ABV of the remaining 66 brands.
 5. The sample SD of ABV of the remaining 66 brands is 0.55 percent. Explain intuitively why this value is smaller than the sample SD of all 67 brands.
 6. Compute the 95% confidence interval based on the sample with O'Doul's removed. Compare to the original interval, both in terms of center of the CI and its width. Explain briefly.
 7. Based on the interval based on the sample with O'Doul's removed, is 4.5% a plausible value of the parameter? Explain briefly.
 8. Which of the analyses is more appropriate: with or without O'Doul's? Explain your reasoning.

Solution

1. The SE of $0.754/\sqrt{67} = 0.092$ measures the sample-to-sample variability of sample mean ABV over many samples of 67 brands of beer each. The 95% CI is $4.61 \pm 2 \times 0.092 \Rightarrow 4.61 \pm 0.184 \Rightarrow [4.42, 4.79]$.
2. We estimate with 95% confidence that the population mean ABV for brands of beer is between 4.42 and 4.79 percent.
3. 4.5 percent is in the confidence interval, so yes it is a plausible value of the parameter (if 95% confidence represents our criteria for plausibility.)
4. The original sample mean of 67 values is 4.61, so the sum of the 67 values is $67 \times 4.61 = 308.87$. Removing the outlier leaves 66 values with a sum of $308.87 - 0.4 = 308.47$, so the sample mean of the 66 values is $308.47/66 = 4.67$. The outlier pulled the mean down, so the mean is larger without it.
5. The sample SD of ABV was larger with O'Doul's; the outlier pulled up the average distance from the mean.
6. O'Doul's was bringing down the mean, so removing it increase the sample mean and increases the center of the confidence interval. Removing O'Douls results in a smaller sample SD, and a narrower CI.

7. Technically, no since the value 4.5 percent lies outside the CI. (Though if we really want to assess the plausibility of a particular value like 4.5 we would want to compute a p-value to measure the strength of the evidence; here we're basically just saying that the p-value is less than 0.05.)
8. In general, EXCLUDING OUTLIERS WITHOUT A COMPELLING REASON IS BAD. However, in this case the population we are interested in is brands of beer, probably meaning beer with actual alcohol in it. So it makes sense to only consider a sample of alcoholic beers, and so I think the more appropriate analysis with the non-alcoholic beer excluded.

Problem 3

Do non-smokers have better lungs than smokers? One measure of lung function is forced expiratory volume (FEV), the amount of air (in liters) an individual can exhale in the first second of forceful breath. Larger values of FEV are indicative of better functioning lungs. In a study, FEV was measured for a group of 654 subjects, along with whether they smoked and other variables like age (years). Note: I'm purposely not yet giving you much information about how the sample was collected.

1. **Coding required.** Use Python to summarize the [FEV data](#). Summarize the distribution of FEV both overall and separately for smokers and non-smokers.
2. Suppose you want to fill in the blanks in the following sentence: 95% of FEV values in the population are between [blank] and [blank]. Assuming this sample is representative, provide reasonable values that fill in the blanks, and explain your reasoning. Hint: it is better to give a correct rough estimate than a precise but incorrect answer, so think before trying to compute anything. (Since you don't have much information about the sample, you don't have to clarify yet what "population" is appropriate.)
3. Using appropriate summary statistics, compute by hand a 95% confidence for the appropriate difference in population means.
4. **Coding required.** Use Python to compute the confidence interval and compare to what you computed by hand.
5. Write a clearly worded sentence interpreting the confidence interval in context.
6. Assuming this is a representative sample, are we reasonably confident that one group (smokers or non-smokers) tends to have better FEV? Why? Which group?
7. **Coding required.** You should have observed something surprising in the previous parts. Investigate some of the other variables in the data set and produce a few appropriate plots or summaries that provide an explanation for the surprising result. Write a clearly worded sentence or two summarizing your explanation.

Solution

1. See Colab notebook for results. FEV for smokers tends to be *higher* than for non-smokers (which is the opposite of what you would expect.)
2. This is NOT a confidence interval question. Find the 2.5th and 97.5th percentile of values: between 1.3 and 4.6. But just eyeballing from the histogram, it seems like most values are between about 1 and 5 liters.
3. Observed difference in means $3.28 - 2.56 = 0.72$. The SE is $\sqrt{\frac{0.85^2}{589} + \frac{0.75^2}{65}} = 0.1$. The endpoints of the 95% CI are $0.72 \pm 2 \times 0.1 = 0.72 \pm 0.2$, an interval of $[0.51, 0.91]$.
4. See Colab notebook; results are comparable.
5. We estimate with 95% confidence that the population mean FEV for smokers is between 0.51 and 0.91 liters greater than the population mean FEV for smokers. (We'll clarify "population" below.)
6. Smokers because of the previous part.
7. See Colab notebook for some plots. The 654 subjects in this study were all children! Their ages varied from 3 to 19 years old. This would certainly limit the "population" we would be willing to generalize to.

Regarding why smokers tend to have higher FEV:

- Smokers tend to be older.
- Older kids tend to have larger and better lung capacities.
- In statistical terms, age is a *confounding variable*. (We could use regression to *control* for age.)

Problem 4

(I had much longer instructions written, but I think they weren't helpful. If you have questions about how the applet is working, don't hesitate to ask.)

You are going to use a simulation applet that randomly generates confidence intervals for a population mean to help you understand some ideas. The applet calculates a confidence interval for each set of randomly generated data.

- In the box for "Statistic" select means "Means".
- In the box next to "Method" select "t".
- Start with "Number of intervals" equal to 1 at first to see what is happening, but then change it to lots.

Experiment with different simulations; be sure to change

- Distribution
- Population Mean
- Population SD
- Sample size
- Confidence level

Then write a paragraph or two or some bullet points of your main observations. Based on this applet, what can you say about how confidence intervals work? How does changing the inputs affect the confidence intervals and the Results?

Solution

- Simulated coverage level is generally close to confidence level, unless sample size is small and population is not Normal
- Increasing population SD makes the intervals wider (but you would have no control over this in practice)
- Increasing the sample size make the intervals narrower (that is, more precise estimates for the same level of confidence)
- Increasing the confidence level makes the intervals wider
- Since sample means are unbiased estimators of the population mean, when the population mean changes, the sample means follow it.

Homework 9

See code and results in [Colab notebook](#).

Problem 1

The following table displays the time (measured continuously in minutes) until the first goal was scored in each of 20 professional hockey games. The sample mean is 12.1 minutes, the sample median is 8.9 minutes, and the sample SD is 13.1 minutes.

4.4	14.9	2.8	1.4	1.1	8.1	11.7	3.9	2.4	15.7
8.8	0.6	5.1	10.4	9.1	13.7	27.8	9.0	46.0	44.7

1. You wish to estimate the population *median* time until the first goal is scored. Describe in detail in words how you could use the sample data and simulation to find an appropriate 95% bootstrap percentile confidence interval.
2. **Coding required.** Code and run the simulation from the previous part using the [hockey data](#). Summarize the results; provide a histogram of the bootstrap distribution, the value of the bootstrap SE, and a 95% bootstrap percentile confidence interval.
3. The following summarizes the bootstrap distribution of the sample median based on the sample data and an appropriate simulation.

Min	2.5th percentile	Median	97.5th percentile	Max	Mean	SD
1.25	4.15	8.90	12.70	19.75	8.52	2.10

Using the above information, compute the endpoints of each of the following bootstrap 95% confidence intervals for the population median.

- a. Normal interval
 - b. Percentile interval
4. Write a clearly worded sentence reporting the bootstrap percentile confidence interval from the previous part in context.

Solution

1. The statistic is the sample median.
 - Simulate a sample of size 20 with replacement from the observed sample and compute the sample median for the simulated sample.
 - Repeat many times to get the bootstrap distribution of sample medians over many bootstrap samples
 - The endpoints of a 95% bootstrap percentile interval would be the 2.5th percentile and 97.5th percentile of the simulated sample medians.
2. See Colab notebook.
3. Normal interval: Observed sample median is 8.9, from the original problem set up based on the actual sample of 20 values. Bootstrap SE is 2.10. The endpoints of the 95% CI are $8.9 \pm 2 \times 2.10$ yielding an interval of [4.7, 13.1].

Percentile interval: From the bootstrap distribution: [4.15, 12.70]
4. We estimate with 95% confidence that the population median time until the first goal was scored in professional hockey games is between 4.15 and 12.70 minutes.

Problem 2

Two different machines (A and B) that fill packages of a certain candy are calibrated to a weight of 50 grams. Naturally, the weights of individual packages vary somewhat in the production process, but too much variation is undesirable. A small sample of packages is taken from each machine; the weights (grams) are in the following table.

Machine						Mean	SD
A	47.1	48.7	50.1	50.2	50.5	49.3	1.4
B	48.6	50.5	50.6	51.4	52.0	50.6	1.3

1. Describe in detail in words how you could use the sample data and simulation to find a 95% bootstrap percentile confidence interval for the *ratio of the variances* of packages weights for the two machines.
2. **Coding required.** Write code to implement the procedure from the previous part, and run the simulation to find a 95% bootstrap percentile confidence interval. Include your code and output.
3. Write a clearly worded sentence reporting the confidence interval from the previous part in context.

Solution

1. The statistic is the ratio of sample variances
 - Simulate a sample of size 5 with replacement from the observed sample A and compute the sample variance (square of the sample SD)
 - Simulate a sample of size 5 with replacement from the observed sample B and compute the sample variance (square of the sample SD)
 - Compute the ratio of the two sample variances; this is the result of one repetition
 - Repeat many times to get the bootstrap distribution of the ratio of sample variances over many bootstrap samples
 - The endpoints of a 95% bootstrap percentile interval would be the 2.5th percentile and 97.5th percentile of the simulated ratio of sample variances.
2. See Colab notebook
3. We estimate with 95% that the variance of weights from machine A is between 0.01 and 14.4 times that for machine B (This interval contains 1 so there isn't any evidence that the ratios are different. But because the sample size is so small, the interval is not very precise.)

Problem 3

Continuing Problem 2.

Describe in detail how you could use simulation to approximate the p-value of the permutation test which uses the ratio of variances as the test statistic.

Solution

Note carefully the differences between this simulation and the bootstrap simulation

- Write each of the 10 observed values on a card.
- Shuffle and deal 5 without replacement to represent A, and the other 5 to represent B.
- Compute the variance of the values in the A pile, the variance of values in the B pile, and the ratio of these two variances (A/B)
- Repeat the above steps many times to generate many hypothetical values of the ratio of variances (A/B) assuming the null hypothesis of no difference
- Find the proportion of simulated repetitions for which the simulated ratio of variance is greater than $1.42/1.32$, the observed ratio of variances. This proportion approximates the p-value

Problem 4

Scores are on a certain standardized test (e.g. GRE) are approximately normally distributed with mean 525 and standard deviation 100 (assume that the scale is 200-800). An agency claims that students who take their test preparation class have a higher mean score than the national average. To test the validity of the claim we consider

$$H_0 : \mu = 525 \quad H_a : \mu > 525,$$

where μ is the mean test score of all students who take the preparation class. Suppose that we take a simple random sample of n students who have taken the class, and let $\bar{y}$ be the mean test score for the sample. You can assume that $\sigma = 100$ and just use the usual empirical rule (that is, don't worry about t distributions.)

1. Compute the p -value of the test if $n = 100$ and $\bar{y} = 541.4$.
2. Compute the p -value of the test if $n = 100$ and $\bar{y} = 541.5$.
3. Compute the p -value of the test if $n = 100,000$ and $\bar{y} = 526$.
4. Write a short paragraph explaining why (1) we should not rely on cutoffs like $\alpha = 0.05$ strictly; (2) why the term statistically “significant” is a poor choice of words which should not be used. Use the results from the previous parts and this context to support your explanation.

Solution

The sample mean (whether you call it $\bar{X}$ or $\bar{Y}$) has an approximate Normal distribution with mean μ and SD $\sigma/\sqrt{n} = 100/\sqrt{n}$. So if the null hypothesis is true then $\mu = 525$ and $\bar{Y}$ has a Normal(525, $100/\sqrt{n}$) distribution.

1. If $n = 100$ then $SD(\bar{Y}) = 100/\sqrt{100} = 10$ so $\bar{Y}$ has a Normal(525, 10) distribution if the null hypothesis is true. The observed sample mean $\bar{y} = 541.4$ is $\frac{541.4-525}{10} = 1.64$ SDs above the hypothesized mean of 525. For a Normal distribution, the probability of observing a value more than 1.64 SDs above the mean is 0.0505. That is, the p -value is 0.0505.
2. If $n = 100$ then $SD(\bar{Y}) = 100/\sqrt{100} = 10$ so $\bar{Y}$ has a Normal(525, 10) distribution if the null hypothesis is true. The observed sample mean $\bar{y} = 541.5$ is $\frac{541.5-525}{10} = 1.65$ SDs above the hypothesized mean of 525. For a Normal distribution, the probability of observing a value more than 1.65 SDs above the mean is 0.0495. That is, the p -value is 0.0495.
3. If $n = 100000$ then $SD(\bar{Y}) = 100/\sqrt{100000} = 0.316$ so $\bar{Y}$ has a Normal(525, 0.316) distribution if the null hypothesis is true. The observed sample mean $\bar{y} = 526$ is $\frac{526-525}{0.316} = 3.16$ SDs above the hypothesized mean of 525. For a Normal distribution, the probability

of observing a value more than 3.16 SDs above the mean is 0.0008. That is, the p-value is 0.0008.

4. In the first two parts, in this context 541.5 and 541.5 are essentially the same sample means (scale of the scores is 200-800) so it makes no sense to reject the null hypothesis in one case and not in the other. But that is exactly what would happen if you treated 0.05 strictly. Rather than adhering to a strict threshold like 0.05, it is better to report the p-value so that the strength of evidence can be interpreted. The strength of evidence would be the same in both of the first two parts; in either case there is some but not strong evidence to reject the null hypothesis.

The p-value in part 3 is very small. However, this just means that there is strong evidence to reject the null hypothesis, i.e. to conclude that the mean of the students who take the class is greater than the national average. But the hypothesis test does not tell you anything else about what the true mean actually is; it could be just slightly greater than the national average, in which case there is no practical difference in the scores for students who take the class versus the national average. In order to get a better idea of the “plausible values” for the true mean based on the data, it is better to report a confidence interval. But in this case the observed sample mean is 526 so there is no practical difference between the observed mean and 525, even though the p-value is small. Especially when the sample size is very large, even a small and meaningless observed difference can result in a small p-value.

Problem 5

Psychologists have shown that we are often able to “chunk” information which allows us to remember more information (typically about seven chunks). A study was performed in previous statistics classes to investigate this idea. Students were given 20 seconds to memorize a sequence of 30 letters. After 20 seconds, everyone was asked to write down, in order, as many of the letters as they could. Every student saw the same sequence of 20 letters, JFKCIAF-BIUSASATGPABFFLOLNBACPR, but students were randomly assigned to see the letters presented (chunked) in one of two different ways

- a “meaningful” grouping: JFK-CIA-FBI-USA-SAT-GPA-BFF-LOL-NBA-CPR
- a “not meaningful” grouping: JFKC-IAF-BIU-SASA-TGP-ABF-FLO-LN-BAC-PR

Each person’s score was the number of correct letters in a row before the first mistake. (For example, JFKCIABIUSAT yields a score of 6 because of missing the F.) We want to know whether those given the meaningful “JFK” sequence (with the letters grouped into familiar acronyms) would tend to remember more letters in sequence, on average, than those given the not meaningful “JFKC” sequence.

1. **Coding required.** Use Python to summarize the [letters data](#).

2. State the null and alternative hypothesis in words and symbols
3. Explain in detail how, in principle, you would use index cards to conduct an appropriate simulation and use the simulation results to compute the p-value.
4. Compute by hand the t-statistic and the p-value.
5. **Coding required.** Use Python to conduct the hypothesis test, and compare the results to what you computed by hand.
6. Write a clearly worded sentence reporting the conclusion of the hypothesis test in context.
7. Compute by hand an appropriate 95% confidence interval.
8. **Coding required.** Use Python to compute the confidence interval, and compare the results to what you computed by hand.
9. Write a clearly worded sentence reporting the conclusion of the confidence interval in context.

Solution

1. See Colab notebook. The observed difference in sample means is $\bar{x}_M - \bar{x}_N = 16.25 - 12.71 = 3.54$
2. $H_0 : \mu_M - \mu_N = 0$, there is no difference in treatment mean letter scores between the meaningful and not meaningful group; $H_a : \mu_M - \mu_N > 0$, the treatment mean letter score is greater for the meaningful group than for the not meaningful group
3. We would use 97 cards
 - On each card write an observed value of letter score
 - Shuffle the cards well and deal them without replacement into 2 piles
 - Deal 52 to represent meaningful
 - Deal 47 to represent not meaningful
 - Compute the hypothetical value of $\bar{x}_M - \bar{x}_N$ for this shuffle
 - Repeat the above process many times (thousands) to simulate the null distribution of $\bar{x}_M - \bar{x}_N$
 - Count the number of repetitions for which $\bar{x}_M - \bar{x}_N$ statistic is greater than the observed value 3.54 and divide by the total number of repetitions simulated to approximate the p-value.
4. The SE is $\sqrt{\frac{7.92^2}{52} + \frac{4.28^2}{45}} = 1.27$. The observed difference in means, 3.54, is

$$\frac{\text{observed} - \text{hypothesized}}{\text{SE}} = \frac{3.54 - 0}{1.27} = 2.78$$

SDs above the mean. Use the empirical rule; p-value is about 0.007.

5. See Colab notebook; results are similar.
6. With a p-value of 0.007 there is moderate evidence to reject the null hypothesis and conclude that the treatment mean letter score for people who would see the meaningful letter grouping is greater than the treatment mean letter score for people who would see the not meaningful letter grouping.

7. $3.54 \pm 2 \times 1.27$ for an interval of [1.0, 6.1]
8. See Colab notebook.
9. We estimate with 95% confidence that the treatment mean letter score for the people who would see the meaningful letter grouping is between 1.0 and 6.1 letters greater than the treatment mean letter score for people who would see the not meaningful letter grouping.

Problem 6

The data in this exercise comes from the study: [Singh R, Meier T, Kuplicki R, Savitz J, et al., “Relationship of Collegiate Football Experience and Concussion With Hippocampal Volume and Cognitive Outcome,” JAMA, 311\(18\), 2014.](#)

The study included 3 groups, with 25 cases in each group. The control group consisted of healthy individuals with no history of brain trauma who were comparable to the other groups in age, sex, and education. The second group consisted of NCAA Division 1 college football players with no history of concussion, while the third group consisted of NCAA Division 1 college football players with a history of concussion. High resolution MRI was used to collect brain hippocampus volume (microliters).

1. **Coding required.** Use Python to summarize the [brain data](#).
2. Compute by hand the ANOVA F statistic.
3. Describe in full detail how (in principle) you could use index cards to simulate the null distribution of the ANOVA F statistic and how you would use the simulation results to find the p-value. (You don’t have to compute the p-value; just explain how you could find it after you performed the simulation.)
4. Optional: use this [applet to perform the permutation test and approximate the p-value](#).
5. **Coding required.** Use Python to conduct the hypothesis test, and compare the results to what you computed by hand.
6. Write a clearly worded sentence containing the conclusion of the hypothesis test in the context of the problem.
7. **Coding required.** Use Python to compute Tukey pairwise 95% confidence intervals.
8. Which of the confidence intervals contain 0, and which do not? Explain what this means.
9. Write a clearly worded sentence interpreting each of the three confidence intervals in context.

Solution

1. See Colab notebook.

2. The mean of the group variances is $\frac{1074^2+779.7^2+593.4^2}{3} = 704510$. The variance of the group means is $941.8^2 = 886987$. The F statistic is

$$\frac{\text{group size} \times \text{variance of group means}}{\text{mean of group variances}} = \frac{25 \times 886987}{704510} = 31.5$$

3. We would use 75 cards

- On each card write an observed value of hippocampus volume
- Shuffle the cards well and deal them without replacement into 3 piles
- Deal 25 to represent control
- Deal 25 to represent football, no concussion
- Deal 25 to represent football with concussion history
- Treat the shuffled cards as if they were the real data For each pile of 25 find the mean and SD of hippocampus volumes and compute the hypothetical value of the F statistic for the shuffle
- Repeat the above process many times (thousands) to simulate the null distribution of the F statistic
- Count the number of repetitions for which the F statistic is greater than the observed 31.5 and divide by the total number of repetitions simulated to approximate the p-value.

4. Optional: use this [applet to perform the permutation test and approximate the p-value](#).

5. See Colab notebook; the F statistic is 31.5 and the p-value is essentially 0.

6. With a p-value < 0.0001 there is very strong evidence to reject the null hypothesis and conclude that there is an association between hippocampus volume and football/concussion history. In particular, we can conclude that the population mean hippocampus volume is not the same for all 3 concussions histories.

7. See Colab notebook.

8. None of the confidence intervals contains 0. This means there is evidence that the mean hippocampus volumes for all three groups are different from one another.

9. We estimate with simultaneous 95% confidence that

- Mean hippocampus volume is between 670 and 1616 microliters greater for those with no football/concussion history than for those who have football experience but no concussion history
- Mean hippocampus volume is between 1394 and 2341 microliters greater for those with no football/concussion history than for those who have football experience and a history of concussion
- Mean hippocampus volume is between 251 and 1198 microliters greater for those who have football experience but no concussion history than for those who have football experience and a history of concussion

Summary of Some Common Distributions

Name and parameters	Possible values	pmf/pdf	cdf	Mean	SD
Discrete					
Poisson(μ)	$x = 0, 1, 2, \dots$	$e^{-\mu} \frac{\mu^x}{x!}$	No closed form	μ	$\sqrt{\mu}$
Binomial(n, p)	$x = 0, 1, 2, \dots, n$	$\binom{n}{x} p^x (1-p)^{n-x}$	No closed form	np	$\sqrt{np(1-p)}$
Geometric(p)	$x = 1, 2, 3, \dots$	$p(1-p)^{x-1}$	$1 - (1-p)^x$	$\frac{1}{p}$	$\sqrt{\frac{1-p}{p^2}}$
Continuous					
Uniform(a, b)	$a < x < b$	$\frac{1}{b-a}$	$\frac{x-a}{b-a}$	$\frac{a+b}{2}$	$\frac{b-a}{\sqrt{12}}$
Exponential(λ)	$x > 0$	$\lambda e^{-\lambda x}$	$1 - e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda}$
Normal(μ, σ)	$-\infty < x < \infty$	$\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$	No closed form (use tables)	μ	σ

References